

# Dummit and Foote Notes

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# Chapter 1

## Group Theory

### 1.1 Introduction to Groups

#### 1.1.1 Introduction

**Definition 1.1.1.** A group is a set  $G$  and a binary operation  $\cdot$  on  $G$  such that

- (Closure)  $a \cdot b \in G$  for all  $a, b \in G$ .
- (Associativity)  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$  for all  $a, b, c \in G$ .
- (Identity) There is some  $e \in G$  so that  $e \cdot a = a \cdot e = a$  for all  $a \in G$ .
- (Inverse) For all  $a \in G$ , there is some  $a^{-1} \in G$  so that  $a^{-1} \cdot a = a \cdot a^{-1} = e$ .

**Definition 1.1.2.** A group  $G$  is said to be abelian if  $a \cdot b = b \cdot a$  for all  $a, b \in G$ .

*Example 1.1.3.* The integers  $\mathbb{Z}$ , the rationals  $\mathbb{Q}$ , the reals  $\mathbb{R}$ , and the complexes  $\mathbb{C}$  all form groups under addition.

*Example 1.1.4.* The rationals without 0 form a group under multiplication. The integers without 0 do not.

*Example 1.1.5.* The set  $\mathbb{Z}/n\mathbb{Z} = \{\bar{0} := \{a : a \equiv 0 \pmod{n}\}, \bar{1}, \dots, \overline{n-1}\}$  forms a group under the set  $\bar{a} \cdot \bar{b} = \overline{a+b}$ .

*Example 1.1.6.* The set  $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} : \gcd(a, n) = 1\}$  forms a group under  $\bar{a} \cdot \bar{b} = \overline{ab}$ .

*Example 1.1.7.* If  $A$  is a group under  $*$ , and  $B$  is a group under  $\diamond$ , the their Cartesian Product  $A \times B$  forms a group under  $(a_1, b_1) \cdot (a_2, b_2) = (a_1 * a_2, b_1 \diamond b_2)$ .

**Notation.** If  $G$  is a group and  $a, b \in G$ , we will write  $ab$  for  $a \cdot b$ .

**Proposition 1.1.8.** Let  $G$  be a group. Then,

- The identity is unique.

- For each  $a \in G$ ,  $a^{-1}$  is unique.
- $(a^{-1})^{-1} = a$  for all  $a \in G$ .
- $(ab)^{-1} = b^{-1}a^{-1}$  for all  $a, b \in G$ .

*Proof.*

- Let  $h$  and  $k$  both be identities. Then  $h = hk = k$ .
- Let  $h$  and  $k$  both be inverses of  $a$ . Then  $h = h(ak) = (ha)k = k$ .
- Note that  $aa^{-1} = a^{-1}a = e$ , so (because inverses are unique),  $a = (a^{-1})^{-1}$ .
- Note that  $(ab)(b^{-1}a^{-1}) = (b^{-1}a^{-1})(ab) = e$ , so  $(ab)^{-1} = b^{-1}a^{-1}$ .

■

**Notation.** We will write 1 for the identity, and  $x^n$  for  $\underbrace{xx \cdots x}_{n \text{ times}}$  (for positive  $n$ ). We also write  $x^0 = 1$ , and  $x^{-n} = \underbrace{x^{-1}x^{-1} \cdots x^{-1}}_{n \text{ times}}$  (for positive  $n$ ).

**Proposition 1.1.9.** Let  $G$  be a group, and  $a, b, u, v \in G$ . Then  $au = av$ , if and only if  $u = v$ , and  $ub = vb$ , if and only if  $u = v$ .

*Proof.* Multiply the first equation on the left by  $a^{-1}$ , and the second equation on the right by  $b^{-1}$ . Undo these to get the other direction. ■

**Proposition 1.1.10.** Let  $G$  be a group and  $a, b \in G$ . Then  $ab = 1$  if and only if  $b = a^{-1}$  (and similarly for  $ba$ ), and  $ab = a$  if and only if  $b = 1$  (and similarly for  $ba$ ).

*Proof.* The reverse directions are both clear. For the other direction, multiply on the left (right for  $ba$ ) by  $a^{-1}$  in both cases. ■

**Definition 1.1.11.** Let  $G$  be a group and  $x \in G$ . The order of  $x$  is the smallest positive integer  $n$  so that  $x^n = 1$ . We write  $|x|$  for this number. If no such number exists, we say that  $|x| = \infty$ .

## 1.1.2 Examples

**Definition 1.1.12.**  $D_{2n}$  is the group of symmetries of a regular  $n$ -gon. It forms a group under composition. We call this the dihedral group of order  $2n$ .

**Proposition 1.1.13.**  $|D_{2n}| = 2n$ .

*Proof.* Choose a vertex. This has  $n$  places to go. The vertex next to it has 2 choices. ■

**Notation.** Fix an  $n$ -gon centered at the origin, and label the vertices  $1, 2, \dots, n$ . Place vertex 1 at  $(1, 0)$ . We write  $r$  for the rotation by  $2\pi/n$  about the origin, and  $s$  for the reflection over the  $x$ -axis. We write 1 for the element of doing nothing. Note that  $|r| = n$ ,  $|s| = 2$ ,  $s \neq r^i$  for any  $i$ , and that  $rs = sr^{-1}$ . Note further that  $r^i s = sr^{-i}$  for any  $i$ , meaning that any element of  $D_{2n}$  can be written as  $s^i r^j$  where  $0 \leq i \leq 1$  and  $0 \leq j \leq n - 1$ .

**Notation.** We write  $G = \langle S \rangle$  if  $S \subseteq G$  with the property that every element in  $G$  can be written as a finite product of elements in  $S$  and their inverses. We say that  $G$  is generated by  $S$ , or that  $S$  generates  $G$ , and call the elements of  $S$  the generators of  $G$ .

**Definition 1.1.14.** If  $G$  is a group, then any equations that its generators satisfy are called relations.

*Example 1.1.15.* The group  $D_{2n}$  is generated by the set  $\{r, s\}$ , with relations  $r^n = s^2 = 1$  and  $rs = sr^{-1}$ . Note that all other relations can be deduced from these 3.

**Notation.** If  $G$  is generated by some  $S \subseteq G$ , and there is some collection of relations  $R_1, R_2, \dots, R_m$  (which are equations in elements of  $S \cup \{1\}$ ) such that any relation among the elements of  $S$  can be deduced from these, we call these generators and relations a presentation of  $G$  and write

$$G = \langle S | R_1, R_2, \dots, R_m \rangle.$$

*Example 1.1.16.*

$$D_{2n} = \langle r, s | r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

**Definition 1.1.17.** Let  $\Omega$  be a nonempty set. We write  $S_\Omega$  for the set of all bijections  $f : \Omega \rightarrow \Omega$ . This forms a group under function composition. We call this the symmetric group on the set  $\Omega$ . In the case where  $\Omega = \{1, 2, \dots, n\}$ , we write  $S_n$  for  $S_\Omega$ .

**Proposition 1.1.18.**  $|S_n| = n!$ .

**Notation.** Let  $a_1, a_2, \dots, a_m$  be integers. The cycle  $(a_1 a_2 \cdots a_m)$  is the permutation with  $a_i \mapsto a_{i+1}$  for  $1 \leq i \leq m-1$  and  $a_m \mapsto a_1$ . In general, for  $\sigma \in S_n$ , the numbers from 1 to  $n$  can be rearranged and grouped into cycles of the form

$$(a_1 a_2 \cdots a_m)(a_{m+1} a_{m+2} \cdots a_{m_1}) \cdots (a_{m_{k-1}+1} a_{m_{k-1}+2} \cdots a_{m_k}),$$

where  $\sigma(a_{m_i+j}) = a_{m_i+j+1}$ , unless  $j = 0$ , in which case it is mapped to  $a_{m_{i-1}+1}$ . This is called the cycle decomposition of  $\sigma$ .

**Definition 1.1.19.** The length of a cycle is the number of integers that appear in it. A cycle of length  $t$  is called a  $t$ -cycle. Two cycles are disjoint if they have no numbers in common. We typically do not write 1-cycles in the cycle decomposition of  $\sigma \in S_n$ .

**Proposition 1.1.20.** For  $n \geq 3$ ,  $S_n$  is not abelian.

*Proof.* Consider  $\sigma \in S_n$  with cycle decomposition (12), and  $\tau \in S_n$  with cycle decomposition (13). Then  $\sigma \circ \tau$  has cycle decomposition (132), while  $\tau \circ \sigma$  has cycle decomposition (123). ■

**Definition 1.1.21.** Let  $F$  be a field (defined in more detail later, for now, imagine being able to add, subtract, multiply, and divide by nonzero elements). We define  $\text{GL}_n(F)$  to be the set of all  $n \times n$  matrices with nonzero determinant and coefficients in  $F$ . This forms a group under matrix multiplication (as the determinant is nonzero), and is called the general linear group of degree  $n$ .

**Definition 1.1.22.** The quaternion group  $Q_8$  is defined by

$$Q_8 = \{1, -1, i, -i, j, -j, k, -k\},$$

with operation defined as

$$1 \cdot a = a \cdot 1 = a \quad (-1) \cdot a = a \cdot (-1) = -a \quad \forall a \in Q_8$$

$$(-1) \cdot (-1) = 1$$

$$i \cdot i = j \cdot j = k \cdot k = i \cdot j \cdot k = -1.$$

All operations can be deduced from these. Note that  $Q_8$  is nonabelian.

### 1.1.3 Homomorphisms and Isomorphisms

**Definition 1.1.23.** Let  $G$  be a group under  $*$ , and  $H$  be a group under  $\diamond$ . A homomorphism from  $G$  to  $H$  is a map  $\phi : G \rightarrow H$  such that, for all  $x, y \in G$ ,

$$\phi(x * y) = \phi(x) \diamond \phi(y).$$

*Remark 1.1.24.* If the group operations are not explicitly written, the condition becomes

$$\phi(xy) = \phi(x)\phi(y).$$

However, it is important to remember that the product on the left is computed in  $G$ , while the one on the right is computed in  $H$ . Intuitively, a homomorphism respects the group structures of its domain and codomain.

*Example 1.1.25.* Let  $G$  and  $H$  be groups. The function  $\phi : G \rightarrow H$  defined by  $\phi(g) = 1_H$  for all  $g$  is a homomorphism. This is called the trivial homomorphism.

**Definition 1.1.26.** Let  $\phi : G \rightarrow H$  be a homomorphism. Then  $\phi$  is said to be an isomorphism if it is a bijection. In this case, we say that  $G$  is isomorphic to  $H$ , and write  $G \cong H$ .

*Remark 1.1.27.* Intuitively,  $G$  and  $H$  are isomorphic if they are the same group with different names for the elements and operations. Thus, any property of  $G$  depending only on its group structure is also true in  $H$ .

**Proposition 1.1.28.** *Isomorphism forms an equivalence relation.*

*Proof.* Let  $G_1, G_2$ , and  $G_3$  be groups. Observe that  $\text{id}_{G_1}$  is an isomorphism from  $G_1$  to itself, so  $G_1 \cong G_1$ . Now, if there is an isomorphism  $\phi : G_1 \rightarrow G_2$ , we may use  $\phi^{-1} : G_2 \rightarrow G_1$  as an isomorphism the other way. Finally, if  $\phi : G_1 \rightarrow G_2$  and  $\varphi : G_2 \rightarrow G_3$  are isomorphisms, then  $\varphi \circ \phi : G_1 \rightarrow G_3$  is an isomorphism. ■

*Example 1.1.29.* Let  $\Omega$  and  $\Delta$  be sets with  $|\Omega| = |\Delta|$ . Then  $S_\Omega \cong S_\Delta$ . To see this, note we must have some bijection  $\rho : \Omega \rightarrow \Delta$ . Let  $\sigma \in S_\Omega$ . Consider the map  $\theta : S_\Omega \rightarrow S_\Delta$  defined by  $\sigma \mapsto \rho \circ \sigma \circ \rho^{-1}$ . This is a bijection, as it clearly has an inverse, and is a homomorphism as  $\theta(\sigma \circ \tau) = \rho \circ \sigma \circ \tau \circ \rho^{-1} = \rho \circ \sigma \circ \rho^{-1} \circ \rho \circ \tau \circ \rho^{-1} = \theta(\sigma) \circ \theta(\tau)$ . Alternatively suppose that  $S_\Omega \cong S_\Delta$ . Then,  $|S_\Omega| = |S_\Delta|$ , so  $|\Omega|! = |\Delta|!$ . Thus,  $|\Omega| = |\Delta|$ . Thus, if  $|\Omega| = n$ ,  $S_\Omega \cong S_n$ .



**Proposition 1.1.30.** *Let  $G$  and  $H$  be groups, and  $\phi : G \rightarrow H$  a homomorphism. Then  $\phi(1_G) = 1_H$ .*

*Proof.* We have that  $\phi(1_G) = \phi(1_G 1_G) = \phi(1_G)\phi(1_G)$ . Thus,  $\phi(1_G) = 1_H$ . ■

**Proposition 1.1.31.** *Let  $G$  and  $H$  be groups, and  $\phi : G \rightarrow H$  a homomorphism. Then,  $\phi(x^n) = \phi(x)^n$  for all integers  $n$ .*

*Proof.* This is clear for nonnegative integers. Now, note that  $\phi(x^{-1})\phi(x) = \phi(x^{-1}x) = \phi(1_G) = 1_H$ . It is now obviously clear for negative  $n$  as well. ■

### 1.1.4 Group Actions

**Definition 1.1.32.** A group action of a group  $G$  on a set  $A$  is a map  $\cdot : G \times A \rightarrow A$  so that

- $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$  for all  $g_1, g_2 \in G$  and  $a \in A$ .
- $1 \cdot a = a$  for all  $a \in A$ .

*Remark 1.1.33.* This is actually a left action, as the group elements appear on the left of the set elements. There is similarly the notion of a right action. Note that the map from left actions to right actions given by  $g \cdot a \mapsto a \cdot g$  does not work when  $G$  is nonabelian, as  $g_1 \cdot (g_2 \cdot a)$  maps to  $(a \cdot g_2) \cdot g_1$ , while  $(g_1 g_2) \cdot a$  maps to  $a \cdot (g_1 g_2)$ .

**Notation.** We will write  $ga$  rather than  $g \cdot a$ .

*Remark 1.1.34.* Note that, for each  $g \in G$ , there is a map  $\sigma_g : A \rightarrow A$  defined by  $\sigma_g(a) = g \cdot a$ .

**Proposition 1.1.35.** *For each  $g \in G$ ,  $\sigma_g$  is a permutation of  $A$ . Furthermore, the map from  $G$  to  $S_A$  defined by  $g \mapsto \sigma_g$  is a homomorphism.*

*Proof.* To prove that  $\sigma_g$  is a permutation, we show that it has a 2-sided inverse. Note that

$$(\sigma_g \circ \sigma_{g^{-1}})(a) = g \cdot (g^{-1} \cdot a) = (gg^{-1}) \cdot a = 1 \cdot a = a,$$

and that we may interchange  $g$  and  $g^{-1}$  to once again get the identity map. Therefore,  $\sigma_{g^{-1}}$  is a 2-sided inverse of  $\sigma_g$ , so  $\sigma_g$  is a permutation. Now, let  $\phi(g) = \sigma_g$ . Then,

$$\phi(g_1 g_2)(a) = (g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = (\phi(g_1) \circ \phi(g_2))(a),$$

so it is a homomorphism. ■

*Remark 1.1.36.* Note that for each homomorphism  $\phi : G \rightarrow S_A$ , the map from  $G \times A$  to  $A$  defined by

$$g \cdot a = \phi(g)(a)$$

is a group action of  $G$  on  $A$ . Thus, there is a bijection between the homomorphisms from  $G$  to  $S_A$ , and the actions of  $G$  on  $A$ .

*Example 1.1.37.* Let  $N$  be the set of vertices of a regular  $n$ -gon. Then,  $D_{2n}$  acts on  $N$  (in the obvious way).

*Example 1.1.38.* Any group  $G$  acts on itself by  $g \cdot h = gh$ .

*Example 1.1.39.* Let  $G$  be a group and  $A$  a set. Then the map defined by  $ga = a$  for all  $g \in G$ ,  $a \in A$  is a group action. This is called the trivial action, and  $G$  is said to act trivially on  $A$ .

## 1.2 Subgroups

### 1.2.1 Definition

**Definition 1.2.1.** Let  $G$  be a group. A subset  $H$  of  $G$  is a subgroup of  $G$  if  $H$  is nonempty, and whenever  $x, y \in H$ ,  $xy \in H$  and  $x^{-1} \in H$ . In this case, we write  $H \leq G$ .

*Remark 1.2.2.* Subgroups form groups by themselves. Closure and inverse are in the definition, and associativity is inherited from  $G$ . To see identity, note that if  $x \in H$ , then  $x^{-1} \in H$ , so  $xx^{-1} = 1 \in H$ . Whenever we talk about subgroups, we will always mean that the operation on  $H$  is the operation on  $G$  restricted to  $H$ . Thus, for example,  $(\mathbb{Z}/n\mathbb{Z})^\times \not\leq \mathbb{Z}/n\mathbb{Z}$ , even though  $(\mathbb{Z}/n\mathbb{Z})^\times \subset \mathbb{Z}/n\mathbb{Z}$ , and  $(\mathbb{Z}/n\mathbb{Z})^\times$  forms a group.

*Example 1.2.3.*  $\mathbb{Z} \leq \mathbb{Q}$  under addition.

*Example 1.2.4.* Let  $G$  be a group. Then the groups  $G$  and  $\{1\}$  are subgroups of  $G$ . The latter is called the trivial subgroup.

**Proposition 1.2.5** (The Subgroup Criterion). *Let  $G$  be a group, and  $H \subseteq G$ . Then,  $H$  is a subgroup of  $G$  if and only if  $H \neq \emptyset$ , and  $xy^{-1} \in H$  for all  $x, y \in H$ . Furthermore, if  $H$  is finite, it suffices to check that  $H$  is nonempty and closed under multiplication.*

*Proof.* If  $H \leq G$ , then this is clearly true. Now assume that this is true. Letting  $x = y$  means that  $1 \in H$ , and letting  $x = 1$  means that  $y^{-1} \in H$  for all  $y \in H$ . Finally, let  $y = z^{-1}$  for some  $x \in H$ . Then,  $xz \in H$  for all  $x, z \in H$ . Now, if  $H$  is finite and closed under multiplication, then there are finitely many distinct elements in the set  $\{x, x^2, x^3, \dots\}$ . Thus,  $x^a = x^b$  for some integers  $a, b$  (where we may assume  $b > a$ ). If  $n = b - a$ , then  $x^n = 1$ , so  $x^{n-1} = x^{-1} \in H$ . ■

### 1.2.2 Centralizers and Normalizers

**Definition 1.2.6.** Let  $G$  be a group, and  $A \subseteq G$  nonempty. We define the centralizer of  $A$  in  $G$  to be the set  $C_G(A) = \{g \in G : gag^{-1} = a \text{ for all } a \in A\}$ . Note that this is the set of elements of  $G$  that commute with every element of  $A$ .

**Proposition 1.2.7.** *Let  $G$  be a group, and  $A \subseteq G$  nonempty. Then,  $C_G(A) \leq G$ .*

*Proof.* Note that it is nonempty, as  $1 \in C_G(A)$ . Note also that if  $xax^{-1} = a$  for all  $a \in A$ , then  $a = x^{-1}ax = x^{-1}a(x^{-1})^{-1}$  for all  $a \in A$ , so  $x^{-1} \in C_G(A)$ . Lastly, if  $x, y \in C_G(A)$ , notice that  $a = xax^{-1} = xyay^{-1}x^{-1} = (xy)a(xy)^{-1}$ , so  $xy \in C_G(A)$ . ■

**Notation.** In the case when  $A = \{a\}$ , we will write  $C_G(a)$  rather than  $C_G(\{a\})$ .

**Definition 1.2.8.** Let  $G$  be a group. The center of  $G$  is the set  $Z(G) = \{g \in G : gh = hg \text{ for all } h \in G\} = C_G(G)$ . Note that  $Z(G) \leq G$ .

**Definition 1.2.9.** Let  $G$  be a group, and  $A \subseteq G$ . The normalizer of  $A$  in  $G$  is the set  $N_G(A) = \{g \in G : gag^{-1} \in A \text{ for all } a \in A\}$ .

**Proposition 1.2.10.** *Let  $G$  be a group, and  $A \subseteq G$ . Then,  $N_G(A) \leq G$ .*

*Proof.* Nearly identical to that of  $C_G(A)$ , and so we omit it. ■

*Remark 1.2.11.* Notice that  $C_G(A) \leq N_G(A)$ .

*Remark 1.2.12.* From here on, we will refrain from things of the form “let  $G$  be a group”, when it is clear from context.

### 1.2.3 Stabilizers and Kernels

**Definition 1.2.13.** Let  $G$  act on a set  $S$ . The set  $G_s = \{g \in G : gs = s\}$  is called the stabilizer of  $s$  in  $G$ .

**Proposition 1.2.14.**  $G_s \leq G$ .

*Proof.* Note that  $1 \in G_s$ . Note also that  $s = 1s = (y^{-1}y)s = y^{-1}(ys) = y^{-1}s$  whenever  $y \in G_s$ , so  $y^{-1} \in G_s$  as well. Finally, note that if  $x, y \in G_s$ , then  $(xy)s = x(ys) = xs = s$ , so  $xy \in G_s$ . ■

**Definition 1.2.15.** The kernel of an action of  $G$  on  $S$  is the set  $\{g \in G : gs = s \text{ for all } s \in S\}$ .

**Proposition 1.2.16.** The kernel is a subgroup of  $G$ .

*Proof.* It is the intersection of the stabilizers. ■

*Remark 1.2.17.* The fact that centralizers and normalizers are subgroups is a special case of the fact that stabilizers and kernels are subgroups. To see this, let  $S = \mathcal{P}(G)$  be the set of all subsets of  $G$ . Let  $G$  act on  $S$  by conjugation. In other words,  $g : B \mapsto gBg^{-1}$ . Then the stabilizer of  $A \in S$  in  $G$  is exactly  $N_G(A)$ , so  $N_G(A) \leq G$ . Now, let  $N_G(A)$  act on  $A$  with  $g : a \mapsto gag^{-1}$ . Then the kernel of this action is exactly  $C_G(A)$ , meaning that  $C_G(A) \leq N_G(A)$ , and so  $C_G(A) \leq G$ .

**Definition 1.2.18.** An action is said to be faithful if its kernel is the identity.

### 1.2.4 Cyclic Groups and Subgroups

**Definition 1.2.19.** A group  $H$  is said to be cyclic if there is some  $x \in H$  such that  $H = \{x^n : n \in \mathbb{Z}\}$ . We say that  $H$  is generated by  $x$ , and write  $H = \langle x \rangle$ .

*Remark 1.2.20.* A group can have multiple generators. For example,  $\langle x \rangle = \langle x^{-1} \rangle$ .

*Example 1.2.21.*  $\mathbb{Z} = \langle 1 \rangle$  (the integer 1).

**Proposition 1.2.22.** If  $H = \langle x \rangle$ , then  $|H| = |x|$ . More specifically,

- If  $|H| = n < \infty$ , then  $|x| = n$ , and  $1, x, x^2, \dots, x^{n-1}$  are all the distinct elements of  $H$ .
- If  $|H| = \infty$ , then  $x^n \neq 1$  for all  $n \neq 0$ , and  $x^a \neq x^b$  for all  $a \neq b \in \mathbb{Z}$ .

*Proof.* Let  $|x| = n < \infty$ . Then, the elements  $1, x, x^2, \dots, x^{n-1}$  are distinct. Now, if  $x^t$  is any power of  $x$ , we may reduce it mod  $n$ , so that it is an element of  $\{1, x, x^2, \dots, x^{n-1}\}$ , so these are all the elements. Now, suppose that  $|x| = \infty$ . Then, if  $x^a = x^b$  (where we may assume  $b > a$ ),  $x^{b-a} = 1$ , a contradiction. Distinct powers of  $x$  are distinct elements of  $H$ , so  $|H| = \infty$ . ■

**Proposition 1.2.23.** *Let  $G$  be a group with  $x \in G$ . If  $x^n = x^m = 1$  for some  $m, n \in \mathbb{Z}$ , then  $x^d = 1$ , where  $d = \gcd(m, n)$ . In particular, if  $x^m = 1$  for some  $m \in \mathbb{Z}$ , then  $|x|$  divides  $m$ .*

*Proof.* There are integers  $r$  and  $s$  so that  $d = mr + ns$ . Thus,

$$x^d = x^{mr+ns} = x^{mr}x^{ns} = (x^m)^r(x^n)^s = 1^r1^s = 1.$$

Now, suppose that  $x^m = 1$ , and let  $n = |x|$ . If  $m = 0$ , this is obviously true, so assume  $m \neq 0$ . Note that  $n \leq m < \infty$ . Let  $d = \gcd(m, n)$ . Then,  $x^d = 1$ . Since  $0 < d \leq n$ , and  $n$  is the smallest power of  $x$  that gives 1,  $d = n$ , so  $n|m$ . ■

**Theorem 1.2.24.** *Any two groups of the same order are isomorphic. In particular,*

- *If  $n \in \mathbb{Z}^+$ , and  $\langle x \rangle$  and  $\langle y \rangle$  are both cyclic groups of order  $n$ , then the map  $\phi : \langle x \rangle \rightarrow \langle y \rangle$  defined by  $x^k \mapsto y^k$  is well defined and an isomorphism.*
- *If  $\langle x \rangle$  is an infinite cyclic group, then the map  $\phi : \mathbb{Z} \rightarrow \langle x \rangle$  defined by  $k \mapsto x^k$  is well defined and an isomorphism.*

*Proof.* Suppose that  $\langle x \rangle$  and  $\langle y \rangle$  are both cyclic groups of order  $n$ . We show that if  $x^r = x^s$ , then  $\phi(x^r) = \phi(y^s)$ . If  $x^r = x^s$ , then  $n|r - s$ . Thus,  $\phi(x^r) = \phi(x^{tn+s}) = y^{tn+s} = y^{tn}y^s = y^s = \phi(x^s)$ . Thus,  $\phi$  is well defined. It is clear that  $\phi$  is a homomorphism. Note that  $\phi$  is surjective, as any  $y^k$  is mapped to by  $x^k$ . Because both groups have the same finite order, it must also be injective, and therefore bijective. Thus,  $\phi$  is an isomorphism. Now suppose that  $\langle x \rangle$  is an infinite cyclic group. The map  $\phi$  is well defined, and clearly a homomorphism. Since  $x^a \neq x^b$  for distinct  $a, b$ ,  $\phi$  is injective. By the definition of a cyclic group,  $\phi$  is also surjective. Thus,  $\phi$  is bijective and therefore an isomorphism. ■

**Notation.** For each  $n \in \mathbb{Z}^+$ , let  $Z_n$  be the cyclic group of order  $n$  (written multiplicatively). Note that  $Z_n \cong \mathbb{Z}/n\mathbb{Z}$ .

**Proposition 1.2.25.** *Let  $G$  be a group,  $x \in G$ , and  $a \in \mathbb{Z} \setminus \{0\}$ . Then,*

- *If  $|x| = \infty$ , then  $|x^a| = \infty$ .*
- *If  $|x| = n < \infty$ , then  $|x^a| = \frac{n}{\gcd(a, n)}$ .*

*Proof.* Suppose that  $|x| = \infty$ , but  $|x^a| = m < \infty$ . Then,  $x^{am} = (x^a)^m = 1 = 1^{-1} = x^{-am}$ . One of  $am$  and  $-am$  must be positive, as  $a, m \neq 0$ . Thus,  $|x|$  is finite, which is a contradiction. Now, let

$$y = x^a \quad d = \gcd(a, n) \quad n = db \quad a = dc.$$

Note that  $\gcd(b, c) = 1$ . Note that

$$y^b = (x^a)^b = x^{acb} = (x^{db})^c = (x^n)^c = 1,$$

so  $k = |y|$  divides  $b$ . Note also that

$$x^{ak} = y^k = 1,$$

so  $n|ak$ , meaning  $db|dck$ , so  $b|ck$ . Since  $b$  and  $c$  are relatively prime,  $b|k$ , so  $b = k$ . ■

**Proposition 1.2.26.** *Let  $H = \langle x \rangle$ .*

- *Let  $|x| = \infty$ . Then  $H = \langle x^a \rangle$  if and only if  $a = \pm 1$ .*
- *Let  $|x| = n < \infty$ . Then,  $H = \langle x^a \rangle$  if and only if  $\gcd(a, n) = 1$ .*

*Proof.* Suppose that  $H = \langle x^a \rangle$ . Then, there is some  $s$  so that  $(x^a)^s = x$ . Thus,  $x^{as-1} = 1$ . Thus,  $as - 1 = 0$ , so  $as = 1$ . Because  $a$  and  $s$  are integers, the only possible solutions are  $a = s = \pm 1$ . Now, suppose  $|x| = n < \infty$ . Then  $|\langle x^a \rangle| = |x^a|$ . This is all of  $H$  if and only if  $|x^a| = |x|$ . Thus,

$$|x^a| = |x| \iff \frac{n}{\gcd(a, n)} = n \iff \gcd(a, n) = 1.$$
■

**Theorem 1.2.27.** *Let  $H = \langle x \rangle$ .*

- (1) *Every subgroup of  $H$  is cyclic. More precisely, if  $K \leq H$ , then either  $K = \{1\}$ , or  $K = \langle x^d \rangle$ , where  $d$  is the smallest positive integer so that  $x^d \in K$ .*
- (2) *If  $|H| = \infty$ , then for any distinct nonnegative integers  $a, b$ ,  $\langle x^a \rangle \neq \langle x^b \rangle$ . Furthermore, for every integer  $m$ ,  $\langle x^m \rangle = \langle x^{|m|} \rangle$ , so that the nontrivial subgroups correspond bijectively with  $1, 2, 3, \dots$*
- (3) *If  $|H| = n < \infty$ , then for each positive integer  $a|n$ , there is a unique subgroup of  $H$  with order  $a$ . Namely,  $\langle x^{n/a} \rangle$ . Furthermore, for every integer  $m$ ,  $\langle x^m \rangle = \langle x^{\gcd(n, m)} \rangle$ , so that the subgroups of  $H$  correspond bijectively with the positive divisors of  $n$ .*

*Proof.*

- (1) Let  $K \leq H$ . If  $K = \{1\}$ , then this is true, so let  $K \neq \{1\}$ . Then there is some  $a \neq 0$  so that  $x^a \in K$ . If  $a < 0$ , then  $x^{-a} = (x^a)^{-1} \in K$ , so we may assume  $a > 0$ . Let

$$\mathcal{P} = \{b \in \mathbb{Z}^+ : x^b \in K\}.$$

This is nonempty, so we may choose a minimum element  $d$ . Then,  $\langle x^d \rangle \leq K$ . For each  $a \in \mathcal{P}$ , write  $a = qd + r$  for  $0 \leq r < d$ . Then,  $x^r = x^{a-qd} = x^a(x^d)^{-q} \in K$ . By the minimality of  $d$ ,  $r = 0$ . Thus,  $x^a \in \langle x^d \rangle$ , so  $K \leq \langle x^d \rangle$ . Thus,  $K = \langle x^d \rangle$ .

- (2) For the first part, suppose without loss of generality that  $a < b$ . Now, assume that  $\langle x^a \rangle = \langle x^b \rangle$ . Then, there is some  $q \in \mathbb{Z}$  so that  $(x^b)^q = x^a$ , meaning that  $x^{qb-a} = 1$ . Thus,  $qb - a = 0$ , or  $qb = a$ . Both  $a$  and  $b$  are nonnegative, so  $q$  must be also. In the case when  $a \neq 0$ , then this is impossible. If  $a = 0$ , then  $b \neq 0$ , so  $x^b \neq 1 \in \langle x^b \rangle$ , meaning that  $\langle x^a \rangle \neq \langle x^b \rangle$ . For the second, notice that this is true when  $m \geq 0$ , and when  $m < 0$ , then  $(x^m)^k = (x^{|m|})^{-k}$ .
- (3) Let  $|H| = n < \infty$ , and that  $a|n$ . Let  $d = n/a$ . Then,  $\langle x^d \rangle$  has order  $a$ . To show uniqueness, let  $K \leq H$  with  $|K| = a$ . We may write  $K = \langle x^b \rangle$ , where  $b$  is the smallest positive integer with  $x^b \in K$ . Then,

$$\frac{n}{d} = a = |K| = |x^b| = \frac{n}{\gcd(n, b)},$$

so  $\gcd(n, b) = d$ . Thus,  $d|b$ , so  $K = \langle x^b \rangle \leq \langle x^d \rangle$ . Since  $|\langle x^d \rangle| = a = |K|$ , this is equality. Finally, note that  $\langle x^m \rangle \leq \langle x^{\gcd(n, m)} \rangle$ , and that

$$|\langle x^m \rangle| = \frac{n}{\gcd(n, m)} = |\langle x^{\gcd(n, m)} \rangle|.$$

■

### 1.2.5 Subgroups Generated By Subsets

**Proposition 1.2.28.** *Let  $\mathcal{A}$  be any nonempty collection of subgroups of  $G$ . Then the intersection of all elements of  $\mathcal{A}$  is also a subgroup of  $G$*

*Proof.* Let

$$K = \bigcap_{H \in \mathcal{A}} H.$$

Note that  $1 \in K$ , so  $K \neq \emptyset$ . Note that if  $a, b \in K$ , then  $a, b \in H$  for all  $H$ , so  $ab^{-1} \in H$ , meaning that  $ab^{-1} \in K$ . ■

**Definition 1.2.29.** Let  $A \subseteq G$ . Define

$$\langle A \rangle = \bigcap_{\substack{A \subseteq H \\ H \leq G}} H.$$

This is called the subgroup of  $G$  generated by  $A$ .

*Remark 1.2.30.* Note that  $\langle A \rangle \leq G$ , and that  $A \subseteq \langle A \rangle$ . Note also that if  $H \leq G$  with  $A \subseteq H$ , then  $\langle A \rangle \leq H$ .

**Notation.** If  $A = \{a_1, a_2, \dots, a_n\}$ , we will write  $\langle a_1, a_2, \dots, a_n \rangle$  for  $\langle A \rangle$ . If  $A, B \subseteq G$ , we will write  $\langle A, B \rangle$  rather than  $\langle A \cup B \rangle$ .

**Definition 1.2.31.** Let  $A \subseteq G$ . We define

$$\overline{A} = \{a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n} : n \in \mathbb{Z}_{\geq 0}, a_i \in A, \epsilon_i = \pm 1\}.$$

If  $A = \emptyset$ , we define  $\overline{A} = \{1\}$ .

**Proposition 1.2.32.**  $\overline{A} = \langle A \rangle$ .

*Proof.* We first prove  $\overline{A} \leq G$ . Note that  $\overline{A} \neq \emptyset$ . If  $a, b \in A$  with  $a = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$ , and  $b = b_1^{\delta_1} b_2^{\delta_2} \cdots b_m^{\delta_m}$ , then  $ab^{-1} = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n} b_m^{-\delta_m} b_{m-1}^{-\delta_{m-1}} \cdots b_1^{-\delta_1} \in \overline{A}$ . Therefore,  $\overline{A} \leq G$ . Note that  $A \subseteq \overline{A}$ , so  $\langle A \rangle \subseteq \overline{A}$ . Note also that, since  $\langle A \rangle$  is closed under the operation and under inverses, every element of the form  $a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$  is in  $\langle A \rangle$ . Thus,  $\overline{A} \subseteq \langle A \rangle$ , so  $\overline{A} = \langle A \rangle$ . ■

*Remark 1.2.33.* Note that we may simplify  $\underbrace{a \cdot a \cdots a}_{n \text{ times}}$  as  $a^n$ , so we may write

$$\overline{A} = \langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} : n \in \mathbb{Z}^+, a_i \in A, \alpha_i \in \mathbb{Z}, a_i \neq a_{i+1}\}.$$

## 1.3 Quotient Groups and Homomorphisms

### 1.3.1 Definition

**Definition 1.3.1.** Let  $G$  and  $H$  be groups, and  $\phi : G \rightarrow H$  a homomorphism. We define the kernel of  $\phi$  as

$$\ker(\phi) = \{x \in G : \phi(x) = 1_H\},$$

and the image of  $\phi$  as

$$\text{im}(\phi) = \{h \in H : \exists x \in G, \phi(x) = h\}.$$

**Proposition 1.3.2.** Let  $G$  and  $H$  be groups, and  $\phi : G \rightarrow H$  a homomorphism. Then,  $\phi$  is injective if and only if  $\ker(\phi) = \{1_G\}$ , and  $\phi$  is surjective if and only if  $\text{im}(\phi) = H$ .

*Proof.* The second statement is obviously true. Now, suppose that  $\phi$  is injective. Then, as  $\phi(1_G) = 1_H$ ,  $\ker(\phi) = \{1_G\}$ . Now, suppose that  $\ker(\phi) = \{1_G\}$ , and that  $\phi(a) = \phi(b)$ . Then,  $\phi(ab^{-1}) = 1_H$ , so  $ab^{-1} \in \ker(\phi)$ . Thus,  $ab^{-1} = 1_G$ , meaning that  $a = b$ . ■

**Proposition 1.3.3.** Let  $G$  and  $H$  be groups, and  $\phi : G \rightarrow H$  a homomorphism. Then  $\ker(\phi) \leq G$ , and  $\text{im}(\phi) \leq H$ .

*Proof.* Suppose that  $a, b \in \ker(\phi)$ . Then,  $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = 1_H 1_H^{-1} = 1_H$ , so  $ab^{-1} \in \ker(\phi)$ . Now, suppose that  $a = \phi(a_0), b = \phi(b_0) \in \text{im}(\phi)$ . Then,  $ab^{-1} = \phi(a_0 b_0^{-1}) \in \text{im}(\phi)$ . ■

**Definition 1.3.4.** Let  $\phi : G \rightarrow H$  be a homomorphism with  $\ker \phi = K$ . The quotient group  $G/K$  (read  $G \bmod K$ ) is the group whose elements are the fibers of  $\phi$  with group operation defined as  $X_a \cdot X_b = X_{ab}$ , where  $X_a$  is the fiber of  $a$  ( $\phi^{-1}(a)$ ).

**Proposition 1.3.5.** Let  $\phi : G \rightarrow H$  be a homomorphism with  $\ker \phi = K$ . Let  $X \in G/K$  be the fiber of  $a$ . Then, for any  $u \in X$ ,  $X = uK = Ku$ .

*Proof.* Let  $u \in X$  and  $k \in K$ . Then,

$$\begin{aligned} \phi(uk) &= \phi(u)\phi(k) \\ &= a1 \\ &= a, \end{aligned}$$

so  $uk \in X$ . Similarly,  $ku \in X$ . Thus,  $uK, Ku \subseteq X$ . Now, suppose that  $g \in X$ , and  $k = u^{-1}g$ . Note that  $k \in K$ . Thus,  $g = uk \in uK$ , so  $X \subseteq uK$ . Similarly for  $Ku$ . ■

**Definition 1.3.6.** Let  $N \leq G$ , and  $g \in G$ . We define

$$gN = \{gn : n \in N\} \quad Ng = \{ng : n \in N\},$$

called a left coset of  $N$  and a right coset of  $N$  in  $G$  respectively. Any element of a coset is called a representative for that coset.

*Remark 1.3.7.* The fibers of the homomorphism are the left (and right) cosets of the kernel.

**Theorem 1.3.8.** Let  $G$  be a group, and let  $K$  be the kernel of some homomorphism. Then the set of left cosets of  $K$  form a group ( $G/K$ ) under

$$uK \cdot vK = (uv)K.$$

*In particular, this is well defined. The same is true of right cosets.*

*Proof.* Let  $X, Y \in G/K$ , and  $Z = XY$ . Then, these are left cosets of  $K$ . Let  $\phi : G \rightarrow H$  with  $\ker \phi = K$ , and let  $X = \phi^{-1}(a)$ ,  $Y = \phi^{-1}(b)$  for some  $a, b \in H$ . Then,  $Z = \phi^{-1}(ab)$ . Let  $u$  and  $v$  be representatives of  $X$  and  $Y$ , so  $X = uK$ , and  $Y = vK$ . Then,  $\phi(uv) = \phi(u)\phi(v) = ab$ , so  $uv \in Z$ . Thus,  $Z = uvK$ . Note also that any  $z \in Z$  can be written as  $uv$  for some  $u \in X$ ,  $v \in Y$ . The second part follows as  $uK = Ku$ . ■

**Proposition 1.3.9.** Let  $N \leq G$ . Then the set of left cosets of  $N$  partition  $G$ . Furthermore, for all  $u, v \in G$ ,  $uN = vN$  if and only if  $v^{-1}u \in N$ .

*Proof.* Note that  $1 \in N$ , so  $g \in gN$  for all  $g \in G$ . Thus,  $G = \bigcup_{g \in G} gN$ . Now, suppose that  $uN \cup vN \neq \emptyset$ . Let  $x$  be in their intersection. Then  $x = un = vm$  for some  $n, m \in N$ . Thus,  $u = vmn^{-1} = vm_1$ , where  $m_1 = mn^{-1} \in N$ . Then for  $ut \in uN$  ( $t \in N$ ),  $ut = (vm_1)t = v(m_1t) = vm_2$ , where  $m_2 = m_1t \in N$ . Thus,  $uN \subseteq vN$ . Similarly,  $vN \subseteq uN$ , so  $vN = uN$ . Now,  $uN = vN$  if and only if  $u \in vN$ , so we can write  $u = vn$  for some  $n \in N$ . Then,  $v^{-1}u = n \in N$ . ■

**Proposition 1.3.10.** Let  $G$  be a group and  $N$  a subgroup.

- The operation of left cosets of  $N$  described by  $uN \cdot vN = (uv)N$  is well defined if and only if  $gng^{-1} \in N$  for all  $g \in G, n \in N$ .
- If the operation is well defined, then it makes the set of left cosets into a group.

*Proof.* Assume that this operation is well defined so that if  $u, u_1 \in uN$  and  $v, v_1 \in vN$ , then  $uvN = u_1v_1N$ . Let  $g \in G$ , and  $n \in N$ . If  $u = 1$ ,  $u_1 = n$ ,  $v = v_1 = g^{-1}$  then  $1g^{-1}N = ng^{-1}N \iff g^{-1}N = ng^{-1}N$ . Now,  $ng^{-1} \in ng^{-1}N = g^{-1}N$ , so there is some  $n_1 \in N$  with  $ng^{-1} = g^{-1}n_1$ . Thus,  $gng^{-1} = n_1 \in N$ . Now assume that  $gng^{-1} \in N$  for all  $g \in G, n \in N$ . Let  $u, u_1 \in uN$  and  $v, v_1 \in vN$ . We write  $u_1 = un$  and  $v_1 = vm$  for some  $n, m \in N$ . Then,  $u_1v_1 = unvm = uvv^{-1}nvm = (uv)(v^{-1}nv)(m)$ . Note that the right 2 terms are in  $N$ , so  $u_1v_1 \in uvN$ . Thus,  $u_1v_1N = uvN$ . We check now that this forms a group. Associativity and closure are clear. Note that  $1N \cdot gN = (1 \cdot g)N = gN$ , so  $1N$  is the identity. Note also that  $g^{-1}N \cdot gN = (g^{-1}g)N = 1N$ , so  $g^{-1}N$  is the inverse of  $gN$ . ■



**Definition 1.3.11.** An element  $g$  is said to normalize  $N$  if  $gNg^{-1} = N$ . A subgroup  $N$  of  $G$  is called normal if  $gNg^{-1} = N$  for all  $g \in G$ . In this case, we write  $N \trianglelefteq G$ .

**Theorem 1.3.12.** Let  $N \leq G$ . The following are equivalent.

- (1)  $N \trianglelefteq G$
- (2)  $N_G(N) = G$
- (3)  $gN = Ng$  for all  $g \in G$
- (4) The operation of left cosets described above makes the set of left cosets into a group
- (5)  $gNg^{-1} \subseteq N$  for all  $g \in G$ .

*Proof.* That (1)  $\iff$  (2) is clear, and we have already proven (1)  $\iff$  (4). If  $gN = Ng$ , then  $gn = n_1g$  for some  $n_1 \in N$ . Thus,  $gng^{-1} = n_1 \in N$ . Similarly, if  $gng^{-1} = n_1 \in N$ , then  $gn = n_1g$ , so (1)  $\iff$  (3). Finally, note that the map  $a \mapsto gag^{-1}$  is injective, so if  $gNg^{-1} \subseteq N$  for all  $g$ , then  $gNg^{-1} = N$  for all  $g$ , and vice versa. ■

**Proposition 1.3.13.** Let  $G$  be a group. Then  $N \trianglelefteq G$  if and only if  $N = \ker \phi$  for some homomorphism  $\phi$ .

*Proof.* If  $N$  is the kernel of some homomorphism  $\phi$ , then  $\phi(gng^{-1}) = \phi(g)\phi(n)\phi(g)^{-1} = \phi(g)\phi(g)^{-1} = 1$  for some  $g \in G$  and  $n \in N$ , so  $gng^{-1} \in N$ . Now, let  $N \trianglelefteq G$ . Let  $G/N$  denote the group of cosets of  $N$ , and define  $\pi : G \rightarrow G/N$  by  $\pi(g) = gN$ . Then,  $\pi(g_1g_2) = (g_1g_2)N = g_1Ng_2N = \pi(g_1)\pi(g_2)$ , so this is a homomorphism. Furthermore,  $\ker \pi = N$ . ■

*Remark 1.3.14.* Let  $N \trianglelefteq G$  (or let  $N = \ker \phi$ ). Then, we can think of  $G/N$  as being the group of cosets of  $N$ , or the group of fibers of  $\phi$ .

**Definition 1.3.15.** Let  $N \trianglelefteq G$ . The natural projection homomorphism  $\pi : G \rightarrow G/N$  is defined by  $\pi(g) = gN$ . If  $H \leq G/N$ , the complete preimage of  $H$  in  $G$  is the preimage of  $H$  under the natural projection homomorphism.

### 1.3.2 More on Cosets and Lagrange's Theorem

**Theorem 1.3.16** (Lagrange's Theorem). If  $G$  is a finite group with  $H \leq G$ , then  $|H|$  divides  $|G|$ , and the number of left cosets of  $H$  in  $G$  is  $|G|/|H|$ .

*Proof.* Note that the map  $\psi : H \rightarrow gH$  defined by  $h \mapsto gh$  is bijective, so  $|H| = |gH|$ . We have seen that cosets partition  $G$ , so we have a way to divide  $G$  into disjoint subsets of cardinality  $|H|$ . Thus,  $|H|$  divides  $|G|$ . If  $k$  is the number of cosets of  $H$  in  $G$ , then  $|G| = k|H|$ , so  $k = |G|/|H|$ . ■

**Corollary 1.3.17.** Let  $G$  be a finite group and  $x \in G$ . Then  $|x|$  divides  $|G|$ . In particular,  $x^{|G|} = 1$  for all  $x$ .

*Proof.* We have that  $|x| = |\langle x \rangle|$ , and that the latter divides  $|G|$  by Lagrange's Theorem. The second statement is clear, as  $|G|$  is a multiple of  $|x|$ . ■

**Corollary 1.3.18** (Euler's Totient Theorem). *Let  $\gcd(a, n) = 1$ . Then  $a^{\phi(n)} \equiv 1 \pmod{n}$ , where  $\phi$  is Euler's Totient Function. In particular, if  $p$  is prime and  $p \nmid a$ , then  $a^{p-1} \equiv 1 \pmod{p}$ .*

*Proof.* For the first part, apply the above corollary to  $(\mathbb{Z}/n\mathbb{Z})^\times$ . Letting  $n$  be prime gives the second. ■

**Corollary 1.3.19.** *If  $G$  is a group of prime order  $p$ , then  $G$  is cyclic.*

*Proof.* Let  $x \neq 1 \in G$ . Then  $|\langle x \rangle| > 1$ , and  $|\langle x \rangle|$  divides  $|G|$ . Because  $|G|$  is prime,  $|\langle x \rangle| = |G|$ , so  $G = \langle x \rangle$ . ■

**Definition 1.3.20.** Let  $G$  be a group and  $H \leq G$ . The number of cosets of  $H$  in  $G$  is called the index of  $H$  in  $G$  and is denoted  $[G : H]$ .

*Example 1.3.21.* Let  $G$  be a group and  $H \leq G$  with  $[G : H] = 2$ . We show that  $H \trianglelefteq G$ . Let  $g \in G \setminus H$ , so that the cosets are  $H$  and  $gH$ . We must also have that  $gH = G \setminus H$ . We must also have that  $Hg = G \setminus H$ , so  $gH = Hg$ . Thus,  $H \trianglelefteq G$ .

*Remark 1.3.22.* If  $G$  is a finite group and  $n$  divides  $|G|$ ,  $G$  need not have a subgroup of order  $n$ . For example, let  $A$  be the group of symmetries of a regular tetrahedron. Then,  $|A| = 12$ . Suppose that  $A$  had a subgroup of order 6 (which is therefore normal). Because the quotient has order 2, the square of each of its elements is the identity. Thus, for all  $g \in A$ ,  $(gH)^2 = H$ , so  $g^2 \in H$ . If  $|g| = 3$ , then  $g = g^4 = (g^2)^2 \in H$ . However, there are 8 elements of order 3, a contradiction.

**Theorem 1.3.23** (Cauchy's Theorem). *Suppose that  $G$  is a (finite) group, and a prime  $p$  divides  $|G|$ . Then  $G$  contains an element of order  $p$ .*

*Proof.* Consider the set

$$S = \{(g_1, g_2, \dots, g_p) : g_1, g_2, \dots, g_p \in G, g_1 g_2 \cdots g_p = 1\}.$$

This has order  $|G|^{p-1}$  (as we can choose the first  $p-1$  elements arbitrarily). Now, consider the map

$$f : (g_1, g_2, \dots, g_p) \mapsto (g_p, g_1, \dots, g_{p-1}).$$

Let us write  $\langle (g_1, g_2, \dots, g_p) \rangle$  for the set of images of  $(g_1, g_2, \dots, g_p)$  under repeated applications of  $f$ . Let us write also  $I$  for the set of these images. Suppose that  $|\langle (g_1, g_2, \dots, g_p) \rangle| = n$ . Note that this must divide  $p$ . Thus,  $n = 1$  (occurring exactly when  $g_1 = g_2 = \dots = g_p$ ) or  $n = p$ . Now, the number of elements  $(g_1, g_2, \dots, g_p)$  with  $|\langle (g_1, g_2, \dots, g_p) \rangle| = 1$  must be congruent to 0 mod  $p$ , as

$$|G|^{p-1} = |S| = \sum_{i \in I} |i| = \sum_{\substack{i \in I \\ |i|=1}} |i| + \sum_{\substack{i \in I \\ |i|=p}} |i|,$$

and we may reduce mod  $p$ . Note that if  $g_1 = g_2 = \dots = g_p = 1$ , then  $|\langle (g_1, g_2, \dots, g_p) \rangle| = 1$ . Thus, there must be at least one more element  $g \in G$  with  $|\langle (g, g, \dots, g) \rangle| = 1$ . Thus,  $g^p = 1$ , so  $|g| = 1$  or  $p$ . We chose  $g \neq 1$ , so  $|g| = p$ . ■

**Definition 1.3.24.** Let  $H, K \leq G$ . We define

$$HK = \{hk : h \in H, k \in K\}.$$

**Proposition 1.3.25.** If  $H, K \leq G$  are finite, then

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

*Proof.* Notice that

$$HK = \bigcup_{h \in H} hK.$$

Now, we find the number of distinct cosets of the form  $hK$  where  $h \in H$ . Note that  $h_1K = h_2K$  if and only if  $h_2^{-1}h_1 \in K$ . Thus,  $h_2^{-1}h_1 \in H \cap K$ , so  $h_1(H \cap K) = h_2(H \cap K)$ . Therefore, the number of distinct cosets of the form  $hK$  is the same as the number of distinct cosets of the form  $h(H \cap K)$ . The latter number is  $\frac{|H|}{|H \cap K|}$ . Thus,

$$|HK| = \frac{|H|}{|H \cap K|} |K| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

■

**Proposition 1.3.26.** If  $H, K \leq G$ , then  $HK \leq G$  if and only if  $HK = KH$ .

*Proof.* Assume that  $HK = KH$ , and let  $a, b \in HK$ . Let  $a = h_1k_1$  and  $b = h_2k_2$ . Then,  $ab^{-1} = h_1k_1k_2^{-1}h_2^{-1}$ . Let  $k_3 = k_1k_2^{-1}$ , and  $h_3 = h_2^{-1}$ . Then,  $ab^{-1} = h_1k_3h_3$ . Because  $HK = KH$ ,  $k_3h_3 = h_4k_4$  for some  $h_4 \in H$ ,  $k_4 \in K$ . Thus,  $ab^{-1} = h_1h_4k_4 \in HK$ . Now, assume that  $HK \leq G$ . Since  $H, K \leq HK$ ,  $KH \subseteq HK$ . Now, let  $hk \in HK$ , and  $hk = a^{-1}$  for some  $a$ . If  $a = h_1k_1$ , then  $hk = (h_1k_1)^{-1} = k_1^{-1}h_1^{-1} \in KH$ , so  $HK \subseteq KH$ . ■

**Corollary 1.3.27.** If  $H, K \leq G$  and  $H \leq N_G(K)$ , then  $HK \leq G$ . In particular, if  $K \trianglelefteq G$ , then  $HK \leq G$  for any  $H \leq G$ .

*Proof.* Let  $h \in H$ ,  $k \in K$ . Then  $hkh^{-1} \in K$ , so  $hk = (hkh^{-1})h$ , so  $HK \subseteq KH$ . Similarly,  $kh = h(h^{-1}kh) \in HK$ , so  $KH \subseteq HK$ . ■

*Remark 1.3.28.* We have worked primarily with left cosets. The same combinatorial results can be proved equally well with right cosets. For example, if  $H \leq G$ , then the number of right cosets of  $H$  in  $G$  is  $|G|/|H|$ .

### 1.3.3 The Isomorphism Theorems

**Theorem 1.3.29** (The First Isomorphism Theorem). Let  $\phi : G \rightarrow H$  be a homomorphism. Then,  $G/\ker \phi \cong \text{im } \phi$ .

*Proof.* From our original definition,  $G/\ker \phi$  is the set of fibers of  $\phi$ . Define  $\psi : G/\ker \phi \rightarrow \text{im } \phi$  by  $\psi(\phi^{-1}(a)) = a$ . If we write  $X_a$  for the fiber of  $a$ , then  $\psi(X_a X_b) = \psi(X_{ab}) = ab = \psi(X_a)\psi(X_b)$ , so  $\psi$  is a homomorphism. Note that  $\ker \psi = X_1 = \ker \phi$ . However, this is princely the identity in  $G/\ker \phi$ . Note also that this is surjective:  $\psi^{-1}(a) = X_a$ . Thus,  $\psi$  is an isomorphism. ■

**Theorem 1.3.30** (The Second or Diamond Isomorphism Theorem). *Let  $G$  be a group,  $A, B \leq G$ , and assume that  $A \leq N_G(B)$ . Then  $AB \leq G$ ,  $B \trianglelefteq AB$ ,  $A \cap B \trianglelefteq A$ , and  $AB/B \cong A/A \cap B$ .*

*Proof.* We have already proved the first part. Since we assume that  $A \leq N_G(B)$ , and  $B \leq N_G(B)$ ,  $AB \leq N_G(B)$ , proving the second statement. Now, the quotient group  $AB/B$  is well defined. We define  $\phi : A \rightarrow AB/B$  by  $a \mapsto aB$ . It is easy to see that  $\phi$  is a homomorphism. Note also that  $\phi$  is surjective. The kernel is the set of elements  $a \in A$  so that  $aB = B$ , which occurs when  $a \in B$ . Thus,  $\ker \phi = A \cap B$ . Therefore,  $A \cap B \trianglelefteq A$ , and the First Isomorphism Theorem implies that  $A/A \cap B \cong AB/B$ . ■

**Theorem 1.3.31** (The Third Isomorphism Theorem). *Let  $G$  be a group and  $H, K \trianglelefteq G$  with  $H \leq K$ . Then  $K/H \trianglelefteq G/H$ , and  $(G/H)/(K/H) \cong G/K$ .*

*Proof.* For  $g \in G$  and  $k \in K$ , we have that  $gH \cdot kH \cdot g^{-1}H = (gkg^{-1})H$ . Because  $K \trianglelefteq G$ , this is an element of  $K/H$ , so  $K/H \trianglelefteq G/H$ . Now define  $\phi : G/H \rightarrow G/K$  with  $gH \mapsto gK$ . We must show that this is well defined. Suppose that  $g_1H = g_2H$ , so that  $g_1 = g_2h$  for some  $h \in H$ . Since  $H \leq K$ ,  $h \in K$  so  $g_1K = g_2K$ . It is easy to check that this is a homomorphism. It is also clearly surjective. Finally, the kernel is the set of  $gH$  with  $gK = K$ . In other words,  $\ker \phi = \{gH : g \in K\}$ , which is simply  $K/H$ . By the First Isomorphism Theorem,  $(G/H)/(K/H) \cong G/K$ . ■

**Theorem 1.3.32** (The Fourth or Lattice Isomorphism Theorem). *Let  $G$  be a group and  $N \trianglelefteq G$ . Then there is a bijection from the set  $\{A \leq G : N \leq A\}$  to the set  $\{H \leq G/N\}$ . Furthermore, for all  $A, B \leq G$  with  $N \leq A, B$*

- (1)  $A \leq B$  if and only if  $A/N \leq B/N$
- (2) If  $A \leq B$  then  $[B : A] = [B/N : A/N]$
- (3)  $\langle A, B \rangle/N = \langle A/N, B/N \rangle$
- (4)  $(A \cap B)/N = A/N \cap B/N$
- (5)  $A \trianglelefteq G$  if and only if  $A/N \trianglelefteq G/N$ .

*Proof.* Consider the map  $\phi(A) = A/N$ . This is clearly injective, so we show only that it is surjective. Suppose that  $H \leq G/N$ . Let  $H$  be the set of cosets of  $h_1, h_2, \dots, h_k$  (where if  $h_iN = h_jN$ , both  $h_i$  and  $h_j$  appear in this list). Then,  $h_1, h_2, \dots, h_k$  form a group. Writing  $\mathcal{H} = \{h_1, h_2, \dots, h_k\}$ ,  $H = \mathcal{H}/N$ . Thus, the map is surjective. We now prove the 5 properties.

- (1) Suppose that  $A \leq B$ . Then it is clear that  $A/N \leq B/N$ . Conversely, suppose that  $A/N \leq B/N$ . Then, for every  $aN \in A/N$ ,  $aN \in B/N$ , so  $aN = bN$  for some  $b \in B$ . In particular,  $a \in bN$ , so  $a = bn$  for some  $n \in N \leq B$ . Thus,  $A \subseteq B$ , and so  $A \leq B$ .

- (2) This is a simple computation:

$$[B/N : A/N] = \frac{|B/N|}{|A/N|} = \frac{|B|/|N|}{|A|/|N|} = \frac{|B|}{|A|} = [B : A].$$

- (3) Note that  $\langle A, B \rangle/N$  is the set of cosets of the form  $a_1b_1 \cdots a_kb_kN$ , which are the same as cosets of the form  $(a_1N)(b_1N) \cdots (a_kN)(b_kN)$  (possibly ending with an element of  $A$  or starting with an element of  $B$  or both). However, the second form is simply  $\langle A/N, B/N \rangle$ .
- (4) We can see that  $(A \cap B)/N$  has elements of the form  $aN$  where  $a \in A, B$ . Also,  $aN \in A/N$  and  $aN \in B/N$ , so  $(A \cap B)/N \subseteq A/N \cap B/N$ . Now, if  $aN \in A/N \cap B/N$ , then  $a \in A, B$ , so  $aN \in (A \cap B)/N$ , proving the reverse inclusion.
- (5) If  $A \trianglelefteq G$ , it is clear that  $A/N \trianglelefteq G/N$ . Suppose that  $A/N \trianglelefteq G/N$ , so that  $(gN)(aN)(g^{-1}N) \in A/N$ . Then,  $(gag^{-1})N = a_1N$  for some  $a_1 \in A$ . Thus,  $gag^{-1} \in a_1N$ , so we have  $gag^{-1} = a_1n$  for some  $n \in N \leq A$ . Therefore,  $A \trianglelefteq G$ . ■

**Proposition 1.3.33.** *Let  $N \trianglelefteq G$ , and  $\phi : G/N \rightarrow H$  be a homomorphism defined by  $\phi(gN) = \Phi(g)$ . Then  $\phi$  is well defined if and only if  $N \leq \ker \Phi$ .*

*Proof.* Suppose that if  $g_1N = g_2N$ , then  $\Phi(g_1) = \phi(g_1N) = \phi(g_2N) = \Phi(g_2)$ . Then,  $\Phi(g_2^{-1}g_1) = 1$ . However,  $g_2^{-1}g_1 \in N$ . We may choose  $g_1$  to go over all representatives of the coset. Thus,  $N \leq \ker \Phi$ . Now suppose that  $N \leq \ker \Phi$ . If  $g_1N = g_2N$ , then  $\Phi(g_2^{-1}g_1) = 1$ , so  $\Phi(g_1) = \Phi(g_2)$ . Thus,  $\phi(g_1N) = \phi(g_2N)$ . ■

### 1.3.4 Composition Series

**Proposition 1.3.34.** *If  $G$  is a finite abelian group and  $p$  is a prime dividing  $|G|$ , then  $G$  contains an element of order  $p$ .*

*Proof.* Although this follows from Cauchy's Theorem, we will prove this by induction on  $|G|$ . Since  $|G| > 1$ , there is some  $x \in G$  with  $x \neq 1$ . If  $|G| = p$ , then  $|x| = p$  by Lagrange's Theorem. Suppose that  $|G| > p$ . If  $|x| = pn$ , then  $|x^n| = p$ . Thus, we assume that  $p \nmid |x|$ . Let  $N = \langle x \rangle$ . Then  $N \trianglelefteq G$ , and  $|G/N| = |G|/|N|$ . Since  $|N| \neq 1$ ,  $|G/N| < |G|$ . Since  $p \nmid |N|$ ,  $p$  divides  $|G/N|$ . By the induction hypothesis, there is some  $yN \in G/N$  with order  $p$ . Therefore,  $y \notin N$ , but  $y^p \in N$ . Thus  $\langle y^p \rangle \neq \langle y \rangle$ . Since  $\langle y^p \rangle \subseteq \langle y \rangle$ , we have that  $|y^p| < |y|$ . Therefore,  $p$  divides  $|y|$ . We have already seen what to do in this case, so we are done. ■

*Remark 1.3.35.* A clear obstruction to this approach for nonabelian groups is that we must produce a normal subgroup  $N \trianglelefteq G$  where  $N \neq 1, G$ .

**Definition 1.3.36.** A group  $G$  is called simple if  $|G| > 1$  and the only normal subgroups of  $G$  are  $1$  and  $G$ .

**Proposition 1.3.37.** Let  $G$  be an abelian simple group. Then  $G \cong Z_p$  for some prime  $p$ .

*Proof.* Since  $G$  is abelian, all subgroups of  $G$  are normal. Thus, the only subgroups of  $G$  are  $1$  and  $G$ . Let  $x \neq 1 \in G$ . Then,  $\langle x \rangle \leq G$ , and therefore  $G = \langle x \rangle$ . If  $|x| = \infty$ , then this group is  $\mathbb{Z}$ , which is not simple ( $2\mathbb{Z} \leq \mathbb{Z}$ ). If  $|x| = n$ , then  $n$  must be prime, as if  $p|n$  is prime, then there is a subgroup of order  $p$  of  $\langle x \rangle$ , contradicting its simplicity. Thus,  $|x| = p$  for some prime  $p$ , and  $G \cong Z_p$ . ■

**Definition 1.3.38.** Let  $G$  be a group. A sequence of subgroups

$$1 = N_0 \leq N_1 \leq N_2 \leq \cdots \leq N_{k-1} \leq H_k = G$$

is called a composition series if  $N_i \trianglelefteq N_{i+1}$  and  $N_{i+1}/N_i$  is a simple group for  $0 \leq i \leq k-1$ . If this is a composition series, the quotient groups  $N_{i+1}/N_i$  are called composition factors of  $G$ .

**Theorem 1.3.39** (Jordan-Hölder). Let  $G$  be a finite group with  $G \neq 1$ . Then,

- (1)  $G$  has a composition series.
- (2) If  $1 = N_0 \leq N_1 \leq \cdots \leq N_r = G$  and  $1 = M_0 \leq M_1 \leq \cdots \leq M_s = G$  are two composition series for  $G$ , then  $r = s$  and there is some permutation  $\pi$  of  $\{1, 2, \dots, r\}$  such that  $M_{\pi(i)}/M_{\pi(i)-1} \cong N_i/N_{i-1}$  for  $1 \leq i \leq r$ .

*Proof.*

- (1) We prove this by induction on  $|G|$ . If  $|G| = 2$ , then  $G \cong Z_2$ , so  $1 \leq G$  is a composition series. Now, suppose that this is true for all groups of order less than  $|G|$ . If  $G$  is simple, then  $1 \leq G$  is a composition series. Assume that  $G$  is not simple. Take  $N \trianglelefteq G$  with  $N \neq G$  of maximal order. We claim that  $G/N$  is simple. Suppose that  $H \trianglelefteq G/N$ . Write  $H = \{g_1N, g_2N, \dots, g_kN\}$ . Let  $\mathcal{H} = \{g_in : 1 \leq i \leq k, n \in N\}$ . Then,  $N \trianglelefteq \mathcal{H} \trianglelefteq G$ , which is a contradiction. Since  $|N| < |G|$ ,  $N$  has a composition series, and so  $G$  does too.
- (2) We prove this by induction on  $|G|$ . If  $|G| = 2$ , then this is trivial. Now, suppose that this is true for all groups of order less than  $|G|$ . Take  $1 = N_0 \leq N_1 \leq \cdots \leq N_r = G$  and  $1 = M_0 \leq M_1 \leq \cdots \leq M_s = G$  to be composition series. The theorem is true for  $N := N_{r-1}$  and  $M := M_{s-1}$ . If  $N = M$ , then we are done. Suppose that  $N \neq M$ , and let  $L = N \cap M$ . Then  $L$  has some composition series  $1 = L_0 \leq L_1 \leq \cdots \leq L_t = L$ . We thus have two composition series for  $N$ :  $1 = N_0 \leq N_1 \leq \cdots \leq N$  and  $1 = L_0 \leq L_1 \leq \cdots \leq L \leq N$  (since  $N/L = N/(N \cap M) = NM/M$  by the Second Isomorphism Theorem, and  $M \trianglelefteq NM \trianglelefteq G$ , so  $NM/M \trianglelefteq G/M$  by the Third Isomorphism Theorem, meaning that  $NM/M = G/M$  as  $NM \neq M$ ). Similarly, we have two composition series for  $M$ . By induction,  $r-1 = t+1$  and  $s-1 = t+1$ . Thus,  $r = s$ . Now, we

say that  $(a_1, a_2, \dots, a_n) \sim (b_1, b_2, \dots, b_n)$  if the  $b_i$ 's are some reordering of the  $a_i$ 's. By induction,

$$(N_1/N_0, N_2/N_1, \dots, N/N_{r-2}) \sim (L_1/L_0, L_2/L_1, \dots, L/L_{t-1}, N/L)$$

$$(M_1/M_0, M_2/M_1, \dots, M/M_{r-2}) \sim (L_1/L_0, L_2/L_1, \dots, L/L_{t-1}, M/L).$$

We may append  $G/N$  and  $G/M$  to get

$$O_N := (N_1/N_0, N_2/N_1, \dots, N/N_{r-2}, G/N) \sim$$

$$(L_1/L_0, L_2/L_1, \dots, L/L_{t-1}, N/L, G/N) := I_N$$

$$I_M := (L_1/L_0, L_2/L_1, \dots, L/L_{t-1}, M/L, G/M) \sim$$

$$(M_1/M_0, M_2/M_1, \dots, M/M_{r-2}, G/M) := O_M.$$

We wish to show that  $O_N \sim O_M$ . Note that  $N/L = N/N \cap M = NM/M = G/M$  for reasons presented above. Similarly,  $M/L = M/N \cap M = NM/N = G/N$ . Thus,  $I_N \sim I_M$ . Therefore,  $O_N \sim O_M$ . ■

**Definition 1.3.40.** A group  $G$  is said to be solvable if there is a chain of subgroups

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \dots \trianglelefteq G_s = G$$

where  $G_{i+1}/G_i$  is abelian for  $0 \leq i \leq s-1$ .

**Proposition 1.3.41.** Let  $G$  be a group and  $N \trianglelefteq G$ . If  $N$  and  $G/N$  are solvable, then  $G$  is solvable.

*Proof.* Let  $\overline{G} = G/N$ . Let the series for  $N$  be  $1 = N_0 \trianglelefteq N_1 \trianglelefteq \dots \trianglelefteq N_n = N$ , and let the series for  $\overline{G}$  be  $\overline{1} = \overline{G}_0 \trianglelefteq \overline{G}_1 \trianglelefteq \dots \trianglelefteq \overline{G}_m = \overline{G}$ . By the Fourth Isomorphism Theorem, there are  $N \leq G_i \leq G$  such that  $G_i/N = \overline{G}_i$  and  $G_i \trianglelefteq G_{i+1}$ . By the Third Isomorphism Theorem,

$$\overline{G_{i+1}}/\overline{G_i} = (G_{i+1}/N)/(G_i/N) \cong G_{i+1}/G_i.$$

Thus,

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \dots \trianglelefteq N_n = N = G_0 \trianglelefteq G_1 \trianglelefteq \dots \trianglelefteq G_m = G$$

is a chain of subgroups allowing us to prove that  $G$  is solvable. ■

**Proposition 1.3.42.** Let  $G$  be a finite group and  $H \trianglelefteq G$ . Then there is a composition series of  $G$  containing  $H$ .

*Proof.* We prove this by induction on  $|G|$ . The base case,  $|G| = |H|$  is obvious. Now, suppose that all groups containing  $H$  or order less than  $|G|$  have a composition series containing  $H$ . Let  $M = \mathcal{M}/H \neq G/H \trianglelefteq G/H$  be maximal. We claim that  $G/\mathcal{M}$  is simple. Note that we may show that  $(G/H)/M$  is simple instead. Suppose there was  $A = \mathcal{A}/M \trianglelefteq (G/H)/M$ . Then,  $M \trianglelefteq \mathcal{A} \trianglelefteq G/H$ , contradicting the maximality of  $M$ . Thus,  $G/\mathcal{M}$  is simple. Take the composition series of  $\mathcal{M}$  containing  $H$  to be  $1 = M_0 \leq M_1 \leq \dots \leq M_n = \mathcal{M}$ . Then, the series  $1 = M_0 \leq M_1 \leq \dots \leq M_n = \mathcal{M} \leq G$  is a composition series of  $G$  containing  $H$ . ■

**Proposition 1.3.43.** *Let  $G$  be a finite group. The following are equivalent.*

- (1)  $G$  is solvable.
- (2) There is a chain of subgroups  $1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_s = G$  so that  $H_{i+1}/H_i$  is cyclic.
- (3) All composition factors of  $G$  are of prime order.
- (4)  $G$  has a chain of subgroups  $1 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_t = G$  such that  $N_i \trianglelefteq G$  and  $N_{i+1}/N_i$  is abelian.

*Proof.* We show that (1)  $\Rightarrow$  (3)  $\Rightarrow$  (2)  $\Rightarrow$  (1) and that (3)  $\Rightarrow$  (4)  $\Rightarrow$  (1). Suppose that  $G$  is solvable, with chain  $1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$ . Since  $G_i/G_{i-1}$  is abelian, it is either not simple, or isomorphic to  $Z_p$  for some prime  $p$ . If it is not simple, then take some  $N \trianglelefteq G_k/G_{k-1}$ . Then  $N = \mathcal{N}/G_{k-1}$  for some  $G_{k-1} \trianglelefteq \mathcal{N} \trianglelefteq G$ , by the Fourth Isomorphism Theorem. Now, we have the series  $1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_{k-1} \trianglelefteq \mathcal{N} \trianglelefteq G_k = G$  (note that  $G/\mathcal{N} \cong (G/G_{k-1})/(\mathcal{N}/G_{k-1})$  is abelian). If  $G/\mathcal{N}$  is not simple, then we repeat this process. Once we reach a stage where  $1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_\ell = G$  and  $G_\ell/G_{\ell-1}$  is simple, we repeat this process with  $G_{\ell-1}$  at the head. Once we arrive at a series  $1 = G'_0 \trianglelefteq G'_1 \trianglelefteq \cdots \trianglelefteq G'_p = G$  where  $G_i/G_{i-1}$  is simple for all  $i$ , since they are abelian, we must have  $G_i/G_{i-1} \cong Z_{p_i}$  for some prime  $p_i$ . This completes the proof that (1)  $\Rightarrow$  (3). Now, assume (3). Since all groups of prime order are cyclic, (2) follows. Similarly, since all cyclic groups are abelian, (1) follows from (2). Now, assume (3). We first show that the minimal nontrivial normal subgroup  $M \trianglelefteq G$  is abelian. We know that  $G$  has a composition series with  $M$  in it. Since all composition factors have prime order, take  $N \leq M$  with  $[M : N] = p$ . Let  $x, y \in M$ . Since  $|M/N| = p$ , it is cyclic and therefore abelian, we have that  $x^{-1}y^{-1}xyN = x^{-1}xy^{-1}yN = N$ , meaning that  $x^{-1}y^{-1}xy \in N$ . Now, consider  $gNg^{-1}$  for any  $g \in G$ . Since  $M$  is normal, this is contained in  $M$ , and therefore a subgroup of  $M$ . We claim that this is a normal subgroup. Note that  $(g^{-1}mg)n(g^{-1}m^{-1}g) \in N$  for  $n \in N$ ,  $m \in M$ , and  $g \in G$ . Thus,  $m(gng^{-1})m^{-1} \in gNg^{-1}$ , so this is a normal subgroup. By the same argument as above,  $x^{-1}y^{-1}xy \in gNg^{-1}$  for all  $g \in G$ . Now, if  $I = \bigcap_{g \in G} gNg^{-1}$  were nontrivial, then it would

be normal (if  $x \in I$ , then  $x \in N$  so  $gxg^{-1} \in I$ ), contradicting the minimality of  $M$ . Thus,  $x^{-1}y^{-1}xy = 1$ , meaning that  $M$  is abelian. Now, let  $M_1$  be a minimal normal subgroup of  $G$ , so that we have the chain  $1 \trianglelefteq M_1 \trianglelefteq G$ . Now, take the minimal normal subgroup of  $G/M_1$  to be  $M_2/M_1$ , so that we have  $M_1/M_1 \trianglelefteq M_2/M_1 \trianglelefteq G/M_1$ . We therefore have  $1 \trianglelefteq M_1 \trianglelefteq M_2 \trianglelefteq G$ , where  $M_2/M_1$  is abelian. We may continue this process (which must eventually terminate) to get a chain  $1 \trianglelefteq M_1 \trianglelefteq M_2 \trianglelefteq \cdots \trianglelefteq M_i \trianglelefteq G$  with  $M_j \trianglelefteq G$  and  $M_j/M_{j-1}$  abelian for  $1 \leq j \leq i$ . All that remains to check is that  $G/M_i$  is abelian. Since the process terminates here,  $G/M_i$  must be simple, and thus have prime order, making it abelian. The proof of (4)  $\Rightarrow$  (1) is trivial.  $\blacksquare$

**Proposition 1.3.44.** *Suppose  $G$  is solvable, and let  $H \leq G$ . Then  $H$  is solvable.*

*Proof.* Let  $G$  have a sequence  $1 = G_0 \leq G_1 \leq \cdots \leq G_n = G$ . Consider the sequence

$$1 = H \cap G_0 \leq H \cap G_1 \leq \cdots \leq H \cap G_n = H.$$



Note that  $H \cap G_i = (H \cap G_{i+1}) \cap G_i$ . Now, by the Second Isomorphism Theorem,  $(H \cap G_{i+1}) \cap G_i = H \cap G_i \trianglelefteq G \cap G_{i+1}$ , and

$$(H \cap G_{i+1}) / (H \cap G_{i+1}) \cap G_i \cong (H \cap G_{i+1}) G_i / G_i \leq G_{i+1} / G_i,$$

is abelian. Thus,  $H$  is solvable. ■

**Proposition 1.3.45.** *Let  $G$  be a finite abelian group, and let  $n$  divide  $|G|$ . Then there is some  $H \leq G$  with  $|H| = n$ .*

*Proof.* We prove this by induction on  $|G|$ . If  $|G| = 1$ , this is trivial. Write  $n = kp$ , where  $p$  is prime. Then, there is some  $K \leq G$  of order  $p$ . Take  $G/K$ , which has order less than  $|G|$ . By the induction hypothesis, it has a subgroup  $H/K$  of order  $k$ . Thus,  $|H| = pk$ , and  $H \leq G$ . ■

### 1.3.5 Transpositions and the Alternating Group

**Definition 1.3.46.** Let  $\sigma \in S_n$  be a 2-cycle. Then it is called a transposition.

**Proposition 1.3.47.** *Every  $\sigma \in S_n$  may be written as a product of transpositions.*

*Proof.* We show that  $m$ -cycles may be written as the product of transpositions. We have that

$$(a_1 a_2 \cdots a_m) = (a_1 a_m)(a_1 a_{m-1}) \cdots (a_1 a_2).$$
■

**Definition 1.3.48.** Let  $\sigma \in S_n$ , and let

$$\Delta = \prod_{1 \leq i < j \leq n} (x_i - x_j).$$

Let

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}).$$

Notice that  $\sigma(\Delta) = \pm \Delta$ . Define the sign of  $\sigma$  to be

$$\epsilon(\sigma) = \begin{cases} +1 & \sigma(\Delta) = \Delta \\ -1 & \sigma(\Delta) = -\Delta \end{cases}.$$

If  $\epsilon(\sigma) = 1$ , then  $\sigma$  is called an even permutation, and if  $\epsilon(\sigma) = -1$ , then  $\sigma$  is called an odd permutation.

**Proposition 1.3.49.** *The map  $\epsilon : S_n \rightarrow \{\pm 1\} \cong Z_2$  is a homomorphism.*

*Proof.* We have that

$$(\tau\sigma)(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\tau\sigma(i)} - x_{\tau\sigma(j)}).$$

Suppose that  $\sigma(\Delta)$  has  $k$  factors of the form  $x_j - x_i$  with  $i < j$ , so that  $\epsilon(\sigma) = (-1)^k$ . Then,  $(\tau\sigma)(\Delta)$  has  $k$  factors of the form  $x_{\tau(j)} - x_{\tau(i)}$  with  $i < j$ . We may therefore pull these out and get a sign change of  $(-1)^k = \epsilon(\sigma)$ . This gives us that

$$(\tau\sigma)(\Delta) = \epsilon(\sigma) \prod_{1 \leq p < q \leq n} (x_{\tau(p)} - x_{\tau(q)}) = \epsilon(\sigma)\epsilon(\tau)\Delta.$$

Since  $(\tau\sigma)(\Delta) = \epsilon(\tau\sigma)\Delta$ , we have that  $\epsilon(\tau\sigma) = \epsilon(\sigma)\epsilon(\tau) = \epsilon(\tau)\epsilon(\sigma)$ . ■

**Proposition 1.3.50.** *Transpositions are odd permutations.*

*Proof.* It is not hard to see that  $\epsilon((1, 2)) = -1$ . We now reduce the general case to this. Let  $\lambda = (1i)(2j)$ . Then,  $(ij) = \lambda(12)\lambda$ . Thus,  $\epsilon((ij)) = \epsilon(\lambda(12)\lambda) = \epsilon((12))\epsilon(\lambda)^2 = -1$ . ■

**Corollary 1.3.51.** *If  $\sigma \in S_n$  can be written as the product of  $k$  transpositions and the product of  $\ell$  transpositions, then  $k \equiv \ell \pmod{2}$ .*

*Proof.* Since transpositions are odd, and since  $\epsilon$  is a homomorphism, we have that  $(-1)^k = \epsilon(\sigma) = (-1)^\ell$ , so  $k \equiv \ell \pmod{2}$ . ■

**Definition 1.3.52.** The alternating group of degree  $n$ , denoted  $A_n$ , is the kernel of the homomorphism  $\epsilon$ , or the set of even permutations.

**Proposition 1.3.53.**  $|A_n| = n!/2$ .

*Proof.* By the First Isomorphism Theorem,  $S_n/A_n \cong \epsilon(S_n) = \{\pm 1\}$ . Thus,  $|A_n| = |S_n|/2 = n!/2$ . ■

**Proposition 1.3.54.** *Let  $\sigma \in S_n$ . Then  $\sigma$  is odd if and only if the number of cycles of even length in its cycle decomposition is odd.*

*Proof.* Note that we have seen a  $m$ -cycle may be written as the product of  $m - 1$  transpositions. Thus, if  $m$  is odd, then a  $m$ -cycle does not contribute to the sign, and if  $m$  is even then the sign is multiplied by  $-1$ . Therefore, in order to have an odd permutation, the number of cycles of even length must be odd. Similarly, if the number of cycles of even length is odd, then  $\sigma$  must be an odd permutation. ■

## 1.4 Group Actions

### 1.4.1 Group Actions and Permutation Representations

**Definition 1.4.1.** Let  $G$  act on a set  $A$ . Recall that for each  $g \in G$ , there is a map  $\sigma_g : A \rightarrow A$  defined by  $a \mapsto g \cdot a$ . Recall further that there is a homomorphism  $\phi : G \rightarrow S_A$  defined by  $\phi(g) = \sigma_g$ . This homomorphism is called the permutation representation associated to the action.

*Remark 1.4.2.* If  $G$  acts on  $A$ , then the kernel of the action is the same as the kernel of the associated permutation representation. Note also that two group elements induce the same permutation on  $A$  if and only if they lie in the same coset of the kernel. We may also view a group action  $G$  on  $A$  as a faithful action of the quotient group  $G/\ker \phi$  on  $A$ .

**Proposition 1.4.3.** *For any group  $G$  and nonempty set  $A$  there is a bijection between the actions of  $G$  on  $A$  and the homomorphisms from  $G$  to  $S_A$ .*

*Proof.* We have already seen that each action induces a homomorphism. Now, let  $\phi : G \rightarrow S_A$  be a homomorphism. Then we can get an action with  $g \cdot a = \phi(g)(a)$ . We claim these are inverses. Given a homomorphism  $\phi$ , the permutation representation associated with its associated action is  $\phi(g)$ , so these are inverses. ■

**Proposition 1.4.4.** *Let  $G$  be a group acting on  $A$ . The relation on  $A$  defined by*

$$a \sim b \iff \exists g \in G, a = g \cdot b$$

*is an equivalence relation. For each  $a \in A$ , the number of elements in the equivalence class containing  $a$  is  $[G : G_a]$ .*

*Proof.* We have that  $a = 1 \cdot a$ , so  $a \sim a$ . If  $a \sim b$  with  $a = g \cdot b$ , then  $g^{-1} \cdot a = g^{-1} \cdot (g \cdot b) = (g^{-1}g) \cdot b = 1 \cdot b = b$ , so  $b \sim a$ . Finally, suppose  $a \sim b$  and  $b \sim c$  with  $a = g \cdot b$  and  $b = h \cdot c$ . Then,  $a = g \cdot (h \cdot c) = (gh) \cdot c$ , so  $a \sim c$ . We now prove a bijection between the left cosets of  $G_a$  in  $G$  and the elements of the equivalence class of  $a$ . Let  $\mathcal{C}_a$  be the equivalence class of  $a$ . Suppose that  $b = g \cdot a \in \mathcal{C}_a$ . Then  $gG_a$  is a left coset of  $G_a$  in  $G$ . The map  $b = g \cdot a \mapsto gG_a$  is a map from  $\mathcal{C}_a$  to the set of left cosets of  $G_a$  in  $G$ . This is clearly surjective. Suppose that  $g \cdot a$  and  $h \cdot a$  map to the same element, so that  $gG_a = hG_a$ . Then,  $h^{-1}g \in G_a$ , so  $g \cdot a = h \cdot a$ . ■

**Definition 1.4.5.** Let  $G$  act on a set  $A$ . The equivalence class  $\{g \cdot a : g \in G\}$  is called the orbit of  $G$  containing  $a$ .

**Definition 1.4.6.** Let  $G$  act on  $A$ . Then the action is said to be transitive if there is only one orbit, so that if  $a, b \in A$  then there is some  $g \in G$  so that  $a = g \cdot b$ .

**Proposition 1.4.7.** *Let  $\sigma \in S_n$ . Then its decomposition into disjoint cycles is unique (up to rearrangement and cyclic permutations of the elements in a cycle).*

*Proof.* Let  $A = \{1, 2, \dots, n\}$ , and let  $G = \langle \sigma \rangle$ . Then  $G$  acts on  $A$ , and so it partitions  $A$  into a unique set of disjoint orbits. Let  $\mathcal{O}$  be one of these orbits, and let  $x \in \mathcal{O}$ . Then, there is a bijection between the left cosets of  $G_x$  in  $G$  and elements of  $\mathcal{O}$ , given by  $\sigma^i x \mapsto \sigma^i G_x$ . Since  $G$  is cyclic,  $G_x \trianglelefteq G$ . We claim that  $G/G_x$  is cyclic: it must have elements of the form  $\sigma^i G_x$ , and if  $d$  is the smallest positive integer for which  $\sigma^d \in G_x$ , then these are  $G_x, \sigma G_x, \dots, \sigma^{d-1} G_x$ . Thus, it is cyclic of order  $d$ . We also have  $d = [G : G_x] = |\mathcal{O}|$ , and the distinct elements of  $\mathcal{O}$  are  $x, \sigma x, \dots, \sigma^{d-1} x$ . Thus, on an orbit of size  $d$ ,  $\sigma$  acts as a  $d$ -cycle. Now, the orbits of  $G$  are uniquely determined by  $\sigma$ . Also, within each orbit  $\mathcal{O}$ , we may begin with any element, which simply produces a cyclic permutation of terms in the original list. ■

**Definition 1.4.8.** Subgroups of symmetric groups are called permutation groups.

**Proposition 1.4.9.** *Let  $G$  act on  $A$ . If  $a, b \in A$  and  $b = g \cdot a$  for some  $g \in G$ , then  $G_b = gG_a g^{-1}$ . In particular, if  $G$  acts transitively on  $A$ , then the kernel of the action is  $\bigcap_{g \in G} gG_a g^{-1}$ .*

*Proof.* Suppose that  $h \in G_b$ , so that  $h \cdot b = b$ . We then have that  $h \cdot (g \cdot a) = g \cdot a$ , so that  $(hg) \cdot a = g \cdot a$ . Thus,  $(g^{-1}hg) \cdot a = a$ , so  $h \in gG_ag^{-1}$ . Now, suppose that  $h \in gG_ag^{-1}$ , with  $h = gh'g^{-1}$ . Then,  $h \cdot b = h \cdot (g \cdot a) = (gh'g^{-1}) \cdot (g \cdot a) = g \cdot (h' \cdot a) = g \cdot a = b$ . Now, the kernel is the intersection of the stabilizers, so if for every  $b \in A$ ,  $b = g \cdot a$  for some  $g \in G$ , we have that  $\bigcap_{b \in A} G_b = \bigcap_{g \in G} gG_ag^{-1}$ . ■

**Proposition 1.4.10.** *Let  $G$  be a permutation group on the set  $A$  ( $G \leq S_A$ ), let  $\sigma \in G$ , and let  $a \in A$ . Then  $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$ . In particular, if  $G$  acts transitively on  $A$ , then  $\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1$ .*

*Proof.* The first statement follows immediately from the last Proposition:  $\sigma(a) = \sigma \cdot a$ , so  $G_{\sigma(a)} = \sigma G_a \sigma^{-1}$ . Note that  $G$  must act faithfully on  $A$ , since  $S_A$  does. Thus, if  $G$  acts transitively on  $A$ , then the intersection given is just the kernel, which is 1. ■

**Proposition 1.4.11.** *Let  $G \leq S_A$  be abelian and transitive. Then  $\sigma(a) \neq a$  for all  $\sigma \in G \setminus \{1\}$  and all  $a \in A$ . In particular,  $|G| = |A|$ .*

*Proof.* Suppose that  $\sigma(a) = a$  for some  $a$ . Since  $G$  is transitive, for any  $b \in A$ ,  $b = g \cdot a$  for some  $g \in G$ . Thus,  $\sigma \cdot b = (\sigma g) \cdot a = (g\sigma) \cdot a = g \cdot a$ , so  $\sigma \in G_b = gG_ag^{-1}$ . Therefore,  $\sigma \in \bigcap_{g \in G} gG_ag^{-1} = 1$ , so  $\sigma = 1$ . Now, suppose that  $\sigma(a) = \tau(a)$  for some  $a \in A$  and  $\sigma, \tau \in G$ . Then,  $\sigma\tau^{-1} = 1$ , so  $\sigma = \tau$ . Since  $G$  acts transitively, each element  $b \in A$  corresponds at least element of  $G$ : an element  $g \in G$  so that  $b = g \cdot a$ . We have just seen that this  $g$  must be unique, and if  $b$  and  $c$  have the same corresponding element, then  $b = c$ , so the correspondence is injective. It is also surjective, since  $g \cdot a$  corresponds to  $g$ . Thus, there is a bijection between elements of  $A$  and elements of  $G$ , so  $|G| = |A|$ . ■

**Definition 1.4.12.** Let  $G$  be a transitive permutation group on the finite set  $A$ . A block is a nonempty subset  $B \subseteq A$  so that for all  $\sigma \in G$ , either  $\sigma(B) = B$  or  $\sigma(B) \cap B = \emptyset$ .

**Proposition 1.4.13.** *Let  $G$  act on  $A$ , and let  $B$  be a block with  $a \in B$ . Then  $G_B = \{\sigma \in G : \sigma(B) = B\}$  is a subgroup of  $G$  containing  $G_a$ .*

*Proof.* Suppose that  $\sigma, \tau \in G_B$ . Then,  $(\sigma\tau)(B) = \sigma(\tau(B)) = \sigma(B) = B$ . We also have that  $\sigma^{-1}(B) = \sigma^{-1}(\sigma(B)) = B$ . Now, if  $\sigma(a) = a$ , then  $a \in \sigma(B) \cap B$ , meaning that  $\sigma \in G_B$ , so  $G_a \leq G_B$ . ■

**Proposition 1.4.14.** *Let  $B$  be a block with distinct images  $\sigma_1(B), \sigma_2(B), \dots, \sigma_n(B)$ . Then these partition  $A$ .*

*Proof.* Suppose that  $a \in \sigma_i(B) \cap \sigma_j(B)$ . Then,  $\sigma_i^{-1} \circ \sigma_j(B) = B$ , as  $a \in \sigma_j(B)$ , and  $\sigma_i^{-1}(a) \in B$ . Thus, for  $b \in B$ ,  $\sigma_i^{-1} \circ \sigma_j(b) = b$ , so  $\sigma_j(b) = \sigma_i(b)$ . ■

**Definition 1.4.15.** A transitive group  $G$  acting on a set  $A$  is said to be primitive if the only blocks in  $A$  have size 1, or are  $A$ .

**Proposition 1.4.16.** *Let  $G$  act on  $A$  transitively. Then  $G$  is primitive if and only if, for each  $a \in A$ , the only subgroups of  $G$  containing  $G_a$  are  $G_a$  and  $G$ .*

*Proof.* Suppose that there is some  $G_a \leq H \leq G$ . We claim that  $H(a) = \{h \cdot a : h \in H\}$  is a block. Suppose that  $\sigma(h \cdot a) = h' \cdot a$  for some  $h, h' \in H$  and  $\sigma \in G$ . Then,  $h'^{-1}\sigma h \cdot a = a$ , so  $h'^{-1}\sigma h \in G_a \leq H$ , so  $\sigma \in H$ . Thus,  $\sigma(H(a)) = H(a)$ , meaning that  $H(a)$  is a block. Now, suppose there is a nontrivial block  $B$ . Take  $G_B$ . We have seen that  $G_a \leq G_B$ . Note that if  $G_B = G$ , then  $G$  would not act transitively, so  $G_B \neq G$ . This completes the proof. ■

**Proposition 1.4.17.** *Let  $G$  act transitively on  $A$  with the property that  $G_a$  acts transitively on  $A \setminus \{a\}$  for all  $a \in A$ . Then the action is primitive.*

*Proof.* Suppose that there is a nontrivial block  $B$ . Take  $a \in B$ . Then  $G_a \leq G_B$ . Thus, for  $b \in B \setminus \{a\}$  and  $g \in G_a$ ,  $gb \in B$ . Since we chose  $B$  to be nontrivial, there is at least one  $c \in A \setminus B$ , meaning that  $G_a$  does not act transitively on  $A \setminus \{a\}$ . ■

## 1.4.2 Groups Acting on Themselves by Left Multiplication

**Definition 1.4.18.** A group  $G$  is said to act on itself by left multiplication if  $G$  acts on the set  $A = G$  with  $g \cdot a = ga$  for  $g, a \in G$ . Note that any group  $G$  may act on itself by left multiplication.

**Proposition 1.4.19.** *Let  $G$  act on itself by left multiplication. Then it is transitive, faithful, and the stabilizer of any point is the group 1.*

*Proof.* We have that  $(hg^{-1}) \cdot g = h$  for any  $g, h \in G$ . Now, if  $gh = h$  for some  $g, h \in G$  (so that  $g \in G_h$ ), then  $g = 1$ . ■

*Remark 1.4.20.* If  $H \leq G$  and  $A$  is the set of left cosets of  $H$  in  $G$ , then  $G$  may act on  $A$  by  $g \cdot aH = gaH$ . In the case  $H = 1$ , this is the same as  $G$  acting on itself by left multiplication.

**Theorem 1.4.21.** *Let  $G$  be a group,  $H \leq G$ ,  $A$  the set of left cosets of  $H$ , and let  $G$  act on  $A$  as described above. Let  $\pi_H$  be the associated permutation representation of this action. Then,*

- (1)  $G$  acts transitively on  $A$ .
- (2) The stabilizer in  $G$  of  $H = 1H \in A$  is  $H$ .
- (3) The kernel of the action is  $\bigcap_{x \in G} xHx^{-1}$ , and  $\ker \pi_H$  is the largest normal subgroup of  $G$  contained in  $H$ .

*Proof.*

- (1) Note that  $(hg^{-1}) \cdot gH = hH$  for any  $g, h \in G$ .
- (2) If  $h \in H$ , then  $h$  clearly stabilizes  $H$ . If  $aH = H$ , then  $a \in H$ .

(3) Recall that the kernel of the action is just  $\ker \pi_H$ . We have

$$\begin{aligned} \ker \pi_H &= \{g \in G : \forall x \in G, gxH = xH\} \\ &= \{g \in G : \forall x \in G, x^{-1}gxH = H\} \\ &= \{g \in G : \forall x \in G, x^{-1}gx \in G\} \\ &= \{g \in G : \forall x \in G, g \in xHx^{-1}\} \\ &= \bigcap_{x \in G} xHx^{-1}. \end{aligned}$$

Now, note that  $\ker \pi_H \trianglelefteq G$ , and that  $\ker \pi_H \leq H$ . Now, if  $N \trianglelefteq G$  is contained in  $H$ , then  $N = xNx^{-1} \leq xHx^{-1}$  for all  $x \in G$ , so  $N \leq \bigcap_{x \in G} xHx^{-1} = \ker \pi_H$ . ■

**Corollary 1.4.22** (Cayley's Theorem). *Every group is isomorphic to a subgroup of some symmetric group. If  $G$  is a group with  $|G| = n$ , then  $G$  is isomorphic to a subgroup of  $S_n$ .*

*Proof.* Let  $H = 1$ , and use the preceding theorem to obtain a homomorphism from  $G$  to  $S_G$ . Since its kernel is contained in  $H = 1$ , the homomorphism must be injective, meaning that  $G$  is isomorphic to its image. ■

**Definition 1.4.23.** Let  $G$  be a group. The left regular representation of  $G$  is the permutation representation from left multiplication on the elements of  $G$  (or cosets of  $H = 1$ ).

**Corollary 1.4.24.** *If  $G$  is a finite group of order  $n$ , and  $p$  is the smallest prime dividing  $|G|$ . Let  $H \leq G$  with  $[G : H] = p$ . Then  $H \trianglelefteq G$ .*

*Proof.* Let  $\pi_H$  be the permutation representation from multiplication on the set of left cosets of  $H$  in  $G$ . Let  $K = \ker \pi_H$ , and let  $[H : K] = k$ . Then  $[G : K] = [G : H][H : K] = pk$ . Now,  $G/K$  is isomorphic to the image of  $G$  under  $\pi_H$ , which is a subgroup of  $S_p$ . Thus,  $pk|p|$ , meaning that  $k|(p-1)!$ . All prime divisors of this are less than  $p$ , and  $p$  is minimal, so  $k = 1$ . Thus,  $H = K \trianglelefteq G$ . ■

### 1.4.3 Groups Acting on Themselves by Conjugation

**Definition 1.4.25.** A group  $G$  is said to act on itself by conjugation if it acts on the set  $A = G$  by  $g \cdot a = gag^{-1}$  for all  $g, a \in G$ .

**Definition 1.4.26.** Two elements  $a, b \in G$  are said to be conjugate in  $G$  if there is some  $g \in G$  such that  $b = gag^{-1}$  (which is equivalent to them being in the same orbit of  $G$  acting on itself by conjugation). The orbits of  $G$  acting on itself by conjugation are called conjugacy classes of  $G$ .

*Example 1.4.27.* If  $G$  acts on itself by conjugation, then it is not transitive, as 1 always maps to itself. In particular, if we let  $G$  be abelian, then the action of  $G$  on itself by conjugation is trivial.

*Remark 1.4.28.* If  $S \subseteq G$ , then this can be generalized by mapping  $S$  to  $gSg^{-1}$ . This acts  $G$  on the set  $\mathcal{P}(G)$ . Note that the action of  $G$  on subsets of  $\mathcal{P}(G)$  with cardinality 1 is analogous to  $G$  acting on itself by conjugation, so this action may be considered an extension of the action of  $G$  on itself by conjugation.

**Definition 1.4.29.** Two subsets  $S, T \subseteq G$  are said to be conjugate in  $G$  if there is some  $g \in G$  with  $T = gSg^{-1}$  (which is the same as them being in the same orbit of  $G$  acting on  $\mathcal{P}(G)$  by conjugation).

**Proposition 1.4.30.** *The number of conjugates of  $S \subseteq G$  is  $[G : N_G(S)]$ . In particular, the number of conjugates of an element  $s$  is  $[G : C_G(s)]$ .*

*Proof.* Let  $G$  act on  $\mathcal{P}(G)$  by conjugation. Recall, by Proposition 1.4.4, that the size of the orbit of  $S \in \mathcal{P}(G)$  is  $[G : G_S] = [G : N_G(S)]$ , which proves the first part. Now,  $N_G(\{s\}) = C_G(s)$ , proving the second part. ■

**Theorem 1.4.31** (The Class Equation). *Let  $G$  be a finite group, and let  $g_1, g_2, \dots, g_r$  be representatives of the distinct conjugacy classes of  $G$  not contained in the center  $Z(G)$  of  $G$ . Then,*

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)].$$

*Proof.* Note that  $\{x\}$  is a conjugacy class of size 1 if and only if  $x \in Z(G)$ . Let  $Z(G) = \{1, z_2, \dots, z_m\}$ , and let  $\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_r$  be the conjugacy classes of  $G$  not contained in the center. Let  $g_i$  be a representative of  $\mathcal{K}_i$  for each  $i$ . Let  $\mathcal{C}$  denote the set of conjugacy classes of  $G$ . Then these partition  $G$ , so

$$\begin{aligned} |G| &= \sum_{C \in \mathcal{C}} |C| \\ &= \sum_{C \in Z(G)} 1 + \sum_{C \in \mathcal{C} \setminus Z(G)} |C| \\ &= |Z(G)| + \sum_{i=1}^r |\mathcal{K}_i| \\ &= |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)]. \end{aligned}$$

■

*Remark 1.4.32.* Note that all of the summands on the right hand side of the class equation divide  $|G|$ .

**Theorem 1.4.33.** *Let  $p$  be a prime, and  $P$  be a group of order  $p^\alpha$  for some  $\alpha \geq 1$ . Then  $P$  has a nontrivial center.*

*Proof.* By the class equation,

$$|P| = |Z(P)| + \sum_{i=1}^r [P : C_P(g_i)],$$

where  $g_1, g_2, \dots, g_r$  are representatives of the distinct conjugacy classes not contained in the center. Since  $C_P(g_i) \neq P$ ,  $p \nmid [P : C_P(g_i)]$ . Since  $p \mid |P|$  as well, it follows that  $p \mid |Z(P)|$ . Since  $1 \in Z(P)$ , we have that  $|Z(P)| > 0$ , meaning that the center must be nontrivial. ■

**Corollary 1.4.34.** *If  $|P| = p^2$  for some prime  $p$ , then  $P$  is abelian. More precisely,  $P$  is isomorphic to either  $Z_{p^2}$  or  $Z_p \times Z_p$ .*

*Proof.* Since  $Z(P) \neq 1$ ,  $P/Z(P)$  is cyclic. Let  $Z = Z(P)$ , and let  $xZ$  be the generator of  $P/Z$ . Then, for any  $g \in P$ , there is some  $z \in Z$  and  $n \in \mathbb{N}$  so that  $gz = x^n$ . Take  $g, h \in P$  and write  $gz_1 = x^n$ ,  $hz_2 = x^m$ . Then,

$$ghz_1z_2 = gz_1hz_2 = x^n x^m = x^m x^n = hz_2gz_1 = hgz_1z_2,$$

so  $gh = hg$ , meaning that  $P$  is abelian. If  $g \in P$  has order  $p^2$ , then  $P$  is cyclic. Assume that every  $g \neq 1 \in P$  has order  $p$ . Take  $x \neq 1 \in P$  and  $y \neq 1 \in P \setminus \langle x \rangle$ . Then  $|\langle x, y \rangle| > |\langle x \rangle|$ , so  $P = \langle x, y \rangle$ . We also have that  $\langle x \rangle \times \langle y \rangle \cong Z_p \times Z_p$ . Now, since  $xy = yx$ , the map  $(x^a, y^b) \mapsto x^a y^b$  is an isomorphism from  $\langle x \rangle \times \langle y \rangle$  to  $P$ . ■

**Proposition 1.4.35.** *Let  $\sigma, \tau \in S_n$ , and suppose that  $\sigma$  has cycle decomposition*

$$(a_1 a_2 \cdots a_{k_1})(b_1 b_2 \cdots b_{k_2}) \cdots$$

*Then  $\tau\sigma\tau^{-1}$  has cycle decomposition*

$$(\tau(a_1)\tau(a_2) \cdots \tau(a_{k_1}))(\tau(b_1)\tau(b_2) \cdots \tau(b_{k_2})) \cdots$$

*Proof.* If  $\sigma(i) = j$ , then  $(\tau\sigma\tau^{-1})(\tau(i)) = \tau(j)$ . Thus, if the ordered pair  $i, j$  appears in the cycle decomposition of  $\sigma$ , then the ordered pair  $\tau(i), \tau(j)$  appears in the cycle decomposition of  $\tau\sigma\tau^{-1}$ . ■

**Definition 1.4.36.** If  $\sigma \in S_n$  is the product of disjoint cycles of lengths  $n_1, n_2, \dots, n_r$  with  $n_1 \leq n_2 \leq \cdots \leq n_r$  (including cycles of length 1), then the integers  $n_1, n_2, \dots, n_r$  are called the cycle type of  $\sigma$ .

**Definition 1.4.37.** Let  $n \in \mathbb{Z}^+$ . A partition of  $n$  is any nondecreasing sequence of positive integers whose sum is  $n$ .

**Proposition 1.4.38.** *Two elements of  $S_n$  are conjugate if and only if they have the same cycle type. The number of conjugacy classes of  $S_n$  equals the number of partitions of  $n$ .*

*Proof.* By the previous proposition, conjugate permutations have the same cycle type. Now, suppose that  $\sigma_1, \sigma_2 \in S_n$  have the same cycle type. Order the cycles in nondecreasing length, including 1-cycles. If we ignore parenthesis, we get a list in which all the integers from 1 to  $n$  appear exactly once. Define  $\sigma_1^i$  to be the  $i$ th number in the list obtained from the cycle decomposition of  $\sigma_1$ , and similarly define  $\sigma_2^i$ . Let  $\tau : \sigma_1^i \mapsto \sigma_2^i$ . Then  $\tau \in S_n$  and  $\tau\sigma_1\tau^{-1} = \sigma_2$ . Now, the number of conjugacy classes is equal to the number of cycle types, which is just the number of partitions of  $n$ . ■



**Proposition 1.4.39.** *Let  $\sigma \in S_n$  be an  $m$ -cycle. Then  $C_{S_n}(\sigma) = \{\sigma^i \tau : 0 \leq i \leq m-1, \tau \in S_{n-m}\}$ , where  $S_{n-m}$  denotes the subgroup of  $S_n$  fixing all integers that  $\sigma$  permutes.*

*Proof.* Let us denote the group by  $A$ . Then, it is clear that  $A \subseteq C_{S_n}(\sigma)$ . Now, it is not hard to see that the number of conjugates of  $\sigma$  is

$$\frac{n \cdot (n-1) \cdot \dots \cdot (n-m+1)}{m}.$$

This is equal to  $[S_n : C_{S_n}(\sigma)] = \frac{|S_n|}{|C_{S_n}(\sigma)|}$ . Thus,

$$|C_{S_n}(\sigma)| = \frac{m|S_n|}{n \cdot (n-1) \cdot \dots \cdot (n-m+1)} = m \cdot (n-m)!.$$

It is not hard to check that  $|A| = m \cdot (n-m)!$ , so this completes the proof. ■

**Proposition 1.4.40.** *Let  $H \trianglelefteq G$ , and let  $\mathcal{K}$  be an arbitrary conjugacy class of  $G$ . Then either  $\mathcal{K} \subseteq H$  or  $\mathcal{K} \cap H = \emptyset$ .*

*Proof.* Let  $x \in \mathcal{K} \cap H$ . Then,  $gxg^{-1} \in gHg^{-1} = H$  for all  $g \in G$ . Thus,  $\mathcal{K} \subseteq H$ . ■

**Theorem 1.4.41.**  *$A_5$  is simple.*

*Proof.* We will first find the conjugacy classes of  $A_5$  and their orders. The conjugacy classes of  $S_5$  have the following cycle types.

|               |                |
|---------------|----------------|
| 1, 1, 1, 1, 1 | In $A_5$       |
| 1, 1, 3       | In $A_5$       |
| 5             | In $A_5$       |
| 1, 2, 2       | In $A_5$       |
| 1, 1, 1, 2    | Not in $A_5$   |
| 1, 4          | Not in $A_5$   |
| 2, 3          | Not in $A_5$ . |

The representatives of those in  $A_5$  can be chosen to be 1, (123), (12345) and (12)(34). The centralizer of a 3-cycle and a 5-cycle in  $S_5$  has been determined above. Intersecting with  $A_5$  gives

$$C_{A_5}((123)) = \langle (123) \rangle \quad \text{and} \quad C_{A_5}((12345)) = \langle (12345) \rangle.$$

These groups have orders 3 and 5 respectively, so there  $60/3 = 20$  conjugates of (123) and  $60/5 = 12$  conjugates of (12345) in  $A_5$ . Now, there are  $2\binom{5}{3} = 20$  3-cycles in  $S_5$ , so all of these are conjugate in  $A_5$ . However, there are  $4! = 24$  5-cycles in  $S_5$  (and so in  $A_5$ ), meaning that there is some 5-cycle  $\sigma$  that is not conjugate to (12345) in  $A_5$ . Similar to above,  $\sigma$  has 12 distinct conjugates in  $A_5$ . Thus, there are 2 conjugacy classes of 5-cycles in  $A_5$ . Since 3-cycles and 5-cycles account for all nonidentity elements with odd order, and since  $A_5$  has no elements with order 4, the remaining 15 nonidentity elements of  $A_5$  must have order 2, and thus have cycle type 2, 2. It is not hard to see that (12)(34) commutes with (13)(24), and that it does not commute with any element of odd order in  $A_5$ . Thus,  $|C_{A_5}((12)(34))| = 4$

(if it was a higher divisor of 60, then it would contain either an element of order 3 or one of order 5). Therefore, all 15 elements of order 2 in  $A_5$  are conjugate to  $(12)(34)$ . Put together, the conjugacy classes in  $A_5$  have orders 1, 14, 20, 12, 12. If  $H \trianglelefteq A_5$ , then it would be the union of conjugacy classes of  $A_5$  (and thus be the sum of the elements some subset of  $\{1, 12, 12, 15, 20\}$ ), and its order would have to divide 60. A check shows that the only possibilities are  $|H| = 1$  and  $|H| = 60$ , so  $A_5$  has no proper nontrivial normal subgroups. ■

*Remark 1.4.42.* Recall that we may define a right group action of  $G$  on  $A$  as a map from  $A \times G$  to  $A$ , denoted  $a \cdot g$ , so that  $(a \cdot g_1) \cdot g_2 = a \cdot (g_1 g_2)$ , and that  $a \cdot 1 = a$ . Many times, conjugation is written as a right group action with  $a^g = g^{-1} a g$ , and similarly  $S^g = g^{-1} S g$ .

It is also not hard to check that a left action can be transformed into a right action by  $a \cdot g = g^{-1} \cdot a$ , and that a right action can be transformed into a left action similarly. Thus, two elements or subsets of a group are “left conjugate” if and only if they are “right conjugate”. More generally, it is also not hard to see that, for any corresponding left and right actions, the orbits are the same.

### 1.4.4 Automorphisms

**Definition 1.4.43.** Let  $G$  be a group. An isomorphism from  $G$  to itself is called an automorphism of  $G$ . The set of all automorphisms of  $G$  is denoted  $\text{Aut}(G)$ .

**Proposition 1.4.44.** Let  $G$  be a group. Then  $\text{Aut}(G)$  is a group under composition.

*Proof.* Compositions of isomorphisms, which are bijective, are isomorphisms. Function composition is associative, and the map  $f : g \mapsto g$  is in  $\text{Aut}(G)$ . ■

*Remark 1.4.45.* Note that automorphisms of  $G$  permute the elements of  $G$ , so  $\text{Aut}(G) \leq S_G$ .

**Proposition 1.4.46.** Let  $H \trianglelefteq G$ . Then the action of  $G$  on  $H$  by conjugation is defined as  $g \cdot h = ghg^{-1}$ . For each  $g \in G$ , conjugation by  $g$  is an automorphism of  $H$ . The permutation representation from this action is a homomorphism from  $G$  onto  $\text{Aut}(H)$  with kernel  $C_G(H)$ .

*Proof.* Let  $\varphi_g$  denote conjugation by  $g$ . It is clear that  $\varphi_1 = 1$ , and that  $\varphi_a \circ \varphi_b = \varphi_{ab}$ . Since  $\varphi_g$  has a 2-sided inverse  $\varphi_{g^{-1}}$ , it is bijective. Further,  $ghkg^{-1} = ghg^{-1}gkg^{-1}$ , so  $\varphi_g$  is a homomorphism, and thus an automorphism. Now, we have already proved that the map  $\phi : G \rightarrow S_H$  defined by  $\phi(g) = \varphi_g$  (which we have shown is a homomorphism) has an image contained in  $\text{Aut}(H) \leq S_H$ . It is clear that the kernel is  $C_G(H)$ . ■

**Corollary 1.4.47.** If  $K \leq G$  and  $g \in G$ , then  $K \cong gKg^{-1}$ . In particular, conjugate elements and conjugate subgroups have the same order.

*Proof.* Let  $G = H$  in the previous proposition. Since isomorphisms send subgroups to subgroups, this proves the first part. To see that conjugate elements have the same order, let  $K = \langle x \rangle$ , and notice that  $g\langle x \rangle g^{-1} = \langle xg^{-1} \rangle$ . ■

**Corollary 1.4.48.** Let  $H \leq G$ . Then  $N_G(H)/C_G(H)$  is isomorphic to a subgroup of  $\text{Aut}(H)$ . In particular,  $G/Z(G)$  is isomorphic to a subgroup of  $\text{Aut}(G)$ .

*Proof.* Since  $H \trianglelefteq N_G(H)$ , the proposition (applied with  $N_G(H)$  rather than with  $G$ ) implies the first part. For the second part, let  $H = G$ . ■

**Definition 1.4.49.** Let  $G$  be a group and  $g \in G$ . Conjugation by  $g$  is called an inner automorphism of  $G$ , and the subgroup of  $\text{Aut}(G)$  consisting of all inner automorphisms is denoted by  $\text{Inn}(G)$ .

**Proposition 1.4.50.**  $\text{Inn}(G) \cong G/Z(G)$ .

*Proof.* Let  $\phi : G \rightarrow \text{Aut}(G)$  be defined by  $g \mapsto (h \mapsto ghg^{-1})$ . The kernel of this is  $Z(G)$  and the image is  $\text{Inn}(G)$ , so  $G/Z(G) \cong \text{Inn}(G)$ . ■

**Definition 1.4.51.** A subgroup  $H$  of  $G$  is called characteristic in  $G$ , denoted  $H \text{ char } G$ , if  $\sigma(H) = H$  for all  $\sigma \in \text{Aut}(G)$ .

**Proposition 1.4.52.** *The following are true.*

- (1) *If  $H \text{ char } G$ , then  $H \trianglelefteq G$ .*
- (2) *If  $H$  is the unique subgroup of  $G$  with order  $|H|$ , then  $H \text{ char } G$ .*
- (3) *If  $K \text{ char } H$  and  $H \trianglelefteq G$ , then  $K \trianglelefteq G$ .*

*Proof.*

- (1) Since  $\text{Inn}(G) \leq \text{Aut}(G)$ ,  $gHg^{-1} = H$  for all  $g \in G$ .
- (2) Isomorphisms map subgroups to subgroups of the same order. Thus, if  $H$  is the only subgroup with its order, it must be mapped to itself by any automorphism.
- (3) The map  $h \mapsto ghg^{-1}$  for  $h \in H$  and  $g \in G$  is an automorphism of  $H$ . Thus,  $gKg^{-1} = K$ , so  $K \trianglelefteq G$ . ■

**Proposition 1.4.53.**  $\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ .

*Proof.* Let  $x$  be a generator of  $Z_n$ . If  $\psi \in \text{Aut}(Z_n)$ , then  $\psi(x) = x^a$  for some  $a \in \mathbb{Z}$ . This integer  $a$  uniquely determines  $\psi$ . We denote this as  $\psi_a$ . Since  $|x| = n$ , the integer  $a$  is only defined mod  $n$ . Now, we must have  $|x| = |x^a|$ , so  $\gcd(a, n) = 1$ . Further, for every  $a$  that is relatively prime to  $n$ , the map  $x \mapsto x^a$  is an automorphism of  $Z_n$ . Thus, we have a surjective map  $\Psi : \text{Aut}(Z_n) \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times$  defined by  $\psi_a \mapsto a \pmod n$ . It is not hard to check that  $\Psi$  is a homomorphism:

$$\psi_a \circ \psi_b(x) = \psi_a(x^b) = (x^b)^a = x^{ab} = \psi_{ab}(x).$$

Since this is clearly injective,  $\Psi$  is an isomorphism. ■

**Proposition 1.4.54.** *Let  $|G| = pq$  where  $p$  and  $q$  are primes with  $p \leq q$  and  $p \nmid q-1$ . Then  $G$  is abelian.*

*Proof.* If  $Z(G) \neq 1$ , then  $G/Z(G)$  is cyclic, meaning that  $G$  is abelian. Now, assume that  $Z(G) = 1$ . If every nonidentity element has order  $p$ , then their centralizers have index  $q$ , so the class equation states that  $pq = 1 + kq$  for some  $k$ . This is clearly impossible, so there is some  $x \in G$  with  $|x| = q$ . Let  $H = \langle x \rangle$ . Since  $[G : H] = p$ , and  $p$  is the smallest prime dividing  $|G|$ ,  $H \trianglelefteq G$ . Now,  $H \leq C_G(H) \leq G$ , and  $H$  is the maximal proper subgroup. If  $C_G(H) = G$ , then  $x \in Z(G)$ , so  $C_G(H) = H$ . Thus,  $G/H = N_G(H)/C_G(H)$  is a group of order  $p$  that is isomorphic to a subgroup of  $\text{Aut}(H)$ . However,  $|\text{Aut}(H)| = \varphi(q) = q - 1$ , meaning that  $p|q - 1$ , a contradiction. ■

### 1.4.5 Sylow's Theorem

**Definition 1.4.55.** Let  $G$  be a group and  $p$  be a prime. A group of order  $p^\alpha$  for some  $\alpha \geq 1$  is called a  $p$ -group. Subgroups of  $G$  which are  $p$ -groups are called  $p$ -subgroups.

**Definition 1.4.56.** Let  $p$  be a prime, and let  $G$  be a group of order  $p^\alpha m$ , where  $p \nmid m$ . Then, a subgroup of order  $p^\alpha$  is called a Sylow  $p$ -subgroup of  $G$ . The set of Sylow  $p$ -subgroups of  $G$  is denoted  $\text{Syl}_p(G)$ , and  $|\text{Syl}_p(G)|$  is denoted  $n_p(G)$  (or just  $n_p$  when  $G$  is clear).

**Lemma 1.4.57.** Let  $P \in \text{Syl}_p(G)$ . If  $Q \leq G$  is a  $p$ -subgroup, then  $Q \cap N_G(P) = Q \cap P$ .

*Proof.* Let  $H = Q \cap N_G(P)$ . Since  $P \leq N_G(P)$ ,  $P \cap Q \leq H$ . Now, we show that  $H \leq P \cap Q$ . Since  $H \leq Q$ , we must show that  $H \leq P$ . We show that  $PH \leq G$  is a  $p$ -subgroup containing both  $P$  and  $H$ . The maximality of  $P$  implies that  $PH = P$ , so  $H \leq P$ .

Since  $H \leq N_G(P)$ ,  $PH \leq G$  by Corollary 1.3.27. We have that

$$|PH| = \frac{|P| \cdot |H|}{|P \cap H|}.$$

Both  $|P|$  and  $|H|$  are powers of  $p$ , and  $|P \cap H| \leq |H|$ , so  $|PH|$  is a power of  $p$ , meaning that  $PH$  is a  $p$ -subgroup. Since  $1 \in P, H$ ,  $P, H \leq PH$ . ■

**Theorem 1.4.58** (Sylow's Theorem). Let  $G$  be a group of order  $p^\alpha m$ , where  $p$  is a prime not dividing  $m$ . Then,

- (1) Sylow  $p$ -subgroups of  $G$  exist. In other words,  $\text{Syl}_p(G) \neq \emptyset$ .
- (2) If  $P$  is a Sylow  $p$ -subgroup of  $G$ , and  $Q$  is any  $p$ -subgroup of  $G$ , then there is some  $g \in G$  so that  $Q \leq gPg^{-1}$ . In particular, any two Sylow  $p$ -subgroups of  $G$  are conjugate in  $G$ .
- (3) The number of Sylow  $p$ -subgroups of  $G$  is of the form  $1 + kp$ . In other words,  $n_p \equiv 1 \pmod{p}$ . Further,  $n_p = [G : N_G(P)]$  for any Sylow  $p$ -subgroup  $P$ , and  $n_p | m$ .

*Proof.*

- (1) We prove this by induction on  $|G|$ . If  $|G| = 1$  there is nothing to prove. Now suppose that this is true for all groups of order less than  $|G|$ . If  $p$  divides  $|Z(G)|$ , then  $Z(G)$  has a subgroup  $N$  of order  $p$ . Let  $\overline{G} = G/N$ . Then,  $|\overline{G}| = p^{\alpha-1}m$ . By induction, there is some  $\overline{P} \leq \overline{G}$  with  $|\overline{P}| = p^{\alpha-1}$ . Let  $N \leq P \leq G$  with  $\overline{P} = P/N$ . Then  $|P| = |P/N| \cdot |N| = p^\alpha$ , so  $P$  is a Sylow  $p$ -subgroup of  $G$ . Now suppose that  $p$  does not divide  $|Z(G)|$ . If  $g_1, g_2, \dots, g_r$  are representatives of the distinct conjugacy classes not contained in the center of  $G$ , then

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)].$$

Since  $p$  divides  $|G|$ , it cannot divide  $[G : C_G(g_i)]$  for all  $i$ , as this would force  $p$  to divide  $|Z(G)|$ . Take  $i$  so that  $p \nmid [G : C_G(g_i)]$ . Let  $H = C_G(g_i)$ . Then  $|H| = p^\alpha k$ , where  $p \nmid k$ . Since  $g_i \notin Z(G)$ , we must have that  $|H| < |G|$ . By induction, there is a Sylow  $p$ -subgroup  $P$  of  $H$ . This is also a subgroup of  $G$ , and since  $|P| = p^\alpha$ , it must be a Sylow  $p$ -subgroup of  $G$ .

- (2) Given a Sylow  $p$ -subgroup  $P \leq G$ , let  $\mathcal{S} = \{P_1, P_2, \dots, P_r\}$  be the set of all conjugates of  $P$ . Let  $Q$  be a  $p$ -subgroup of  $G$ . Then  $G$ , and hence  $Q$ , acts on  $\mathcal{S}$  by conjugation. We write  $\mathcal{S} = \mathcal{O}_1 \cup \mathcal{O}_2 \cup \dots \cup \mathcal{O}_s$ , where the  $\mathcal{O}_i$ 's are the disjoint orbits under this action by  $Q$ . Note that  $r = |\mathcal{O}_1| + |\mathcal{O}_2| + \dots + |\mathcal{O}_s|$ . By reordering the elements of  $\mathcal{S}$ , we may assume that  $P_i \in \mathcal{O}_i$  for  $1 \leq i \leq s$ . Then, by Proposition 1.4.4,  $|\mathcal{O}_i| = [Q : N_Q(P_i)]$ . Now,  $N_Q(P_i) = N_G(P_i) \cap Q$ , which is equal to  $P_i \cap Q$  by our lemma. Thus,  $|\mathcal{O}_i| = [Q : P_i \cap Q]$ . Let  $Q = P_1$ , so that  $|\mathcal{O}_1| = 1$ . For all  $i > 1$ ,  $P_1 \neq P_i$ , so  $P_1 \cap P_i < P_1$ . Thus,  $|\mathcal{O}_i| = [P_1 : P_1 \cap P_i] > 1$  for  $2 \leq i \leq s$ . Since  $P_1$  is a  $p$ -group,  $[P_1 : P_1 \cap P_i]$  must be a power of  $p$ , meaning that  $p \mid |\mathcal{O}_i|$  for  $2 \leq i \leq s$ . Thus,  $r = |\mathcal{O}_1| + |\mathcal{O}_2| + \dots + |\mathcal{O}_s| \equiv 1 \pmod{p}$ .

Now, let  $Q$  be a  $p$ -subgroup of  $G$ . Suppose that  $Q \not\leq P_i$  for any  $1 \leq i \leq r$ . Then  $Q \cap P_i < Q$  for all  $i$ , so  $|\mathcal{O}_i| = [Q : Q \cap P_i] > 1$ . Thus,  $p$  divides  $|\mathcal{O}_i|$  for all  $i$ , meaning that  $p$  divides their sum, which is  $r$ . This contradicts the fact that  $r \equiv 1 \pmod{p}$ , so  $Q \leq P_i$  for some  $i$ , meaning that  $Q \leq gPg^{-1}$  for some  $g \in G$ . To see that all Sylow  $p$ -subgroups of  $G$  are conjugate, let  $Q$  be a Sylow  $p$ -subgroup of  $G$ . Then  $Q \leq gPg^{-1}$  for some  $g \in G$ . Since  $|Q| = |gPg^{-1}|$ , we must have that  $Q = gPg^{-1}$ .

- (3) Using the terminology of the proof of (2), we have that all Sylow  $p$ -subgroups are conjugate, so  $\mathcal{S} = \text{Syl}_p(G)$ . Thus,  $r = |\mathcal{S}| = |\text{Syl}_p(G)| = n_p$ , and so  $n_p = r \equiv 1 \pmod{p}$ . This proves the first part. For the second, notice that Proposition 1.4.30 implies that  $n_p = |\mathcal{S}| = [G : N_G(P)]$  for any  $P \in \text{Syl}_p(G)$ .

■

**Corollary 1.4.59.** *Let  $G$  be a group,  $p$  a prime, and  $P, Q \in \text{Syl}_p(G)$ . Then  $P \cong Q$ .*

*Proof.* We must have  $Q = gPg^{-1}$  for some  $g \in G$ . Define  $\phi : P \rightarrow Q$  by  $\phi(p) = gpg^{-1}$ . This is clearly bijective, and it is a homomorphism, as  $gp_1p_2g^{-1} = gp_1g^{-1}gp_2g^{-1}$ . ■

**Corollary 1.4.60.** *Let  $P$  be a Sylow  $p$ -subgroup of  $G$ . Then the following are equivalent.*

- (1)  $P$  is the unique Sylow  $p$ -subgroup of  $G$ .
- (2)  $P$  is normal in  $G$ .
- (3)  $P$  is characteristic in  $G$ .
- (4) If  $X \subseteq G$  so that  $|x|$  is a power of  $p$  for all  $x \in X$ , then  $\langle x \rangle$  is a  $p$ -group.

*Proof.* If (1) is true, then, as  $gPg^{-1} \in \text{Syl}_p(G)$ ,  $gPg^{-1} = P$ . Thus, (1)  $\Rightarrow$  (2). Conversely, if  $P \trianglelefteq G$  and  $Q \in \text{Syl}_p(G)$ , then there is some  $g \in G$  so that  $Q = gPg^{-1} = P$ , meaning that  $\text{Syl}_p(G) = \{P\}$ . We have already seen that (3)  $\Rightarrow$  (2). Conversely, if  $P \trianglelefteq G$ , then  $P$  is the unique subgroup of  $G$  of order  $p^\alpha$ , so  $P \text{ char } G$ . We have therefore shown that (1)  $\iff$  (2)  $\iff$  (3). Finally, assume (1), and suppose  $X \subseteq G$  satisfies the conditions. For each  $x \in X$ ,  $\langle x \rangle$  is a  $p$ -group, so there is some  $g \in G$  with  $\langle x \rangle \leq gPg^{-1} = P$ . In particular,  $x \in P$ . Thus,  $X \subseteq P$ , and  $\langle X \rangle \leq P$ , meaning that  $\langle X \rangle$  is a  $p$ -group. Conversely, assume (4), and let  $X = \bigcup_{P \in \text{Syl}_p(G)} P$ . For any Sylow  $p$ -subgroup  $P$ , we have that  $P \leq \langle X \rangle$ . Since  $\langle X \rangle$  is a  $p$ -group, by assumption, we have that  $P = \langle X \rangle$ , so (1) holds. ■

**Proposition 1.4.61.** *Let  $p$  and  $q$  be primes with  $p < q$ , and let  $G$  be a group of order  $pq$ . Let  $P \in \text{Syl}_p(G)$  and  $Q \in \text{Syl}_q(G)$ . Then  $Q \trianglelefteq G$ , and if  $P \trianglelefteq G$ , then  $G$  is cyclic.*

*Proof.* We know that  $n_q = 1_k q$  for some  $k \geq 0$ . Since  $n_q | p$  and  $p < q$ , we have  $k = 0$ , and  $n_q = 1$ . Thus,  $Q \trianglelefteq G$ .

Now, suppose that  $P \trianglelefteq G$ . Write  $P = \langle x \rangle$  and  $Q = \langle y \rangle$ . We have that  $G/C_G(P) = N_G(P)/C_G(P)$  is isomorphic to a subgroup of  $\text{Aut}(P) \cong \text{Aut}(Z_p)$ , the order of which is  $p-1$ . Since  $p, q \nmid p-1$ ,  $G = C_G(P)$ . Thus,  $x \in P \leq Z(G)$ , so  $xy = yx$ . Therefore,  $|xy| = pq$ , and so  $G$  is cyclic. In particular, notice that, since  $n_p | q$ , either  $n_p = 1$  or  $n_p = q$ , if  $p \nmid q-1$ , then  $n_p \neq q$ , so  $P \trianglelefteq G$ . ■

**Proposition 1.4.62.** *Let  $G$  be a group with  $|G| = 30$ . Then there is some  $H \trianglelefteq G$  with  $H \cong Z_{15}$ .*

*Proof.* Note that any subgroup of order 15 is normal, as it has index 2. Note also that it is cyclic, by the preceding result. Thus, we show that there is some  $H \leq G$  with  $|H| = 15$ . Let  $P \in \text{Syl}_5(G)$  and  $Q \in \text{Syl}_3(G)$ . If one of  $P$  or  $Q$  is normal in  $G$ , then  $PQ \leq G$ , and  $|PQ| = 15$ . Note also that if one of  $P$  and  $Q$  is normal in  $G$ , then  $P \text{ char } PQ$  and  $Q \text{ char } PQ$ , and since  $PQ \trianglelefteq G$ , both of  $P$  and  $Q$  are normal in  $G$ . Thus, assume that neither of  $P$  and  $Q$  are normal. Then, we must have  $n_5 = 6$  and  $n_3 = 10$ . Now, each element of order 5 is in a Sylow 5-subgroup, and so each Sylow 5-subgroup must contain 4 nonidentity elements, and they must intersect in the identity. Thus, the number of elements of order 5 is  $4 \cdot 6 = 24$ . Similarly, the number of elements of order 3 is  $2 \cdot 10 = 20$ . However,  $24 + 20 = 44 > 30$ , which is a contradiction. ■

**Proposition 1.4.63.** *Let  $G$  be a group of order 12. Then either  $G$  has a normal Sylow 3-subgroup, or  $G \cong A_4$ , in which case it has a normal Sylow 2-subgroup.*

*Proof.* Suppose that  $n_3 \neq 1$ , and let  $P \in \text{Syl}_3(G)$ . Since  $n_3|4$  and  $n_3 \equiv 1 \pmod{3}$ , we have that  $n_3 = 4$ . Since distinct Sylow 3-subgroups intersect in the identity,  $G$  contains  $2 \cdot 4 = 8$  elements of order 3. Since  $[G : N_G(P)] = n_3 = 4$ ,  $N_G(P) = P$ . Now,  $G$  acts by conjugation on its four Sylow 3-subgroups, so we have a permutation representation  $\phi : G \rightarrow S_4$ . The kernel  $K$  of this action normalizes all Sylow 3-subgroups, so  $K \leq N_G(P) = P$ . Since  $P \not\trianglelefteq G$ , and  $K \trianglelefteq G$ , we must have that  $K = 1$ . Thus,  $G \cong \phi(G) \leq S_4$ . Since  $G$  contains 8 elements of order 3 and there are 8 elements of order 3 in  $S_4$ , all contained in  $A_4$ ,  $|\phi(G) \cap A_4| \geq 8$ . Since both groups have order 12,  $\phi(G) = A_4$ , and  $G \cong A_4$ . Now, let  $V \in \text{Syl}_2(A_4)$ . Then  $|V| = 4$ , so it contains all of the remaining elements of  $A_4$ . In particular, there cannot be another Sylow 2-subgroup. ■

**Proposition 1.4.64.** *Let  $p$  and  $q$  be distinct primes, and let  $G$  be a group with order  $p^2q$ . Then  $G$  has a normal Sylow subgroup.*

*Proof.* Let  $P \in \text{Syl}_p(G)$  and  $Q \in \text{Syl}_q(G)$ . If  $p > q$ , then  $n_p = 1$ , as  $n_p|q$ . Thus,  $P \trianglelefteq G$ . Now, consider when  $p < q$ , and assume that  $n_q > 1$ . We have that  $n_q = p$  or  $p^2$ . Since  $q > p$ , we have  $n_q = p^2$ . Writing  $n_q = 1 + tq$ , we have that  $tq = p^2 - 1 = (p+1)(p-1)$ . Thus, either  $q|p-1$  or  $q|p+1$ . Since  $q > p$ ,  $q|p+1$ . We must also have  $q = p+1$ , forcing  $p = 2$  and  $q = 3$ . Then  $|G| = 12$ , and the result follows from the previous proposition. ■

**Proposition 1.4.65.** *If  $|G| = 60$  and  $G$  has more than one Sylow 5-subgroup, then  $G$  is simple.*

*Proof.* Suppose that  $|G| = 60$ ,  $n_5 > 1$  and there is some  $H \neq 1, G$  with  $H \trianglelefteq G$ . We must have that  $n_5 = 6$ , and for any  $P \in \text{Syl}_5(G)$ ,  $|N_G(P)| = 10$ . If  $5||H|$ , then  $H$  contains a Sylow 5-subgroup of  $G$ , and therefore all of them, as  $H$  is normal. In particular,  $|H| \geq 1+6 \cdot 4 = 25$ , so  $|H| = 30$ . We have already seen that any group of order 30 has a normal Sylow 5-subgroup. Thus,  $5 \nmid |H|$  for any  $H < G$ . If  $|H| = 6$  or  $|H| = 12$ , then  $H$  has a normal, and hence characteristic, Sylow subgroup, which is thus also normal in  $G$ . We may replace  $H$  with this to assume that  $|H| = 2, 3$ , or  $4$ . Let  $\overline{G} = G/H$ , so that  $|\overline{G}| = 30, 20$ , or  $15$ . By previous results, there is some  $\overline{P} \trianglelefteq \overline{G}$  with  $|\overline{P}| = 5$ . Let  $H_1$  be so that  $\overline{P} = H_1/H$ . Then  $H_1 \trianglelefteq G$  with  $H_1 \neq G$ , and  $5||H_1|$ . This is a contradiction. ■

**Corollary 1.4.66.**  *$A_5$  is simple.*

*Proof.*  $\langle (12345) \rangle \neq \langle (13245) \rangle$ . ■

**Proposition 1.4.67.** *If  $G$  is a simple group of order 60, then  $G \cong A_5$ .*

*Proof.* If  $G$  is a simple group of order 60, then  $n_2 = 3, 5$ , or  $!5$ . Let  $P \in \text{Syl}_2(G)$ , and let  $N = N_G(P)$ , so  $[G : N] = n_2$ .

Note that  $G$  cannot have a proper subgroup  $H$  with  $[G : H] < 5$ . Suppose that there was one of index 4, 3, or 2. Then, by Theorem 1.4.21, there is some normal subgroup  $K$  of  $G$  contained in  $H$  with  $G/K$  isomorphic to a subgroup of  $S_4, S_3$ , or  $S_2$ . Since  $K \neq G$ , we must have that  $K = 1$ , but  $|G| = 600 \nmid 4!$ . In particular,  $n_2 \neq 3$ .

Suppose that  $n_2 = 5$ . Then  $N$  has index 5 in  $G$ , so the action of  $G$  by left multiplication on the set of left cosets of  $N$  gives a permutation representation of  $G$  onto  $S_5$ . Since the kernel of this is a proper normal subgroup, it must be 1, and  $G$  is isomorphic to a subgroup

of  $S_5$ . We may identify group with this isomorphic copy, and assume  $G \leq S_5$ . If  $G \not\leq A_5$ , then  $S_5 = GA_5$ , so the Second Isomorphism Theorem implies that  $[G : A_5 \cap G] = 2$ , a contradiction. Thus,  $G \leq A_5$ . Since  $|G| = |A_5|$ , we have that  $G \cong A_5$ .

Finally, assume that  $n_2 = 15$ . If for every  $P \neq Q \in \text{Syl}_2(G)$   $P \cap Q = 1$ , then the number of nonidentity elements in the Sylow 2-subgroups of  $G$  would be  $(4-1) \cdot 15 = 45$ , but  $n_5 = 6$  so the number of elements of order 5 in  $G$  is  $(5-1) \cdot 6 = 24$ , accounting for 69 elements. Thus, there are two distinct Sylow 2-subgroups  $P$  and  $Q$  with  $|P \cap Q| = 2$ . Let  $M = N_G(P \cap Q)$ . Since  $G$  is simple,  $M \neq G$ . Thus,  $4 \nmid |M|$ , and  $|M| > 4$  (otherwise  $P = M = Q$ ). The only possibility is  $|M| = 12$ , meaning that  $[G : M] = 5$ . The argument of the previous paragraph with  $M$  replacing  $N$  means that  $G \cong A_5$ . Computation gives that  $n_2(A_5) = 5$ , so this is a contradiction. ■

### 1.4.6 The Simplicity of $A_n$

**Theorem 1.4.68.**  $A_n$  is simple for all  $n \geq 5$ .

*Proof.* We prove this by induction on  $n$ . This has already been shown for  $n = 5$ , so assume  $n \geq 6$  and let  $G = A_n$ . Suppose that there is some  $H \trianglelefteq G$  with  $H \neq 1, G$ . For each  $i \in \{1, 2, \dots, n\}$ , let  $G_i$  be the stabilizer of  $i$  in the natural action of  $G$  on  $\{1, 2, \dots, n\}$ . Then  $G_i \leq G$ , and  $G_i \cong A_{n-1}$ . By induction,  $G_i$  is simple for  $1 \leq i \leq n$ .

Now, suppose that there is some  $\tau \in H$  with  $\tau \neq 1$  and  $\tau(i) = i$  for some  $i \in \{1, 2, \dots, n\}$ . Since  $\tau \in H \cap G_i$ , and  $H \cap G_i \trianglelefteq G_i$  by the Second Isomorphism Theorem, we must have that  $H \cap G_i = G_i$ , so  $G_i \leq H$ . Note that  $\sigma G_i \sigma^{-1} = G_{\sigma(i)}$ , so  $\sigma G_i \sigma^{-1} \leq \sigma H \sigma^{-1} = H$  for all  $i$ . Thus,  $G_j \leq H$  for all  $1 \leq j \leq n$ . Each  $\lambda \in A_n$  can be written as a product of  $2t$  transpositions for  $t \in \mathbb{N}$ , so write  $\lambda = \lambda_1 \lambda_2 \cdots \lambda_t$ , where  $\lambda_k$  is the product of two transpositions. Since  $n > 4$ , for each  $k$ ,  $\lambda_k \in G_j$  for some  $j$ . Thus,  $G = \langle G_1, G_2, \dots, G_n \rangle \leq H$ , which is a contradiction. Thus, if  $\tau \neq 1 \in H$ , then  $\tau(i) \neq i$  for all  $i$ . Thus, if  $\tau_1, \tau_2 \in H$  with  $\tau_1(i) = \tau_2(i)$  for some  $i$ , then  $\tau_1 = \tau_2$ , as  $\tau_2^{-1} \tau_1(i) = i$ .

Now, suppose that  $\tau \in H$  has a cycle decomposition with a cycle of length greater than 2, say

$$\tau = (a_1 a_2 a_3 \cdots)(b_1 b_2 \cdots) \cdots.$$

Let  $\sigma \in G$  have  $\sigma(a_1) = a_1$ ,  $\sigma(a_2) = a_2$ , but  $\sigma(a_3) = a_3$  (such an element exists, since  $n \geq 5$ ). Then

$$\tau_1 := \sigma \tau \sigma^{-1} = (a_1 a_2 \sigma(a_3) \cdots)(\sigma(b_1) \sigma(b_2) \cdots) \cdots.$$

Thus,  $\tau \neq \tau_1$ , but  $\tau(a_1) \tau_1(a_1)$ , which is a contradiction. Thus, only 2-cycles can appear in the cycle decomposition of nonidentity elements of  $H$ . Let  $\tau \neq 1 \in H$  have a cycle decomposition

$$(a_1 a_2)(a_3 a_4)(a_5 a_6) \cdots,$$

where we use the assumption that  $n \geq 6$ . Let  $\sigma = (a_1 a_2)(a_3 a_5) \in G$ . Then,

$$\tau_1 := \sigma \tau \sigma^{-1} = (a_1 a_2)(a_5 a_4)(a_3 a_6) \cdots.$$

Thus,  $\tau \neq \tau_1 \in H$  with  $\tau(a_1) = \tau_1(a_1)$ , a contradiction. ■



## 1.5 Direct and Semidirect Products of Abelian Groups

### 1.5.1 Direct Products

**Definition 1.5.1.** Let  $G_1, G_2, \dots, G_n$  be groups with operations  $\star_1, \star_2, \dots, \star_n$  respectively. Their direct product  $G_1 \times G_2 \times \dots \times G_n$  is the set of  $n$ -tuples  $(g_1, g_2, \dots, g_n)$ , where  $g_i \in G_i$ , with operation

$$(g_1, g_2, \dots, g_n) \cdot (h_1, h_2, \dots, h_n) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots, g_n \star_n h_n).$$

Similarly, if  $G_1, G_2, \dots$  are groups with operations  $\star_1, \star_2, \dots$  respectively, then their direct product  $G_1 \times G_2 \times \dots$  is the set of sequences  $(g_1, g_2, \dots)$ , where  $g_i \in G_i$ , with operation

$$(g_1, g_2, \dots) \cdot (h_1, h_2, \dots) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots).$$

**Notation.** Although the operations may be different in each of the factors, we will write abstract groups multiplicatively, so that, for example, the operation of  $G_1 \times G_2 \times \dots \times G_n$  becomes  $(g_1, g_2, \dots, g_n)(h_1, h_2, \dots, h_n) = (g_1 h_1, g_2 h_2, \dots, g_n h_n)$ .

**Proposition 1.5.2.** Let  $G_1, G_2, \dots, G_n$  be groups, then their direct product is a group of order  $|G_1||G_2| \dots |G_n|$ .

*Proof.* Let  $G$  denote  $G_1 \times G_2 \times \dots \times G_n$ . If  $1_i$  denotes the identity of  $G_i$ , then the identity of  $G$  is  $(1_1, 1_2, \dots, 1_n)$ . The inverse of  $(g_1, g_2, \dots, g_n)$  is  $(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$ . Associativity and closure follow from the fact that the  $G_i$ 's are groups. The formula for the order of  $G$  is clear. ■

**Proposition 1.5.3.** Let  $G_1, G_2, \dots, G_n$  be groups, and  $\sigma \in S_n$ . Then  $G_1 \times G_2 \times \dots \times G_n \cong G_{\sigma(1)} \times G_{\sigma(2)} \times \dots \times G_{\sigma(n)}$ .

*Proof.* Let  $G = G_1 \times G_2 \times \dots \times G_n$ , and let  $G' = G_{\sigma(1)} \times G_{\sigma(2)} \times \dots \times G_{\sigma(n)}$ . Define  $\phi : G \rightarrow G'$  by  $\phi((g_1, g_2, \dots, g_n)) = (g_{\sigma(1)}, g_{\sigma(2)}, \dots, g_{\sigma(n)})$ . This is clearly an isomorphism. ■

**Proposition 1.5.4.** Let  $G_1, G_2, \dots, G_n$  be groups and let  $G = G_1 \times G_2 \times \dots \times G_n$ .

(1) For each fixed  $i$ ,

$$G_1 \cong \{(1, 1, \dots, 1, g_i, 1, \dots, 1 : g_i \in G_i\},$$

where  $g_i$  appears in the  $i$ th position. If we identify  $G_i$  with this subgroup of  $G$ , then  $G_i \trianglelefteq G$  and

$$G/G_i \cong G_1 \times G_2 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n.$$

(2) For each fixed  $i$ , the map  $\pi_i : G \rightarrow G_i$  defined by  $\pi_i((g_1, g_2, \dots, g_n)) = g_i$  is a surjective homomorphism with

$$\ker \pi_i = \{(g_1, g_2, \dots, g_{i-1}, 1, g_{i+1}, \dots, g_n : \forall j \neq i, g_j \in G_j\}$$

$$\cong G_1 \times G_2 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n,$$

where the 1 appears in position  $i$ .

(3) Under the identifications in (1), if  $x \in G_i$  and  $y \in G_j$  for  $i \neq j$ , then  $xy = yx$ .

*Proof.*

- (1) Since the operation is defined component-wise, the first part of this is clear. Identifying  $G_i$  with its subgroup in  $G$ , define

$$\phi : G \rightarrow G_1 \times G_2 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n$$

defined by

$$\phi((g_1, g_2, \dots, g_n)) = (g_1, g_2, \dots, g_{i-1}, g_{i+1}, \dots, g_n).$$

This is clearly a surjective homomorphism with kernel

$$\{(g_1, g_2, \dots, g_n) : \forall j \neq i, g_j = 1\} = G_i.$$

- (2) It is clear that  $\pi_i$  is a surjective homomorphism with a kernel where  $g_i = 1$ . The isomorphism is  $\phi$  as defined in the proof of (1), which is injective because of the added condition that  $g_i = 1$ .

- (3) We have that

$$xy = (1, 1, \dots, 1, g_i, 1, \dots, 1, g_j, 1, \dots, 1) = yx.$$

■

## 1.5.2 The Fundamental Theorem of Finitely Generated Abelian Groups

**Definition 1.5.5.** A group  $G$  is said to be finitely generated if there is a finite subset  $A \subseteq G$  so that  $G = \langle A \rangle$ .

**Definition 1.5.6.** For each  $r \in \mathbb{N} \cup \{0\}$ , let  $\mathbb{Z}^r = \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{r \text{ times}}$ , where  $\mathbb{Z}^0 = 1$ . The group  $\mathbb{Z}^r$  is called the free abelian group of rank  $r$ .

**Theorem 1.5.7** (Fundamental Theorem of Finitely Generated Abelian Groups). *Let  $G$  be a finitely generated abelian group. Then,*

- (1)  $G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s}$  for some integers  $r, n_1, n_2, \dots, n_s$  satisfying  $r \geq 0$ ,  $n_j \geq 2$ , and  $n_{i+1} | n_i$  for  $1 \leq i \leq s-1$ .
- (2) The expression is unique in that if  $G \cong \mathbb{Z}^t \times Z_{m_1} \times Z_{m_2} \times \cdots \times Z_{m_u}$ , where  $t$  and  $m_1, m_2, \dots, m_u$  satisfy the conditions outlined, then  $t = r$ ,  $u = s$  and  $m_i = n_i$  for all  $i$ .

*Proof.* We will derive this later, as a consequence of a more general classification theorem. ■

**Definition 1.5.8.** The integer  $r$  is called the free rank or Betti number of  $G$ , and the integers  $n_1, n_2, \dots, n_s$  are called the invariant factors of  $G$ . The breakdown of  $G$  is called the invariant factor decomposition of  $G$ .

**Definition 1.5.9.** If  $G$  is a finite abelian group with invariant factors  $n_1, n_2, \dots, n_s$ , where  $n_{i+1} | n_i$ , then  $G$  is said to be of type  $(n_1, n_2, \dots, n_s)$ .

**Proposition 1.5.10.** *Let  $G$  be a finite abelian group of order  $n$ . Then every prime divisor of  $n$  must divide the first invariant factor  $n_1$ .*

*Proof.* Note that  $n = n_1 n_2 \cdots n_s$ . Thus, if  $p$  is prime divides  $n$ , then  $p | n_j$  for some  $j$ . Since  $n_j | n_{j-1}$ , we also have that  $p | n_{j-1}$ . Continuing,  $p | n_1$ . ■

**Corollary 1.5.11.** *Let  $n$  be squarefree. Then, up to isomorphism, the only abelian group of order  $n$  is  $Z_n$ .*

*Proof.* Write  $n = p_1 p_2 \cdots p_q$ . Then  $p_i | n_1$ , so  $n | n_1$ . Since  $n_1 | n$ , we have  $n = n_1$ , and so any abelian group of order  $n$  is  $Z_n$ . ■

**Theorem 1.5.12.** *Let  $G$  be an abelian group of order  $n > 1$  and let the unique factorization of  $n$  into distinct prime powers be*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}.$$

*Then,*

(1)  $G \cong A_1 \times A_2 \times \cdots \times A_k$ , where  $|A_i| = p_i^{\alpha_i}$ .

(2) For each  $A \in \{A_1, A_2, \dots, A_k\}$  with  $|A| = p^\alpha$ ,

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_t}}$$

with  $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_t \geq 1$  and  $\beta_1 + \beta_2 + \cdots + \beta_t = \alpha$ .

(3) The decomposition in (1) and (2) is unique.

*Proof.*

(1) Notice that the  $A_i$ 's are the Sylow  $p_i$ -subgroups of  $G$ . Now, we claim that if  $A, B \leq G$  with  $A \cap B = 1$ , then  $A \times B \cong AB$ . If we define  $\phi : A \times B \rightarrow AB$  by  $\phi((a, b)) = ab$ , then  $\phi((a_1, b_1)(a_2, b_2)) = \phi((a_1 a_2, b_1 b_2)) = a_1 a_2 b_1 b_2 = a_1 b_1 a_2 b_2 = \phi((a_1, b_1))\phi((a_2, b_2))$ . Thus,  $\phi$  is a homomorphism. Since  $\phi^{-1}(ab) = (a, b)$ ,  $\phi$  is surjective, and if  $\phi((a_1, b_1)) = \phi((a_2, b_2))$ , then  $a_1 b_1 = a_2 b_2$ , so  $a_2 a_1^{-1} = b_2 b_1^{-1}$ . Since the left hand side is in  $A$ , and the right hand side is in  $B$ , they must both be equal to 1. Thus,  $\phi$  is an isomorphism. Now,  $A_1 \times A_2 \times \cdots \times A_k$  has order  $|A_1| \cdot |A_2| \cdots |A_k| = |G|$ . Thus, since  $A_1 A_2 \cdots A_k \leq G$ ,  $A_1 \times A_2 \times \cdots \times A_k \cong G$ .

(2) We show that if  $A$  is a finite abelian  $p$ -group and if  $B \leq A$  is a cyclic subgroup of  $A$  of maximal order, then there is some  $H \leq A$  with  $B \cap H = 1$  so that  $A \cong B \times H$ . First, we will show that if  $A$  has a unique subgroup of order  $p$ , then it is cyclic. To do so, we use induction, with the base case  $|A| = p$  being trivial. Now, suppose that  $B \leq A$  is unique with order  $p$ , and let  $\phi : A \rightarrow A$  be defined by  $\phi(a) = a^p$ . Then  $K = \ker \phi$  contains all elements of order 1 or  $p$ , meaning that  $B = K$ . Thus,  $A/K \cong \text{im } \phi$ . This is a  $p$ -group, and thus has a subgroup  $C$  of order  $p$ . Since  $C \leq A$ , it is unique. Since

$|\text{im } \phi| < |A|$ , the inductive hypothesis implies that  $\text{im } \phi$  is cyclic, so that  $A/K = \langle aK \rangle$  for some  $a$ . Now, for every coset  $c$  of  $A/K$ , there is some  $m_c$  so that  $a^{m_c} \in c$ . Now,  $\langle a \rangle \leq A$  has a subgroup of order  $p$ , which must be  $K$ , so  $K \leq \langle a \rangle$ . Thus, every element can be written as  $a^{m_c} \cdot k$  for some  $m_c$  and  $k \in K$ , meaning that  $A = \langle a \rangle$ . Now, we show our result. We do this by induction on  $A$ . If  $A$  is cyclic (which covers the case  $|A| = p$ ), then this is trivial. Now, suppose that  $A$  is not cyclic, and let  $B \leq A$  be a cyclic subgroup of  $A$  of maximal order. It is not hard to check that  $B$  has a unique cyclic subgroup of order  $p$ . Furthermore,  $A$  must have at least two subgroups of order  $p$ . Let  $C \leq A$  be a subgroup of order  $p$  not contained in  $B$ . Then  $B \cap C = 1$ . Now, since  $B$  is maximal,  $|a| \leq |B|$  for all  $a \in A$ . In particular,  $|aC| \leq |B|$ , where  $aC \in A/C$ . We also have that  $BC/C \cong B/(B \cap C) = B/1 \cong B$ , so  $BC/C$  is cyclic. This must be the maximal cyclic subgroup of  $A/C$ . Since  $|A/C| < |A|$ , there must be some  $H' \leq A/C$  so that  $A/C \cong (BC)/C \times H'$ . Take  $H \leq A$  so that  $H/C = H'$ , so that  $A/C \cong BC/C \times H/C$ . This is isomorphic to  $BC/CH/C$ , which is clearly equal to  $(BC)H/C$ . By order considerations,  $A = (BC)H = B(CH)$ . Since  $C \leq H$ , this is equal to  $BH$ . Since  $H \cap BC = C$ , we have  $H \cap B = 1$ , so  $A \cong B \times H$ .

- (3) That the decomposition in (1) is unique is clear. Suppose that a finite abelian  $p$ -group has two decompositions, as described in (2). Suppose that the largest elementary divisor (see below) in the first decomposition is  $p^a$ , and that the largest in the second is  $p^b$  with  $a > b$ . Then the first decomposition has an element of order  $a$ , while the second cannot. Thus,  $a = b$ . Letting the element in this slot be the identity, we may repeat this process to show that the decompositions are the same. ■

**Definition 1.5.13.** The integers  $p^{\beta_j}$  described in the preceding theorem are called the elementary divisors of  $G$ . The description of  $G$  in the theorem is called the elementary divisor decomposition of  $G$ .

**Proposition 1.5.14.** Let  $m, n \in \mathbb{Z}^+$ .

- (1)  $Z_m \times Z_n \cong Z_{mn}$  if and only if  $\gcd(m, n) = 1$ .  
 (2) If  $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ , then  $Z_n \cong Z_{p_1^{\alpha_1}} \times Z_{p_2^{\alpha_2}} \times \cdots \times Z_{p_k^{\alpha_k}}$ .

*Proof.*

- (1) Let  $Z_m = \langle x \rangle$  and  $Z_n = \langle y \rangle$  and let  $l = \text{lcm}(m, n)$ . Then  $l = mn$  if and only if  $\gcd(m, n) = 1$ . Let  $x^a y^b$  be an element of  $Z_m \times Z_n$ . Then  $(x^a y^b)^l = x^{la} y^{lb} = 1$ , so if  $\gcd(m, n) \neq 1$ , then no element of  $Z_m \times Z_n$  has order  $mn$ , meaning that it cannot be isomorphic to  $Z_{mn}$ . Conversely, if  $\gcd(m, n) = 1$ , then  $|xy| = mn$ , so  $Z_m \times Z_n$  is cyclic.  
 (2) This follows easily from (1). ■

**Definition 1.5.15.** If  $G$  is a finite abelian group of type  $(n_1, n_2, \dots, n_t)$ , the integer  $t$  is called the rank of  $G$ .

**Definition 1.5.16.** If  $G$  is any group, the exponent of  $G$  is the smallest positive integer  $n$  so that  $x^n = 1$  for all  $x \in G$ . If no such  $n$  exists, the exponent of  $G$  is  $\infty$ .

### 1.5.3 Recognizing Direct Products

**Definition 1.5.17.** Let  $G$  be a group with  $x, y \in G$  and  $\emptyset \neq A, B \subseteq G$ .

- The commutator of  $x$  and  $y$  is  $[x, y] = x^{-1}y^{-1}xy$ .
- We define  $[A, B] = \langle [a, b] : a \in A, b \in B \rangle$ .
- The commutator subgroup of  $G$  is  $G' = [G, G] = \langle [x, y] : x, y \in G \rangle$ .

**Proposition 1.5.18.** Let  $G$  be a group with  $x, y \in G$  and  $H \leq G$ .

- (1)  $xy = yx[x, y]$ . In particular,  $x$  and  $y$  commute if and only if  $[x, y] = 1$ .
- (2)  $H \trianglelefteq G$  if and only if  $[H, G] \leq H$ .
- (3) For any  $\sigma \in \text{Aut}(G)$ ,  $\sigma[x, y] = [\sigma(x), \sigma(y)]$ . In particular,  $G' \text{ char } G$ , and  $G/G'$  is abelian.
- (4) If  $H \trianglelefteq G$  and  $G/H$  is abelian, then  $G' \leq H$ . Conversely, if  $G' \leq H$ , then  $H \trianglelefteq G$  and  $G/H$  is abelian.
- (5) If  $\phi : G \rightarrow A$  is any homomorphism of  $G$  to an abelian group  $A$ , then  $G' \leq \ker \phi$ , and if  $\pi : G \rightarrow G/G'$  is the projection homomorphism, and  $\psi : G/G' \rightarrow A$  is defined as  $\psi(gG') = \phi(g)$ , then  $\phi = \psi \circ \pi$ .

*Proof.*

- (1) This is clear from the definition.
- (2) We have that  $H \trianglelefteq G$  if and only if  $g^{-1}hg \in H$  for all  $g \in G$  and  $h \in H$ . This occurs if and only if  $h^{-1}g^{-1}hg \in H$ , so that  $H \trianglelefteq G$  if and only if  $[h, g] \in H$  for all  $h \in H$  and  $g \in G$ . Thus,  $H \trianglelefteq G$  if and only if  $[H, G] \leq H$ .
- (3) We have that  $\sigma[x, y] = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) = [\sigma(x), \sigma(y)]$ . Thus, for every  $[x, y] \in G'$ ,  $\sigma([x, y]) \in G'$ . Since  $\sigma$  is bijective, and since the commutators are a generating set for  $G'$ ,  $\sigma(G') = G'$ , so  $G' \text{ char } G$ . Now, let  $x, y \in G/G'$ . Then,  $(xG')(yG') = (xy)G' = (yx[x, y])G' = (yx)G' = (yG')(xG')$ , as  $[x, y] \in G'$ .
- (4) Suppose  $H \trianglelefteq G$  and  $G/H$  is abelian. Then for all  $x, y \in G$  we have that  $(xH)(yH) = (yH)(xH)$ , so  $1H = (xH)^{-1}(yH)^{-1}(xH)(yH) = [x, y]H$ . Thus,  $[x, y] \in H$  for all  $x, y \in G$ , so  $G' \leq H$ . Now, if  $G' \leq H$ , then, as  $G/G'$  is abelian, every subgroup of  $G/G'$  is normal. Thus,  $H/G' \trianglelefteq G/G'$ . By the Fourth Isomorphism Theorem,  $H \trianglelefteq G$ . We also have that  $G/H \cong (G/G')/(H/G')$ , which is abelian.
- (5) Since  $\ker \phi \trianglelefteq G$ ,  $G' \leq \ker \phi$ . Note that this makes sure that  $\psi$  is well defined. The fact that  $\phi = \psi \circ \pi$  is trivial. ■

**Proposition 1.5.19.** Let  $H, K \leq G$ . The number of distinct ways of writing each element of the set  $HK$  in the form  $hk$  for  $h \in H$  and  $k \in K$  is  $|H \cap K|$ . In particular, if  $H \cap K = 1$ , then each element of  $HK$  can be written uniquely as a product  $hk$  for some  $h \in H$  and  $k \in K$ .

*Proof.* Suppose that the number of distinct ways of writing each element is  $n$ . Then,  $n|HK| = |H| \cdot |K|$ . Since  $|HK| = \frac{|H| \cdot |K|}{|H \cap K|}$ , the proposition follows. ■

**Theorem 1.5.20.** *Suppose  $G$  is a group with  $H, K \trianglelefteq G$  and  $H \cap K = 1$ . Then  $HK \cong H \times K$ .*

*Proof.* Note that  $H, K \trianglelefteq G$  means that  $HK \leq G$ . Let  $h \in H$  and  $k \in K$ . Then  $k^{-1}hk \in H$ , so  $h^{-1}(k^{-1}hk) \in H$ . Similarly,  $(h^{-1}k^{-1}h)k \in K$ . Thus,  $h^{-1}k^{-1}hk = 1$ , so that  $hk = kh$ . Now consider the map  $\phi : HK \rightarrow H \times K$  defined by  $\phi(hk) = (h, k)$ . Since  $H \cap K = 1$ , this is well defined. Now, note that  $\phi(h_1k_1h_2k_2) = \phi(h_1h_2k_1k_2) = (h_1h_2, k_1k_2) = (h_1, k_1)(h_2, k_2) = \phi(h_1k_1)\phi(h_2k_2)$ , so  $\phi$  is a homomorphism. Since it is clearly surjective, and since  $|HK| = |H \times K|$ ,  $\phi$  is a bijection. ■

**Definition 1.5.21.** Let  $G$  be a group, and let  $H, K \trianglelefteq G$  with  $H \cap K = 1$ . We call  $HK$  the internal direct product of  $H$  and  $K$ , and we call  $H \times K$  the external direct product of  $H$  and  $K$ .

### 1.5.4 Semidirect Products

**Theorem 1.5.22.** *Let  $H$  and  $K$  be groups, and let  $\phi : K \rightarrow \text{Aut}(H)$  be a homomorphism. Let  $\cdot$  denote the (left) action of  $K$  on  $H$  determined by  $\phi$ . Let  $G$  be the set of ordered pairs  $(h, k)$  with  $h \in H$  and  $k \in K$ , and define multiplication on  $G$  as*

$$(h_1, k_1)(h_2, k_2) = (h_1(k_1 \cdot h_2), k_1k_2).$$

- (1)  $G$  is a group of order  $|H| \cdot |K|$ .
- (2) The sets  $\{(h, 1) : h \in H\}$  and  $\{(1, k) : k \in K\}$  are subgroups of  $G$ , and the maps  $h \mapsto (h, 1)$  for  $h \in H$  and  $k \mapsto (1, k)$  for  $k \in K$  are isomorphisms of these subgroups with  $H$  and  $K$  respectively.
- (3) Identifying  $H$  with its isomorphic copy in  $G$  as described above,  $H \trianglelefteq G$ .
- (4) Identifying  $H$  and  $K$  with their isomorphic copies in  $G$  as described above,  $H \cap K = 1$ .
- (5) Identifying  $H$  and  $K$  with their isomorphic copies in  $G$  as described above, for all  $h \in H$  and  $k \in K$ ,  $khk^{-1} = k \cdot h = \phi(k)(h)$ .

*Proof.*

- (1) Note that  $1 \in K$  must map to the identity in  $\text{Aut}(H)$  under  $\phi$ , so that  $(1, 1)$  is the identity. It is also straightforward to check that the inverse of  $(h, k)$  is  $(k^{-1} \cdot h^{-1}, k^{-1})$ . Closure is clear. The associative property is as follows:

$$\begin{aligned} ((a, x)(b, y))(c, z) &= (a(x \cdot b), xy)(c, z) \\ &= (a(x \cdot b)((xy) \cdot c), xyz) \\ &= (a(x \cdot b)x \cdot (y \cdot c), xyz) \\ &= (a(x \cdot (b(y \cdot c))), xyz) \\ &= (a, x)(b(y \cdot c), yz) \\ &= (a, x)((b, y)(c, z)). \end{aligned}$$

That  $|G| = |H| \cdot |K|$  is clear.

- (2) Let  $\tilde{H}$  and  $\tilde{K}$  denote the two sets. Note that  $(a, 1)(b, 1) = (ab, 1)$  and that  $(1, x)(1, y) = (1, xy)$ . Note also that  $(a, 1)^{-1} = (a^{-1}, 1)$  and that  $(1, x)^{-1} = (1, x^{-1})$ , so  $\tilde{H}, \tilde{K} \leq G$ . This also shows that the maps are isomorphisms.
- (3) We rely on (5) that  $K \leq N_G(H)$ . Since  $(h, 1)(1, k) = (h, k)$ , we have that  $G = HK$ . Now,  $H \leq N_G(H)$ , and so  $G = HK \leq N_G(H)$ . Thus,  $N_G(H) = G$ , so  $H \trianglelefteq G$ .
- (4) This is clear.
- (5) We have that

$$\begin{aligned} (1, k)(h, 1)(1, k)^{-1} &= (k \cdot h, k)(1, k^{-1}) \\ &= ((k \cdot h)(k \cdot 1), kk^{-1}) \\ &= (k \cdot h, 1). \end{aligned}$$

Thus, under the identifications,  $khk^{-1} = k \cdot h$ . ■

**Definition 1.5.23.** Let  $H$  and  $K$  be groups, and let  $\phi : K \rightarrow \text{Aut}(H)$  be a homomorphism. The group described in the above theorem is called the semidirect product of  $H$  and  $K$  with respect to  $\phi$ , and is denoted  $H \rtimes_{\phi} K$  (or simply  $H \rtimes K$ , when there is no danger of confusion).

**Proposition 1.5.24.** Let  $H$  and  $K$  be groups, and let  $\phi : K \rightarrow \text{Aut}(H)$  be a homomorphism. Then the following are equivalent:

- (1) The map between  $H \rtimes_{\phi} K$  and  $H \times K$  given by  $(h, k) \mapsto (h, k)$  is a group homomorphism (and therefore an isomorphism).
- (2)  $\phi$  is the trivial homomorphism.
- (3)  $K \trianglelefteq H \rtimes_{\phi} K$ .

*Proof.* We show that  $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$ . Now, we have that

$$(h_1, k_1)(h_2, k_2) = (h_1(k_1 \cdot h_2), k_1 k_2).$$

Assuming (1), we must also have that  $(h_1, k_1)(h_2, k_2) = (h_1 h_2, k_1 k_2)$ . Thus,  $k_1 \cdot h_2 = h_2$  for all  $k_1 \in K$  and  $h_2 \in H$ . Thus,  $\phi(k_1) = 1 \in \text{Aut}(H)$  for all  $k_1 \in K$ . Now, assume (2). Then  $k \cdot h = h$  for all  $h \in H$  and  $k \in K$ . Thus,  $khk^{-1} = h$ , so  $hk = kh$ . In particular,  $H \leq N_G(K)$ . Since  $K \leq N_G(K)$ ,  $G = HK \leq N_G(K)$ . Now, assume (3). Then,  $[h, k] = h^{-1}k^{-1}hk = (h^{-1}k^{-1}h)k = h^{-1}(k^{-1}hk)$  is in both  $H$  and  $K$  for all  $h \in H$  and  $k \in K$ . Thus,  $[h, k] \in H \cap K = 1$ , so  $hk = kh$ , meaning  $k \cdot h = khk^{-1} = h$ . Now, we have that  $(h_1, k_1)(h_2, k_2) = (h_1 h_2, k_1 k_2)$ , so the map is an isomorphism. ■

**Theorem 1.5.25.** Suppose  $G$  is a group with  $H \trianglelefteq G$  and  $K \leq G$  so that  $H \cap K = 1$ . Let  $\phi : K \rightarrow \text{Aut}(H)$  be defined by  $k \mapsto (h \mapsto khk^{-1})$ . Then  $HK \cong H \rtimes K$ . In particular, if  $G = HK$  with  $H$  and  $K$  satisfying the necessary conditions, then  $G$  is the semidirect product of  $H$  and  $K$ .

*Proof.* Notice that  $HK \leq G$ , and that every element of  $HK$  can be written uniquely in the form  $hk$  for some  $h \in H$  and  $k \in K$ , since  $H \cap K = 1$ . Thus, the map  $hk \mapsto (h, k)$  is a set bijection from  $HK$  to  $H \rtimes K$ . Note that  $h_1 k_1 h_2 k_2 = h_1 h'_2 k_1 k_2$  for some  $h'_2 \in H$ . In particular,  $k_1 \cdot h_2 = k_1 h_2 k_1^{-1} = h'_2$ , so  $h_1 k_1 h_2 k_2$  maps to  $(h_1(k_1 \cdot h_2), k_1 k_2)$ , meaning that this map is a homomorphism. ■

**Definition 1.5.26.** Let  $G$  be a group and  $H \leq G$ . A subgroup  $K \leq G$  is called a complement for  $H$  in  $G$  if  $G = HK$  and  $H \cap K = 1$ .

## 1.6 Further Topics in Group Theory

### 1.6.1 $p$ -groups, Nilpotent Groups, and Solvable Groups

**Definition 1.6.1.** A maximal subgroup of a group  $G$  is a proper subgroup  $M$  so that there are no subgroups  $H$  of  $G$  with  $M < H < G$ .

**Theorem 1.6.2.** Let  $p$  be a prime and let  $P$  be a group of order  $p^a$  for  $a \geq 1$ .

- (1) The center of  $P$  is nontrivial.
- (2) If  $H \trianglelefteq P$  is nontrivial, then  $H \cap Z(P) \neq 1$ . In particular, every normal subgroup of order  $p$  is contained in the center.
- (3) If  $H \trianglelefteq G$ , then there is some  $K \leq H$  with  $K \trianglelefteq P$  and  $|K| = p^b$  for each  $p^b$  that divides  $|H|$ . In particular,  $P$  has a normal subgroup of order  $p^b$  for every  $0 \leq b \leq a$ .
- (4) If  $H < P$ , then  $H < N_P(H)$ .
- (5) Every maximal subgroup of  $P$  is of index  $p$ , and is normal in  $P$ .

*Proof.*

- (1) We have proved this previously.
- (2) For each conjugacy class  $\mathcal{C}$  of  $P$ , either  $\mathcal{C} \subseteq H$  or  $\mathcal{C} \cap H = \emptyset$ , as  $H$  is normal. Pick representatives of the conjugacy classes of  $P$  to be  $a_1, a_2, \dots, a_r$ , with  $a_1, a_2, \dots, a_k \in H$  and  $a_{k+1}, a_{k+2}, \dots, a_r \notin H$ . Let  $\mathcal{C}_i$  be the conjugacy class of  $a_i$  in  $P$ . Reordering if necessary, assume  $|\mathcal{C}_1| = |\mathcal{C}_2| = \dots = |\mathcal{C}_s| = 1$ , and that  $|\mathcal{C}_{s+1}|, |\mathcal{C}_{s+2}|, \dots, |\mathcal{C}_k| \neq 1$ . Since  $H$  is the disjoint union of these, we have

$$|H| = \sum_{i=1}^k \frac{|P|}{|C_P(a_i)|} = |H \cap Z(P)| + \sum_{i=s+1}^k \frac{|P|}{|C_P(a_i)|}.$$

Now,  $p$  divides  $|H|$ , and  $p$  divides  $\sum_{i=s+1}^k \frac{|P|}{|C_P(a_i)|}$ , so  $p$  divides  $|H \cap Z(P)|$ . Thus,  $H \cap Z(P) \neq 1$ . In particular, if  $|H| = p$ , then  $H \cap Z(P) = H$ , so  $H \leq Z(P)$ .



- (3) We prove this by induction on  $a$ . If  $a \leq 1$ , or if  $H = 1$ , the result is trivial. Suppose that  $a > 1$  and that  $H \neq 1$ . Then  $H \cap Z(P) \neq 1$ , so it contains a (normal) subgroup  $Z$  of order  $p$ . We have that  $H/Z \trianglelefteq P/Z$ , and that  $|P/Z| = p^{a-1}$ . For every nonnegative integer  $b$  with  $p^b \parallel |H|$ , we have that  $p^{b-1} \parallel |H/Z|$ . Thus, there is some  $\bar{K} \leq H/Z$  of order  $p^{b-1}$  that is normal in  $P/Z$ . Write  $\bar{K} = K/Z$ . Then  $|K| = p^b$  and  $K \trianglelefteq P$ . The second assertion comes by letting  $H = P$ .
- (4) We prove this by induction on  $|P|$ . If  $P$  is abelian, then the result is trivial. Thus, we may assume that  $|P| > p$ . Let  $H < P$ . Note that  $Z(P)$  normalizes every subgroup of  $P$ . We know that  $Z(P) \neq 1$ . If  $Z(P) \not\leq H$ , then  $H < \langle H, Z(P) \rangle \leq N_P(H)$ . Now, assume that  $Z(P) \leq H$ . Since  $|P/Z(P)| < |P|$ , we have that  $H/Z(P) < N_{P/Z(P)}(H/Z(P)) = N_P(H)/Z(P)$ . Thus,  $H < N_P(H)$ .
- (5) Let  $M$  be a maximal subgroup of  $P$ . Then  $M < P$ , so  $M < N_P(M)$ . Thus,  $N_P(M) = P$ , so  $M \trianglelefteq P$ . Now, for every  $p^b$  dividing  $|P/M|$ , there is a subgroup of order  $p^b$  of  $P/M$ . The Fourth Isomorphism Theorem implies that  $|P/M| = p$ , so that  $[P : M] = p$ .

■

**Definition 1.6.3.** Let  $G$  be a group. We define  $Z_0(G) = 1$ ,  $Z_1(G) = Z(G)$ , and  $Z_{i+1}(G)$  to be the subgroup of  $G$  containing  $Z_i(G)$  such that

$$Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G)).$$

The chain of subgroups  $Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$  is called the upper central series of  $G$ .

**Definition 1.6.4.** A group  $G$  is called nilpotent if  $Z_c(G) = G$  for some  $c \in \mathbb{Z}$ . The smallest such  $c$  is called the nilpotence class of  $G$ .

**Proposition 1.6.5.** For all  $i$ ,  $Z_i(G) \text{ char } G$ .

*Proof.* We prove this by induction on  $i$ . If  $i = 0$ , this is trivial. If  $i = 1$ , let  $g \in Z(G)$ , and let  $\sigma \in \text{Aut}(G)$ . Then, for any  $h \in G$ ,  $\sigma(g)\sigma(h) = \sigma(h)\sigma(g)$ , so  $\sigma(g) \in Z(G)$ . Now, let  $\sigma \in \text{Aut}(G)$ . We have that  $Z_{i-1}(G) \leq Z_i(G) \cap \sigma(Z_i(G))$ . We also have an induced isomorphism  $\phi : Z_i(G) \rightarrow \sigma(Z_i(G))$  defined by  $\phi(g) = \sigma(g)$ . Since  $Z_i(G) \cong \sigma(Z_i(G))$ , we have that  $Z_i(G)/Z_{i-1}(G) \cong \sigma(Z_i(G))/Z_{i-1}(G) \leq G/Z_{i-1}(G)$ . The left hand side is  $Z(G/Z_{i-1}(G))$ . Since  $Z(H) \text{ char } H$  for any group  $H$ , we have that the isomorphism between them, which extends to an automorphism of  $G/Z_{i-1}(G)$ , must fix  $Z(G/Z_{i-1}(G))$ . Thus,  $\sigma(Z_i(G))/Z_{i-1}(G) = Z(G/Z_{i-1}(G))$ , so  $\sigma(Z_i(G)) = Z_i(G)$ . ■

*Remark 1.6.6.* If  $G$  is abelian, then  $G$  is nilpotent (of class 1 if  $|G| > 1$ ). In the hierarchy of structure, we have that all cyclic groups are abelian groups, all abelian groups are nilpotent groups, all nilpotent groups are solvable groups, and all solvable groups are groups.

*Remark 1.6.7.* If  $G$  is finite, there must be an integer  $n$  such that  $Z_n(G) = Z_{n+1}(G) = Z_{n+2}(G) = \cdots$  by order considerations. If  $G$  is not nilpotent, then  $Z_n(G)$  is a proper subgroup of  $G$ .

**Proposition 1.6.8.** *Let  $p$  be a prime and let  $P$  be a group of order  $p^a$  for  $a \geq 0$ . Then  $P$  is nilpotent of nilpotence class at most  $a - 1$ .*

*Proof.* For each  $i \geq 0$ ,  $P/Z_i(P)$  is a  $p$ -group, so if it is nontrivial, then it has a nontrivial center. Thus, if  $Z_i(P) \neq G$ , then  $|Z_{i+1}(P)| \geq p|Z_i(P)|$ , as  $|Z(P/Z_i(P))| \geq p$ . Thus,  $|Z_{i+1}(P)| \geq p^{i+1}$ , so  $P = Z_a(P)$ . Thus,  $P$  is nilpotent of class at most  $a$ . If it had class equal to  $a$ , then  $|Z_i(P)| = p^i$  for all  $i$ . Then  $[P : Z_{a-2}(P)] = 2$ , so  $|P/Z_{a-2}(P)| = p^2$ , and is thus abelian. Then, it would be its own center, and so  $Z_{a-1}(P)$  would equal  $P$ , a contradiction. Thus, it cannot have class  $a$ , and so it has class less than  $a$ . ■

**Theorem 1.6.9.** *Let  $G$  be a finite group, let  $p_1, p_2, \dots, p_s$  be the distinct primes dividing  $|G|$ , and let  $P_i \in \text{Syl}_{p_i}(G)$ . Then the following are equivalent.*

- (1)  $G$  is nilpotent.
- (2) If  $H < G$ , then  $H < N_G(H)$ .
- (3)  $P_i \trianglelefteq G$  for every  $i$ .
- (4)  $G \cong P_1 \times P_2 \times \dots \times P_s$ .

*Proof.* That (1)  $\Rightarrow$  (2) is the same argument as for  $p$ -groups, assuming that if  $G$  is nilpotent then so is  $G/Z(G)$ , so we only show this. We have that  $Z_1(G/Z(G)) = Z(G/Z(G)) = Z_2(G)/Z(G)$ . We also have that

$$\begin{aligned} Z_2(G/Z(G))/Z_1(G/Z(G)) &= Z(G/Z(G))/(Z_2(G)/Z(G)) = Z(G/Z_2(G)) \\ &= Z_3(G)/Z_2(G) = (Z_3(G)/Z(G))/(Z_2(G)/Z(G)). \end{aligned}$$

Thus,  $Z_2(G/Z(G)) = Z_3(G)/Z(G)$ . Continuing,  $Z_i(G/Z(G)) = Z_{i+1}(G)/Z(G)$ , so if  $G$  is nilpotent of class  $n$ , then  $G/Z(G)$  is nilpotent of class  $n - 1$ .

Now, assume (2). Let  $P = P_i$  for some  $i$ , and let  $N = N_G(P)$ . Since  $P \trianglelefteq N$ , we must have  $P \text{ char } N$  by Corollary 1.4.60. Since  $P \text{ char } N \trianglelefteq N_G(N)$ , we have that  $P \trianglelefteq N_G(N)$ , so that  $N_G(N) \leq N$ . Thus,  $N_G(N) = N$ . By (2), we must therefore have  $N = G$ , so that  $P \trianglelefteq G$ .

We now prove that (3)  $\Rightarrow$  (4). We show that, for any  $1 \leq t \leq s$ ,  $P_1 P_2 \dots P_t \cong P_1 \times P_2 \times \dots \times P_t$ . Note that  $P_i \trianglelefteq G$ , so  $P_1 \dots P_t \leq G$ . Let  $H = P_1 \dots P_{t-1}$ , and  $K = P_t$ , so that  $H \cong P_1 \times P_2 \times \dots \times P_{t-1}$  by induction. In particular,  $|H| = |P_1| \cdot |P_2| \cdot \dots \cdot |P_{t-1}|$ . Since  $|K| = |P_t|$  is relatively prime to  $|H|$ , we have that  $H \cap K = 1$ . By definition,  $P_1 \dots P_t = HK$ , and so  $HK \cong H \times K = (P_1 \times \dots \times P_{t-1}) \times P_t \cong P_1 \times P_2 \times \dots \times P_t$ . Now, take  $t = s$  to obtain (4).

Finally, we show that (4)  $\Rightarrow$  (1). Note that

$$Z(P_1 \times \dots \times P_s) \cong Z(P_1) \times \dots \times Z(P_s),$$

and that

$$\text{Inn}(P_1 \times \dots \times P_s) \cong \text{Inn}(P_1) \times \dots \times \text{Inn}(P_s),$$

so

$$(P_1 \times \dots \times P_s)/Z(P_1 \times \dots \times P_s) \cong (P_1/Z(P_1)) \times \dots \times (P_s/Z(P_s)).$$

Thus, (4) is also true for  $G/Z(G)$ . If  $P_t \neq 1$ , then  $Z(P_t) \neq 1$ , so if  $G \neq 1$ , then  $|G/Z(G)| < |G|$ . We may use induction, and therefore assume that  $G/Z(G)$  is nilpotent by the inductive hypothesis. Similar to above, we have that  $G$  is nilpotent. ■

**Corollary 1.6.10.** *A finite abelian group is the direct product of its Sylow subgroups.*

*Proof.* Abelian groups are nilpotent. ■

**Proposition 1.6.11.** *If  $G$  is a finite group such that for all positive integers  $n$  dividing  $|G|$ ,  $G$  contains at most  $n$  elements  $x$  satisfying  $x^n = 1$ , then  $G$  is cyclic.*

*Proof.* Let  $|G| = p_1^{\alpha_1} \cdots p_s^{\alpha_s}$ , and let  $P_i$  be a Sylow  $p_i$ -subgroup of  $G$ . Since  $p_i^{\alpha_i} \mid |G|$  and  $|P_i| = p_i^{\alpha_i}$ , we have that each element of  $P_i$  is a solution of  $x^{p_i^{\alpha_i}} = 1$ . Since there are at most  $p_i^{\alpha_i}$  of these solutions,  $P_i$  must consist of only these solutions. Thus,  $P_i$  is the unique (and hence normal) Sylow  $p_i$ -subgroup of  $G$ . Thus,  $G$  is isomorphic to the direct product of its Sylow subgroups. We also know that each  $P_i$  must possess a normal  $M_i$  subgroup of index  $p_i$ . Thus,  $|M_i| = p_i^{\alpha_i - 1}$ . Since  $G$  has at most  $p_i^{\alpha_i - 1}$  solutions to  $x^{p_i^{\alpha_i - 1}} = 1$ ,  $M_i$  must contain all of them. Thus any element of  $P_i$  not contained in  $M_i$  satisfies  $x^{p_i^{\alpha_i}} = 1$ , but  $x^{p_i^{\alpha_i - 1}} \neq 1$ , meaning that they have order  $p_i^{\alpha_i}$ . Thus,  $P_i$  is cyclic for all  $i$ , so  $G$  is the direct product of cyclic groups of relatively prime order. Hence,  $G$  is cyclic. ■

**Proposition 1.6.12** (Frattini's Argument). *Let  $G$  be a finite group,  $H \trianglelefteq G$ , and  $P \in \text{Syl}_p(H)$ . Then  $G = HN_G(P)$  and  $[G : H]$  divides  $|N_G(P)|$ .*

*Proof.* Note that  $HN_G(P) \leq G$ , and that  $HN_G(P) = N_G(P)H$ , since  $H \trianglelefteq G$ . Let  $g \in G$ . Recall that we write  $S^g$  for  $g^{-1}Sg$ . With this notation, we have that  $P^g \leq H^g = H$ , so  $P$  and  $P^g$  are Sylow  $p$ -subgroups of  $H$ . Thus, there is some  $x \in H$  so that  $P^g = P^h$ . Thus,  $gx^{-1} \in N_G(P)$ , and so  $g \in N_G(P)x$ . Since  $g$  was arbitrary,  $G \leq N_G(P)H = HN_G(P)$ , so  $G = HN_G(P)$ . By the Second Isomorphism Theorem,

$$[G : H] = [N_G(P) : N_G(P) \cap H],$$

so  $[G : H]$  divides  $|N_G(P)|$ . ■

**Proposition 1.6.13.** *A finite group is nilpotent if and only if every maximal subgroup is normal.*

*Proof.* Let  $G$  be a finite nilpotent group, and let  $M$  be a maximal subgroup. Then  $M < N_G(M)$ , so  $N_G(M) = G$ , meaning that  $M \trianglelefteq G$ . Now, suppose that every maximal subgroup of  $G$  is normal, and let  $P \in \text{Syl}_p(G)$ . We show  $P \trianglelefteq G$ , so that  $G$  is nilpotent. If  $P$  is not normal, let  $M$  be a maximal subgroup of  $G$  containing  $N_G(P)$ . Then  $M \trianglelefteq G$ , and Frattini's Argument implies that  $G = MN_G(P)$ . Since  $N_G(P) \leq M$ , we have that  $MN_G(P) = M$ , a contradiction. ■

**Definition 1.6.14.** For any group  $G$ , define  $G^0 = G$ ,  $G^1 = [G, G]$ , and  $G^{i+1} = [G, G^i]$ . The chain of groups

$$G^0 \geq G^1 \geq G^2 \geq \cdots$$

is called the lower central series of  $G$ .

**Proposition 1.6.15.** *For all  $i$ ,  $G^i \text{ char } G$ .*

*Proof.* We prove this by induction on  $i$ , with  $i = 0$  being trivial. Now, for  $x^{-1}y^{-1}xy \in G^{i+1}$  and  $\sigma \in \text{Aut}(G)$ , we have that  $\sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y)$ . Since  $\sigma(x) \in G$  and  $\sigma(y) \in G^i$ , we have that  $G^{i+1} \text{ char } G$ . ■

**Theorem 1.6.16.** *A group  $G$  is nilpotent if and only if  $G^n = 1$  for some  $n \geq 0$ . More precisely,  $G$  is nilpotent of class  $c$  if and only if  $c$  is the smallest nonnegative integer so that  $G^c = 1$ . We also have that*

$$Z_i(G) \leq G^{c-i-1} \leq Z_{i+1}(G)$$

for  $0 \leq i \leq c-1$ .

*Proof.* We prove this by induction on  $c$ , with the case  $c = 0$  being trivial. Now, suppose that  $G$  is nilpotent of class  $c$ . Then  $G/Z(G)$  is nilpotent of class  $c-1$ , so  $(G/Z(G))^{c-1} = 1$ . Note that this is equal to  $G^{c-1}/Z(G)$ . Thus,  $G^{c-1} = Z(G)$ , so  $G^c = [G, G^{c-1}] = [G, Z(G)] = 1$ . Now, suppose that  $G^c = 1$ . Notice that  $H := G^{c-1} \leq Z(G)$ . We also have that  $(G/H)^{c-1} = G^{c-1}/H = 1$ . Thus  $G/H$  is nilpotent of class  $c-1$ . We have that  $Z_0(G/H) = Z_0(G)/H$ , and that (by induction)

$$\begin{aligned} Z_{i+1}(G/H)/Z_i(G/H) &= Z(G/Z_i(G/H)) = Z((G/H)/(Z_i(G)/H)) \\ &= Z(G/Z_i(G)) = Z_{i+1}(G)/Z_i(G) = (Z_{i+1}(G)/H)/(Z_i(G)/H), \end{aligned}$$

so that  $Z_{i+1}(G/H) = Z_{i+1}(G)/H$ . In particular,  $G/H = Z_c(G/H) = Z_c(G)/H$ , so that  $Z_c(G) = G$ . Now, the second statement is also true when  $c = 1$ . Now, suppose that it is true for  $c-1$ , and that  $G$  is nilpotent of class  $c$ . Then,

$$Z_i(G/Z(G)) \leq (G/Z(G))^{c-i-2} \leq Z_{i+1}(G/Z(G))$$

for all  $i$ . Thus,

$$Z_{i+1}(G)/Z(G) \leq G^{c-(i+1)-1}/Z(G) \leq Z_{(i+1)+1}(G)/Z(G)$$

for  $0 \leq i \leq c-2$ . The Fourth Isomorphism Theorem implies the statement for  $1 \leq i \leq c-1$ . Finally, when  $i = 0$ , this is trivial. ■

**Definition 1.6.17.** Let  $G$  be a group. We define  $G^{(0)} = G$ ,  $G^{(1)} = [G, G]$  and  $G^{(i+1)} = [G^{(i)}, G^{(i)}]$  for all  $i \geq 1$ . This series of subgroups is called the derived or commutator series of  $G$ .

**Notation.** We may also write  $G^{(1)} = G'$ ,  $G^{(2)} = G''$ , and so on.

*Remark 1.6.18.* For each  $i$ ,  $G^{(i)} \text{ char } G$ . This result is trivial, so we will not prove it here. It is also important to note that in general  $G^{(i)} \leq G^i$ , since the former does not take the commutator with the whole group  $G$ . It is true, however, that  $G^{(i)} \leq G^i$ , for that reason.

**Theorem 1.6.19.** *A group  $G$  is solvable if and only if  $G^{(n)} = 1$  for some  $n \geq 0$ .*

*Proof.* Suppose that  $G$  is solvable, and that  $1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_s = G$  so that  $H_{i+1}/H_i$  is abelian. We prove that  $G^{(i)} \leq H_{s-i}$  by induction, with the case  $i = 0$  being trivial. Now, assume that  $G^{(i)} \leq H_{s-i}$ . Then,  $G^{(i+1)} = [G^{(i)}, G^{(i)}] \leq [H_{s-i}, H_{s-i}]$ . Since  $H_{s-i}/H_{s-i-1}$  is abelian, part 4 of Proposition 1.5.18, we have that  $[H_{s-i}, H_{s-i}] \leq H_{s-i-1}$ . Now, since  $H_0 = 1$ , we have that  $G^{(s)} = 1$ .

Now, suppose that  $G^{(n)} = 1$  for some  $n \geq 0$ . By part 4 of Proposition 1.5.18, if  $H_i = G^{(n-i)}$ , then  $H_i \trianglelefteq H_{i+1}$  and  $H_{i+1}/H_i$  is abelian. ■

**Definition 1.6.20.** Let  $G$  be solvable. The smallest nonnegative integer  $n$  for which  $G^{(n)} = 1$  is called the solvable length of  $G$ .

*Remark 1.6.21.* The derived series is a series of shortest length whose successive quotients are abelian, with the property that it consists of subgroups that are characteristic in the whole group.

**Proposition 1.6.22.** Let  $G$  and  $K$  be groups,  $H \leq G$ , and  $\phi : G \rightarrow K$  be a surjective homomorphism.

- (1)  $H^{(i)} \leq G^{(i)}$  for all  $i \geq 0$ . In particular, if  $G$  is solvable, then  $H$  is as well (with solvable length less than or equal to that of  $G$ ).
- (2)  $\phi(G^{(i)}) = K^{(i)}$ . In particular, quotient groups of solvable groups are solvable (with solvable length at most that of the domain group).
- (3) If  $N \trianglelefteq G$  and  $N$  and  $G/N$  are solvable, then  $G$  is solvable.

*Proof.*

- (1) Since  $H \leq G$ , we have that  $[H, H] \leq [G, G]$ , and by induction,  $H^{(i)} \leq G^{(i)}$ . In particular, if  $G^{(n)} = 1$ , then  $H^{(n)} = 1$ .
- (2) We have that  $\phi([x, y]) = [\phi(x), \phi(y)]$ , so  $\phi(G^{(1)}) = K^{(1)}$ , and induction implies that  $\phi(G^{(i)}) = K^{(i)}$ . Again, if  $G^{(n)} = 1$  for some  $n$ , then  $K^{(n)} = 1$  as well.
- (3) Suppose that  $G/N$  and  $N$  are solvable of lengths  $n$  and  $m$  respectively. Let  $\phi : G \rightarrow G/N$  be the natural projection. Then,  $\phi(G^{(n)}) = (G/N)^{(n)} = N$ , meaning that  $G^{(n)} \leq N$ . Thus,  $G^{(n+m)} = (G^{(n)})^{(m)} \leq N^{(m)} = 1$ , meaning that  $G$  is solvable. ■



# Chapter 2

## Ring Theory

### 2.1 Introduction to Rings

#### 2.1.1 Introduction

**Definition 2.1.1.** A ring  $R$  is a set together with two binary operations  $+$  and  $\times$  (called addition and multiplication) so that the following are true.

- (1)  $R$  forms an abelian group under  $+$ .
- (2)  $(a \times b) \times c = a \times (b \times c)$  for all  $a, b, c \in R$ .
- (3) For all  $a, b, c \in R$ ,  $(a + b) \times c = (a \times c) + (b \times c)$  and  $a \times (b + c) = (a \times b) + (a \times c)$ .

**Definition 2.1.2.** A ring  $R$  is said to be commutative if  $a \times b = b \times a$  for all  $a, b \in R$ .

**Definition 2.1.3.** A ring  $R$  is said to have an identity or contain a 1 if there is some  $1 \in R$  with  $1 \times a = a \times 1 = a$  for all  $a \in R$ .

**Notation.** We denote the additive identity of  $R$  as 0, and the additive inverse of  $a$  as  $-a$ . We will also normally write  $ab$  instead of  $a \times b$ .

*Remark 2.1.4.* If  $R$  contains an identity, then commutativity of addition follows from the other axioms  $(a + a + b + b = (1 + 1)(a + b) = a + b + a + b)$ .

**Definition 2.1.5.** A ring  $R$  with identity  $1 \neq 0$  is called a division ring or a skew field if, for every  $a \neq 0 \in R$ , there is some  $b \in R$  with  $ab = ba = 1$ . A commutative division ring is called a field.

*Example 2.1.6.* Take  $R$  to be an abelian group with identity  $0$ , and define  $a \times b = 0$ . This makes  $R$  into a commutative ring. If  $R = \{0\}$ , then  $R$  is called the zero ring, and is denoted  $R = 0$ . This is the only ring where  $1 = 0$ .

*Example 2.1.7.* The integers  $\mathbb{Z}$  form a ring. Further,  $\mathbb{Q}$ ,  $\mathbb{R}$ , and  $\mathbb{C}$  are all fields.

*Example 2.1.8.* The group  $\mathbb{Z}/n\mathbb{Z}$  is a commutative ring with identity  $(1 \bmod n)$  under addition and multiplication  $\bmod n$ .

*Example 2.1.9.* The quaternions  $\mathbb{H}$  are a division ring with identity. However, they are not a field.

*Example 2.1.10.* Let  $X \neq \emptyset$  be a set and  $A$  be a ring. The set of functions  $f : X \rightarrow A$  form a ring under  $(f + g)(x) = f(x) + g(x)$  and  $(fg)(x) = f(x)g(x)$ . This is commutative if and only if  $A$  is commutative, and it contains an identity if and only if  $A$  does as well.

*Example 2.1.11.* The set  $2\mathbb{Z}$  of even integers is a ring that does not contain an identity.

*Example 2.1.12.* The set  $\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} : a, b \in \mathbb{Q}\}$  is a ring, where  $D \in \mathbb{Q}$  is not a square. In fact, it is a field.

**Proposition 2.1.13.** *Let  $R$  be a ring.*

- (1)  $0a = a0 = 0$  for all  $a \in R$ .
- (2)  $(-a)b = a(-b) = -(ab)$  for all  $a, b \in R$ .
- (3)  $(-a)(-b) = ab$  for all  $a, b \in R$ .
- (4) If  $R$  has an identity  $1$ , then it is unique and  $-a = (-1)a$ .

*Proof.* (1) We have that  $0a = (0 + 0)a = 0a + 0a$ , meaning that  $0a = 0$ . Similarly,  $a0 = a(0 + 0) = a0 + a0$ , so  $a0 = 0$ .

(2) We have that  $ab + (-a)b = (a - a)b = 0b = 0$ . Since inverses in a group are unique,  $(-a)b = -(ab)$ . Similarly, we have that  $ab + a(-b) = a(b - b) = a0 = 0$ , so that  $a(-b) = -(ab)$ .

(3) We have that  $(-a)(-b) = a(- - b) = ab$ .

(4) Suppose that  $1$  and  $1'$  are identities. Then,  $1 = 11' = 1'$ , so the identity is unique. Now,  $(-1)a = 1(-a) = -a$ . ■

**Definition 2.1.14.** Let  $R$  be a ring. An element  $a \neq 0 \in R$  is called a zero divisor if there is some  $b \neq 0 \in R$  with either  $ab = 0$  or  $ba = 0$ . If  $R$  has an identity  $1 \neq 0$ , an element  $u \in R$  is called a unit in  $R$  if there is some  $v \in R$  so that  $uv = vu = 1$ . The set of units in  $R$  is denoted  $R^\times$ .

**Proposition 2.1.15.** *Suppose that  $R$  is a ring and  $a \in R$ . If  $a$  is a zero divisor, then it is not a unit, and vice versa.*



*Proof.* Suppose that  $a$  is a zero divisor with  $ab = 0$  and that it is a unit with  $ac = ca = 1$ . Then,  $0 = c0 = cab = 1b = b$ , which is a contradiction. The case with  $ba = 0$  is similar. ■

**Corollary 2.1.16.** *Fields do not contain zero divisors.*

**Definition 2.1.17.** Let  $R$  be a commutative ring with identity  $1 \neq 0$ . Then  $R$  is called an integral domain if it has no zero divisors.

**Proposition 2.1.18** (Cancellation Law). *Suppose that  $R$  is a ring with  $a, b, c \in R$ . Suppose further that  $a$  is not a zero divisor. Then, if  $ab = ac$ , either  $a = 0$  or  $b = c$ . In particular, if  $R$  is an integral domain and  $ab = ac$ , then either  $a = 0$  or  $b = c$ .*

*Proof.* If  $ab = ac$ , then  $ab - ac = a(b - c) = 0$ , so either  $a = 0$  or  $b - c = 0$ , meaning  $b = c$ . The second statement follows from the definition of an integral domain. ■

**Corollary 2.1.19.** *Any finite integral domain is a field.*

*Proof.* Let  $R$  be a finite integral domain, and let  $a \neq 0 \in R$ . Then, the map  $x \mapsto ax$  is an injective function by the previous proposition. Since  $R$  is finite, this is surjective, meaning that there is some  $b \in R$  with  $ab = 1$ . Thus,  $a$  is a unit. Since  $a$  was arbitrary, any nonzero element of  $R$  is a unit. ■

**Definition 2.1.20.** Let  $R$  be a ring. A subring of  $R$  is a subgroup of  $R$  that is closed under multiplication. In other words, a  $S \subseteq R$  is a subring of  $R$  if addition and multiplication restricted to  $S$  give  $S$  a ring structure.

*Example 2.1.21.* If  $D$  is a squarefree integer, then  $\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} : a, b \in \mathbb{Z}\}$  is a subring of  $\mathbb{Q}(\sqrt{D})$ . If  $D \equiv 1 \pmod{4}$ , then  $\mathbb{Z}\left[\frac{1+\sqrt{D}}{2}\right] = \{a + b\frac{1+\sqrt{D}}{2} : a, b \in \mathbb{Z}\}$  is also a subring (closure under multiplication follows from  $(a + b\frac{1+\sqrt{D}}{2})(c + d\frac{1+\sqrt{D}}{2}) = (ac + bd\frac{D-1}{4}) + (ad + bc + bd)\frac{1+\sqrt{D}}{2}$ ). We define

$$\mathcal{O} = \mathcal{O}_{\mathbb{Q}(\sqrt{D})} = \mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\},$$

where

$$\omega = \begin{cases} \sqrt{D} & D \equiv 2, 3 \pmod{4} \\ \frac{1+\sqrt{D}}{2} & D \equiv 1 \pmod{4}. \end{cases}$$

This is called the ring of integers in the quadratic field  $\mathbb{Q}(\sqrt{D})$ , and is called a Quadratic Integer Ring when standing alone. The field norm  $N : \mathbb{Q}(\sqrt{D}) \rightarrow \mathbb{Q}$  is defined by  $N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \in \mathbb{Q}$ , which is nonzero if  $a + b\sqrt{D} \neq 0$ . It is easy to see that  $N$  is multiplicative. On the subring  $\mathcal{O}$ , the field norm is given by

$$N(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = \begin{cases} a^2 - Db^2 & D \equiv 2, 3 \pmod{4} \\ a^2 + ab + \frac{1-D}{4}b^2 & D \equiv 1 \pmod{4}, \end{cases}$$

where

$$\bar{\omega} = \begin{cases} -\sqrt{D} & D \equiv 2, 3 \pmod{4} \\ \frac{1-\sqrt{D}}{2} & D \equiv 1 \pmod{4}. \end{cases}$$

This is an integer for every  $a \in \mathcal{O}$ . It is not hard to see that  $\alpha \in \mathcal{O}$  is a unit if and only if  $N(\alpha) = \pm 1$ .

## 2.1.2 Polynomial Rings, Matrix Rings and Group Rings

### Polynomial Rings

**Definition 2.1.22.** Let  $R$  be a commutative ring with identity, and let  $x$  be an indeterminate. The sum  $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$  with  $n \geq 0$  and  $a_i \in R$  is called a polynomial in  $x$  with coefficients  $a_i$  in  $R$ . If  $a_n \neq 0$ , then the polynomial is said to be of degree  $n$ ,  $a_n x^n$  is called the leading term, and  $a_n$  is called the leading coefficient (where the leading coefficient of 0 is 0). The polynomial is monic if  $a_n = 1$ . The set of all such polynomials is called the ring of polynomials in the variable  $x$  with coefficients in  $R$ , and is denoted  $R[x]$ . Addition is defined componentwise, and multiplication is defined as  $(ax^i)(bx^j) = abx^{i+j}$ , and extending this using the distributive laws. The ring  $R$  appears in  $R[x]$  as the constant polynomials, and  $R[x]$  is a commutative ring with identity.

**Proposition 2.1.23.** Let  $R$  be an integral domain, and let  $p(x), q(x) \in R[x]$  be nonzero. Let  $\deg p(x)$  denote the degree of  $p(x)$ , and similarly for  $q(x)$ .

- (1)  $\deg p(x)q(x) = \deg p(x) + \deg q(x)$ .
- (2) The units in  $R[x]$  are just the units in  $R$ .
- (3)  $R[x]$  is an integral domain.

*Proof.*

- (1) Suppose that the leading terms of  $p(x)$  and  $q(x)$  are  $a_n x^n$  and  $b_m x^m$  respectively. Then, the leading term of  $p(x)q(x)$  is  $a_n b_m x^{n+m}$ .
- (2) Suppose that  $p(x)$  is a unit with  $p(x)q(x) = 1$ . Then,  $0 = \deg p(x)q(x) = \deg p(x) + \deg q(x)$ . Thus,  $p(x), q(x) \in R$ , and are units in  $R$ .
- (3) From the proof of (1), the leading coefficient of  $p(x)q(x)$  is nonzero if  $p(x)$  and  $q(x)$  are both nonzero. ■

### Matrix Rings

**Definition 2.1.24.** Let  $R$  be a ring and  $n \in \mathbb{Z}^+$ . Then  $M_n(R)$  denotes the set of all  $n \times n$  matrices with entries from  $R$ .  $M_n(R)$  is a ring under the usual rules of matrix addition and multiplication.

*Remark 2.1.25.* If  $R$  is nontrivial, then  $M_n(R)$  is not commutative, even if  $R$  is. To see this, take  $a, b \in R$  so that  $ab \neq 0$ . Let  $A$  have  $a$  in position 1, 1 and zeros everywhere else, and  $B$  have  $b$  in position 1, 2 and zeroes everywhere else. Then  $AB$  is in position 1, 2 of  $AB$ , while  $BA$  is the zero matrix.  $M_n(R)$  also has zero divisors whenever  $n \geq 2$ .

**Definition 2.1.26.** An element  $(a_{ij}) \in M_n(R)$  is called a scalar matrix if there is some  $a \in R$  with  $a_{ii} = a$  for all  $i$  and  $a_{ij} = 0$  for all  $i \neq j$ . The set of scalar matrices is a subring of  $M_n(R)$ . If  $R$  is commutative, the set of scalar matrices commute, and if  $R$  has a 1, then the matrix with 1's down the diagonal is the 1 of  $M_n(R)$ . In this case, the units of  $M_n(R)$  are the invertible  $n \times n$  matrices, and the group of units is called the general linear group of degree  $n$  over  $R$ , denoted  $GL_n(R)$ .

## Group Rings

**Definition 2.1.27.** Let  $R$  be a commutative ring with identity  $1 \neq 0$ , and let  $G = \{g_1, g_2, \dots, g_n\}$  be a finite group with operation written multiplicatively. We define the group ring  $RG$  of  $G$  with coefficients in  $R$  to be the set of all formal sums  $a_1g_1 + a_2g_2 + \dots + a_ng_n$  where  $a_i \in R$ . If  $g_1$  is the identity of  $G$ , we write  $a_1g_1$  as  $a_1$ , as we write  $1g$  as  $g$  for  $g \in G$ . We define addition componentwise as

$$(a_1g_1 + \dots + a_ng_n) + (b_1g_1 + \dots + b_ng_n) = (a_1 + b_1)g_1 + (a_2 + b_2)g_2 + \dots + (a_n + b_n)g_n.$$

We perform multiplication by extending the operation  $(ag_i)(bg_j) = (ab)g_k$ , where  $g_k = g_i g_j$  in  $G$  with the distributive laws. This makes  $RG$  into a ring. This is commutative if and only if  $G$  is abelian.

*Remark 2.1.28.* If  $|G| > 1$ , then  $RG$  contains zero divisors. For example, if  $g \in G$  has order  $m > 1$ , then  $(1 - g)(1 + g + \dots + g^{m-1}) = 1 - g^m = 1 - 1 = 0$ , meaning that  $1 - g$  is a zero divisor.

### 2.1.3 Ring Homomorphisms and Quotient Rings

**Definition 2.1.29.** Let  $R$  and  $S$  be rings. A ring homomorphism is a map  $\phi : R \rightarrow S$  satisfying  $\phi(a + b) = \phi(a) + \phi(b)$  and  $\phi(ab) = \phi(a)\phi(b)$  for all  $a, b \in R$ .

**Definition 2.1.30.** Let  $R$  and  $S$  be rings, and  $\phi : R \rightarrow S$  be a (ring) homomorphism. The kernel of  $\phi$ , denoted  $\ker \phi$  is the set of elements in  $R$  that map to 0 in  $S$  under  $\phi$ .

**Definition 2.1.31.** A bijective ring homomorphism is called an isomorphism. If there is an isomorphism between rings  $R$  and  $S$ , we write  $R \cong S$ .

**Proposition 2.1.32.** Let  $R$  and  $S$  be rings, and let  $\phi : R \rightarrow S$  be a homomorphism. Then the image of  $\phi$  is a subring of  $S$ , and the kernel of  $\phi$  is a subring of  $R$ . Further, if  $a \in \ker \phi$ , then  $ra, ar \in \ker \phi$  for every  $r \in R$ .

*Proof.* If  $s_1 = \phi(r_1)$  and  $s_2 = \phi(r_2)$ , then  $\phi(r_1 - r_2) = s_1 - s_2$ , and  $\phi(r_1 r_2) = s_1 s_2$ , so  $\text{im } \phi$  is a subring of  $S$ . Similarly, if  $r_1, r_2 \in \ker \phi$ , then  $\phi(r_1 - r_2) = \phi(r_1) - \phi(r_2) = 0$ , and  $\phi(r_1 r_2) = 0$ , meaning that  $\ker \phi$  is a subring of  $R$ . Now,  $\phi(ar) = \phi(a)\phi(r) = 0\phi(r) = 0$ , and  $\phi(ra) = \phi(r)\phi(a) = \phi(r)0 = 0$ . ■

*Remark 2.1.33.* Suppose that  $R$  and  $S$  are rings,  $\phi : R \rightarrow S$  is a homomorphism, and  $\ker \phi = I$ . Since  $R$  and  $S$  are additive groups, since  $\phi$  is a homomorphism of these groups, and since the fibers of  $\phi$  are the additive cosets  $r + I$ , the fibers have a structure of a ring that is naturally isomorphic to the image of  $\phi$ : if  $X_a$  denotes the fiber over  $a$ , then the fiber over  $a + b$  is  $X_a + X_b$ , and the fiber over  $ab$  is  $X_a X_b$ . In terms of the cosets of  $I$ , we have that  $(r + I) + (s + I) = (r + s) + I$ , and that  $(r + I) \times (s + I) = (rs) + I$ . It is not hard to check that these define a ring structure. This ring of cosets is called the quotient ring of  $R$  by  $I$ , denoted  $R/I$ .

We can consider whether we can define a ring structure on the set of cosets of  $I$  for an arbitrary subgroup  $I$  of  $R$ . Note that addition is well defined on these cosets, as  $R$  is an

additive abelian group, so  $I$  must be a normal subgroup. To check that multiplication is well defined, take  $\alpha, \beta \in I$ . Then, we must have that  $(r + \alpha)(s + \beta) + I = rs + I$  for all  $r, s \in R$  (and all  $\alpha, \beta \in I$ ). If we take  $r = s = 0$ , we have that  $I$  must be closed under multiplication, and so must be a subring of  $R$ . Now, if  $s = 0$  and  $r$  is arbitrary, we have that  $r\beta \in I$  for all  $r \in R$  and  $\beta \in I$ , meaning that  $I$  must be closed under left multiplication by elements of  $R$  (in other words,  $rI \subseteq I$  for all  $r \in R$ ). Similarly, letting  $r = 0$ , we have that  $I$  must also be closed under right multiplication by elements of  $R$ . Conversely, if  $rI$  and  $Ir$  are subsets of  $I$  for all  $r \in R$  (in particular implying that  $I$  is closed under multiplication), then the multiplication is well defined, so this is a necessary and sufficient condition. It is not hard to check that this makes the additive quotient group  $R/I$  into a ring.

**Definition 2.1.34.** Let  $R$  be a ring.

- (1)  $I \subseteq R$  is called a left ideal of  $R$  if  $I$  is a subring of  $R$ , and if  $rI \subseteq I$  for all  $r \in R$ .
- (2)  $I \subseteq R$  is called a right ideal of  $R$  if  $I$  is a subring of  $R$  and  $Ir \subseteq I$  for all  $r \in R$ .
- (3) A subset  $I \subseteq R$  that is both a left ideal and a right ideal is called an ideal (or a two-sided ideal) of  $R$ .

*Remark 2.1.35.* If  $R$  has an identity, an equivalent definition of a left ideal is that  $I$  is nonempty, closed under addition and left multiplication by  $R$  (multiplying by 0 gives the additive identity, and multiplying by  $-1$  gives additive inverses, and this guarantees multiplicative closure). In the case that  $R$  does not have an identity, we may replace closure under addition with closure under subtraction, by the subgroup criterion. Similarly for right ideals.

**Proposition 2.1.36.** Let  $R$  be a ring and  $I$  be an ideal of  $R$ . Then the additive quotient group  $R/I$  is a ring under the binary operations

$$(r + I) + (s + I) = (r + s) + I \quad \text{and} \quad (r + I) \times (s + I) = (rs) + I.$$

Conversely, if  $I$  is any subgroup of  $R$  such that the above operations are well defined, then  $I$  is an ideal of  $R$ .

**Definition 2.1.37.** Let  $R$  be a ring and  $I$  an ideal. Then the ring  $R/I$  is called the quotient ring of  $R$  by  $I$ .

**Theorem 2.1.38** (The First Isomorphism Theorem for Rings). Let  $\phi : R \rightarrow S$  be a homomorphism of rings. Then  $R/\ker \phi \cong \text{im } \phi$  as a ring (recall that we proved that  $\ker \phi$  is an ideal previously).

*Proof.* Let  $\psi : R/\ker \phi \rightarrow \text{im } \phi$  be defined as  $\psi(r + \ker \phi) = \phi(r)$ . It is clear that  $\psi$  is a homomorphism. Since  $\psi$  is clearly bijective, it is an isomorphism. ■

**Theorem 2.1.39.** If  $I$  is an ideal of  $R$ , then the map  $R \rightarrow R/I$  defined by  $r \mapsto r + I$  is a surjective ring homomorphism with kernel  $I$ . This homomorphism is called the natural projection of  $R$  onto  $R/I$ .

*Proof.* Since  $I$  is an ideal,  $R/I$  is a ring, and the map  $\pi : r \mapsto r + I$  is a group homomorphism with kernel  $I$ . It is clear that  $\pi$  is a ring homomorphism. ■

**Notation.** We may write  $\bar{r}$  for  $r + I$  when reducing mod  $I$ . We then have that  $\bar{r} + \bar{s} = \overline{r + s}$  and  $\bar{r} \cdot \bar{s} = \overline{rs}$ .

*Example 2.1.40.* The trivial ideal, denoted  $0$ , is  $\{0\}$ .

*Example 2.1.41.*  $n\mathbb{Z}$  is an ideal of  $\mathbb{Z}$  for any  $n \in \mathbb{Z}$ . This gives a quotient ring of  $\mathbb{Z}/n\mathbb{Z}$ .

**Theorem 2.1.42.** *Let  $R$  be a ring.*

- (1) *(The Second Isomorphism Theorem for Rings) Let  $A$  be a subring of  $R$  and  $B$  an ideal. Then  $A + B = \{a + b : a \in A, b \in B\}$  is a subring of  $R$ ,  $A \cap B$  is an ideal of  $A$ , and  $(A + B)/B \cong A/(A \cap B)$ .*
- (2) *(The Third Isomorphism Theorem for Rings) Let  $I$  and  $J$  be ideals of  $R$  with  $I \subseteq J$ . Then  $J/I$  is an ideal of  $R/I$ , and  $(R/I)/(J/I) \cong R/J$ .*
- (3) *(The Fourth or Lattice Isomorphism Theorem for Rings) Let  $I$  be an ideal of  $R$ . The correspondence  $A \leftrightarrow A/I$  is an inclusion preserving bijection between the set of subrings  $A$  of  $R$  containing  $I$  and the set of subrings of  $R/I$ . Further,  $A$  (a subring containing  $I$ ) is an ideal of  $R$  if and only if  $A/I$  is an ideal of  $R/I$ .*

*Proof.*

- (1) Since  $A, B \leq R$ , and  $R$  is an additive abelian group, we have that  $A + B \leq R$ , so we only need to check that it is closed under multiplication. For this, we have that  $(a + b)(a' + b') = aa' + ab' + ba' + bb'$ , where the last 3 terms are in  $B$ , and the first is in  $A$ . Again by the Second Isomorphism Theorem (for groups), we have that  $A \cap B \leq A$ , so we only need to check that it is closed under multiplication by elements of  $A$ , which is clear. Now, define  $\phi : A \rightarrow (A + B)/B$  by  $a \mapsto a + B$ . This is clearly a (ring) homomorphism. Note also that it is surjective. The kernel of  $\phi$  is exactly  $A \cap B$ , so  $A/A \cap B \cong AB/B$  (as rings).
- (2) By the Third Isomorphism Theorem (for groups), we have that  $J/I \leq R/I$ . We also have that  $rI(J/I) \subseteq J/I$  and  $(J/I)rI \subseteq J/I$ , since  $J$  is an ideal, meaning that  $J/I$  is an ideal of  $R/I$ . Now, define  $\phi : R/I \rightarrow R/J$  by  $gI \mapsto gJ$ . Since  $I \subseteq J$ , this is well defined. It is clearly a surjective homomorphism. It has kernel  $J/I$ , meaning that  $(R/I)/(J/I) \cong R/J$ . (Much of the reasoning here is identical to that in the proof of the Third Isomorphism Theorem for groups).
- (3) We essentially wish to show that every subring of  $R/I$  is of the form  $A/I$ , where  $A$  is a subring of  $R$ , that  $A/I$  is a subring of  $R/I$  for each  $A$  that is a subring of  $R$  (which we only state for completeness, as it is clear), that  $A$  is a subring of  $B$  (where  $I \subseteq A, B$ ) if and only if  $A/I$  is a subring of  $B/I$ , and that  $A$  is an ideal of  $R$  if and only if  $A/I$  is an ideal of  $R/I$ .

Consider the map  $\phi$  from the set of subrings of  $R$  containing  $I$  to the set of subrings of  $R/I$  defined by  $\phi(A) = A/I$ . This is injective, so we show that it is surjective.

Let  $A$  be a subring of  $R/I$ , and let it be the set  $\{a_1 + I, a_2 + I, \dots, a_n + I\}$ , where every possible choice of representative for each coset appears as an  $a_i$ . Then the set  $\mathcal{A} := \{a_1, a_2, \dots, a_n\}$  is a subring of  $R$ , and  $A = \mathcal{A}/I$ . Thus,  $\phi$  is surjective. Now, if  $A$  is a subring of  $B$ , then it is clear that  $A/I$  is a subring of  $B/I$ . Now, if  $A/I$  is a subring of  $B/I$ , then  $a + I \in A/I \subseteq B/I$ , meaning that  $a + I = b + I$  for some  $b \in B$ . Since  $0 = 0I \subseteq I$ , we have in particular that  $a \in b + I$ . Thus, for some  $i \in I \subseteq B$ , we have that  $a = b + i \in B$ , meaning that  $A \subseteq B$ . Thus,  $A$  is a subring of  $B$ . If  $A$  is an ideal of  $R$ , it is clear that  $A/I$  is an ideal of  $R/I$ . Now suppose that  $A/I$  is an ideal of  $R/I$ , so that  $(r + I)(a + I) \in A/I$ . Then,  $(ra) + I \in A/I$ , so  $ra \in A$ . Similarly,  $(a + I)(r + I) \in A/I$ , meaning that  $(ar) + I \in A/I$ , so  $ar \in A$ . Thus,  $A$  is an ideal of  $R$ . ■

**Definition 2.1.43.** Let  $I$  and  $J$  be ideals of a ring  $R$ .

- (1) The sum of  $I$  and  $J$  is  $I + J = \{a + b : a \in I, b \in J\}$ .
- (2) The product of  $I$  and  $J$ , denoted  $IJ$ , is the set of all finite sums of elements of the form  $ab$  with  $a \in I$  and  $b \in J$ .
- (3) For any  $n \geq 1$ , the  $n$ th power of  $I$ , denoted  $I^n$ , is defined inductively as  $I^1$  and  $I^n = II^{n-1}$ .

*Remark 2.1.44.* If  $I$  and  $J$  are ideals, then  $I + J$  is the smallest ideal of  $R$  containing  $I$  and  $J$ . It is also worth noting that  $IJ \subseteq I \cap J$ .

### 2.1.4 Properties of Ideals

*Remark 2.1.45.* In this subsection,  $R$  is a ring with identity  $1 \neq 0$ .

**Definition 2.1.46.** Let  $A \subseteq R$ .

- (1) The ideal generated by  $A$  is the smallest ideal of  $R$  containing  $A$ , and is denoted  $(A)$ .
- (2) We write  $RA$  for the set of all finite sums of elements of the form  $ra$  with  $a \in A$  and  $r \in R$ . Similarly,  $AR$  contains all finite sums of elements of the form  $ar$  with  $a \in A, r \in R$ , and  $RAR$  contains all finite sums of elements of the form  $r_1ar_2$  with  $a \in A$  and  $r_1, r_2 \in R$ .
- (3) An ideal generated by a single element is called a principle ideal.
- (4) An ideal generated by a finite set is called finitely generated.

**Notation.** If  $A = \{a_1, a_2, \dots, a_n\}$ , then we write  $(A) = (a_1, a_2, \dots, a_n)$ , rather than  $(A) = (\{a_1, a_2, \dots, a_n\})$ .

**Proposition 2.1.47.** *We have that*

$$(A) = \bigcap_{\substack{I \text{ an ideal} \\ A \subseteq I}} I.$$

*Proof.* It is not hard to see that the intersection of ideals is an ideal. Since  $A \subseteq R$ , and  $R$  is an ideal of itself, we can say this. ■

**Definition 2.1.48.** Let  $A \subseteq R$ . The left ideal generated by  $A$  is the intersection of all left ideals of  $R$  containing  $A$ . The right ideal generated by  $A$  is defined similarly.

**Proposition 2.1.49.** Let  $A \subseteq R$ .

- (1) The left ideal generated by  $A$  is  $RA$ .
- (2) The right ideal generated by  $A$  is  $AR$ .
- (3) The ideal generated by  $A$  is  $RAR$ .

*Proof.*

- (1) Let  $(A)_L$  denote the left ideal generated by  $A$ . It is not hard to check that  $RA$  is a left ideal, and that  $A \subseteq RA$ . Thus,  $(A)_L \subseteq RA$ . Conversely, if  $I$  is a left ideal containing  $A$ , then it contains  $rA$  for  $r \in R$ , and all sums of these elements, meaning that  $RA \subseteq I$ . Thus,  $RA \subseteq (A)_L$ , so  $(A)_L = RA$ .
- (2) Let  $(A)_R$  denote the right ideal generated by  $A$ . It is not hard to check that  $AR$  is a right ideal, and that  $A \subseteq AR$ . Thus,  $(A)_R \subseteq AR$ . Conversely, if  $I$  is a right ideal containing  $A$ , then it contains  $Ar$  for  $r \in R$ , and all sums of these elements, meaning that  $AR \subseteq I$ . Thus,  $AR \subseteq (A)_R$ , so  $(A)_R = AR$ .
- (3) It is not hard to check that  $RAR$  is an ideal, and that  $A \subseteq RAR$ . Thus,  $(A) \subseteq RAR$ . Conversely, if  $I$  is an ideal containing  $A$ , then it contains  $rAr'$  for  $r, r' \in R$ , and all sums of these elements, meaning that  $RAR \subseteq I$ . Thus,  $RAR \subseteq (A)$ , so  $(A) = RAR$ . ■

**Proposition 2.1.50.** Let  $I$  be an ideal of  $R$ .

- (1)  $I = R$  if and only if  $I$  contains a unit.
- (2) Assume that  $R$  is commutative. Then  $R$  is a field if and only if its only ideals are  $0$  and  $R$ .

*Proof.*

- (1) If  $I = R$ , then it contains the unit  $1$ . If  $I$  contains a unit  $u$  with inverse  $v$ , then  $vu = 1 \in I$ , meaning that  $r \cdot 1 = r \in I$  for all  $r \in R$ . Thus,  $R = I$ .
- (2) Suppose that  $R$  is a field, and let  $I$  be an ideal. Unless  $I = 0$ ,  $I$  contains a nonzero element, which is a unit, since  $R$  is a field. Thus,  $I = R$ . Now suppose that  $0$  and  $R$  are the only ideals of  $R$ . Take  $a \in R$ . If  $a = 0$ , then  $(a) = 0$ . Otherwise,  $(a) \neq 0$ , so  $(a) = R$ . In particular,  $1 \in (a)$ . Note that  $(a) = Ra$ , since  $R$  is commutative. Thus, there is some  $b \in R$  with  $1 = ba$ , meaning that  $a$  is a unit. Thus,  $R$  is a field. ■

**Corollary 2.1.51.** *If  $R$  is a field then any nonzero ring homomorphism from  $R$  into another ring is an injection.*

*Proof.* If the homomorphism is nonzero, its kernel is not  $R$ . Since the kernel is an ideal, it must be 0. ■

**Definition 2.1.52.** An ideal  $M$  in an arbitrary ring  $S$  is called a maximal ideal if  $M \neq S$  and the only ideals containing  $M$  are  $M$  and  $S$ .

*Remark 2.1.53.* Our next proof requires Zorn's Lemma, so we state it with the appropriate terminology here.

*Definition 2.1.54.* A partial order on a nonempty set  $A$  is a relation  $\leq$  on  $A$  satisfying

- (1)  $x \leq x$  for all  $x \in A$  (reflexive)
- (2) If  $x \leq y$  and  $y \leq x$  then  $x = y$  for all  $x, y \in A$  (antisymmetric)
- (3) If  $x \leq y$  and  $y \leq z$  then  $x \leq z$  for all  $x, y, z \in A$  (transitive).

*Definition 2.1.55.* Let  $A \neq \emptyset$  be partially ordered by  $\leq$ .

- (1) A subset  $B \subseteq A$  is called a chain if, for all  $x, y \in B$ , either  $x \leq y$  or  $y \leq x$ .
- (2) An upper bound of  $B \subseteq A$  is an element  $u \in A$  so that  $b \leq u$  for all  $b \in B$ .
- (3) A maximal element of  $A$  is an element  $m$  such that if  $m \leq x$  for any  $x \in A$ , then  $m = x$ .

Zorn's Lemma states that if  $A$  is a nonempty partially ordered set in which every chain has an upper bound, then  $A$  contains a maximal element. We shall not prove this here.

**Proposition 2.1.56.** *Let  $R$  be a ring with identity. Then every proper ideal of  $R$  is contained in a maximal ideal.*

*Proof.* Let  $I$  be a proper ideal of  $R$ . Let  $\mathcal{S}$  be the set of all proper ideals of  $R$  containing  $I$ . Then  $\mathcal{S}$  is nonempty ( $I \in \mathcal{S}$ ), and is partially ordered by inclusion. If  $\mathcal{C}$  is a chain in  $\mathcal{S}$ , let  $J = \bigcup_{A \in \mathcal{C}} A$ . We show that  $J$  is an ideal. Since  $\mathcal{C}$  is nonempty,  $J$  is as well ( $0 \in J$ ). If  $a, b \in J$ , then there are  $A, B \in \mathcal{C}$  with  $a \in A$  and  $b \in B$ . Either  $A \subseteq B$  or  $B \subseteq A$ . In either case,  $a - b \in J$ , so  $J$  is closed under subtraction. Similarly,  $ab \in J$ , so  $J$  is an ideal. If  $J$  is not a proper ideal, then  $1 \in J$ , meaning that  $1 \in A$  for some  $A \in \mathcal{C}$ . Since  $A$  must be a proper ideal, this is a contradiction. Thus,  $J \in \mathcal{S}$ . Note that  $J$  is an upper bound of  $\mathcal{C}$ , meaning that we may apply Zorn's Lemma to get a maximal element of  $\mathcal{S}$ . This is a maximal ideal containing  $I$ . ■

**Proposition 2.1.57.** *Assume that  $R$  is commutative. The ideal  $M$  is a maximal ideal of and only if  $R/M$  is a field.*



*Proof.* Suppose that  $R/M$  is a field. Take  $a \in R \setminus M$ . Then, there is some  $b \in R$  so that  $(a + M)(b + M) = 1 + M$ . Thus,  $ab - 1 \in M$ . Write  $ab = 1 + m$  for  $m \in M$ . Now, consider  $(M \cup \{a\})$ . This contains  $a$ , and thus  $ab = 1 + m$ . Since it is closed under subtraction,  $1 + m - m = 1$  is in it as well, so  $(M \cup \{a\}) = R$ . Thus,  $M$  is maximal. Now, suppose that  $M$  is maximal. Then  $1 \in (M \cup \{a\})$ , and can be written as  $ab + m$  for  $b \in R \setminus M$  and  $m \in M$ . Thus,  $(a + M)(b + M) = (ab) + M = 1 + M$ , so  $R/M$  is a field. ■

**Definition 2.1.58.** Assume that  $R$  is commutative. An ideal  $P$  is called a prime ideal if  $P \neq R$ , and whenever  $ab \in P$  for  $a, b \in R$ , then at least one of  $a$  and  $b$  is an element of  $P$ .

*Remark 2.1.59.* This is in fact a generalization of the primes in  $\mathbb{Z}$ . An ideal  $n\mathbb{Z}$  is prime if, whenever  $ab \in n\mathbb{Z}$ , then  $a \in n\mathbb{Z}$  or  $b \in n\mathbb{Z}$ , and  $n \neq 1$ . Since  $n\mathbb{Z}$  contains all multiples of  $n$ , this happens exactly when  $n$  is prime.

**Proposition 2.1.60.** Suppose that  $R$  is commutative. Then the ideal  $P$  is prime in  $R$  if and only if  $R/P$  is an integral domain.

*Proof.* Suppose that  $P$  is prime. Take  $a + P, b + P \in R/P$  so that  $(a + P)(b + P) = 0 + P$ . Then,  $ab \in P$ . By the definition of a prime ideal, either  $a \in P$  or  $b \in P$ . If  $a \in P$ , then  $a + P = 0 + P$ , and similarly for  $b$ . Now, suppose that  $R/P$  is an integral domain. If  $ab \in P$ , then  $ab + P = 0 + P$ . Thus,  $a + P = 0 + P$  or  $b + P = 0 + P$ . In other words, at least one of  $a$  and  $b$  is in  $P$ . ■

**Corollary 2.1.61.** Suppose that  $R$  is commutative. Every maximal ideal of  $R$  is also a prime ideal.

*Proof.* If  $M$  is maximal, then  $R/M$  is a field. Since all fields are integral domains,  $M$  is prime. ■

## 2.1.5 Rings of Fractions

*Remark 2.1.62.* In this subsection,  $R$  is a commutative ring.

**Theorem 2.1.63.** Let  $R$  be a commutative ring, and let  $D \subseteq R$  be nonempty with the property that it does not contain any zero divisors, does not contain 0, and is closed under multiplication. Then there is a commutative ring  $Q$  with 1 such that  $Q$  contains  $R$  as a subring and every element of  $D$  is a unit in  $Q$ . The following are also true.

- (1) Every element of  $Q$  is of the form  $rd^{-1}$  for some  $r \in R$  and  $d \in D$ . In particular, if  $D = R \setminus \{0\}$  then  $Q$  is a field.
- (2) The ring  $Q$  is the smallest ring with this property in the following sense. If  $S$  is a commutative ring with identity and  $\phi : R \rightarrow S$  is an injective ring homomorphism such that  $\phi(d)$  is a unit in  $S$  for all  $d \in D$ , then there is an injective homomorphism  $\Phi : Q \rightarrow S$  so that  $\Phi|_R = \phi$ . In other words, a ring containing an isomorphic copy of  $R$  in which all elements of  $D$  become units also contains an isomorphic copy of  $Q$ .

*Proof.* Let  $\mathcal{F} = \{(r, d) : r \in R, d \in D\}$ , and define the relation  $\sim$  on  $F$  by

$$(r, d) \sim (s, e) \iff re = sd.$$

This is clearly reflexive and symmetric. If  $(r, d) \sim (s, e)$  and  $(s, e) \sim (t, f)$ , then  $re - sd = sf - te = 0$ . Thus,  $fre - fsd = 0$  and  $dsf - dte = 0$ , so  $fre - dte = (rf - td)e = 0$ . Since  $e \in D$ , meaning that it is not 0 or a zero divisor, we have that  $rf - td = 0$ , meaning that  $(r, d) \sim (t, f)$ . Thus,  $\sim$  is an equivalence relation. Denote the equivalence class of  $(r, d)$  by  $\frac{r}{d}$ . Let  $\mathcal{Q}$  be the set of equivalence classes under  $\sim$ . Note that, in particular,  $\frac{r}{d} = \frac{re}{de}$  for all  $e \in D$ . We now define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}.$$

We claim that  $\mathcal{Q}$  is a commutative ring with identity. We first show that these operations are well defined. If  $(a, b) \sim (a_0, b_0)$  and  $(c, d) \sim (c_0, d_0)$ , then

$$(ad + bc)(b_0d_0) = adb_0d_0 + bcb_0d_0 = (ab_0)(dd_0) + (cd_0)(bb_0) = a_0bdd_0 + c_0dbb_0 = (a_0d_0 + b_0c_0)(bd),$$

so  $(ad + bc, bd) \sim (a_0d_0 + b_0c_0, b_0d_0)$ , meaning that addition is well defined. Similarly, multiplication is well defined. Now, for any  $\frac{r}{d}$ , we have that  $\frac{r}{d} + \frac{0}{e} = \frac{re}{de} = \frac{r}{d}$  for any  $e \in D$ . It is also easy to check that  $(0, d) \sim (0, e)$  for any  $d, e \in D$ , meaning that  $\frac{0}{e}$  is an additive identity in  $\mathcal{Q}$  for any  $e \in D$ . Now,  $\frac{a}{d} + \frac{-a}{d} = \frac{ad - ad}{dd} = \frac{0}{dd}$ , meaning that each element of  $\mathcal{Q}$  has an additive inverse. Closure follows from the fact that  $D$  is closed under multiplication. Finally,

$$\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{ad + bc}{bd} + \frac{e}{f} = \frac{adf + bcf + bde}{bdf} = \frac{a}{b} + \frac{cf + de}{df} = \frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right).$$

Since  $R$  is commutative, this addition is also commutative, so  $\mathcal{Q}$  is an abelian group under  $+$ . That multiplication is associative and commutative is clear. We have that

$$\frac{a}{b} \left(\frac{c}{d} + \frac{e}{f}\right) = \frac{a}{b} \times \frac{cf + de}{df} = \frac{acf + ade}{bdf} = \frac{acf}{bdf} + \frac{ade}{bdf} = \frac{a}{c} \times \frac{c}{d} + \frac{a}{c} \times \frac{e}{f}.$$

The other distributive law is verified similarly. Finally, for any  $d, d_0 \in D$ ,  $(d, d) \sim (d_0, d_0)$ , and  $\frac{d}{d} \frac{a}{d_0} = \frac{a}{d_0} \frac{d}{d} = \frac{a}{d_0}$ , so  $\frac{d}{d}$  is an identity for the ring  $\mathcal{Q}$ . Now we embed  $R$  into  $\mathcal{Q}$  by defining

$$\begin{aligned} \iota : R &\rightarrow \mathcal{Q} \\ r &\mapsto \frac{rd}{d} \quad d \in D. \end{aligned}$$

This is well defined, since  $\frac{rd}{d} = \frac{re}{e}$  for all  $d, e \in D$ . Now,  $\iota(rs) = \frac{rsd^2}{d^2} = \frac{rd}{d} \frac{sd}{d} = \iota(r)\iota(s)$  for some  $d \in D$ , and  $\iota(r + s) = \frac{rd^2 + sd^2}{d^2} = \frac{rd}{d} + \frac{sd}{d} = \iota(r) + \iota(s)$  for  $d \in D$ . Thus  $\iota$  is a ring homomorphism. Now,  $\iota$  is injective, as

$$\iota(r) = 0 \iff \frac{rd}{d} = \frac{0}{d} \iff rd^2 = 0 \iff r = 0,$$

since  $d^2 \in D$  is not a zero divisor or 0. Thus,  $\iota(R)$  is a subring of  $\mathcal{Q}$  that is isomorphic to  $R$ . From here on, we identify  $r \in R$  with  $\iota(r)$  and consider  $R$  as a subring of  $\mathcal{Q}$ . Now,

for  $d, e \in D$ , we have that  $d \in R$  is represented by  $\frac{de}{e}$ . This has multiplicative inverse  $\frac{e}{de}$ , meaning that each element of  $D$  is a unit. Further, for  $r \in R$  and  $d \in D$ , we have that  $\frac{r}{d} = \frac{re^2}{de^2} = \frac{re}{e} \times \frac{e}{de}$ , meaning that every element of  $\mathcal{Q}$  is of the form  $rd^{-1}$  for some  $r \in R$  and  $d \in D$ . In particular, if  $D = R \setminus \{0\}$ , then  $\frac{r}{d}$  has multiplicative inverse  $\frac{d}{r}$  (assuming  $r \neq 0$ , in which case  $\frac{r}{d}$  is the 0 of  $\mathcal{Q}$ ), meaning that  $\mathcal{Q}$  is a field. Finally, we must show that  $\mathcal{Q}$  is unique in the sense described. Assume that  $\phi : R \rightarrow S$  is an injective ring homomorphism, so that  $\phi(d)$  is a unit in  $S$  for all  $d \in D$ . Extend  $\phi$  to the map  $\Phi : \mathcal{Q} \rightarrow S$  by defining  $\Phi(rd^{-1}) = \phi(r)\phi(d)^{-1}$  for all  $r \in R$  and  $d \in D$ . This is well defined, as if  $rd^{-1} = se^{-1}$ , then  $re = sd$ , so  $\phi(r)\phi(e) = \phi(s)\phi(d)$ , meaning that  $\phi(r)\phi(d)^{-1} = \phi(s)\phi(e)^{-1}$ . Note that  $\frac{r}{d} = rd^{-1}$ . Now,

$$\begin{aligned} \Phi\left(\frac{r}{d} + \frac{s}{e}\right) &= \phi(re + sd)\phi(de)^{-1} = \phi(re)\phi(de)^{-1} + \phi(sd)\phi(de)^{-1} \\ &= \Phi\left(\frac{re}{de}\right) + \Phi\left(\frac{sd}{de}\right) = \Phi\left(\frac{r}{d}\right) + \Phi\left(\frac{s}{e}\right), \end{aligned}$$

and

$$\Phi\left(\frac{r}{d} \times \frac{s}{e}\right) = \phi(rs)\phi(de)^{-1} = \phi(r)\phi(d)^{-1}\phi(s)\phi(e)^{-1} = \Phi\left(\frac{r}{d}\right)\Phi\left(\frac{s}{e}\right).$$

Finally  $\Phi$  is injective, because if  $rd^{-1} \in \ker \Phi$ , then  $\phi(r)\phi(d)^{-1} = 0$ . Since  $\phi(d)^{-1}$  is a unit, it cannot be a zero divisor, meaning that  $\phi(r) = 0$ , or  $r \in \ker \phi$ . Since  $\phi$  is injective,  $r = 0$ , and so  $rd^{-1} = 0$ . Finally, since  $r$  is represented by  $\frac{re}{e} = (re)e^{-1}$ , we have that  $\Phi(r) = \phi(re)\phi(e)^{-1} = \phi(r)$ , meaning that  $\Phi|_R = \phi$ . ■

**Definition 2.1.64.** Let  $R$ ,  $D$ , and  $\mathcal{Q}$  be as in the previous theorem.

- (1) The ring  $\mathcal{Q}$  is called the ring of fractions of  $D$  with respect to  $R$ , and is denoted  $D^{-1}R$ .
- (2) If  $R$  is an integral domain and  $D = R \setminus \{0\}$ ,  $\mathcal{Q}$  is called the field of fractions or quotient field of  $R$ , and is denoted  $\text{Frac}(R)$ .

**Definition 2.1.65.** Let  $F$  be a field and  $A \subseteq F$ . The subfield generated by  $A$  is the intersection of all the subfields of  $F$  containing  $A$ . This is the smallest subfield of  $F$  containing  $A$ .

**Corollary 2.1.66.** Let  $R$  be an integral domain and let  $\mathcal{Q}$  be the field of fractions of  $R$ . If a field  $F$  contains a subring  $R'$  isomorphic to  $R$ , then the subfield of  $F$  generated by  $R'$  is isomorphic to  $\mathcal{Q}$ .

*Proof.* Let  $\phi : R \cong R' \subseteq F$  be a ring isomorphism from  $R$  to  $R'$ . In particular,  $\phi : R \rightarrow F$  is an injective homomorphism from  $R$  into  $F$ . Let  $\Phi : \mathcal{Q} \rightarrow F$  be the extension of  $\phi$  to  $\mathcal{Q}$ . By the previous theorem,  $\Phi$  is injective, so  $\Phi(\mathcal{Q})$  is an isomorphic copy of  $\mathcal{Q}$  in  $F$  containing  $\phi(R) = R'$ . Now, any subfield of  $F$  containing  $R' = \phi(R)$  contains the elements  $\phi(r_1)\phi(r_2)^{-1} = \phi(r_1r_2^{-1})$  for all  $r_1, r_2 \in R$ . Since every element of  $\mathcal{Q}$  is of this form, any subfield of  $F$  containing  $R'$  contains  $\Phi(\mathcal{Q})$ , so that  $\Phi(\mathcal{Q})$  is the subfield of  $F$  generated by  $R'$ . ■

*Example 2.1.67.* If  $R$  is a field, then  $\text{Frac}(R) = R$ . To see this, note that  $R \subseteq \text{Frac}(R)$ , so if we show that  $|\text{Frac}(R)| = |R|$ , then this will be true. If  $m$  denotes the size of the equivalence class  $\frac{r}{d}$ , then  $|\text{Frac}(R)| = \frac{|R|(|R|-1)}{m}$ . Note that if  $\frac{r}{d} = \frac{s}{e}$ ,  $s$  is completely determined by  $e$ , so that  $m = |R| - 1$ .

*Example 2.1.68.*  $\text{Frac}(\mathbb{Z}) = \mathbb{Q}$ .

*Example 2.1.69.* If  $\mathcal{O}$  is a quadratic integer ring (which is an integral domain), then  $\text{Frac}(\mathcal{O}) = \mathbb{Q}(\sqrt{D})$ .

*Example 2.1.70.* If  $R$  is a commutative ring with identity and  $d \neq 0$  is not a zero divisor in  $R$ , we may form  $R[1/d]$  by setting  $D = \{1, d, d^2, \dots\}$ , and defining  $R[1/d] = D^{-1}R$ . In this case,  $R$  is the subring of elements of the form  $\frac{r}{1}$ . In this way, a nonzero element of  $R$  that is not a zero divisor can be inverted in a larger ring containing  $R$ .

## 2.1.6 The Chinese Remainder Theorem

*Remark 2.1.71.* In this subsection, all rings are commutative with an identity  $1 \neq 0$  unless otherwise stated.

**Definition 2.1.72.** Given an arbitrary collection of rings (not necessarily commutative or with identity), their (ring) direct product is defined to be their direct product as abelian groups, made into a ring by defining multiplication componentwise. In particular, if  $R_1$  and  $R_2$  are two rings, we denote their direct product as rings by  $R_1 \times R_2$ .

*Remark 2.1.73.* Let us suppose we wish to extend the idea of two integers being relatively prime to a general ring  $R$ . Note that  $m, n \in \mathbb{Z}$  are relatively prime if and only if  $nx + my = 1$  has an integer solution. This is equivalent to saying that  $n\mathbb{Z} + m\mathbb{Z} = \mathbb{Z}$  as ideals.

**Definition 2.1.74.** The ideals  $A$  and  $B$  of the ring  $R$  are said to be comaximal if  $A + B = R$ .

**Theorem 2.1.75** (Chinese Remainder Theorem). *Let  $A_1, A_2, \dots, A_k$  be ideals in  $R$  (commutative with an identity  $1 \neq 0$ ). The map*

$$R \rightarrow R/A_1 \times R/A_2 \times \cdots \times R/A_k \quad \text{defined by} \quad r \mapsto (r + A_1, r + A_2, \dots, r + A_k)$$

*is a ring homomorphism with kernel  $A_1 \cap A_2 \cap \cdots \cap A_k$ . If, for each  $i, j \in \{1, 2, \dots, k\}$  with  $i \neq j$ , the ideals  $A_i$  and  $A_j$  are comaximal, then this map is surjective, and  $A_1 \cap A_2 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$ , so*

$$R/(A_1 A_2 \cdots A_k) = R/(A_1 \cap A_2 \cap \cdots \cap A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k.$$

*Proof.* We first prove this for  $k = 2$ . Let  $A = A_1$  and  $B = A_2$ . Let  $\phi$  denote the map from  $R$  to  $R/A \times R/B$  defined by  $r \mapsto (r + A, r + B)$ . This is a ring homomorphism, since  $\phi$  is just the natural projection for the two components. The kernel of  $\phi$  is all elements in  $R$  that are in  $A$  and  $B$ , so  $\ker \phi = A \cap B$ . Now, suppose that  $A$  and  $B$  are comaximal. Since  $A + B = R$ , there are  $x \in A$  and  $y \in B$  so that  $x + y = 1$ . Thus,  $\phi(x) = (0, 1)$  and  $\phi(y) = (1, 0)$ , since  $x + A = 0 + A$  and  $x + B = 1 - y + B = 1 + B$ , and similarly for  $y$ . Now, if  $(r_1 + A, r_2 + B)$  is an arbitrary element of  $R/A \times R/B$ , then  $\phi(r_2 x + r_1 y) = (r_1 + A, r_2 + B)$ . Thus,  $\phi$  is

surjective. Now,  $AB \subseteq A \cap B$ , so we show that  $A \cap B \subseteq AB$  if they are comaximal. If  $x$  and  $y$  are as above, then for any  $c \in A \cap B$ ,  $c = c1 = cx + cy \in AB$ , so that  $A \cap B \subseteq AB$ , and so  $A \cap B = AB$ . This completes the proof when  $k = 2$ .

We now prove the general statement by induction. The first part of the statement is clear. The second statement follows easily by taking  $A = A_1$  and  $B = A_2A_3 \cdots A_k$ , after showing that  $A_1$  and  $A_2A_3 \cdots A_k$  are comaximal. By hypothesis, for each  $i \in \{2, \dots, k\}$ , there are  $x_i \in A_1$  and  $y_i \in A_i$  so that  $x_i + y_i = 1$ . Thus,  $1 = (x_2 + y_2) \cdots (x_k + y_k)$ . Since  $(x_i + y_i) + A_1 = y_i + A_1$ ,  $(x_2 + y_2) \cdots (x_k + y_k) + A_1 = y_2 \cdots y_k + A_1$ , meaning that  $1 = (x_2 + y_2) \cdots (x_k + y_k) \in A_1 + (A_2 \cdots A_k)$ . ■

*Example 2.1.76.* If  $m$  and  $n$  are relatively prime, then  $(m\mathbb{Z})(n\mathbb{Z}) = (mn)\mathbb{Z}$ , and  $m\mathbb{Z}$  and  $n\mathbb{Z}$  are comaximal. Thus,  $\mathbb{Z}/(mn)\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ .

**Corollary 2.1.77.** *If  $n$  is a positive integer with prime factorization  $p_1^{\alpha_1} \cdots p_k^{\alpha_k}$ , then*

$$\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})$$

*as rings. In particular, there is an isomorphism of multiplicative groups*

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})^\times.$$

*Proof.* The first statement follows from the previous example. The second comes from the fact that an isomorphism of rings induces an isomorphism on the group of units. ■

## 2.2 Domains

*Remark 2.2.1.* All rings in this section are commutative.

### 2.2.1 Euclidean Domains

**Definition 2.2.2.** Let  $R$  be an integral domain. Any function  $N : R \rightarrow \mathbb{Z}^+ \cup \{0\}$  with  $N(0) = 0$  is called a norm on  $R$ . If  $N(a) > 0$  for  $a \neq 0$ ,  $N$  is said to be a positive norm.

**Definition 2.2.3.** The integral domain  $R$  is said to be a Euclidean Domain (or possess a Division Algorithm) if there is a norm  $N$  on  $R$  such that for any two elements  $a$  and  $b$  of  $R$  with  $b \neq 0$ , there exist elements  $q$  and  $r$  in  $R$  with  $a = qb + r$ , where  $r = 0$  or  $N(r) < N(b)$ . The element  $q$  is called the quotient and the element  $r$  the remainder of the division.

*Example 2.2.4.* Fields are Euclidean Domains, where any norm will work, since this is always possible with  $r = 0$ .

*Example 2.2.5.* The integers  $\mathbb{Z}$  are a Euclidean Domain with  $N(a) = |a|$ . The existence is verified as follows. Suppose that  $b > 0$ . The intervals  $[nb, (n+1)b)$  for  $n \in \mathbb{Z}$  partition the real line, and so  $a$  is in one of them, say  $a \in [kb, (k+1)b)$ . Take  $q = k$ . Then  $a - qb = r \in [0, |b|)$ , so  $a = qb + r$  with  $|r| < |b|$ . If  $b < 0$ , there is an integer  $q$  such that  $a = q(-b) + r$  with either  $r = 0$  or  $r < |-b| = |b|$ . Thus,  $a = (-q)b + r$  satisfies the requirements.

*Example 2.2.6.* If  $F$  is a field, then  $F[x]$  is a Euclidean Domain with  $N(p(x)) = \deg p(x)$ . The Division Algorithm is long division of polynomials. We shall prove this later.

*Example 2.2.7.* The quadratic integer rings  $\mathcal{O}$  are integral domains with norm defined by the absolute value of the field norm, but they are not integral domains in general.

**Proposition 2.2.8.** *If  $I$  is any nonzero ideal in the Euclidean Domain  $R$ , then  $I = (d)$ , where  $d \neq 0 \in I$  has minimum norm.*

*Proof.* If  $I = 0$ , there is nothing to prove. Otherwise, let  $d \neq 0 \in I$  have minimum norm (such a  $d$  exists, since the set  $\{N(a) : a \in I\} \subseteq \mathbb{Z}^+ \cup \{0\}$  has a minimum element, since  $\mathbb{Z}^+ \cup \{0\}$  does). Clearly  $(d) \subseteq I$ , so we show the reverse inclusion. Let  $a \in I$ , and write  $a = qd + r$  with the division algorithm (in particular,  $r = 0$  or  $N(r) < N(d)$ ). Then  $r = a - qd \in I$ . Since  $d$  has minimal norm,  $r = 0$ , meaning that  $a = qd \in (d)$ . Thus,  $I = (d)$ . ■

*Example 2.2.9.* We claim that  $R = \mathbb{Z}[x]$  is not a Euclidean Domain. We show that the ideal  $(2, x)$  is not principal. We have that  $(2, x) = \{2p(x) + xq(x) : p(x), q(x) \in \mathbb{Z}[x]\}$ , and so contains all polynomials with integer coefficients who have an even constant term. In particular, this is a proper ideal. Now, suppose that  $(2, x) = (a(x))$  for some  $a(x) \in \mathbb{Z}[x]$ . Since  $2 \in (a(x))$ , there must be some  $p(x)$  so that  $2 = p(x)a(x)$ . Thus,  $0 = \deg 2 = \deg p(x) + \deg a(x)$ , meaning that both  $p(x)$  and  $a(x)$  must be constant. Since 2 is prime,  $a(x), p(x) \in \{\pm 1, \pm 2\}$ . If  $a(x) = \pm 1$ , then  $(a(x))$  would not be proper, so  $a(x) = \pm 2$ . Now,  $x \in (a(x)) = (2) = (-2)$ , meaning that  $x = 2q(x)$  for some polynomial  $q(x)$  with integer coefficients, which is impossible.

*Example 2.2.10.* We show that the quadratic integer ring  $R = \mathbb{Z}[\sqrt{-5}]$  is not a Euclidean Domain. Let  $N$  be the associated field norm  $N(a + b\sqrt{-5}) = a^2 + 5b^2$ , and consider the ideal  $I = (3, 2 + \sqrt{-5})$ . Suppose that  $I = (a + b\sqrt{-5})$  for  $a, b \in \mathbb{Z}$ . Then we could write  $3 = \alpha(a + b\sqrt{-5})$  and  $2 + \sqrt{-5} = \beta(a + b\sqrt{-5})$ . Since  $a^2 + 5b^2 \in \mathbb{Z}^+$ , it must be 1, 3, or 9. If it is 0, then  $N(\alpha) = 1$ , meaning that  $\alpha = \pm 1$ , so  $a + b\sqrt{-5} = \pm 3$ . However, the coefficients of  $2 + \sqrt{-5}$  are not divisible by 3, so this is not possible. If  $a^2 + 5b^2 = 1$ , then  $a + b\sqrt{-5} = \pm 1$ , and  $I = R$ , meaning that  $1 \in I$ . Then,  $3\gamma + (2 + \sqrt{-5})\delta = 1$  for some  $\gamma, \delta \in R$ . Multiplying both sides by  $2 - \sqrt{-5}$  implies that  $2 - \sqrt{-5}$  is a multiple of 3 in  $R$ , a contradiction. There are no integer solutions to  $a^2 + 5b^2 = 3$ . Thus,  $I$  is not principal, and  $R$  is not a Euclidean Domain.

**Definition 2.2.11.** Let  $R$  be a commutative ring and let  $a, b \in R$  with  $b \neq 0$ .

- (1)  $a$  is said to be a multiple of  $b$  if there exists an element  $x \in R$  with  $a = bx$ . In this case,  $b$  is said to divide  $a$  or be a divisor of  $a$ , written  $b|a$ .
- (2) A greatest common divisor of  $a$  and  $b$  is a nonzero element  $d$  such that  $d|a$ ,  $d|b$ , and if  $d'|a$  and  $d'|b$ , then  $d'|d$ . A greatest common divisor of  $a$  and  $b$  is denoted  $\gcd(a, b)$ .

**Proposition 2.2.12.** *In a ring  $R$ ,  $b|a$  if and only if  $a \in (b)$ , which occurs if and only if  $(a) \subseteq (b)$ . In particular, if  $d|a$  and  $d|b$ , then  $(a, b) \subseteq (d)$ . Further, the conditions that  $(a, b) \subseteq (d)$  and that if  $(a, b) \subseteq (d')$ , then  $(d) \subseteq (d')$  are necessary and sufficient for  $d$  to be a greatest common divisor of  $a$  and  $b$ .*

*Proof.* If  $b|a$ , then  $a = bx \in (b)$ . Conversely, if  $a = bx \in (b)$ , then  $b|a$ . Thus, if  $d|a$  and  $d|b$ , then  $a, b \in (d)$ , meaning that  $(a, b) \subseteq (d)$ . Now, if  $d = \gcd(a, b)$ , then  $(a, b) \subseteq (d)$ , and if  $(a, b) \subseteq (d')$ , then  $d'|a$  and  $d'|b$ , meaning that  $d|d'$ , so  $(d) \subseteq (d')$ . If  $(a, b) \subseteq (d)$  and  $(a, b) \subseteq (d') \implies (d) \subseteq (d')$ , then  $d|a$ ,  $d|b$ , and if  $d'|a$  and  $d'|b$ , then  $d|d'$ . ■

**Proposition 2.2.13.** *If  $a, b \neq 0 \in R$  so that  $(a, b) = (d)$ , then  $d$  is a greatest common divisor of  $a$  and  $b$ .*

*Proof.* It is clear that  $(a, b) \subseteq (d)$ , and if  $(a, b) \subseteq (d')$ , then  $(d) \subseteq (d')$ . ■

**Definition 2.2.14.** An integral domain in which every ideal  $(a, b)$  generated by two elements is principal is called a Bezout Domain.

**Proposition 2.2.15.** *Let  $R$  be an integral domain. If  $(d) = (d')$  for some  $d, d' \in R$ , then  $d' = ud$  for some unit  $u$  in  $R$ . In particular, if  $d$  and  $d'$  are both greatest common divisors of  $a$  and  $b$ , then  $d' = ud$  for some unit  $u$ .*

*Proof.* This is clear if either one of  $d$  or  $d'$  is zero, so we assume they are both nonzero. Since  $d \in (d')$ , there is some  $x \in R$  with  $d = xd'$ . Similarly, there is some  $y \in R$  so that  $d' = yd$ . Thus,  $d = (xy)d$ , and so  $d(1 - xy) = 0$ . Since  $d \neq 0$  and  $R$  is an integral domain,  $1 - xy = 0$ , so that  $xy = 1$ . Thus,  $x$  and  $y$  are units. The second statement follows from the fact that any two greatest common divisors of  $a$  and  $b$  must generate the same principal ideal (they divide each other). ■

**Definition 2.2.16.** Let  $R$  be a Euclidean Domain, and let  $a, b \in R$ . The Euclidean Algorithm on  $a$  and  $b$  is repeating the following.

$$\begin{aligned} a &= q_0b + r_0 \\ b &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\vdots \\ r_{n-2} &= q_nr_{n-1} + r_n \\ r_{n-1} &= q_{n+1}r_n, \end{aligned}$$

where  $r_n$  is the last nonzero remainder. Note that such an  $r_n$  must exist, since  $N(b) > N(r_0) > N(r_1) > \dots > N(r_n)$  is a decreasing sequence of nonnegative integers if the remainders are nonzero, and thus cannot continue indefinitely. There is no guarantee that the elements are unique.

**Theorem 2.2.17.** *Let  $R$  be a Euclidean Domain, and let  $a, b \in R$  be nonzero. Let  $d = r_n$  be the last nonzero remainder in the Euclidean Algorithm for  $a$  and  $b$  as described above.*

- (1)  $d$  is a greatest common divisor of  $a$  and  $b$ .
- (2) The principal ideal  $(d)$  is the ideal generated by  $a$  and  $b$ . In particular,  $d$  can be written as an  $R$ -linear combination of  $a$  and  $b$  (there are  $x, y \in R$  so that  $d = ax + by$ ).

*Proof.* We know that the ideal generated by  $a$  and  $b$  is principal (all ideals in Euclidean Domains are), so  $a$  and  $b$  do have a greatest common divisor (any element which generates  $(a, b)$ ). We first show that  $d|a$  and  $d|b$ . We have that  $r_n|r_n$ , and that  $r_n|r_{n-1}$  by the last equation in the Euclidean Algorithm. Now, if  $r_n$  divides  $r_k$  and  $r_{k+1}$ , then  $r_n$  also divides  $r_{k-1} = q_{k+1}r_k + r_{k+1}$ . Thus,  $r_n$  divides  $r_0$  and  $r_1$ , and so  $r_n|b = q_1r_0 + r_1$ , and  $r_n|a = q_0b + r_0$ . Therefore,  $(a, b) \subseteq (d)$ . Now,  $r_0 = a - q_0b \in (a, b)$ , and  $r_1 = b - q_1r_0 \in (a, b)$ . If  $r_{k-1}, r_k \in (a, b)$ , then  $r_{k+1} = r_{k-1} - q_{k+1}r_k \in (a, b)$ . Thus,  $r_n \in (a, b)$ , so that  $(d) \subseteq (a, b)$ . Therefore,  $(d) = (a, b)$ , meaning that  $d$  is a greatest common divisor of  $a$  and  $b$ . This completes the proof. ■

**Proposition 2.2.18.** *Let  $R$  be a Euclidean Domain. If  $\gcd(a, b) = 1$  and  $a|bc$ , then  $a|c$ . More generally, if  $a|bc$  and  $a, b \neq 0$ , then  $\frac{a}{\gcd(a, b)}$  divides  $c$ .*

*Proof.* Since  $\gcd(a, b) = 1$ , we can write  $ua + vb = 1$  for some  $u, v \in R$ . Thus,  $uca + vbc = c$ . Since  $a$  divides the left, it divides the right. Now, if  $a|bc$ , write  $bc = xa$ . We may factor both sides as  $\frac{b}{\gcd(a, b)}c \cdot \gcd(a, b) = x \frac{a}{\gcd(a, b)} \cdot \gcd(a, b)$ . By the cancellation law,  $\frac{b}{\gcd(a, b)}c = x \frac{a}{\gcd(a, b)}$ . In particular,  $\frac{a}{\gcd(a, b)} | \frac{b}{\gcd(a, b)}c$ . Now, suppose that  $d | \frac{a}{\gcd(a, b)}, \frac{b}{\gcd(a, b)}$ . Then,  $d \gcd(a, b) | a, b$ . Since  $d \gcd(a, b) | \gcd(a, b)$  by definition,  $d$  is a unit, and so  $\gcd(\frac{a}{\gcd(a, b)}, \frac{b}{\gcd(a, b)}) = 1$ . Thus,  $\frac{a}{\gcd(a, b)} | c$ . ■

**Corollary 2.2.19.** *Suppose  $a, b, N \in \mathbb{Z}$  and  $a, b \neq 0$ , and that  $(x_0, y_0)$  is an integer solution to the equation  $ax_0 + by_0 = N$ . The full set of solutions to this equation is given by*

$$\left\{ \left( x_0 + m \frac{b}{\gcd(a, b)}, y_0 - m \frac{a}{\gcd(a, b)} \right) : m \in \mathbb{Z} \right\}.$$

*Proof.* Suppose that  $(x, y)$  is a solution, so that  $ax + by = N$ . Then,  $a(x_0 - x) + b(y_0 - y) = 0$ , so that  $a(x_0 - x) = b(y - y_0)$ . Thus,  $\frac{a}{\gcd(a, b)} | y - y_0$ , meaning that  $y = y_0 + k \frac{a}{\gcd(a, b)}$  for some integer  $k$ . We also have that  $\frac{b}{\gcd(a, b)} | x_0 - x$ , meaning that  $x = x_0 + m \frac{b}{\gcd(a, b)}$  for some integer  $m$ . Plugging these values back into the original equation gives that  $k + m = 0$ , or that  $k = -m$ . ■

**Definition 2.2.20.** Let  $R$  be an integral domain, and let  $\tilde{R}$  denote  $R^\times \cup \{0\}$ . An element  $u \in R \setminus \tilde{R}$  is called a universal side divisor if, for every  $x \in R$ , there is some  $z \in \tilde{R}$  such that  $u|x - z$ . In other words, every  $x$  may be written as  $qu + z$ , where  $z$  is either zero or a unit.

**Proposition 2.2.21.** *Let  $R$  be an integral domain that is not a field. If  $R$  is a Euclidean Domain then there are universal side divisors in  $R$ .*

*Proof.* Suppose that  $R$  is Euclidean with respect to some norm  $N$ , and let  $u \in R \setminus \tilde{R}$  of minimal norm. For  $x \in R$ , write  $x = qu + r$  where  $r = 0$  or  $N(r) < N(u)$ . In either case, the minimality of  $u$  implies  $r \in \tilde{R}$ . Thus,  $u$  is a universal side divisor in  $R$ . ■

*Example 2.2.22.* We show that  $R = \mathbb{Z}[(1 + \sqrt{-19})/2]$  is not a Euclidean Domain. It is checkable that  $R^\times = \{\pm 1\}$ , so that  $\tilde{R} = \{0, \pm 1\}$ . Suppose that  $u$  is a universal side divisor, and let  $N(a + b(1 + \sqrt{-19})/2) = a^2 + ab + 5b^2$ . Note that  $N(ab) = N(a)N(b)$ . If  $a, b \in \mathbb{Z}$



and  $b \neq 0$ , then  $a^2 + ab + 5b^2 \geq 5$ , meaning that the smallest nonzero values of  $N$  on  $R$  are 1 ( $\pm 1$ ) and 4 ( $\pm 2$ ). If  $x = 2$  in the definition of a universal side divisor,  $u|2 - 0$  or  $u|2 \pm 1$  in  $R$ , meaning that  $u$  is a nonunit divisor of 2 or 3 in  $R$ . If  $2 = \alpha\beta$ , then  $4 = N(2) = N(\alpha\beta) = N(\alpha)N(\beta)$ , and the previous remark implies that one of  $\alpha$  and  $\beta$  has norm 1, meaning that it is either 1 or  $-1$ . Thus, the only divisors of 2 in  $R$  are  $\{\pm 1, \pm 2\}$ . Similarly, the only divisors of 3 are  $\{\pm 1, \pm 3\}$ . Thus,  $u \in \{\pm 2, \pm 3\}$ . If  $x = (1 + \sqrt{-19})/2$ , it is easy to see that none of  $x$  and  $x \pm 1$  are divisible by  $\pm 2$  or  $\pm 3$  in  $R$ , so none of these is a universal side divisor.

### 2.2.2 Principal Ideal Domains (PIDs)

**Definition 2.2.23.** A Principal Ideal Domain (PID) is an integral domain in which every ideal is principal.

*Example 2.2.24.* Every Euclidean Domain is a Principle Ideal Domain. (Thus, every result about Principle Ideal Domains also holds for Euclidean Domains)

*Example 2.2.25.* The ring  $\mathbb{Z}[x]$  is not a Principle Ideal Domain, as  $(2, x)$  is not principal.

*Remark 2.2.26.* Recall that if  $(a, b) = (d)$ , then  $d = \gcd(a, b)$ . Thus, greatest common divisors exist in Principle Ideal Domains.

**Proposition 2.2.27.** Let  $R$  be a PID, and let  $a, b \in R$  be nonzero. Let  $d$  be a generator for  $(a, b)$ . Then,

- (1)  $d$  is a greatest common divisor of  $a$  and  $b$ .
- (2) There are  $x, y \in R$  with  $d = ax + by$ .
- (3)  $d$  is unique up to multiplication by a unit of  $R$ .

*Proof.* Proven previously. ■

**Proposition 2.2.28.** Every nonzero prime ideal in a PID is a maximal ideal.

*Proof.* Let  $(p)$  be a nonzero prime ideal in the PID  $R$ , and let  $I = (m)$  be an ideal containing  $p$ . We show that  $I = (p)$  or that  $I = R$ . Since  $p \in (m)$ ,  $p = rm$  for some  $r \in R$ . Since  $p$  is a prime ideal, either  $r \in (p)$  or  $m \in (p)$ . In the first case, write  $r = ps$ . Then  $p = rm = psm$ , so  $sm = 1$ , meaning that  $1 \in I$ , so  $I = R$ . In the latter case,  $I \subseteq (p)$ , so  $I = (p)$ . ■

**Corollary 2.2.29.** If  $R$  is any commutative ring such that  $R[x]$  is a PID (or a Euclidean Domain), then  $R$  is a field.

*Proof.* Assume that  $R[x]$  is a PID. Since  $R$  is a subring of  $R[x]$ , it must be an integral domain. The ideal  $(x)$  is a nonzero prime ideal in  $R[x]$ , since  $R[x]/(x) \cong R$ , which is an integral domain. By the previous proposition,  $(x)$  is maximal, and so  $R \cong R[x]/(x)$  is a field. ■

**Definition 2.2.30.** A positive norm  $N$  is said to be a Dedekind-Hasse norm if for every nonzero  $a, b \in R$ , either  $a \in (b)$  or there is a nonzero element  $c \in (a, b)$  so that  $N(c) < N(b)$ . In other words, either  $b$  divides  $a$  or there exist  $s, t \in R$  with  $0 < N(sa - tb) < N(b)$ .

*Remark 2.2.31.* If it is always possible to satisfy the Dedekind-Hasse condition with  $s = 1$ , then  $R$  is Euclidean with respect to the positive norm  $N$ . Thus, this is a weakening of the Euclidean condition.

**Proposition 2.2.32.** *The integral domain  $R$  is a PID if and only if  $R$  has a Dedekind-Hasse norm.*

*Proof.* We show that having a Dedekind-Hasse norm implies that  $R$  is a PID, and prove the converse later on. Let  $I$  be any nonzero ideal in  $R$  and let  $b$  be a nonzero element of  $I$  with  $N(b)$  minimal. Suppose that  $a \in I$  is nonzero, so that  $(a, b) \subseteq I$ . The Dedekind-Hasse condition on  $N$  and the minimality of  $b$  imply that  $a \in (b)$ . Thus,  $I = (b)$  is principal. ■

*Example 2.2.33.* Let  $R = \mathbb{Z}[(1 + \sqrt{-19})/2]$ . We show that the positive norm  $N(a + b(1 + \sqrt{-19})/2) = a^2 + ab + 5b^2$  defined on  $R$  is a Dedekind-Hasse norm, proving that  $R$  is a PID but not a Euclidean Domain.

Suppose that  $\alpha, \beta \in R$  are nonzero, and  $\alpha/\beta \notin R$ . We must show that there are  $s, t \in R$  with  $0 < N(s\alpha - t\beta) < N(\beta)$ , which is equivalent to  $0 < N(\frac{\alpha}{\beta}s - t) < 1$ . Write  $\frac{\alpha}{\beta} = \frac{a+b\sqrt{-19}}{c} \in \mathbb{Q}[\sqrt{-19}]$  with integers  $a, b, c$  having no common divisor and  $c > 1$ . Since  $a, b, c$  have no common divisor, there are  $x, y, z \in \mathbb{Z}$  with  $ax + by + cz = 1$ . Write  $ay - 19bx = cq + r$  for some quotient  $q$  and remainder  $r$  with  $|r| \leq c/2$ . Let  $s = y + x\sqrt{-19}$  and  $t = q - z\sqrt{-19}$ . Then,

$$0 < N\left(\frac{\alpha}{\beta}s - t\right) = \frac{(ay - 19bx - cq)^2 + 19(ax + by + cz)^2}{c^2} \leq \frac{1}{4} + \frac{19}{c^2}.$$

Thus, this  $s$  and  $t$  work provided that  $c \geq 5$ .

Suppose that  $c = 2$ . Then one of  $a$  and  $b$  is even, and the other is odd. In this case,  $s = 1$  and  $t = \frac{(a-1)+b\sqrt{-19}}{2}$  work.

Suppose that  $c = 3$ . Then  $a^2 + 19b^2$  is not divisible by 3 (if it were then  $3|a, b$  meaning that  $\gcd(a, b, c) \neq 1$ ). Write  $a^2 + 19b^2 = 3q + r$  with  $r = 1$  or  $r = 2$ . Then  $s = a - b\sqrt{-19}$  and  $t = q$  work.

Finally, suppose that  $c = 4$ , so  $a$  and  $b$  are not both even. If one is even and the other is odd, then  $a^2 + 19b^2$  is odd, so we can write  $a^2 + 19b^2 = 4q + r$  for some  $q, r \in \mathbb{Z}$  and  $0 < r < 4$ . Then  $s = a - b\sqrt{-19}$  and  $t = q$  work. If they are both odd, then  $a^2 + 19b^2 \equiv 1 + 3 \pmod{8}$ , so we can write  $a^2 + 19b^2 = 8q + 4$  for some  $q \in \mathbb{Z}$ . Then  $s = \frac{a-b\sqrt{-19}}{2}$  and  $t = q$  work.

**Proposition 2.2.34.** *In a PID,  $(a)$  and  $(b)$  are comaximal if and only if  $\gcd(a, b) = 1$ .*

*Proof.* If  $\gcd(a, b) = 1$ , then there are  $r, s \in R$  so that  $ra + sb = 1$ , meaning that  $1 \in (a) + (b)$ . Thus,  $(a)$  and  $(b)$  are comaximal. If  $(a)$  and  $(b)$  are comaximal, then there are  $r, s \in R$  so that  $ra + sb = 1$ . Let  $d = \gcd(a, b)$ . Then,  $d|ra$  and  $d|sb$ , so  $d|ra + sb = 1$ . Thus,  $1 = du$  for some  $u \in R$ , meaning that  $d$  is a unit, so  $\gcd(a, b) = 1$ . ■

**Proposition 2.2.35.** *The quotient of a PID by a prime ideal is a PID.*

*Proof.* Let  $R$  be a PID and  $P$  a prime ideal. If  $P = 0$ , then  $R/P = R$  is a PID. If  $P \neq 0$ , then  $P$  is maximal, and so  $R/P$  is a field, so the only ideals are  $(0)$  and  $(1)$ , which are both principal. ■

**Proposition 2.2.36.** *If  $R$  is a PID, and  $D \subseteq R$  is multiplicatively closed and not containing 0, then  $D^{-1}R$  is a PID.*

*Proof.* Let  $I$  be an ideal of  $D^{-1}R$ . Note that the numerators of the elements in  $I$  form an ideal of  $R$  (the only nontrivial property is additive closure, which is as follows: if  $r/d$  and  $s/e$  are elements of  $I$ , then so is  $s/e \cdot e/d = s/d$ , and so  $r/d + s/d = (r+s)/d$  is also an element of  $I$ ). Note also that every element of  $D$  appears as a denominator for every element of the set of numerators of  $I$  (which we now denote  $I_n$ ) (this follows from the fact that if  $s/e \in I$ , then  $s/e \cdot e/d = s/d \in I$  for all  $d \in D$ ). Let  $g$  generate  $I_n$  in  $R$ , and let  $d \in D$ . We claim that  $I = (\frac{g}{d})$ . All elements of the numerator clearly appear, and all elements of the denominator appear, because if  $\frac{r}{d} \in I$ , then  $\frac{r}{d} \cdot \frac{d}{e} = \frac{r}{e} \in I$ . ■

### 2.2.3 Unique Factorization Domains (UFDs)

**Definition 2.2.37.** Let  $R$  be an integral domain.

- (1) Suppose  $r \in R$  is nonzero and not a unit. Then  $r$  is called irreducible in  $R$  if, whenever  $r = ab$  with  $a, b \in R$ , at least one of  $a$  or  $b$  must be a unit in  $R$ . Otherwise,  $r$  is said to be reducible.
- (2) The nonzero element  $p \in R$  is called prime in  $R$  if the ideal  $(p)$  is a prime ideal. In other words, a nonzero element  $p$  is prime if it is not a unit and whenever  $p|ab$  for any  $a, b \in R$ , then  $p|a$  or  $p|b$ .
- (3) Two elements  $a, b \in R$  are said to be associate in  $R$  if  $a = ub$  for some unit  $u \in R$ .

**Proposition 2.2.38.** *In an integral domain, a prime element is always irreducible.*

*Proof.* Suppose that  $(p)$  is a nonzero prime ideal and  $p = ab$ . Then  $ab \in (p)$ , so either  $a \in (p)$  or  $b \in (p)$ . Assume that  $a \in (p)$  without loss of generality. Then  $a = pr$  for some  $r$ , so  $p = ab = prb$ , meaning that  $rb = 1$ , so  $b$  is a unit. Thus,  $p$  is irreducible. ■

*Example 2.2.39.* The converse is not necessarily true. For example, consider  $3 \in \mathbb{Z}[\sqrt{-5}]$ . If  $3 = \alpha(a + b\sqrt{-5})$ , then  $9 = N(\alpha) + (a^2 + 5b^2)$ . Since  $a^2 + 5b^2$  is a positive integer, it must be 1, 3, or 9. If it is 9, then  $N(\alpha) = 1$ , so  $\alpha = \pm 1$  is a unit. It cannot be 3, and if it is 1, then  $a + b\sqrt{-5}$  is a unit. Thus, 3 is irreducible. However,  $3|(2 + \sqrt{-5})(2 - \sqrt{-5})$ , while 3 does not divide either of  $2 \pm \sqrt{-5}$ , so 3 is not prime.

**Proposition 2.2.40.** *In a PID, a nonzero element is prime if and only if it is irreducible.*

*Proof.* We only need to show that irreducible implies prime. Let  $p$  be irreducible, and let  $M$  be any ideal containing  $(p)$ . By hypothesis,  $M = (m)$  is principal. Since  $p \in (m)$ ,  $p = rm$  for some  $r$ . Since  $p$  is irreducible, either  $r$  or  $m$  is a unit. Thus,  $(p) = (m)$  or  $(m) = (1)$ . Therefore,  $(p)$  is a maximal ideal. Since maximal ideals are prime ideals,  $p$  is prime. ■

**Definition 2.2.41.** A Unique Factorization Domain (UFD) is an integral domain  $R$  in which every nonzero element  $r \in R$  which is not a unit has the following two properties:

- (1)  $r$  can be written as a finite product of irreducibles  $p_i$  of  $R$  (not necessarily distinct):  

$$r = p_1 p_2 \cdots p_n.$$
- (2) The decomposition in (1) is unique up to associates: if  $r = q_1 q_2 \cdots q_m$  is another factorization of  $r$  into irreducibles, then  $m = n$  and there is some renumbering of the factors so that  $p_i$  is associate to  $q_i$  for  $i = 1, 2, \dots, n$ .

*Example 2.2.42.* A field is a UFD, since every nonzero element is a unit, so there are no elements for which the properties must be verified.

**Proposition 2.2.43.** *In a UFD, a nonzero element is prime if and only if it is irreducible.*

*Proof.* Let  $R$  be a UFD. Since primes of  $R$  are irreducible, we only need to show that each irreducible element is a prime. Let  $p$  be an irreducible in  $R$  and assume  $p|ab$  for some  $a, b \in R$ . Then,  $ab = pc$  for some  $c \in R$ . Writing  $a$  and  $b$  as a product of irreducibles, the uniqueness of the decomposition implies that  $p$  must be associate to one of the irreducibles occurring in either the factorization of  $a$  or in the factorization of  $b$ . Assume that it is  $a$  without loss of generality. Then we can write  $a = (up)p_2 \cdots p_n$  for a unit  $u$  and some (possibly empty) set of irreducibles  $p_2, \dots, p_n$ . Then,  $a = p(up_2 \cdots p_n)$ , so  $p|a$ . ■

*Remark 2.2.44.* In a UFD, we will now use the terms “prime” and “irreducible” interchangeably.

**Proposition 2.2.45.** *Let  $R$  be a UFD, and let  $a, b \in R$  be nonzero. Suppose*

$$a = up_1^{e_1} p_2^{e_2} \cdots p_n^{e_n} \quad b = vp_1^{f_1} p_2^{f_2} \cdots p_n^{f_n}$$

*are prime factorizations for  $a$  and  $b$  where  $u$  and  $v$  are units, the primes  $p_1, \dots, p_n$  are distinct, and  $e_i, f_i \geq 0$  for all  $i$ . Then,*

$$d = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \cdots p_n^{\min(e_n, f_n)}$$

*(where  $d = 1$  if all exponents are 0) is a greatest common divisor of  $a$  and  $b$ .*

*Proof.* Since the exponents of each of the primes occurring in  $d$  are no larger than the exponents occurring in the factorizations of both  $a$  and  $b$ ,  $d|a, b$ . Let  $c$  be a common divisor of  $a$  and  $b$ , and write  $c = q_1^{g_1} q_2^{g_2} \cdots q_m^{g_m}$  as the prime factorization of  $c$ . Since each  $q_i$  divides  $c$ , and therefore  $a$  and  $b$ , each  $q_i$  must divide a  $p_j$ . In particular, up to associates, the primes occurring in  $c$  must be a subset of the primes occurring in  $a$  and  $b$ . Similarly, the exponents for the primes occurring in  $c$  must be no larger than those occurring in  $d$ , so  $c|d$ . ■

**Theorem 2.2.46.** *Every Principal Ideal Domain is a Unique Factorization Domain.*

*Proof.* Let  $R$  be a PID, and let  $r \in R$  be a nonzero element that is not a unit. We must show that  $r$  can be written as a finite product of irreducible elements of  $R$ , and that this is unique up to units.

We first show that  $r$  can be written as a finite product of irreducibles. Suppose that  $r$  is nonzero and not a unit. If it is irreducible, we are done. If not, write  $r = r_1 r_2$ , where neither  $r_1$  or  $r_2$  is a unit. If these are both irreducibles, then we are done. Otherwise, at least one of them, say  $r_1$ , is reducible, and so we may write it as the product of two nonunit elements  $r_1 = r_{11} r_{12}$ . We continue this process to obtain a decomposition of  $r$  as a product of irreducibles. However, we must verify that this terminates. Suppose that this is not the case. From  $r = r_1 r_2$ , we get  $(r) \subset (r_1) \subset R$ , where the first inclusion is proper since  $r_2$  is not a unit, and the second inclusion is proper since  $r_1$  is not a unit. We similarly obtain  $(r) \subset (r_1) \subset (r_{11}) \subset R$ . If our process does not terminate after a finite number of steps, then we obtain an infinite ascending chain of ideals:

$$(r) \subset (r_1) \subset (r_{11}) \subset \cdots \subset R.$$

We now show that any such chain  $I_1 \subseteq I_2 \subseteq \cdots \subseteq R$  in a PID eventually becomes stationary (there is some  $n$  so that  $I_k = I_n$  for all  $k \geq n$ ). In particular, it is not able to have an infinite ascending chain of ideals where all containments are proper. Let  $I = \bigcup_{i=1}^{\infty} I_i$ . Then  $I$  is an ideal. Since  $R$  is a PID, we may say  $I = (a)$ . Since  $I$  is the union of the ideals, we must have  $a \in I_n$  for some  $n$ . Then,  $I_n \subseteq I$ , and  $I = (a) \subseteq I_n$ , so  $I = I_n$ . Thus, the chain becomes stationary at  $I_n$ , and so every nonzero element of  $R$  that is not a unit has some factorization into irreducibles in  $R$ .

Now we prove that the decomposition is essentially unique. We show this by induction on  $n$ , the number of irreducible factors in some factorization of the element  $r$ . If  $n = 0$ , then  $r$  is a unit. If we had  $r = qc$  for some irreducible  $q$ , then  $q$  would divide a unit, and would therefore also be a unit, a contradiction. Now suppose that  $n \geq 1$  and that we have

$$r = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m \quad m \geq n,$$

where the  $p_i$ 's and the  $q_j$ 's are (not necessarily distinct) irreducibles. Since  $p_1$  divides the product on the right,  $p_1$  must divide one of the factors on the right. Renumbering if necessary, we may assume that  $p_1 | q_1$ . Then  $q_1 = p_1 u$  for some  $u \in R$ , which must be a unit since  $q_1$  is irreducible. Thus  $p_1$  and  $q_1$  are associates. Cancelling  $p_1$  (we are in an integral domain, so we may do this), we obtain

$$p_2 \cdots p_n = u q_2 q_2 \cdots q_m = q'_2 q_3 \cdots q_m \quad m \geq n,$$

where  $q'_2 = u q_2$  is again an irreducible. By induction on  $n$ , we conclude that each of the factors on the left matches bijectively with the factors on the far right up to associates, and hence with the factors in the middle (which are the same, up to associates). Since  $p_1$  and  $q_1$  (after the renumbering) are associate, proving the theorem. ■

**Corollary 2.2.47** (The Fundamental Theorem of Arithmetic). *The integers  $\mathbb{Z}$  are a Unique Factorization Domain.*

*Proof.* The integers are a Euclidean Domain, and are therefore a PID, and thus a UFD. ■

**Corollary 2.2.48.** *Let  $R$  be a PID. Then there exists a multiplicative Dedekind-Hasse norm on  $R$ .*

*Proof.* If  $R$  is a PID, then it is a UFD. Define the norm  $N$  by

$$N(r) = \begin{cases} 0 & r = 0 \\ 1 & r \text{ is a unit.} \\ 2^n & r = p_1 p_2 \cdots p_n \text{ where the } p_i \text{'s are irreducibles in } R. \end{cases}$$

Clearly  $N(ab) = N(a)N(b)$ , so  $N$  is positive and multiplicative. Suppose that  $a, b \in R$  are nonzero, and write  $(a, b) = (r)$ . If  $a \notin (b)$ , then  $r \notin (b)$ , meaning that  $r$  is not divisible by  $b$ . Since  $b = xr$  for some  $x \in R$ ,  $x$  is not a unit in  $R$ , and so  $N(b) = N(x)N(r) > N(r)$ . Thus,  $(a, b)$  contains a nonzero element with norm strictly lesser than  $N(b)$ . ■

**Proposition 2.2.49.** *A ring  $R$  is a PID if and only if  $R$  is a UFD that is also a Bezout Domain.*

*Proof.* If  $R$  is a PID, then  $R$  is a UFD, and all ideals are principal. In particular,  $(a, b)$  is principal. Now suppose that  $R$  is a UFD and a Bezout Domain. Take  $I$  to be an ideal of  $R$ , and take  $a \in I$  to have a minimal number of irreducible factors. We claim  $I = (a)$ . If not, there is some  $b \in I$  with  $b \notin (a)$ . Thus, at least one of the irreducible factors appearing in  $a$  must appear less times in  $b$ . Take one such factor to be  $p_1$ . Since  $R$  is a Bezout Domain, we may write  $(a, b) = (d)$ . In particular,  $d \in I$ . Since  $d|b$ , the number of times  $p_1$  appears in  $d$  must be less than or equal to the number of times it appears in  $b$  (in particular, less than the number of times it appears in  $a$ ). Since  $d|a$ , each irreducible must appear no more times in  $d$  than in  $a$ . Thus, the number of irreducible factors of  $d$  is less than that of  $a$ , a contradiction. ■

## 2.3 Polynomial Rings

*Remark 2.3.1.* In this section, the ring  $R$  will always be commutative with identity  $1 \neq 0$ .

### 2.3.1 Basic Properties

**Proposition 2.3.2.** *Let  $I$  be an ideal of the ring  $R$  and let  $(I) = I[x]$  denote the ideal of  $R[x]$  generated by  $I$ . Then,*

$$R[x]/(I) \cong (R/I)[x].$$

*In particular, if  $I$  is a prime ideal of  $R$ , then  $(I)$  is a prime ideal of  $R[x]$ .*

*Proof.* There is a natural map  $\phi : R[x] \rightarrow (R/I)[x]$  given by reducing each of the coefficients of a polynomial modulo  $I$  (in other words,  $a_k x^k \mapsto (a_k + I)x^k$ , and this extends due to the homomorphism property). The definition of addition and multiplication in the two rings show that  $\phi$  is a ring homomorphism. The kernel is the set of polynomials who have coefficients in  $I$ , or  $I[x] = (I)$ . Thus,  $R[x]/(I) \cong (R/I)[x]$ . The second statement follows from Proposition 2.1.23, since if  $I$  is a prime ideal of  $R$ , then  $R/I$  is an integral domain, meaning that  $(R/I)[x]$  is an integral domain, so that  $R[x]/(I)$  is an integral domain. Thus,  $(I)$  is a prime ideal of  $R[x]$ . ■

**Definition 2.3.3.** The polynomial ring in the variables  $x_1, x_2, \dots, x_n$  with coefficients in  $R$ , denoted  $R[x_1, x_2, \dots, x_n]$ , is defined inductively as

$$R[x_1, x_2, \dots, x_n] = R[x_1, x_2, \dots, x_{n-1}][x_n].$$

**Definition 2.3.4.** We may view a nonzero polynomial in  $x_1, x_2, \dots, x_n$  with coefficients in  $R$  as a finite sum of nonzero monomial terms, which are of the form  $ax_1^{d_1}x_2^{d_2}\cdots x_n^{d_n}$ , where  $a \in R$ , and the  $d_i$ 's are nonnegative integers. In this case,  $a$  is called the coefficient of the term, and if  $a = 1$ , the term is called a monomial. The monomial part of  $ax_1^{d_1}x_2^{d_2}\cdots x_n^{d_n}$  is  $x_1^{d_1}x_2^{d_2}\cdots x_n^{d_n}$ . The exponent  $d_i$  is the degree in  $x_i$  of the term, and the sum  $d = d_1 + d_2 + \cdots + d_n$  is called the degree of the term. The ordered  $n$ -tuple  $(d_1, d_2, \dots, d_n)$  is the multidegree of the term. The degree of a nonzero polynomial is the largest degree of any of its monomial terms. A polynomial is called homogeneous or a form if all its terms have the same degree. If  $f$  is a nonzero polynomial in  $n$  variables, the sum of all the monomial terms in  $c$  of degree  $k$  is called the homogeneous component of  $f$  of degree  $k$ . If  $f$  had degree  $d$ , then  $f$  may be written uniquely as the sum  $f_0 + f_1 + \cdots + f_d$ , where  $f_k$  is the homogeneous component of  $f$  of degree  $k$  for  $0 \leq k \leq d$  (where some  $f_k$  may be 0). A polynomial ring in an arbitrary number of variables with coefficients in  $R$  is the set of all finite sums of monomial terms of the type above (but where the variables are not restricted to  $x_1, x_2, \dots, x_n$ ) with the natural addition and multiplication. Alternatively, this ring is the union of all polynomial rings in a finite number of variables.

### 2.3.2 Polynomial Rings Over Fields (I)

**Theorem 2.3.5.** *Let  $F$  be a field. The polynomial ring  $F[x]$  is a Euclidean Domain with norm  $N(p(x)) = \deg p(x)$  (where we set  $N(0) = 0$ ). Specifically, if  $a(x), b(x) \in F[x]$  with  $b(x)$  nonzero, then there are unique  $q(x), r(x) \in F[x]$  such that  $a(x) = q(x)b(x) + r(x)$  with  $r(x) = 0$  or  $\deg r(x) < \deg b(x)$ .*

*Proof.* If  $a(x)$  is the zero polynomial, then take  $q(x) = r(x) = 0$ . Thus, we may assume  $a(x)$  is nonzero. We prove the existence of  $q(x)$  and  $r(x)$  by induction on  $n = \deg a(x)$ . Let  $\deg b(x) = m$ . If  $n < m$ , then we may take  $q(x) = 0$  and  $r(x) = a(x)$ . If  $n \geq m$ , write

$$a(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

and

$$b(x) = b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0.$$

Then,  $a'(x) = a(x) - \frac{a_n}{b_m}x^{n-m}b(x)$  is of degree less than  $n$  (note we may do this, since  $b_m \neq 0$  and we are working in a field). By induction, we may write  $a'(x) = q'(x)b(x) + r(x)$  with  $r(x) = 0$  or  $\deg r(x) < \deg b(x)$ . If  $q(x) = q'(x) + \frac{a_n}{b_m}x^{n-m}$ , then  $a(x) = q(x)b(x) + r(x)$  with  $r(x) = 0$  or  $\deg r(x) < \deg b(x)$ .

Now we show that  $q(x)$  and  $r(x)$  are unique. Suppose that  $q_1(x)$  and  $r_1(x)$  also satisfy the conditions. Then  $a(x) - q(x)b(x)$  and  $a(x) - q_1(x)b(x)$  have degree less than  $m$ . Thus, the difference of these polynomials  $b(x)(q(x) - q_1(x))$  is also of degree less than  $m$ . However,  $F$  is an integral domain, so  $\deg b(x) + \deg(q(x) - q_1(x)) = \deg b(x)(q(x) - q_1(x)) < \deg b(x)$ , meaning that  $\deg(q(x) - q_1(x)) < 0$ , a contradiction. Thus,  $q(x) - q_1(x)$  must be 0 (this is the only time when  $\deg a(x)b(x) = \deg a(x) + \deg b(x)$  does not apply), so that  $q(x) = q_1(x)$ , and  $r(x) = r_1(x)$ . ■

*Example 2.3.6.* The ring  $\mathbb{Z}[x]$  is not a PID or a Euclidean Domain, though we have already shown this. However, the ring  $\mathbb{Q}[x]$  is a PID and a Euclidean Domain.

*Example 2.3.7.* If  $p$  is prime,  $\mathbb{Z}/p\mathbb{Z}[x]$  is a PID. Since this is isomorphic to  $\mathbb{Z}[x]/(p)$ , this shows that the quotient of a ring that is not a PID may be a PID.

*Remark 2.3.8.* Suppose that the field  $F$  is contained in the field  $E$ , and that  $a(x), b(x) \in F[x]$ . Write  $a(x) = Q(x)b(x) + R(x)$  for  $Q(x), R(x) \in E[x]$  satisfying the conditions of the theorem, and write  $a(x) = q(x)b(x) + r(x)$  for  $q(x), r(x) \in F[x]$  satisfying the conditions of the theorem. The uniqueness condition implies that  $Q(x) = q(x)$  and  $R(x) = r(x)$ . In particular,  $b(x)|a(x)$  in  $E[x]$  if and only if  $b(x)|a(x)$  in  $F[x]$ . The greatest common divisor of  $a(x)$  and  $b(x)$  is also the same in the two rings, once it is specified to be monic.

### 2.3.3 Polynomial Rings that are UFDs

**Proposition 2.3.9** (Gauss' Lemma). *Let  $R$  be a UFD with field of fractions  $F$ , and let  $p(x) \in R[x]$ . If  $p(x)$  is reducible in  $F[x]$  then  $p(x)$  is reducible in  $R[x]$ . More precisely, if  $p(x) = A(x)B(x)$  for some nonconstant polynomials  $A(x), B(x) \in F[x]$ , then there are nonzero elements  $r, s \in F$  such that  $rA(x) = a(x) \in R[x]$ ,  $sB(x) = b(x) \in R[x]$ , and  $p(x) = a(x)b(x)$  is a factorization in  $R[x]$ .*

*Proof.* The coefficients of the polynomials on the right hand side of the equation  $p(x) = A(x)B(x)$  are elements in  $F$ , and are therefore quotients of elements from the UFD  $R$ . Multiplying by a common denominator for all those coefficients, we obtain  $dp(x) = a'(x)b'(x)$ , where  $a'(x), b'(x) \in R[x]$  and  $d$  is a nonzero element of  $R$ . If  $d$  is a unit in  $R$ , then this is true with  $a(x) = d^{-1}a'(x)$  and  $b(x) = b'(x)$ . Assume that  $d$  is not a unit, and write  $d$  as a product of irreducibles in  $R$ ,  $d = p_1 \cdots p_n$ . Since  $p_1$  is irreducible in  $R$ ,  $(p_1)$  is prime, and so  $p_1R[x]$  is prime in  $R[x]$ , and  $(R/p_1R)[x]$  is an integral domain. Reducing  $dp(x) = a'(x)b'(x)$  modulo  $p_1$ , we get  $0 = (a'(x) + p_1R[x])(b'(x) + p_1R[x])$ . One of these, say  $a'(x) + p_1R[x]$  must be zero in this ring. Then, all coefficients of  $a'(x)$  are divisible by  $p_1$ , so that  $\frac{1}{p_1}a'(x)$  also has coefficients in  $R$ . Thus, in the equation  $dp(x) = a'(x)b'(x)$ , we may cancel a factor of  $p_1$  from  $d$  on the left and either  $a(x)$  or  $b(x)$  on the right, and still have an equation in  $R[x]$ . Continuing this process, we can cancel all factors of  $d$  into the two polynomials on the right, leaving an equation  $p(x) = a(x)b(x)$  with  $a(x), b(x) \in R[x]$ , which are  $F$ -multiples of  $A(x)$  and  $B(x)$ . ■



**Corollary 2.3.10.** *Let  $R$  be a UFD, let  $F = \text{Frac}(R)$ , and let  $p(x) \in R[x]$ . Suppose that the greatest common divisor of the coefficients of  $p(x)$  is 1. Then  $p(x)$  is irreducible in  $R[x]$  if and only if it is irreducible in  $F[x]$ . In particular, if  $p(x)$  is a monic polynomial that is irreducible in  $R[x]$ , then  $p(x)$  is irreducible in  $F[x]$ .*

*Proof.* If  $p(x)$  is reducible in  $F[x]$ , then it is reducible in  $R[x]$  by Gauss' Lemma. Conversely, if  $p(x)$  is reducible in  $R[x]$ , then  $p(x) = a(x)b(x)$  where neither  $a(x)$  or  $b(x)$  are constant, since the greatest common divisor of the coefficients of  $p(x)$  are 1. This shows that  $p(x)$  is reducible in  $F[x]$  (note that the constants in  $R$  are units in  $F$ , which is why we need this condition). ■

**Theorem 2.3.11.**  *$R$  is a UFD if and only if  $R[x]$  is a UFD.*

*Proof.* If  $R[x]$  is a UFD (and hence an integral domain, forcing  $R$  to be one as well), then each constant element of  $R[x]$  (the elements of  $R$ ) must factor into irreducibles in  $R[x]$ . These irreducibles must have degree 0, as the degree of the product is the sum of the degree. Thus  $R$  is a UFD. Now, suppose that  $R$  is a UFD. Let  $F = \text{Frac}(R)$ , and  $p(x) \in R[x]$  be nonzero. Let  $d$  be the greatest common divisor of the coefficients of  $p(x)$ , so that  $p(x) = dp'(x)$ , where the gcd of the coefficients of  $p'(x)$  is 1. This is unique up to change in  $d$ , and so a unit in  $R$ . Since  $d$  can be factored uniquely into irreducibles in  $R$  (which are also irreducibles in  $R[x]$ ), it suffices to show that  $p'(x)$  can be factored uniquely into irreducibles in  $R[x]$ . Thus, we may assume that the greatest common divisor of the coefficients of  $p(x)$  is 1. We may further assume that  $\deg p(x) > 0$ .

Since  $F[x]$  is a UFD,  $p(x)$  can be factored uniquely into irreducibles in  $F[x]$ . By Gauss' Lemma, such a factorization implies there is a factorization of  $p(x)$  in  $R[x]$ , whose factors are  $F$ -multiples of the factors in  $F[x]$ . Since the greatest common divisor of the coefficients of  $p(x)$  is 1, the gcd of the coefficients in each of these factors in  $R[x]$  must be 1. By the previous corollary, each of these factors is irreducible in  $R[x]$ . Thus,  $p(x)$  can be written as a finite product of irreducibles in  $R[x]$ .

The uniqueness of the factorization of  $p(x)$  follows from the uniqueness in  $F[x]$ . Suppose

$$p(x) = q_1(x) \cdots q_r(x) = q'_1(x) \cdots q'_s(x)$$

are two factorizations of  $p(x)$  into irreducibles in  $R[x]$ . Since the gcd of the coefficients of  $p(x)$  is 1, the same is true for each of the irreducible factors. In particular, each has positive degree. By the previous Corollary, each  $q_i(x)$  and  $q'_j(x)$  are irreducible in  $R[x]$ . By the unique factorization in  $F[x]$ ,  $r = s$ , and (possibly after rearrangement)  $q_i(x)$  and  $q'_i(x)$  are associates in  $F[x]$  for all  $1 \leq i \leq r$ . It remains to show that they are associates in  $R[x]$ . Since the units of  $F[x]$  are the elements of  $F^\times$ , we need to consider when  $q(x) = \frac{a}{b}q'(x)$  for some  $q(x), q'(x) \in R[x]$  and nonzero elements  $a, b \in R$ , where the greatest common divisors of the coefficients of  $q(x)$  and  $q'(x)$  are 1. In this case  $bq(x) = aq'(x)$ . The greatest common divisor of the coefficients on the left hand side is  $b$ , and on the right hand side, it is  $a$ . Since  $R$  is a UFD, the gcd of coefficients of a nonzero polynomial is unique up to units, so that  $a = ub$  for some  $u \in R^\times$ . Thus,  $q(x) = uq'(x)$ , and so  $q(x)$  and  $q'(x)$  are associates in  $R[x]$  as well. ■

**Corollary 2.3.12.** *If  $R$  is a UFD, then a polynomial ring in an arbitrary number of variables with coefficients in  $R$  is also a UFD.*

*Proof.* For finitely many variables, this follows by induction from the previous theorem. The general case follows from the definition of a polynomial ring in an arbitrary number of variables as the union of polynomial rings in finitely many variables. ■

*Example 2.3.13.*  $\mathbb{Z}[x]$ ,  $\mathbb{Z}[x, y]$ , and so on are UFDs. Thus,  $\mathbb{Z}[x]$  is an example of a UFD that is not a PID.

### 2.3.4 Irreducibility Criteria

**Proposition 2.3.14.** *Let  $F$  be a field and let  $p(x) \in F[x]$ . Then  $p(x)$  has a factor of degree one if and only if there is some  $\alpha \in F$  with  $p(\alpha) = 0$ .*

*Proof.* If  $p(x)$  has a factor of degree one, then since  $F$  is a field, we may assume that it is monic, so that it is of the form  $x - \alpha$  for some  $\alpha \in F$ . Then,  $p(\alpha) = 0$ . Conversely, suppose that  $p(\alpha) = 0$ . By the division algorithm in  $F[x]$ , we may write  $p(x) = q(x)(x - \alpha) + r$ , where  $r$  is a constant. Since  $p(\alpha) = 0$ ,  $r = 0$ , meaning that  $(x - \alpha)$  is a factor of  $p(x)$ . ■

**Corollary 2.3.15.** *A polynomial of degree two or three over a field  $F$  is reducible if and only if it has a root in  $F$ .*

*Proof.* Polynomials of degree two can only factor as the product of two linear factors, and polynomials of degree three can only factor as the product of a linear factor and a quadratic factor. ■

**Proposition 2.3.16** (Rational Root Theorem). *Let  $R$  be a UFD, and let  $p(x) \in R[x]$  be of the form  $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ . If  $r/s \in \text{Frac}(R)$  has the property that  $\gcd(r, s) = 1$  and  $p(r/s) = 0$  (extending to  $\text{Frac}(R)[x]$ ), then  $r|a_0$  and  $s|a_n$ . In particular, if  $p(x)$  is monic with  $p(d) \neq 0$  for all  $d \in R$  dividing the constant term of  $p(x)$ , then  $p(x)$  has no roots in  $\text{Frac}(R)$ .*

*Proof.* By hypothesis,  $0 = p(r/s) = a_n(r/s)^n + a_{n-1}(r/s)^{n-1} + \cdots + a_0$ . Multiplying by  $s^n$  gives

$$0 = a_n r^n + a_{n-1} r^{n-1} s + \cdots + a_0 s^n.$$

Thus,  $a_n r^n = s(-a_{n-1} r^{n-1} - \cdots - a_0 s^{n-1})$ , so  $s|a_n r^n$ . Since  $\gcd(r, s) = 1$ ,  $s|a_n$ . Similarly,  $r|a_0$ . ■

**Proposition 2.3.17.** *Let  $I$  be a proper ideal of the integral domain  $R$ , and let  $p(x)$  be a nonconstant monic polynomial in  $R[x]$ . If the image of  $p(x)$  in  $(R/I)[x]$  cannot be factored in  $(R/I)[x]$  into two polynomials of smaller degree, then  $p(x)$  is irreducible in  $R[x]$ .*

*Proof.* Suppose that  $p(x)$  cannot be factored in  $(R/I)[x]$ , but that  $p(x)$  is reducible in  $R[x]$ . Then, there are monic, nonconstant polynomials  $a(x), b(x) \in R[x]$  such that  $p(x) = a(x)b(x)$ . Reducing the coefficients modulo  $I$  gives a factorization in  $(R/I)[x]$  with nonconstant factors, a contradiction. ■

*Remark 2.3.18.* It is not true that if a polynomial is reducible modulo every ideal then it is reducible. For example, the polynomial  $x^4 - 72x^2 + 4 \in \mathbb{Z}[x]$ .

*Example 2.3.19.* The polynomial  $x^2+x+1 \in \mathbb{Z}[x]$  is irreducible, because  $x^2+x+1 \in (\mathbb{Z}/2\mathbb{Z})[x]$  is irreducible.

*Example 2.3.20.* The polynomial  $x^2 + 1$  is irreducible in  $\mathbb{Z}[x]$ , since it is irreducible in  $(\mathbb{Z}/3\mathbb{Z})[x]$  (it has no roots), but it is reducible in  $(\mathbb{Z}/2\mathbb{Z})[x]$ .

**Proposition 2.3.21** (Eisenstein's Criterion). *Let  $P$  be a prime ideal of the integral domain  $R$ , and let  $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$  be an element of  $R[x]$  with  $n \geq 1$ . Suppose that  $a_{n-1}, \dots, a_1, a_0 \in P$ , and suppose that  $a_0 \notin P^2$ . Then  $f(x)$  is irreducible in  $R[x]$ .*

*Proof.* Suppose that  $f(x)$  were reducible, say  $f(x) = a(x)b(x)$  in  $R[x]$ , where  $a(x), b(x)$  are nonconstant polynomials. Reducing modulo  $P$ , and using the assumptions on the coefficients of  $f(x)$ , we get  $x^n = \overline{a(x)} \cdot \overline{b(x)}$  in  $(R/P)[x]$ , where the bar denotes the polynomials with coefficients reduced mod  $P$ . Since  $P$  is prime,  $R/P$  is an integral domain, and so  $\overline{a(x)}$  and  $\overline{b(x)}$  have a constant term of 0, so that the constant terms of  $a(x)$  and  $b(x)$  are in  $P$ . Then, the constant term  $a_0$  of  $f(x)$  is the product of these two, and would therefore be an element of  $P^2$ , a contradiction. ■

**Corollary 2.3.22** (Eisenstein's Criterion for  $\mathbb{Z}$ ). *Let  $p \in \mathbb{Z}$  be prime, and let  $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 + a_0 \in \mathbb{Z}[x]$  with  $n \geq 1$ . Suppose that  $p|a_i$  for  $i \in \{0, 1, \dots, n-1\}$ , but that  $p^2 \nmid a_0$ . Then  $f(x)$  is irreducible in  $\mathbb{Z}[x]$  and  $\mathbb{Q}[x]$ .*

*Proof.* This is a restatement of the previous proposition in the case of the prime ideal  $(p)$  together with Gauss' Lemma. ■

### 2.3.5 Polynomial Rings Over Fields (II)

**Proposition 2.3.23.** *Let  $F$  be a field. The maximal ideals in  $F[x]$  are the ideals  $(f(x))$  generated by irreducible polynomials  $f(x)$ . In particular,  $F[x]/(f(x))$  is a field if and only if  $f(x)$  is irreducible.*

*Proof.* Recall that  $F[x]$  is a Euclidean Domain, and hence a PID. Thus, the nonzero prime ideals are the maximal ideals. We show that all nonzero prime ideals are of the form  $(f(x))$  for some irreducible  $f(x)$ . Since  $F[x]$  is a PID, all prime ideals are of the form  $(f(x))$  for some prime element  $f(x)$ , and the prime elements are the irreducibles in a PID. ■

**Proposition 2.3.24.** *Let  $g(x) \in F[x]$  be nonconstant, and let*

$$g(x) = f_1(x)^{n_1} f_2(x)^{n_2} \cdots f_k(x)^{n_k}$$

*be its factorization into irreducibles, where the  $f_i(x)$  are distinct. Then we have the following isomorphism of rings:*

$$F[x]/(g(x)) \cong F[x]/(f_1(x)^{n_1}) \times F[x]/(f_2(x)^{n_2}) \times \cdots \times F[x]/(f_k(x)^{n_k}).$$

*Proof.* The ideals  $(f_i(x)^{n_i})$  and  $(f_j(x)^{n_j})$  are comaximal if  $i \neq j$ , since they are relatively prime in the Euclidean Domain  $F[x]$ . The rest follows from the Chinese Remainder Theorem. ■

**Definition 2.3.25.** Let  $F$  be a field, and  $f(x) \in F[x]$ . Recall that a root  $\alpha$  corresponds to a factor of  $x - \alpha$  of  $f(x)$ . If  $(x - \alpha)^m | f(x)$ , but  $(x - \alpha)^{m+1} \nmid f(x)$ , then  $\alpha$  is said to be a root of multiplicity  $m$ .

**Proposition 2.3.26.** *If the polynomial  $f(x) \in F[x]$  has roots  $\alpha_1, \alpha_2, \dots, \alpha_k$  in  $F$  (not necessarily distinct), then  $f(x)$  is divisible by  $(x - \alpha_1) \cdots (x - \alpha_k)$ . In particular, a polynomial of degree  $n$  in one variable over a field  $F$  has at most  $n$  roots in  $F$ , even counted with multiplicity.*

*Proof.* The first statement follows easily by induction. The second follows since linear factors are irreducible and since  $F[x]$  is a UFD. ■

**Proposition 2.3.27.** *A finite subgroup of the multiplicative group of a field is cyclic. In particular, if  $F$  is a finite field, then  $F^\times$  is a cyclic group.*

*Proof.* Let this group be  $G$ , and let  $n$  divide  $|G|$ . By the previous proposition, there are at most  $n$  roots of  $x^n - 1$  in  $F^\times$  (0 is not a root), and thus in  $G$ . It has been previously proven that this implies that  $G$  is cyclic (§1.6). ■

**Corollary 2.3.28.** *Let  $p$  be prime. The multiplicative group  $(\mathbb{Z}/p\mathbb{Z})^\times$  is cyclic.*

*Proof.* This is the multiplicative group of the finite field  $\mathbb{Z}/p\mathbb{Z}$ . ■

**Corollary 2.3.29.** *Let  $n \geq 2$  be an integer with factorization  $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$  in  $\mathbb{Z}$ , where  $p_1, p_2, \dots, p_r$  are distinct primes. We have the following isomorphisms of (multiplicative) groups:*

- (1)  $(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})^\times$ .
- (2)  $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$  is the direct product of a cyclic group of order 2 and a cyclic group of order  $2^{\alpha-2}$  for all  $\alpha \geq 2$ .
- (3)  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$  is a cyclic group of order  $p^{\alpha-1}(p-1)$  for all primes  $p$ .

*Proof.* The first isomorphism follows from the Chinese Remainder Theorem. The second follows from the fact that  $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$  is not cyclic (both  $2^\alpha - 1$  and  $2^{\alpha-1} + 1$  have order 2), and that 5 has order  $2^{\alpha-2}$  (this follows from writing  $5 = 1 + 2^2$  and applying the Binomial Theorem) (thus, there must be some group  $H$  of order 2 that intersects the subgroup generated by 5 ( $K$ ) trivially, meaning that the entire group is  $HK$ , which is isomorphic to  $H \times K$ ). Finally, if  $p$  is an odd prime, then  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$  is an abelian group of order  $p^{\alpha-1}(p-1)$ . The element  $1 + p$  has order  $p^{\alpha-1}$  by the Binomial Theorem, and so the Sylow  $p$ -subgroup of  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$  is cyclic. The map

$$\mathbb{Z}/p^\alpha\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z} \quad a + (p^\alpha) \mapsto a + (p)$$

is a ring homomorphism (reduction mod  $p$ ), giving a surjective group homomorphism from  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$  onto  $(\mathbb{Z}/p\mathbb{Z})^\times$ . The latter group is cyclic of order  $p-1$ . The kernel of this map has order  $p^{\alpha-1}$ , and so for all primes  $q \neq p$ , the Sylow  $q$ -subgroup of  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$  maps isomorphically into the cyclic group  $(\mathbb{Z}/p\mathbb{Z})^\times$ . All Sylow subgroups of  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$  are therefore cyclic, and thus their direct product, which is isomorphic to  $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ , is cyclic. ■

**Proposition 2.3.30.** *The additive and multiplicative groups of a field are never isomorphic.*

*Proof.* Let  $F$  denote our field. We shall denote its additive group as  $\mathcal{F}$ . If  $|F|$  is finite, then this is clear, so suppose that  $|F|$  is infinite. Suppose first that  $1 + 1 = 0$  in  $F$ . Then,  $a + a = a(1 + 1) = 0$  for all  $a \in F$ . In order for  $\mathcal{F}$  to be isomorphic to  $F^\times$ , each element of  $F^\times$  must have order 2. In other words,  $a^2 = 1$  for all nonzero  $a \in F$ . Then, the subgroup  $\{1, a, b, ab\}$  of  $F^\times$  is isomorphic to the Klein-4 Group, and is thus not cyclic, a contradiction. Now, suppose that  $1 + 1 \neq 0$  in  $F$ . In particular,  $1 \neq -1$ . Since  $(-1)^2 = 1$ , and  $-1 \neq 1$ ,  $F^\times$  contains an element of order 2. In order for this to be isomorphic to  $\mathcal{F}$ ,  $\mathcal{F}$  must contain some  $a \neq 0$  with  $a + a = 0$ . Thus,  $a(1 + 1) = 0$ , meaning that  $1 + 1 = 0$ , since  $a$  is nonzero, a contradiction. ■

### 2.3.6 Polynomials in Several Variables Over a Field and Gröbner Bases

**Definition 2.3.31.** A commutative ring  $R$  with 1 is called Noetherian if every ideal of  $R$  is finitely generated.

*Example 2.3.32.* All PIDs are Noetherian.

**Theorem 2.3.33** (Hilbert's Basis Theorem). *If  $R$  is a Noetherian ring, then so is the polynomial ring  $R[x]$ .*

*Proof.* Let  $I$  be an ideal in  $R[x]$ , and let  $L$  be the set of all leading coefficients of the elements in  $I$ . We show that  $L$  is an ideal of  $R$ . Since  $0 \in I$ ,  $0 \in L$ . Let  $f = ax^d + \cdots + a_0$  and  $g = bx^e + \cdots + b_0$  be polynomials in  $I$  of degrees  $d$  and  $e$  with leading coefficients  $a, b \in R$ . Then, for any  $r \in R$ , either  $ra - b = 0$  or  $ra - b$  is the leading coefficient of the polynomial  $rx^e f - x^d g$ . Since the latter is in  $I$ ,  $ra - b \in L$ , showing that  $L$  is an ideal of  $R$  (take  $b = 0$  and  $r = 1$  for the two properties). Since  $R$  is Noetherian,  $L$  is finitely generated, say  $L = (a_1, a_2, \dots, a_n)$  for  $a_1, a_2, \dots, a_n \in R$ . For  $1 \leq i \leq n$ , let  $f_i \in I$  have leading coefficient  $a_i$ , let  $e_i$  denote the degree of  $f_i$ , and let  $N = \max(e_1, e_2, \dots, e_n)$ .

For each  $d \in \{0, 1, \dots, N-1\}$ , let  $L_d$  be the set of all leading coefficients of polynomials in  $I$  of degree  $d$ , together with 0. A similar argument shows that  $L_d$  is an ideal of  $R$ , which must be finitely generated. For each  $L_d$  that is nonzero, let  $b_{d,1}, b_{d,2}, \dots, b_{d,n_d}$  be a set of generators for  $L_d$ , and let  $f_{d,i}$  be a polynomial in  $I$  of degree  $d$  with leading coefficient  $b_{d,i}$ .

We show that  $I$  is generated by  $f_1, f_2, \dots, f_n$  together with all the polynomials  $f_{d,i}$ . Let  $I'$  denote the ideal generated by these elements. By construction,  $I' \subseteq I$ , since all its generators are in  $I$ . If  $I' \neq I$ , there is some  $f \in I$  of minimum degree with  $f \notin I'$ . Let  $d = \deg f$  and  $a$  be the leading coefficient of  $f$ . Suppose first that  $d \geq N$ . Since  $a \in L$ , we may write  $a = r_1 a_1 + \cdots + r_n a_n$ . Then,  $g = r_1 x^{d-e_1} f_1 + \cdots + r_n x^{d-e_n} f_n$  is an element of  $I'$  with degree  $d$  and leading coefficient  $a$ . Then,  $f - g \in I$  is a polynomial in  $I$  of smaller degree of  $f$  that is not in  $I'$ , a contradiction. Now suppose that  $d < N$ , meaning that  $a \in L_d$  for some  $d < N$ , so we may write  $a = r_1 b_{d,1} + \cdots + r_{n_d} b_{d,n_d}$  for some  $r_i \in R$ . Then  $g = r_1 f_{d,1} + \cdots + r_{n_d} f_{d,n_d}$  is an element in  $I'$  with the same degree  $d$  and leading coefficient  $a$ . In this case,  $f - g$  is an element of  $I$  but not  $I'$  of lower degree than  $f$ , a contradiction. Thus,  $I = I'$  is finitely generated. ■

**Corollary 2.3.34.** *Let  $F$  be a field. Every ideal in  $F[x_1, x_2, \dots, x_n]$  is finitely generated.*

*Proof.* It is clear that  $F$  is Noetherian, so the rest follows from induction and Hilbert's Basis Theorem. ■

**Definition 2.3.35.** A monomial ordering is a well ordering  $\geq$  on the set of monomials that satisfies  $mm_1 \geq mm_2$  whenever  $m_1 \geq m_2$  for monomials  $m, m_1, m_2$ . Equivalently, a monomial ordering may be specified by defining a well ordering on the  $n$ -tuples  $\alpha = (a_1, \dots, a_n) \in \mathbb{Z}^n$  of multidegrees of monomials  $Ax_1^{a_1} \cdots x_n^{a_n}$  that satisfies  $\alpha + \gamma \geq \beta + \gamma$  if  $\alpha \geq \beta$ .

**Definition 2.3.36.** Fix a monomial ordering on  $F[x_1, x_2, \dots, x_n]$ .

- (1) The leading term of a nonzero polynomial  $f \in F[x_1, x_2, \dots, x_n]$ , denoted  $\text{LT}(f)$ , is the monomial term of maximal order in  $f$ , and the leading term of  $f = 0$  is 0. Define the multidegree of  $f$ , denoted  $\partial(f)$ , to be the multidegree of the leading term of  $f$ .
- (2) If  $I$  is an ideal in  $F[x_1, x_2, \dots, x_n]$ , then ideal of leading terms, denoted  $\text{LT}(I)$ , is the ideal generated by the leading terms of all elements of  $I$ .

**Definition 2.3.37.** A Gröbner basis for an ideal  $I$  in the polynomial ring  $F[x_1, \dots, x_n]$  is a finite set of generators  $\{g_1, \dots, g_m\}$  for  $I$  whose leading terms generate the ideal of all leading terms in  $I$ . In other words,

$$I = (g_1, \dots, g_m) \quad \text{and} \quad \text{LT}(I) = (\text{LT}(g_1), \dots, \text{LT}(g_m)).$$

*Remark 2.3.38.* Fix a monomial ordering on  $F[x_1, \dots, x_n]$  and suppose that  $g_1, \dots, g_m$  is a set of nonzero polynomials in  $F[x_1, \dots, x_n]$ . We give an algorithm of general polynomial division. If  $f \in F[x_1, \dots, x_m]$ , start with a set of quotients  $q_1, \dots, q_m$  and a remainder  $r$  all initially equal to 0, and successively check whether the leading term of  $f$  is divisible by the leading terms of the divisors  $g_1, \dots, g_m$  in that order. We have two cases:

- If  $\text{LT}(f)$  is divisible by  $\text{LT}(g_i)$ , say  $\text{LT}(f) = a_i \text{LT}(g_i)$ , add  $a_i$  to  $q_i$ , and replace  $f$  by  $f - a_i g_i$  (which has a lower order leading term), and reiterate the process.
- If  $\text{LT}(f)$  is not divisible by any of  $\text{LT}(g_1), \dots, \text{LT}(g_m)$ , add the leading term of  $f$  to  $r$ , and replace  $f$  by  $f - \text{LT}(f)$ , and reiterate the process.

This terminates, and results in a set of quotients  $q_1, \dots, q_m$  and a remainder  $r$  with

$$f = q_1 g_1 + \cdots + q_m g_m + r.$$

Each  $q_i g_i$  has multidegree less than or equal to that of  $f$ , and no nonzero term in  $r$  is divisible by any of the leading terms  $\text{LT}(g_1), \dots, \text{LT}(g_m)$ .

**Theorem 2.3.39.** *Fix a monomial ordering on  $R = F[x_1, \dots, x_n]$ , and suppose that  $\{g_1, \dots, g_m\}$  is a Gröbner basis for the nonzero ideal  $I \subseteq R$ .*

- (1) *Every polynomial  $f \in R$  can be written uniquely in the form  $f = f_I + r$ , where  $f_I \in I$  and no nonzero monomial term of the ‘remainder’  $r$  is divisible by any of the terms  $\text{LT}(g_1), \dots, \text{LT}(g_m)$ .*

- (2) Both  $f_I$  and  $R$  can be computed by general polynomial division by  $g_1, \dots, g_m$ , and are independent of the order in which these polynomials are used in the division.
- (3) The remainder  $r$  provides a unique representative for the coset of  $f$  in  $F[x_1, \dots, x_n]/I$ . In particular,  $f \in I$  if and only if  $r = 0$ .

*Proof.*

- (1) If we let  $f_I = \sum_{i=1}^m q_i g_i \in I$  in the general polynomial division of  $f$  by  $g_1, \dots, g_m$ , we get  $f = f_I + r$  for any generators  $g_1, \dots, g_m$ . Now suppose that  $\{g_1, \dots, g_m\}$  is a Gröbner basis, and that  $f = f_I + r = f'_I + r'$ . Then  $r - r' = f'_I - f_I \in I$ , so  $\text{LT}(r - r') \in \text{LT}(I) = (\text{LT}(g_1), \dots, \text{LT}(g_m))$ . Thus,  $r - r'$  is a sum of terms each of which is divisible by one of the  $\text{LT}(g_i)$ . However, both  $r$  and  $r'$ , and hence also  $r - r'$ , are sums of monomial terms none of which are divisible by  $\text{LT}(g_1), \dots, \text{LT}(g_m)$ , which is a contradiction unless  $r - r' = 0$ . Thus  $r = r'$  is unique and so is  $f_I = f - r$ .
- (2) Since we have seen that  $f_I$  and  $r$  can be computed by polynomial division, we only need to show that they are independent of the order of which we use  $g_1, \dots, g_m$ . Since  $f_I$  and  $r$  are unique, they must be independent of the order.
- (3) The first statement follows from the uniqueness in (1). If  $r = 0$ , then  $f = f_I \in I$ . Conversely, if  $f \in I$ , then  $f = f + 0$ , and so the uniqueness of  $r$  implies  $r = 0$ . ■

**Proposition 2.3.40.** Fix a monomial ordering on  $R = F[x_1, \dots, x_m]$ , and let  $I$  be a nonzero ideal of  $R$ .

- (1) If  $g_1, \dots, g_m$  are any elements of  $I$  such that  $\text{LT}(I) = (\text{LT}(g_1), \dots, \text{LT}(g_m))$ , then  $\{g_1, \dots, g_m\}$  is a Gröbner basis for  $I$ .
- (2) The ideal  $I$  has a Gröbner basis.

*Proof.*

- (1) Suppose that  $g_1, \dots, g_m \in I$  with  $\text{LT}(I) = (\text{LT}(g_1), \dots, \text{LT}(g_m))$ . We must show that  $g_1, \dots, g_m$  generates  $I$ . If  $f \in I$ , use general polynomial division to write  $f = \sum_{i=1}^m q_i g_i + r$ , where no nonzero term in the remainder is divisible by any  $\text{LT}(g_i)$ . Since  $f \in I$ ,  $r \in I$ , meaning that  $\text{LT}(r) \in \text{LT}(I)$ . Then  $\text{LT}(r)$  is divisible by one of the  $\text{LT}(g_1), \dots, \text{LT}(g_m)$ , which is a contradiction unless  $r = 0$ . Thus,  $I = (g_1, \dots, g_m)$ .
- (2) Since  $F[x_1, \dots, x_m]$  is Noetherian,  $\text{LT}(I)$  is finitely generated, say by  $h_1, \dots, h_k$ . Since it is also generated by  $S = \{\text{LT}(f) : f \in I\}$ , we may write each  $h_i$  as a combination of finitely many elements of  $S$ . All of these elements together are a finite set that generates  $\text{LT}(I)$ , each of the form  $\text{LT}(f)$  for some  $f \in I$ , and so the existence follows from (1). ■

**Lemma 2.3.41.** *Suppose  $f_1, \dots, f_m \in F[x_1, \dots, x_n]$  are polynomials with the same multidegree  $\alpha$  and that  $h = a_1 f_1 + \dots + a_m f_m$  where  $a_i \in F$  has a strictly smaller multidegree. Then,*

$$h = \sum_{i=2}^m b_i S(f_{i-1}, f_i) \quad b_i \in F,$$

where  $S(f, g)$  is defined to be  $\frac{M}{\text{LT}(f)}f - \frac{M}{\text{LT}(g)}f$ , where  $M$  is the monic least common multiple of  $\text{LT}(f)$  and  $\text{LT}(g)$ .

*Proof.* Write  $f_i = c_i f'_i$ , where  $c_i \in F$  and  $f'_i$  is monic of multidegree  $\alpha$ . Then,

$$h = a_1 c_1 (f'_1 - f'_2) + (a_1 c_1 + a_2 c_2) (f'_2 - f'_3) + \dots$$

$$+ (a_1 c_1 + \dots + a_{m-1} c_{m-1}) (f'_{m-1} - f'_m) + (a_1 c_1 + \dots + a_m c_m) f'_m.$$

Since  $f'_{i-1} - f'_i = S(f_{i-1}, f_i)$ , and since  $h$  and  $f'_{i-1} - f'_i$  has multidegree strictly smaller than  $\alpha$ , we have that  $a_1 c_1 + \dots + a_m c_m = 0$ , so the last term on the right is 0. ■

**Notation.** For a fixed monomial ordering on  $R = F[x_1, \dots, x_n]$  and ordered set of polynomials  $G = \{g_1, \dots, g_m\}$ , write  $f \equiv r \pmod{G}$  if  $r$  is the remainder obtained by general polynomial division of  $f \in R$  by  $g_1, \dots, g_m$ .

**Proposition 2.3.42** (Buchberger's Criterion). *Let  $R = F[x_1, \dots, x_n]$  and fix a monomial ordering on  $R$ . If  $I = (g_1, \dots, g_m)$  is a nonzero ideal in  $R$ , then  $G = \{g_1, \dots, g_m\}$  is a Gröbner basis for  $I$  if and only if  $S(g_i, g_j) \equiv 0 \pmod{G}$  for  $1 \leq i < j \leq m$ .*

*Proof.* If  $\{g_1, \dots, g_m\}$  is a Gröbner basis for  $I$ , then  $S(g_i, g_j) \equiv 0 \pmod{G}$  because  $S(g_i, g_j) \in I$ . Now suppose that  $S(g_i, g_j) \equiv 0 \pmod{G}$  for  $1 \leq i < j \leq m$ , and take  $f \in I$ . We need to show that  $\text{LT}(f) \in (\text{LT}(g_1), \dots, \text{LT}(g_m))$ . Write  $f = \sum_{i=1}^m h_i g_i$  for some  $h_1, \dots, h_m$ . Among all such representations, choose one for which  $\max_{i \in \{1, \dots, m\}} \partial(h_i g_i)$  is minimal, say  $\alpha$ . Then  $\partial(f) \leq \alpha$ . Write

$$\begin{aligned} f &= \sum_{i=1}^m h_i g_i = \sum_{\partial(h_i g_i) = \alpha} h_i g_i + \sum_{\partial(h_i g_i) < \alpha} h_i g_i \\ &= \sum_{\partial(h_i g_i) = \alpha} \text{LT}(h_i) g_i + \sum_{\partial(h_i g_i) = \alpha} (h_i - \text{LT}(h_i)) g_i + \sum_{\partial(h_i g_i) < \alpha} h_i g_i. \end{aligned}$$

Suppose that  $\partial(f) < \alpha$ . Since the multidegree of the two second sums is strictly smaller than  $\alpha$ , the multidegree of the first sum is strictly smaller than  $\alpha$ . If  $a_i \in F$  is the constant coefficient of  $\text{LT}(h_i)$ , then  $\text{LT}(h_i) = a_i h'_i$ , where  $h'_i$  is a monomial. We may apply our lemma to  $\sum a_i (h'_i g_i)$  to write the first sum as  $\sum b_i S(h'_{i-1} g_{i-1}, h'_i g_i)$  with  $\partial(h'_{i-1} g_{i-1}) = \partial(h'_i g_i) = \alpha$ . Let  $\beta_{i-1, i}$  be the multidegree of the monic least common multiple of  $\text{LT}(g_{i-1})$  and  $\text{LT}(g_i)$ . Then,  $S(h'_{i-1} g_{i-1}, h'_i g_i)$  is  $S(g_{i-1}, g_i)$  multiplied by the monomial of multidegree  $\alpha - \beta_{i-1, i}$ . Since  $S(g_{i-1}, g_i)$  has a multidegree less than  $\beta_{i-1, i}$ , and since  $S(g_{i-1}, g_i) \equiv 0 \pmod{G}$ , after general polynomial division of  $S(g_{i-1}, g_i)$  by  $g_1, \dots, g_m$ , each  $S(g_{i-1}, g_i)$  can be written as  $\sum q_j g_j$  with  $\partial(q_j g_j) < \beta_{i-1, i}$ . Thus, each  $S(h'_{i-1} g_{i-1}, h'_i g_i)$  is a sum  $\sum q'_j g_j$  with  $\partial(q'_j g_j) < \alpha$ . Then, because of our lemma, all sums on the right hand side of the equation can be written



as a sum of terms of the form  $p_i g_i$  with  $p_i$  satisfying  $\partial(p_i g_i) < \alpha$ , contradicting the minimality of  $\alpha$ . Thus,  $\partial(f) = \alpha$ . Now if we take the terms of multidegree  $\alpha$ , we get that

$$\text{LT}(f) = \sum_{\partial(h_i g_i) = \alpha} \text{LT}(h_i) \text{LT}(g_i),$$

meaning that  $\text{LT}(f) \in (\text{LT}(g_1), \dots, \text{LT}(g_m))$ . Thus  $G$  is a Gröbner basis.  $\blacksquare$

**Remark 2.3.43.** Using this, we can get an algorithm to find a Gröbner basis for an ideal  $I$ . If  $I = (g_1, \dots, g_m)$  and  $S(g_i, g_j)$  leaves a remainder 0 when divided by  $G = \{g_1, \dots, g_m\}$  using general polynomial division, then  $G$  is a Gröbner bases. Otherwise, it leaves a nonzero remainder  $r$ . Append  $g_{m+1} = r$  to  $G$  to get  $G' = \{g_1, \dots, g_m, g_{m+1}\}$ , and begin again. It is not hard to check that this process terminates.

**Remark 2.3.44.** If  $\{g_1, \dots, g_m\}$  is a Gröbner basis for  $I$  and  $\text{LT}(g_j)$  is divisible by  $\text{LT}(g_i)$  for some  $j \neq i$ , then we do not need  $\text{LT}(g_j)$  as a generator for  $\text{LT}(I)$ . By Proposition 2.3.40, we may therefore delete  $g_j$  and still have a Gröbner basis for  $I$ . We may also assume without loss of generality that each  $\text{LT}(g_i)$  is monic. A Gröbner basis  $\{g_1, \dots, g_m\}$  for  $I$  where each  $\text{LT}(g_i)$  is monic and not divisible by  $\text{LT}(g_j)$  for  $j \neq i$  is called a minimal Gröbner basis.

**Lemma 2.3.45.** Let  $\{g_1, \dots, g_r\}$  and  $\{h_1, \dots, h_s\}$  be two minimal Gröbner bases for the ideal  $I \subseteq R = F[x_1, \dots, x_n]$ . Then  $r = s$ , and (reordering if necessary),  $\text{LT}(g_i) = \text{LT}(h_i)$ .

*Proof.* Let  $G$  and  $H$  denote the sets respectively. Note that the leading terms of the  $g_i$ 's and the  $h_i$ 's must be distinct. Thus, the second statement implies the first. Now, we show that, for each  $i$ ,  $\text{LT}(g_i)$  is the same as  $\text{LT}(h_k)$  for some  $k$ , which implies the statement by symmetry. Since  $H$  is a Gröbner basis, we get some  $h_k$  so that  $\text{LT}(h_k) \mid \text{LT}(g_i)$ . Since  $G$  is a Gröbner basis, there is some  $g_\ell$  such that  $\text{LT}(g_\ell) \mid \text{LT}(h_k)$ , meaning that  $\text{LT}(g_\ell) \mid \text{LT}(g_i)$ . Since  $G$  is minimal,  $\ell = i$ , and so  $\text{LT}(g_i) = c \text{LT}(h_k)$  for some  $c \neq 0$ . From the minimality of  $G$  and  $H$ ,  $c = 1$ , so  $\text{LT}(g_i) = \text{LT}(h_k)$ .  $\blacksquare$

**Definition 2.3.46.** Fix a monomial ordering on  $R = F[x_1, \dots, x_n]$ . A Gröbner basis  $\{g_1, \dots, g_m\}$  for the nonzero idea  $I \subseteq I$  is called a reduced Gröbner basis if  $\text{LT}(g_i)$  is monic for  $i = 1, \dots, m$  and no term in  $g_j$  is divisible by  $\text{LT}(g_i)$  for  $j \neq i$ .

**Theorem 2.3.47.** Fix a monomial ordering on  $R = F[x_1, \dots, x_n]$ . Then there is a unique reduced Gröbner basis for every nonzero ideal  $I$  in  $R$ .

*Proof.* Note that reduced bases are minimal bases, and so two reduced bases thus have the same number of elements and the same leading terms. If  $G = \{g_1, \dots, g_m\}$  and  $G' = \{g'_1, \dots, g'_m\}$  are two reduced bases for the same nonzero ideal  $I$ , then we may assume that  $\text{LT}(g_i) = \text{LT}(g'_i) = h_i$ . Fix  $i$ , and consider  $f_i = g_i - g'_i$ . If this is nonzero, then since  $f_i \in I$ , its leading term must be divisible by some  $h_j$ . For  $j \neq i$ ,  $h_j$  does not divide any of the terms in  $g_i$  or  $g'_i$ , and so cannot divide  $\text{LT}(f_i)$ . However,  $h_i$  does not divide  $\text{LT}(f_i)$ , since all the terms in  $f_i$  have strictly smaller multidegree, forcing  $f_i = 0$ . Thus,  $g_i = g'_i$  for all  $i$ .  $\blacksquare$

**Corollary 2.3.48.** Let  $I$  and  $J$  be two ideals in  $F[x_1, \dots, x_n]$ . Then  $I = J$  if and only if  $I$  and  $J$  have the same reduced Gröbner basis with respect to any fixed monomial ordering on  $F[x_1, \dots, x_n]$ .

*Remark 2.3.49.* We can also use Gröbner bases to solve systems of equations. If  $f_1, \dots, f_m$  is a system of equations in  $n$  variables, where  $f_i(a_1, \dots, a_n) = 0$ , then any  $f \in I = (f_1, \dots, f_m)$  satisfies  $f(a_1, \dots, a_n) = 0$ . It is also easy to see that if  $S = \{g_1, \dots, g_s\}$  is any set of generators, then the set of solutions to  $g_1 = 0, \dots, g_s = 0$  is the same as the set of solutions to  $f_1 = 0, \dots, f_m = 0$ . If  $f_1, \dots, f_m$  are nonlinear polynomials, we use a principle inspired by Gaussian elimination. If there is a nonzero polynomial in  $I$  involving only one of the variables, say  $p(x_n)$ , then  $a_n$  satisfies  $p(x_n) = 0$ . Now, if there is a polynomial involving only  $x_{n-1}$  and  $x_n$ , say  $q(x_{n-1}, x_n)$ , then the coordinate  $a_{n-1}$  is a solution of  $q(x_{n-1}, x_n) = 0$ . We continue this process. If we can successively find polynomials in  $I$  that eliminate  $x_1, x_2, \dots$ , then we will be able to determine all solutions  $(a_1, \dots, a_n)$ .

**Definition 2.3.50.** If  $I$  is an ideal in  $F[x_1, \dots, x_n]$ , then  $I_i = I \cap F[x_{i+1}, \dots, x_n]$  is called the  $i$ th elimination ideal of  $I$  with respect to the ordering  $x_1 > \dots > x_n$ .

**Proposition 2.3.51** (Elimination). *Suppose that  $G = \{g_1, \dots, g_m\}$  is a Gröbner basis for the nonzero ideal  $I \subseteq F[x_1, \dots, x_n]$  with respect to the lexicographic monomial ordering  $x_1 > \dots > x_n$ . Then  $G \cap F[x_{i+1}, \dots, x_n]$  is a Gröbner basis of the  $i$ th elimination ideal  $I_i = I \cap F[x_{i+1}, \dots, x_n]$  of  $I$ . In particular,  $I \cap F[x_{i+1}, \dots, x_n] = 0$  if and only if  $G \cap F[x_{i+1}, \dots, x_n] = \emptyset$ .*

*Proof.* Write  $G_i = G \cap F[x_{i+1}, \dots, x_n]$ . Then,  $G_i \subseteq I_i$ , so it suffices to show that the set of leading terms of  $G_i$ , which we denote by  $\text{LT}(G_i)$ , generate  $\text{LT}(I_i)$ . Clearly,  $(\text{LT}(G_i)) \subseteq \text{LT}(I_i)$ . Now let  $f \in I_i$ . Then  $f \in I$ , and so we can write

$$\text{LT}(f) = a_1(x_1, \dots, x_n) \text{LT}(g_1) + \dots + a_m(x_1, \dots, x_n) \text{LT}(g_m)$$

for some  $a_1, \dots, a_m \in F[x_1, \dots, x_n]$ . Writing the  $a_i$ 's as a sum of monomial terms, we see that  $\text{LT}(f)$  is the sum of monomial terms of the form  $ax_1^{s_1} \dots x_n^{s_n} \text{LT}(g_i)$ . Since  $\text{LT}(f)$  involves only  $x_{i+1}, \dots, x_n$ , the sum of all such terms containing any of  $x_1, \dots, x_i$  must be 0. Thus,  $\text{LT}(f)$  can be written as a  $F[x_{i+1}, \dots, x_n]$ -linear combination of some monomial terms  $\text{LT}(g_t)$ , where  $\text{LT}(g_t)$  does not involve  $x_1, \dots, x_i$ . By the choice of ordering, if  $\text{LT}(g_t)$  does not involve  $x_1, \dots, x_i$ , then neither do any of the other terms in  $g_t$ , so that  $g_t \in F_i$ . Thus,  $\text{LT}(f)$  can be written as a  $F[x_{i+1}, \dots, x_n]$ -linear combination of elements of  $\text{LT}(G_i)$ . ■

*Example 2.3.52.* Let us find the intersections of  $2x^2 + 2xy + y^2 - 2x - 2y = 0$  and  $x^2 + y^2 = 1$ . We may compute a Gröbner basis for  $I = (2x^2 + 2xy + y^2 - 2x - 2y, x^2 + y^2 - 1) \subset \mathbb{R}[x, y]$  using to lexicographic monomial order  $x > y$  to eliminate  $x$ , to get  $g_1 = 2x + y^2 + 5y^3 - 2$  and  $g_2 = 5y^4 - 4y^3$ . Thus,  $5y^4 = 4y^3$ , so that  $y = 0$  or  $y = 4/5$ . Substituting into  $g_1 = 0$ , we find the intersection points are  $(1, 0)$  and  $(-3/5, 4/5)$ .

*Example 2.3.53.* Let us find the solutions in  $\mathbb{C}$  to

$$f_1 = x^3 - 2xy + y^3 = 0 \quad f_2 = x^5 - 2x^2y^2 + y^5 = 0.$$

A Gröbner basis for  $(f_1, f_2)$  with respect to the lexicographic monomial order  $x > y$ , we get

$$\begin{aligned} g_1 &= x^3 - 2xy + y^3 \\ g_2 &= 200xy^2 + 193y^9 + 158y^8 - 45y^7 - 456y^6 + 50y^5 - 100y^4 \\ g_3 &= y^{10} - y^8 - 2y^7 + 2y^6. \end{aligned}$$

Any solution to  $f_1 = f_2 = 0$  satisfies  $g_1 = g_2 = g_3 = 0$ . Since  $g_3 = y^6(y-1)^2(y^2+2y+2)$ , we have  $y = 0$ ,  $y = 1$ , or  $y = -1 \pm i$ . Now,  $g_1(x, 0) = x^3$ , and  $g_2(x, 0) = 0$ , the only solution with  $y = 0$  is  $(0, 0)$ . Since  $g_1(x, 1) = x^3 - 2x + 1$  and  $g_2(x, 1) = 200(x-1)$ , the only solution with  $y = 1$  is  $(1, 1)$ . Finally,

$$\begin{aligned} g_1(x, -1 \pm i) &= x^3 + (2 \mp 2i)x + (2 \pm 2i) \\ g_2(x, -1 \pm i) &= -400i(x + i \pm i), \end{aligned}$$

which have common zero  $x = -1 \mp i$  when  $y = -1 \pm i$  respectively. This gives four solutions to our original equations:

$$(x, y) = (0, 0), \quad (1, 1) \quad (-1 + i, -1 - i) \quad (-1 - i, -1 + i).$$

*Example 2.3.54.* Consider the solutions in  $\mathbb{C}$  to the system of equations

$$\begin{aligned} x + y + z &= 1 \\ x^2 + y^2 + z^2 &= 2 \\ x^3 + y^3 + z^3 &= 3. \end{aligned}$$

The reduced Gröbner basis with respect to the lexicographic ordering  $x > y > z$  is

$$x + y + z - 1 \quad y^2 + yz - y + z^2 - z - 1/2 \quad z^3 - z^2 - (1/2)z - (1/6),$$

so  $z$  is a root of  $t^3 - t^2 - (1/2)t - (1/6)$ . For each of the three roots of this, there are two values of  $y$  (from the second element of the basis), and one value of  $x$  for these two values. This gives six solutions.

*Remark 2.3.55.* Suppose that  $I = (f_1, \dots, f_s)$  and  $H = (h_1, \dots, h_t)$  are two ideals in  $F[x_1, \dots, x_n]$ . Then  $I + J = (f_1, \dots, f_s, h_1, \dots, h_t)$  and  $IJ = (f_1h_1, \dots, f_ih_j, \dots, f_sh_t)$ .

**Proposition 2.3.56.** *If  $I$  and  $J$  are two ideals in  $F[x_1, \dots, x_n]$ , then  $tI + (1-t)J$  is an ideal in  $F[t, x_1, \dots, x_n]$ , and  $I \cap J = (tI + (1-t)J) \cap F[x_1, \dots, x_n]$ . In particular,  $I \cap J$  is the first elimination ideal of  $tI + (1-t)J$  with respect to the ordering  $t > x_1 > \dots > x_n$ .*

*Proof.* Clearly  $tI$  and  $(1-t)J$  are ideals in  $F[t, x_1, \dots, x_n]$ , and so their sum is also an ideal in  $F[t, x_1, \dots, x_n]$ . If  $f \in I \cap J$ , then  $f = tf + (1-t)f$ , which shows that  $I \cap J \subseteq (tI + (1-t)J) \cap F[x_1, \dots, x_n]$ . Now suppose that  $f = tf_1 + (1-t)f_2$  is an element of  $F[x_1, \dots, x_n]$ , where  $f_1 \in I$  and  $f_2 \in J$ . Then  $t(f_1 - f_2) = f - f_2 \in F[x_1, \dots, x_n]$ , meaning that  $f_1 - f_2 = 0$  and  $f = f_2$ . Thus,  $f = f_1 = f_2 \in I \cap J$ . Thus,  $I \cap J = (tI + (1-t)J) \cap F[x_1, \dots, x_n]$ . ■



# Chapter 3

## Modules and Vector Spaces

### 3.1 Introduction to Module Theory

#### 3.1.1 Basic Definitions and Examples

**Definition 3.1.1.** Let  $R$  be a ring (not necessarily commutative or with a 1). A left  $R$ -module or a left module over  $R$  is a set  $M$  together with

- (1) A binary operation  $+$  on  $M$  under which  $M$  is an abelian group.
  - (2) An action of  $R$  on  $M$  (a map  $R \times M \rightarrow M$ ), denoted by  $rm$  for all  $r \in R$  and  $m \in M$ , which satisfies
    - (a) For all  $r, s \in R$  and  $m \in M$ ,  $(r + s)m = rm + sm$
    - (b) For all  $r, s \in R$  and  $m \in M$ ,  $(rs)m = r(sm)$
    - (c) For all  $r \in R$  and  $m, n \in M$ ,  $r(m + n) = rm + rn$
- In the case  $R$  has a 1, we impose
- (d)  $1m = m$  for all  $m \in M$ .

*Remark 3.1.2.* We can similarly define right  $R$ -modules, with the ring elements appearing on the right instead of on the left. If  $R$  is commutative and  $M$  is a left  $R$ -module, we can make  $M$  into a right  $R$ -module by defining  $mr = rm$  for  $m \in M$  and  $r \in R$ . However, if  $R$  is not commutative, this definition does not necessarily work. Unless explicitly mentioned, the term module will always mean left module. Modules satisfying 2(d) are called unital modules, and all of our modules will be unital.

**Definition 3.1.3.** Let  $F$  be a field. A module over  $F$  is called a vector space over  $F$ .

**Definition 3.1.4.** Let  $R$  be a ring and  $M$  be an  $R$ -module. An  $R$ -submodule of  $M$  is a subgroup  $N$  of  $M$  with  $rn \in N$  for all  $r \in R$  and  $n \in N$ .

*Example 3.1.5.* Every  $R$ -module  $M$  has two submodules  $M$  and  $0$  (the latter is called the trivial submodule).

*Example 3.1.6.* Let  $R$  be any ring. Then  $M = R$  is a left  $R$ -module, where the action of a ring element on a module element is just the usual multiplication in  $R$  (similarly  $R$  is a right  $R$ -module over itself). When  $R$  is considered as a left module over itself, the submodules of  $R$  are the left ideals of  $R$  (and if  $R$  is a right  $R$ -module over itself, the submodules are the right ideals).

*Example 3.1.7.* Let  $F$  be a field and  $n \in \mathbb{Z}^+$ . Let  $F^n$  denote the set of  $n$ -tuples of elements of  $F$ . Then  $F^n$  is a vector space over  $F$  with addition and multiplication defined componentwise.

*Example 3.1.8.* Let  $R$  be a ring with 1 and define  $R^n$  to be the set of  $n$ -tuples of  $R$ . This is an  $R$ -module with componentwise addition and multiplication. This is called the free module of rank  $n$  over  $R$ .

*Example 3.1.9.* If  $M$  is an  $R$ -module and for some (2-sided) ideal  $I$  of  $R$ ,  $am = 0$  for all  $a \in I$  and  $m \in M$ , we say that  $M$  is annihilated by  $I$ . In this case, we can make  $M$  into an  $(R/I)$ -module by defining  $(r + I)m = rm$ . It is not hard to check that this is well defined and makes  $M$  into an  $R/I$  module.

*Example 3.1.10.* Let  $R = \mathbb{Z}$  and  $A$  be any abelian group, and write the operation of  $A$  as  $+$ . Make  $A$  into a  $\mathbb{Z}$ -module as follows: for any  $n \in \mathbb{Z}$  and  $a \in A$ , define

$$na = \begin{cases} \underbrace{a + a + \cdots + a}_{n \text{ times}} & n > 0 \\ 0 & n = 0 \\ \underbrace{-a - a - \cdots - a}_{n \text{ times}} & n < 0 \end{cases}.$$

This makes  $A$  into a  $\mathbb{Z}$ -module, and the module axioms imply that this is the only possible action of  $\mathbb{Z}$  on  $A$ . This in fact shows that  $\mathbb{Z}$ -modules are the same as abelian groups. Note also that, if  $|A| = m$ , then  $mx = 0$  for all  $x \in A$ , then  $A$  is a  $\mathbb{Z}/m\mathbb{Z}$ -module.

*Example 3.1.11.* Let  $F$  be a field and let  $R = F[x]$ . Let  $V$  be a vector space over  $F$ , and  $T : V \rightarrow V$  be a linear transformation (i.e. one that satisfies  $T(ax + by) = aT(x) + bT(y)$  for  $a, b \in F$  and  $x, y \in V$ ). We make  $V$  into an  $F[x]$ -module. For a nonnegative integer  $n$ , define  $T^0 = I$  and  $T^n = T \circ T \circ \cdots \circ T$  ( $n$  times), where  $I$  is the identity map from  $V$  to  $V$ . Also, for any two linear transformations  $A, B : V \rightarrow V$ , and  $\alpha, \beta \in F$ , let  $\alpha A + \beta B$  be defined by  $(\alpha A + \beta B)(v) = \alpha A(v) + \beta B(v)$ . It is easy to show that this is a linear transformation. Let  $p(x) = a_n x^n + \cdots + a_1 x + a_0 \in F[x]$  be a polynomial in  $x$ . We define the action of  $p(x)$  on  $v \in V$  by

$$p(x)v = (a_n T^n + a_{n-1} T^{n-1} + \cdots + a_1 T + a_0)v.$$

It is not hard to see that this makes  $V$  into an  $F[x]$ -module.

**Proposition 3.1.12** (The Submodule Criterion). *Let  $R$  be a ring and let  $M$  be an  $R$ -module. A subset  $N \subseteq M$  is a submodule of  $M$  if and only if  $N \neq \emptyset$  and  $x + ry \in N$  for all  $r \in R$  and  $x, y \in N$ .*

*Proof.* If  $N$  is a submodule, then  $N \neq \emptyset$ , as  $0 \in N$ , and  $N$  is closed under addition and multiplication by elements of  $R$ . Now suppose that (1) and (2) hold. Let  $r = -1$  and apply the subgroup criterion to get that  $N \leq M$ . In particular,  $0 \in N$ . Now let  $x = 0$  to get that  $N$  is closed under the action of  $R$ . ■

**Definition 3.1.13.** Let  $R$  be a commutative ring with identity. An  $R$ -algebra is a ring  $A$  with identity together with a ring homomorphism  $f : R \rightarrow A$  mapping  $1_R$  to  $1_A$  such that the subring  $f(R)$  of  $A$  is contained in the center of  $A$ .

*Remark 3.1.14.* If  $A$  is an  $R$ -algebra, then  $A$  has a natural left and right  $R$ -module structure defined by  $r \cdot a = a \cdot r = f(r)a$  (which is the same as  $af(r)$  since  $f(r)$  is in the center of  $A$ ). Unless otherwise stated, this natural module structure on an algebra will be assumed.

**Definition 3.1.15.** If  $A$  and  $B$  are two  $R$ -algebras, an  $R$ -algebra homomorphism (or isomorphism) is a ring homomorphism (isomorphism respectively)  $\phi : A \rightarrow B$  mapping  $1_A$  to  $1_B$  such that  $\phi(r \cdot a) = r \cdot \phi(a)$  for all  $r \in R$  and  $a \in A$ .

*Example 3.1.16.* Any ring with identity is a  $\mathbb{Z}$ -algebra.

*Example 3.1.17.* If  $A$  is a ring with identity and  $R$  is a subring of the center of  $A$  containing 1, then  $A$  is an  $R$ -algebra. In particular, a commutative ring with identity  $A$  is an  $R$ -algebra for any subring  $R$  of  $A$  containing 1.

**Proposition 3.1.18.** Let  $R$  be a commutative ring with identity. A ring with identity  $A$  is an  $R$ -algebra if and only if  $A$  is a unital left module over  $R$  satisfying  $r \cdot (ab) = (r \cdot a)b = a(r \cdot b)$  for all  $r \in R$  and  $a, b \in A$ .

*Proof.* If  $A$  is an  $R$ -algebra, then  $A$  is a unital left  $R$ -module with  $r \cdot a$  defined as  $f(r)a$ . Since  $f(r)$  is assumed to be in the center of  $A$ , the extra property follows. Now, suppose that  $A$  is a unital left module over  $R$  satisfying  $r \cdot (ab) = (r \cdot a)b = a(r \cdot b)$ , and define  $f : R \rightarrow A$  by  $f(r) = r \cdot 1$ . That  $f$  is a homomorphism follows from the module axioms and our second assumption. Since  $A$  is unital,  $f(1_R) = 1_A$ . Finally, if  $r \in R$  and  $a \in A$ , then  $f(r)a = (r \cdot 1)a = r \cdot (1a) = r \cdot (a1) = a(r \cdot 1) = af(r)$ , so  $f(R)$  is in the center of  $A$ . Thus  $A$  is an  $R$ -algebra. ■

### 3.1.2 Quotient Modules and Module Homomorphisms

**Definition 3.1.19.** Let  $R$  be a ring, and let  $M$  and  $N$  be  $R$ -modules. A map  $\phi : M \rightarrow N$  is an  $R$ -module homomorphism if  $\phi(x + y) = \phi(x) + \phi(y)$  for all  $x, y \in M$ , and  $\phi(rx) = r\phi(x)$  for all  $r \in R$  and  $x \in M$ . An  $R$ -module homomorphism is an isomorphism of  $R$ -modules if it is bijective. We write  $M \cong N$ . We write  $\ker \phi = \{m \in M : \phi(m) = 0\}$  and  $\text{im } \phi = \{\phi(m) : m \in M\}$ . We write  $\text{Hom}_R(M, N)$  for the set of all  $R$ -module homomorphisms from  $M$  to  $N$ .

**Proposition 3.1.20.** Let  $M, N$  and  $L$  be  $R$ -modules.

- (1) A map  $\phi : M \rightarrow N$  is an  $R$ -module homomorphism if and only if  $\phi(rx + y) = r\phi(x) + \phi(y)$  for all  $x, y \in M$  and  $r \in R$ .
- (2) Let  $\phi, \psi \in \text{Hom}_R(M, N)$ . If  $(\phi + \psi)(m) = \phi(m) + \psi(m)$ , then  $\phi + \psi \in \text{Hom}_R(M, N)$ , and  $\text{Hom}_R(M, N)$  is an abelian group. If  $R$  is a commutative ring with  $r \in R$ , and if  $(r\phi)(m) = r(\phi(m))$ , then  $r\phi \in \text{Hom}_R(M, N)$ , and this makes  $\text{Hom}_R(M, N)$  into an  $R$ -module.
- (3) If  $\phi \in \text{Hom}(L, M)$  and  $\psi \in \text{Hom}(M, N)$ , then  $\psi \circ \phi \in \text{Hom}(L, N)$ .

- (4) With addition as above, and multiplication defined as composition,  $\text{Hom}_R(M, M)$  is a ring with identity. When  $R$  is commutative,  $\text{Hom}_R(M, M)$  is an  $R$ -algebra.

*Proof.*

- (1) Clearly, if  $\phi$  is an  $R$ -module homomorphism,  $\phi(rx + y) = r\phi(x) + \phi(y)$ . Take  $r = 1$  and  $y = 0$  to get additivity and multiplicity respectively.
- (2) For all  $x, y \in M$  and  $r \in R$ , we have that

$$\begin{aligned} (\phi + \psi)(rx + y) &= \phi(rx + y) + \psi(rx + y) = r\phi(x) + \phi(y) + r\psi(x) + \psi(y) \\ &= r(\phi(x) + \psi(x)) + (\phi(y) + \psi(y)) = r(\phi + \psi)(x) + (\phi + \psi)(y). \end{aligned}$$

Take the zero map for identity, and  $\psi(x) = -\phi(x)$  for inverse. Associativity and commutativity are immediate. Now suppose that  $R$  is commutative. Now, for  $x, y \in M$  and  $s \in R$ ,

$$(r\phi)(sx + y) = r\phi(sx + y) = r(s\phi(x) + \phi(y)) = s(r\phi)(x) + (r\phi)(y).$$

Clearly, all the module axioms are satisfied.

- (3) Let  $x, y \in L$  and  $r \in R$ . Then,

$$(\psi \circ \phi)(rx + y) = \psi(\phi(rx + y)) = \psi(r\phi(x) + \phi(y)) = r\psi(\phi(x)) + \psi(\phi(y)) = r(\psi \circ \phi)(x) + (\psi \circ \phi)(y).$$

- (4) All axioms are clear using the multiplicative identity of  $I(x) = x$  and the additive identity of  $0(x) = 0$ .

■

*Example 3.1.21.* Consider  $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$ . For a map  $\phi$ ,  $\phi$  is completely determined by what it does to  $1 \in \mathbb{Z}/n\mathbb{Z}$ . Further, we must have that  $\phi(0) = 0$ , so that  $n\phi(1) = 0$ . Thus, we can write  $n\phi(1) = hm$ , so that  $\phi(1) = h \cdot m/n = h \cdot (m/\gcd(m, n))/(n/\gcd(m, n))$ . As  $\gcd(m/\gcd(m, n), n/\gcd(m, n)) = 1$ , we have that  $h/(n/\gcd(m, n))$  must be an integer, and that  $\phi(1)$  is an integral multiple of  $m/\gcd(m, n)$ . Further, any integral multiple of  $m/\gcd(m, n)$  is a suitable value for  $\phi(1)$ . Now, notice that  $h/(n/\gcd(m, n))$  can be assumed to be less than  $\gcd(m, n)$ , so as to not repeat values. Now, consider the map from  $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$  to  $\mathbb{Z}/\gcd(m, n)\mathbb{Z}$  given by  $\phi \mapsto \phi(1)/(m/\gcd(m, n))$ . This is clearly a module homomorphism, and so a module isomorphism.

**Proposition 3.1.22.** *Let  $R$  be a commutative ring. Then  $\text{Hom}_R(R, M)$  and  $M$  are isomorphic as left  $R$ -modules.*

*Proof.* Let  $\phi \in \text{Hom}_R(R, M)$ . Then,  $\phi(r) = \phi(r \cdot 1) = r\phi(1)$ , so  $\phi$  is completely determined by  $\phi(1)$ . Further, each element of  $M$  gives a suitable value for  $\phi(1)$ . Consider the bijective map from  $\text{Hom}_R(R, M)$  to  $M$  given by  $\phi \mapsto \phi(1)$ . This is clearly a module homomorphism, and so a module isomorphism. ■



**Definition 3.1.23.** The ring  $\text{Hom}_R(M, M)$  is called the endomorphism ring of  $M$ , denoted  $\text{End}_R(M)$ , or  $\text{End}(M)$  when  $R$  is clear.

*Remark 3.1.24.* When  $R$  is commutative, there is a natural map from  $R$  to  $\text{End}(M)$  given by  $r \mapsto rI$ . If  $R$  has an identity, this makes  $M$  into an  $R$ -algebra.

**Proposition 3.1.25.** *Let  $R$  be a ring and  $M$  an  $R$ -module, and let  $N$  be a submodule of  $M$ . Then the (additive, abelian) quotient group  $M/N$  can be made into an  $R$ -module by defining  $r(x + N) = (rx) + N$ . Further, the natural projection map  $\pi : M \rightarrow M/N$  defined by  $\pi(x) = x + N$  is an  $R$ -module homomorphism with kernel  $N$ .*

*Proof.* Suppose that  $x + N = y + N$ . Then,  $x - y \in N$ , so  $r(x - y) = rx - ry \in N$ , and  $rx + N = ry + N$ . The  $R$ -module axioms follow immediately. Now,  $\pi$  is a group homomorphism with kernel  $N$ , so it remains to show that  $\pi$  is a module homomorphism (the kernel must stay the same). We have that  $\pi(rm) = rm + N = r(m + N) = r\pi(m)$ , so  $\pi$  is a module homomorphism. ■

**Definition 3.1.26.** Let  $A$  and  $B$  be submodules of the  $R$ -module  $M$ . The sum of  $A$  and  $B$  is the set

$$A + B = \{a + b : a \in A, b \in B\}.$$

**Theorem 3.1.27** (Isomorphism Theorems).

- (1) *Let  $M$  and  $N$  be  $R$ -modules and let  $\phi : M \rightarrow N$  be an  $R$ -module homomorphism. Then  $\ker \phi$  is a submodule of  $M$ , and  $M/\ker \phi \cong \phi(M)$ .*
- (2) *Let  $A$  and  $B$  be submodules of the  $R$ -module  $M$ . Then  $(A + B)/B \cong A/(A \cap B)$ .*
- (3) *Let  $M$  be an  $R$ -module and let  $A$  and  $B$  be submodules of  $M$  with  $A \subseteq B$ . Then  $(M/A)/(B/A) \cong M/B$ .*
- (4) *Let  $N$  be a submodule of the  $R$ -module  $M$ . There is a bijection between the submodules of  $M$  which contain  $N$  and the submodules of  $M/N$ . The correspondence is given by  $A \leftrightarrow A/N$  for all  $A \supseteq N$ . This commutes with taking sums and intersections.*

*Proof.*

- (1) Clearly,  $\ker \phi$  is a submodule of  $M$ . Consider the map  $\psi(x + \ker \phi) = \phi(x)$ . This has been shown to be a group isomorphism and well-defined. Now,  $\psi(r(x + \ker \phi)) = \psi(rx + \ker \phi) = \phi(rx) = r\phi(x) = r\psi(x + \ker \phi)$ . Thus, this is a module isomorphism.
- (2) Consider the map  $\phi : A \rightarrow (A + B)/B$  by  $\phi(a) = a + B$ . This is clearly a surjective module homomorphism. Now, this is zero when  $a \in B$ , so that  $\ker \phi = A \cap B$ . By (1),  $(A + B)/B \cong A/(A \cap B)$ .
- (3) Let  $\phi : M/A \rightarrow M/B$  be given by  $a + A \mapsto a + B$ . This has been shown to be a surjective group homomorphism with kernel  $B/A$ . Now,  $\phi(r(a + A)) = \phi(ra + A) = ra + B = r(a + B) = r\phi(a + A)$ , so this is a module homomorphism. Thus,  $(M/A)/(B/A) \cong M/B$ .
- (4) This has all been previously shown, except for the fact that this commutes with sums, which is trivial. ■

### 3.1.3 Generation of Modules, Direct Sums, and Free Modules

*Remark 3.1.28.* Here,  $R$  is a ring with 1.

**Definition 3.1.29.** Let  $M$  be an  $R$ -module, and let  $N_1, \dots, N_n$  be submodules of  $M$ .

- (1) The sum of  $N_1, \dots, N_n$  is the set  $N_1 + \dots + N_n = \{a_1 + \dots + a_n : a_i \in N_i\}$ .
- (2) For any subset  $A$  of  $M$ , let

$$RA = \{r_1 a_1 + r_2 a_2 + \dots + r_m a_m : r_1, \dots, r_m \in R, a_1, \dots, a_m \in A, m \in \mathbb{Z}^+\},$$

where by convention  $R\emptyset = \{0\}$ . If  $A$  is the finite set  $\{a_1, \dots, a_n\}$ , we shall write  $Ra_1 + \dots + Ra_n$  for  $RA$ . We call  $RA$  the submodule of  $M$  generated by  $A$ . If  $N$  is a submodule of  $M$  and  $N = RA$  for some subset  $A$  of  $M$ , we call  $A$  a set of generators or a generating set for  $N$ , and we say  $N$  is generated by  $A$ .

- (3) A submodule  $N$  of  $M$  is finitely generated if there is some finite subset  $A$  of  $M$  such that  $N = RA$ .
- (4) A submodule  $N$  of  $M$  is cyclic if there exists an element  $a \in M$  such that  $N = Ra$ .

*Example 3.1.30.* If  $M$  is any abelian group and  $a \in M$ , then, viewing  $M$  as a  $\mathbb{Z}$ -module,  $\mathbb{Z}a = \langle a \rangle$ .

*Remark 3.1.31.* If  $A \subseteq M$ , then  $RA$  is a submodule of  $M$ .

**Definition 3.1.32.** Let  $M_1, \dots, M_k$  be a collection of  $R$ -modules. The collection of  $k$ -tuples  $(m_1, \dots, m_k)$ , where  $m_i \in M_i$  with addition and action of  $R$  defined componentwise is called the direct product of  $M_1, \dots, M_k$ , denoted  $M_1 \times \dots \times M_k$ .

*Remark 3.1.33.* The direct product of a collection of  $R$ -modules is again an  $R$ -module. This is also sometimes called the external direct sum of  $M_1, \dots, M_k$ , denoted  $M_1 \oplus \dots \oplus M_k$ .

**Proposition 3.1.34.** Let  $N_1, \dots, N_k$  be submodules of the  $R$ -module  $M$ . Then the following are equivalent:

- (1) The map  $\pi : N_1 \times \dots \times N_k \rightarrow N_1 + \dots + N_k$  defined by

$$\pi(a_1, \dots, a_k) = a_1 + \dots + a_k$$

is an isomorphism of  $R$ -modules.

- (2) For all  $1 \leq j \leq k$ ,  $N_j \cap (N_1 + \dots + N_{j-1} + N_{j+1} + \dots + N_k) = 0$ .
- (3) Every  $x \in N_1 + \dots + N_k$  can be written uniquely in the form  $a_1 + \dots + a_k$  with  $a_i \in N_i$ .

*Proof.* Suppose (1), but that there is some  $j$  so that (2) does not hold. Say,  $0 \neq a_j \in N_j \cap (N_1 + \dots + N_{j-1} + N_{j+1} + \dots + N_k)$ . Then, there are  $a_i \in N_i$  with  $a_j = a_1 + \dots + a_{j-1} + a_{j+1} + \dots + a_k$ . Then,  $\pi((a_1, \dots, a_{j-1}, -a_j, a_{j+1}, \dots, a_k)) = 0$ , while  $a_j \neq 0$ , a contradiction. Now suppose (2). If, for  $a_i, b_i \in N_i$ , we have  $a_1 + \dots + a_k = b_1 + \dots + b_k$ , we have that

$$a_j - b_j = (b_1 - a_1) + \dots + (b_{j-1} - a_{j-1}) + (b_{j+1} - a_{j+1}) + \dots + (b_k - a_k).$$

The left hand side is an element of  $N_j$ , and the right hand side is an element of  $N_1 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k$ . Thus,  $a_j - b_j = 0$ , so  $a_j = b_j$  for all  $j$ . Finally, suppose (3). Notice that  $\pi$  is clearly a surjective  $R$ -module homomorphism. Further, by (3),  $\pi$  is injective, and so an isomorphism.  $\blacksquare$

*Remark 3.1.35.* If  $M = N_1 + \cdots + N_k$  for submodules  $N_1, \dots, N_k$  of  $M$  satisfying one of the conditions above, we say that  $M$  is the internal direct sum of  $N_1, \dots, N_k$  and write  $M = N_1 \oplus \cdots \oplus N_k$ .

**Definition 3.1.36.** An  $R$ -module  $F$  is said to be free on the subset  $A$  of  $F$  if, for every  $x \neq 0 \in F$ , there are unique  $0 \neq r_1, \dots, r_n \in R$  and unique  $a_1, \dots, a_n \in A$  such that  $x = r_1 a_1 + \cdots + r_n a_n$  for some  $n \in \mathbb{Z}^+$ . In this case, we say  $A$  is a basis or set of free generators for  $F$ .

**Theorem 3.1.37** (Universal Property of Free Modules). *For any set  $A$ , there is a free  $R$ -module  $F(A)$  on the set  $A$ , and  $F(A)$  satisfies the following universal property: if  $M$  is any  $R$ -module and  $\phi : A \rightarrow M$  is any map of sets, then there is a unique  $R$ -module homomorphism  $\Phi : F(A) \rightarrow M$  such that  $\Phi(a) = \phi(a)$  for all  $a \in A$ . In other words, the following diagram commutes*

$$\begin{array}{ccc} A & \xrightarrow{\text{inclusion}} & F(A) \\ & \searrow \phi & \downarrow \Phi \\ & & M \end{array}$$

When  $A$  is the finite set  $\{a_1, \dots, a_n\}$ ,  $F(A) = Ra_1 \oplus \cdots \oplus Ra_n \cong R^n$ .

*Proof.* Let  $F(A) = \{0\}$  if  $A = \emptyset$ . If  $A$  is nonempty, let  $F(A)$  be the collection of functions  $f : A \rightarrow R$  as sets, such that  $f(a) = 0$  for all but finitely many  $a \in A$ . Make  $F(A)$  into an  $R$ -module by defining  $(f + g)(a) = f(a) + g(a)$  and  $(rf)(a) = r(f(a))$ . Identify  $A$  as a subset of  $F(A)$  by  $a \mapsto f_a$ , where

$$f_a(b) = \begin{cases} 1 & b = a \\ 0 & b \neq a \end{cases}.$$

Now,  $F(A)$  is all finite  $R$ -linear combinations of elements of  $A$ , as if  $f$  is nonzero on  $a_1, \dots, a_n \in A$ , then  $f = f(a_1)a_1 + \cdots + f(a_n)a_n$ . Moreover, each element of  $F(A)$  has a unique expression in this form, so  $F(A)$  is free on  $A$ . Now suppose that  $\phi : A \rightarrow M$  maps the set  $A$  to the  $R$ -module  $M$ . Define  $\Phi : F(A) \rightarrow M$  by

$$\Phi : \sum_{i=1}^n r_i a_i \mapsto \sum_{i=1}^n r_i \phi(a_i).$$

This is well-defined by the uniqueness of expression for the elements of  $F(A)$  as linear combinations of the  $a_i$ . Further,  $\Phi$  is clearly an  $R$ -module homomorphism. By definition,  $\Phi|_A = \phi$ . Now, since  $F(A)$  is generated by  $A$ , given the value of an  $R$ -module homomorphism on  $A$ , we can determine its value on every element of  $F(A)$  uniquely. Thus,  $\Phi$  is the unique extension of  $\phi$  to all of  $F(A)$ . When  $A = \{a_1, \dots, a_n\}$ , part (3) of the previous proposition shows that  $F(A) = Ra_1 \oplus \dots \oplus Ra_n$ . Since  $R \cong Ra_i$  for all  $i$  (using  $r \mapsto ra_i$ ), part (1) of the proposition establishes that the direct sum is isomorphic to  $R^n$ . ■

**Corollary 3.1.38.** *If  $F_1$  and  $F_2$  are free modules on the same set  $A$ , there is a unique isomorphism between  $F_1$  and  $F_2$  which is the identity map on  $A$ . In particular, if  $F$  is a free  $R$ -module with basis  $A$ , then  $F \cong F(A)$  and  $F$  has the same universal property with respect to  $A$  as  $F(A)$ .*

*Proof.* Define  $\phi : F_1 \rightarrow F_2$  given by

$$\phi : \sum_{i=1}^n r_i a_i \mapsto \sum_{i=1}^n r_i a_i.$$

This is a well-defined  $R$ -module homomorphism which restricts to the identity on  $A$ . Now, if  $\psi : F_2 \rightarrow F_1$  is given by

$$\psi : \sum_{i=1}^n r_i a_i \mapsto \sum_{i=1}^n r_i a_i,$$

then  $\psi \circ \phi = \text{id}_{F_1}$  and  $\phi \circ \psi = \text{id}_{F_2}$ , so  $\phi$  is bijective. Thus,  $\phi$  is an isomorphism. It is also the unique isomorphism fixing  $A$ , as any homomorphism is determined by its value on  $A$ . ■

*Remark 3.1.39.* If  $F$  is a free  $R$ -module with basis  $A$ , we shall often define  $R$ -module homomorphism from  $F$  into other  $R$ -modules by specifying their values on  $A$ .

### 3.1.4 Tensor Products of Modules

*Remark 3.1.40.* Suppose that  $R$  is a subring of  $S$  with  $1_R = 1_S$ . Then, any  $S$ -module automatically is an  $R$ -module. However, given an  $R$ -module, it is generally impossible to make it into an  $S$ -module. For example, viewing  $\mathbb{Z}$  as a  $\mathbb{Z}$ -module,  $\mathbb{Z}$  cannot be made into a  $\mathbb{Q}$ -module (if it could, then  $(1/2) \cdot 1 + (1/2) \cdot 1 = 1$ , and  $(1/2) \cdot 1 \in \mathbb{Z}$ , an impossibility). However,  $\mathbb{Z}$  is contained in another  $\mathbb{Q}$  module:  $\mathbb{Q}$  itself. In other words, there is an injection (called an embedding) of the  $\mathbb{Z}$ -module  $\mathbb{Z}$  into the  $\mathbb{Q}$ -module  $\mathbb{Q}$ . We wish to find whether an arbitrary  $R$ -module  $N$  can be embedded as an  $R$ -submodule of some  $S$ -module. More generally, we ask the question of what  $R$ -module homomorphisms exist from  $N$  to  $S$ -modules. For example, if  $N = \mathbb{Z}/2\mathbb{Z}$  is a  $\mathbb{Z}$ -module, there are no nonzero  $\mathbb{Z}$ -module homomorphisms of  $N$  into some  $\mathbb{Q}$ -module, as every nonzero element of a  $\mathbb{Q}$ -module must have infinite (additive) order. Now, given an  $R$ -module  $N$ , we construct an  $S$ -module that is the “best possible” target in which to try to embed  $N$ . If  $N$  is an  $S$ -module, then we can easily “extend” scalars from  $R$  to  $S$ , so we look at the module axioms to see if we can define “products” of the form  $sn$  for  $s \in S$  and  $n \in N$ . We start with an abelian group  $N$  with a map from  $S \times N$  to  $N$ , where the image of  $(s, n)$  is denoted  $sn$ . Now, we consider the free  $\mathbb{Z}$ -module on the

set  $S \times N$ . In other words, we consider the collection of finite commuting sums of elements of the form  $(s_i, n_i)$ , where  $s_i \in S$  and  $n_i \in N$ . This is an abelian group where there are no relations between distinct pairs  $(s, n)$  and  $(s', n')$ . In other words, there are no relations between the formal products  $sn$ . Now, to satisfy the relations needed for an  $S$ -module, and to make sure that this is compatible with the action of  $R$ , we quotient by the subgroup  $H$  generated by all elements of the form

$$(s_1 + s_2, n) - (s_1, n) - (s_2, n), \quad (s, n_1 + n_2) - (s, n_1) - (s, n_2), \quad \text{and} \quad (sr, n) - (s, rn)$$

for  $s, s_1, s_2 \in S$ ,  $n, n_1, n_2 \in N$ , and  $r \in R$  (recall that  $N$  is an  $R$ -module, so  $rn$  is defined).

**Definition 3.1.41.** The quotient group described above is denoted  $S \otimes_R N$  (or just  $S \otimes N$  when  $R$  is clear), and is called the tensor product of  $S$  and  $N$  over  $R$ . For  $s \in S$  and  $n \in N$ , we write  $s \otimes n$  for the coset containing  $(s, n)$  in  $S \otimes_R N$ . We call the elements of  $S \otimes_R N$  tensors.

*Remark 3.1.42.* By the definition of the quotient, we have forced

$$\begin{aligned} (s_1 + s_2) \otimes n &= s_1 \otimes n + s_2 \otimes n \\ s \otimes (n_1 + n_2) &= s \otimes n_1 + s \otimes n_2 \\ sr \otimes n &= s \otimes rn. \end{aligned}$$

*Remark 3.1.43.* Each element of  $S \otimes_R N$  can be written as finite sums of “simple tensors” of the form  $s \otimes n$ .

**Proposition 3.1.44.** *The tensor product  $S \otimes_R N$  is a left  $S$ -module under*

$$s \left( \sum_{\text{finite}} s_i \otimes n_i \right) = \sum_{\text{finite}} (ss_i) \otimes n_i.$$

*Proof.* We first show that this is well defined. If  $s' \in S$ , then

$$(s'(s_1 + s_2), n) - (s's_1, n) - (s's_2, n) = (s's_1 + s's_2, n) - (s's_1, n) - (s's_2, n) \in H,$$

$$(s's, n_1 + n_2) - (s's, n_1) - (s's, n_2) \in H,$$

and

$$(s'(sr), n) - (s's, rn) = ((s's)r, n) - (s's, rn) \in H.$$

It therefore follows that, for any  $\sum (s_i, n_i) \in H$ ,  $\sum (s's_i, n_i) \in H$ . Now, suppose that  $\sum s_i \otimes n_i = \sum s'_i \otimes n'_i$  are two representations for the same element of  $S \otimes_R N$ . Then,  $\sum (s_i, n_i) - \sum (s'_i, n'_i) \in H$ , so  $\sum (ss_i, n_i) - \sum (ss'_i, n'_i) \in H$  for any  $s \in S$  by what we have just seen. Thus,  $\sum ss_i \otimes n_i = \sum ss'_i \otimes n'_i$ , so this is well defined. By the relations established above, this clearly makes  $S \otimes_R N$  into a left  $S$ -module. ■

**Definition 3.1.45.** The module  $S \otimes_R N$  is called the left  $S$ -module obtained by extension of scalars from the left  $R$ -module  $N$ .

**Proposition 3.1.46.** *The map  $\iota : N \rightarrow S \otimes_R N$  defined by  $n \mapsto 1 \otimes n$  is an  $R$ -module homomorphism.*

*Proof.* This is clearly a group homomorphism. Now,

$$\iota(rn) = 1 \otimes (rn) = r \otimes n = r(1 \otimes n) = r\iota(n).$$

■

*Remark 3.1.47.* As we take the quotient,  $\iota$  is not injective in general, so we cannot assume that  $S \otimes_R N$  contains an isomorphic copy of  $N$ . However, we only imposed the minimal relations to obtain an  $S$ -module, so this is still the “best possible”  $S$ -module to serve as a target from an  $R$ -module homomorphism from  $N$ .

**Theorem 3.1.48.** *Let  $R$  be a subring of  $S$ , let  $N$  be a left  $R$ -module, and let  $\iota : N \rightarrow S \otimes_R N$  be the  $R$ -module homomorphism defined by  $\iota(n) = 1 \otimes n$ . Suppose that  $L$  is any left  $S$ -module (and so an  $R$ -module), and that  $\phi : N \rightarrow L$  is an  $R$ -module homomorphism. Then there is a unique  $S$ -module homomorphism  $\Phi : S \otimes_R N \rightarrow L$  such that  $\phi = \Phi \circ \iota$ . Conversely, if  $\Phi : S \otimes_R N \rightarrow L$  is an  $S$ -module homomorphism, then  $\phi = \Phi \circ \iota$  is an  $R$ -module homomorphism from  $N$  to  $L$ .*

*Proof.* Suppose that  $\phi : N \rightarrow L$  is an  $R$ -module homomorphism. By the universal property of free modules, there is a  $\mathbb{Z}$ -module homomorphism from the free  $\mathbb{Z}$ -module  $F$  on the set  $S \times N$  to  $L$  that sends each generator  $(s, n)$  to  $s\phi(n)$ . Since  $\phi$  is an  $R$ -module homomorphism, the generators of  $H$  all map to 0 in  $L$ . Thus, there is a well defined  $\mathbb{Z}$ -module homomorphism  $\Phi : F/H = S \oplus_R N \rightarrow L$  with  $\Phi(s \otimes n) = s\phi(n)$ . Moreover,

$$s'\Phi(s \otimes n) = s'(s\phi(n)) = (s's)\phi(n) = \Phi((s's) \otimes n) = \Phi(s'(s \otimes n))$$

for  $s \in S$ . Since  $\Phi$  is additive, it is an  $S$ -module homomorphism. This proves existence. Now, the module  $S \otimes_R N$  is generated as an  $S$ -module by elements of the form  $1 \otimes n$ , so any  $S$ -module homomorphism is uniquely determined by its values on these elements. Since  $\Phi(1 \otimes n) = \phi(n)$ ,  $\Phi$  is uniquely determined by  $\phi$ . The converse statement is immediate. ■

**Corollary 3.1.49.** *Let  $\iota : N \rightarrow S \otimes_R N$  be the natural  $R$ -module homomorphism. Then  $N/\ker \iota$  is the unique largest quotient of  $N$  that can be embedded in any  $S$ -module. In particular,  $N$  can be embedded as an  $R$ -submodule of some left  $S$ -module if and only if  $\iota$  is injective, in which case  $N$  is isomorphic to the  $R$ -submodule  $\iota(N)$  of the  $S$ -module  $S \otimes_R N$ .*

*Proof.* The quotient  $N/\ker \iota$  is mapped injectively (by  $\iota$ ) into the  $S$ -module  $S \otimes_R N$ . Now suppose that  $\phi$  is an  $R$ -module homomorphism injecting the quotient  $N/\ker \phi$  of  $N$  into an  $S$ -module  $L$ . Then,  $\ker \iota$  is mapped to 0 by  $\phi$ , so  $\ker \iota \subseteq \ker \phi$ . Thus,  $N/\ker \phi$  is a quotient of  $N/\ker \iota$  (quotient by  $\ker \phi/\ker \iota$ ). Thus,  $N/\ker \iota$  is the unique largest quotient of  $N$  that can be embedded in any  $S$ -module. ■

*Example 3.1.50.* If  $N$  is a left  $R$ -module, then  $R \otimes_R N \cong N$ . Take  $\phi$  to be the identity map from  $N$  to itself in the previous theorem. Then  $\iota$  is an isomorphism with inverse  $\Phi$ .

*Example 3.1.51.* Let  $R = \mathbb{Z}$ ,  $S = \mathbb{Q}$ , and  $A$  be a finite abelian group of order  $n$ . We claim that  $\mathbb{Q} \oplus_{\mathbb{Z}} A = 0$ . Firstly, notice that  $1 \oplus 0 = 1 \oplus (0 + 0) = 1 \oplus 0 + 1 \oplus 0$ , so  $1 \oplus 0 = 0$ . Now, for  $q \oplus a$ , we can write  $q = (q/n)n$ . Then, as  $na = 0$  by Lagrange's Theorem,

$$q \oplus a = (q/n \cdot n) \oplus a = (q/n) \oplus (na) = (q/n) \oplus 0 = (q/n)(1 \oplus 0) = 0.$$

Thus,  $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$ . In particular,  $\iota$  is the zero map. By the previous theorem, we see that any homomorphism of a finite abelian group into a rational vector space is the zero map.

*Remark 3.1.52.* Notice that the construction of  $S \otimes_R N$  as an abelian group only required  $S$  to be a right  $R$ -module and  $N$  to be a left  $R$ -module. Now, the  $S$ -module structure on  $S \otimes_R N$  only required a left  $S$ -module structure on  $S$  together with a compatibility relation  $s'(sr) = (s's)r$  between this left  $S$ -module structure and right  $R$ -module structure on  $S$ . We generalize this.

**Definition 3.1.53.** Suppose that  $N$  is a left  $R$ -module and  $M$  is a right  $R$ -module. The quotient of the free  $\mathbb{Z}$ -module on the set  $M \times N$  by the subgroup generated by all elements of the form

$$(m_1 + m_2, n) - (m_1, n) - (m_2, n), \quad (m, n_1 + n_2) - (m, n_1) - (m, n_2) \quad (mr, n) - (m, rn)$$

for  $m, m_1, m_2 \in M$ ,  $n, n_1, n_2 \in N$ , and  $r \in R$  is an abelian group, denoted  $M \otimes_R N$ , or  $M \otimes N$  if  $R$  is clear from context, and is called the tensor product of  $M$  and  $N$  over  $R$ . The elements of  $M \otimes_R N$  are called tensors, and the coset  $m \otimes n$  of  $(m, n)$  in  $M \otimes_R N$  is called a simple tensor.

*Remark 3.1.54.* We have the relations

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n,$$

$$m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2,$$

and

$$mr \otimes n = m \otimes rn.$$

Also, every tensor can be written as a finite sum of simple tensors.

*Remark 3.1.55.* We once again have a map  $\iota : M \times N \rightarrow M \oplus_R N$  with  $\iota(m, n) = m \otimes n$ . This is not in general a group homomorphism. However, it is additive in  $m$  and  $n$  separately, and satisfies  $\iota(mr, n) = mr \otimes n = m \otimes rn = \iota(m, rn)$ .

**Definition 3.1.56.** Let  $M$  be a right  $R$ -module, let  $N$  be a left  $R$ -module, and let  $L$  be an abelian group written additively. A map  $\phi : M \times N \rightarrow L$  is called  $R$ -balanced or middle linear with respect to  $R$  if

$$\phi(m_1 + m_2, n) = \phi(m_1, n) + \phi(m_2, n)$$

$$\phi(m, n_1 + n_2) = \phi(m, n_1) + \phi(m, n_2)$$

$$\phi(m, rn) = \phi(mr, n)$$

for all  $m, m_1, m_2 \in M$ ,  $n, n_1, n_2 \in N$ , and  $r \in R$ .

**Theorem 3.1.57.** *Suppose  $R$  is a ring with 1,  $M$  is a right  $R$ -module, and  $N$  is a left  $R$ -module. Let  $M \otimes_R N$  be the tensor product of  $M$  and  $N$  over  $R$ , and let  $\iota : M \times N \rightarrow M \otimes_R N$  be the natural  $R$ -balanced map.*

- (1) *If  $\Phi : M \otimes_R N \rightarrow L$  is any group homomorphism from  $M \otimes_R N$  to an abelian group  $L$ , then the composite map  $\phi = \Phi \circ \iota$  is an  $R$ -balanced map from  $M \times N$  to  $L$ .*
- (2) *Suppose that  $L$  is an abelian group and  $\phi : M \times N \rightarrow L$  is any  $R$ -balanced map. Then there is a unique group homomorphism  $\Phi : M \otimes_R N \rightarrow L$  such that  $\phi = \Phi \circ \iota$ .*

*Proof.* The proof of (1) is immediate. Now,  $\phi$  gives a unique  $\mathbb{Z}$ -module homomorphism  $\tilde{\phi}$  from the free group on  $M \times N$  to  $L$  such that  $\tilde{\phi}(m, n) = \phi(m, n) \in L$ . Since  $\phi$  is  $R$ -balanced,  $\tilde{\phi}$  maps every generator of the group we quotient by to zero (for example,  $\tilde{\phi}((mr, n) - (m, rn)) = \phi(mr, n) - \phi(m, rn) = 0$ ). Thus, the kernel of  $\tilde{\phi}$  contains this subgroup, so we have a homomorphism  $\Phi : M \otimes_R N \rightarrow L$  with  $\Phi(m \otimes n) = \tilde{\phi}(m, n) = \phi(m, n)$ . In other words,  $\phi = \Phi \circ \iota$ . Now, since the simple tensors generate  $M \otimes_R N$  as an abelian group,  $\Phi$  is uniquely determined by its value on them. ■

**Corollary 3.1.58.** *Suppose that  $D$  is an abelian group, and  $\iota' : M \times N \rightarrow D$  is an  $R$ -balanced map such that*

- (1) *The image of  $\iota'$  generates  $D$  as an abelian group*
- (2) *Every  $R$ -balanced map defined on  $M \times N$  factors through  $\iota'$  as in the previous theorem.*

*Then, there is an isomorphism  $f : M \otimes_R N \cong D$  of abelian groups with  $\iota' = f \circ \iota$ .*

*Proof.* Since  $\iota' : M \times N \rightarrow D$  is balanced, there is a unique homomorphism  $f : M \otimes_R N \rightarrow D$  with  $\iota' = f \circ \iota$ . In particular,  $\iota'(m, n) = f(m \otimes n)$  for every  $m \in M, n \in N$ . By the first assumption, these elements generate  $D$  as an abelian group, so  $f$  is surjective. Now, the second assumption on  $\iota'$  implies there is a unique homomorphism  $g : D \rightarrow M \otimes_R N$  with  $\iota = g \circ \iota'$ . Then,  $m \otimes n = (g \circ f)(m \otimes n)$ . Since the simple tensors generate  $M \otimes_R N$ ,  $g \circ f$  is the identity on  $M \otimes_R N$ , and so  $f$  is injective. ■

**Definition 3.1.59.** Let  $R$  and  $S$  be rings with 1. An abelian group  $M$  is called an  $(S, R)$ -bimodule if  $M$  is a left  $S$ -module, a right  $R$ -module, and  $s(mr) = (sm)r$  for  $s \in S, r \in R$  and  $m \in M$ .

*Example 3.1.60.* Any ring  $S$  is an  $(S, R)$ -bimodule for any subring  $R \subseteq S$  with  $1_R = 1_S$ . More generally, if  $f : R \rightarrow S$  is any ring homomorphism with  $f(1_R) = 1_S$ , then  $S$  can be considered as a right  $R$ -module with  $s \cdot r = sf(r)$ , and this makes  $S$  an  $(S, R)$ -bimodule.

*Example 3.1.61.* If  $R$  is commutative, then a left (right)  $R$ -module  $M$  can be made into a right (left)  $R$ -module by defining  $mr = rm$  ( $rm = mr$ ) for  $m \in M$  and  $r \in R$ . This makes  $M$  into an  $(R, R)$ -bimodule.



**Definition 3.1.62.** Suppose  $M$  is a left (or right)  $R$ -module over the commutative ring  $R$ . Then the  $(R, R)$ -bimodule structure on  $M$  defined by letting the left and right  $R$ -actions coincide ( $mr = rm$  for all  $m \in M$  and  $r \in R$ ) is called the standard  $R$ -module structure on  $M$ .

*Remark 3.1.63.* If  $N$  is a left  $R$ -module and  $M$  is an  $(S, R)$ -bimodule, then this structure on  $M$  implies that

$$s \left( \sum_{\text{finite}} m_i \otimes n_i \right) = \sum_{\text{finite}} (sm_i) \otimes n_i$$

is a well defined action of  $S$  under which  $M \otimes_R N$  is a left  $S$ -module. Similarly, if  $M$  is a right  $R$ -module and  $N$  is an  $(R, S)$ -bimodule, then  $M \otimes_R N$  has the structure of a right  $S$ -module, where

$$\left( \sum_{\text{finite}} m_i \otimes n_i \right) s = \sum_{\text{finite}} m_i \otimes (n_i s).$$

In the case where  $R$  is commutative,  $S = R$ , and  $M$  has the standard  $R$ -modules structure, then  $M \otimes_R N$  always has the structure of a left  $R$ -module. Here, the natural map  $\iota : M \times N \rightarrow M \otimes_R N$  satisfies  $r\iota(m, n) = \iota(rm, n) = \iota(m, rn)$ , as  $r(m \otimes n) = rm \otimes n = mr \otimes n = m \otimes rn$ .

**Definition 3.1.64.** Let  $R$  be a commutative ring with 1, and let  $M, N, L$  be left  $R$ -modules. The map  $\phi : M \times N \rightarrow L$  is called  $R$ -bilinear if it is  $R$ -linear in each factor.

**Corollary 3.1.65.** Suppose  $R$  is a commutative ring. Let  $M$  and  $N$  be two left  $R$ -modules, and give  $M$  the standard  $R$ -module structure. Then,  $M \otimes_R N$  is a left  $R$ -module with  $r(m \otimes n) = (rm) \otimes n = m \otimes (rn)$ , and the map  $\iota : M \times N \rightarrow M \otimes_R N$  is an  $R$ -bilinear map. If  $L$  is any left  $R$ -module, and  $\phi : M \times N \rightarrow L$  is  $R$ -bilinear, then there exists a unique  $R$ -module homomorphism  $\Phi : M \otimes_R N \rightarrow L$  with  $\phi = \Phi \circ \iota$ . Conversely, if  $\Phi : M \otimes_R N \rightarrow L$  is an  $R$ -module homomorphism,  $\phi = \Phi \circ \iota$  is  $R$ -bilinear.

*Proof.* We have already shown that  $M \otimes_R N$  is an  $R$ -module, and that  $\iota$  is bilinear. If  $\phi : M \times N \rightarrow L$  is bilinear, then it is an  $R$ -balanced map, so we can get a corresponding group homomorphism  $\Phi : M \otimes_R N \rightarrow L$ . Moreover, we have that  $\Phi((rm) \otimes n) = \phi(rm, n) = r\phi(m, n) = r\Phi(m \otimes n)$ . Since  $\Phi$  is additive, it is an  $R$ -module homomorphism. Clearly, if  $\Phi$  is an  $R$ -module homomorphism,  $\phi$  is bilinear. ■

*Example 3.1.66.* We have that  $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/3\mathbb{Z}) = 0$ , since  $3a = a$  for  $a \in \mathbb{Z}/2\mathbb{Z}$ , so that

$$a \otimes b = 3a \otimes b = a \otimes 3b = a \otimes 0 = 0$$

(note that  $a \otimes 0 = a \otimes (0 + 0) = a \otimes 0 + a \otimes 0$ ). Thus, there are no nonzero balanced or bilinear maps from  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$  to any abelian group.

Now, the tensor product  $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$  is generated as an abelian group by the elements  $0 \otimes 0 = 1 \otimes 0 = 0 \otimes 1 = 0$  and  $1 \otimes 1$ . We have that  $1 \otimes 1 \neq 0$ , we the map  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$  defined by  $(a, b) \mapsto ab$  is nonzero and bilinear. Since  $2(1 \otimes 1) = 2 \otimes 1 = 0 \otimes 1 = 0$ , the element  $1 \otimes 1$  has order 2, so  $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ .

In general,

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}.$$

To see this, notice that

$$a \otimes b = a \otimes (b \cdot 1) = (ab) \otimes 1 = ab(1 \otimes 1),$$

so  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$  is a cyclic group generated by  $1 \otimes 1$ . We have that  $m(1 \otimes 1) = m \otimes 1 = 0 \otimes 1 = 0$  and  $n(1 \otimes 1) = 1 \otimes n = 1 \otimes 0 = 0$ , so that  $\gcd(m, n)(1 \otimes 1) = 0$ , so the order of the group divides  $\gcd(m, n)$ . Now, the map  $\phi : \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/\gcd(m, n)\mathbb{Z}$  given by  $(a, b) \mapsto ab$  is well defined and  $\mathbb{Z}$ -bilinear. The induced map  $\Phi : \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/\gcd(m, n)\mathbb{Z}$  maps  $1 \otimes 1$  to  $1 \in \mathbb{Z}/\gcd(m, n)\mathbb{Z}$ , which has order  $\gcd(m, n)$ . Thus,  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$  has order at least  $\gcd(m, n)$ , and so exactly  $\gcd(m, n)$ . Hence,  $1 \otimes 1$  is an element of order  $\gcd(m, n)$ , and  $\Phi$  gives an isomorphism  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}$ .

**Theorem 3.1.67.** *Let  $M$  and  $M'$  be right  $R$ -modules, let  $N$  and  $N'$  be left  $R$ -modules, and suppose that  $\phi : M \rightarrow M'$  and  $\psi : N \rightarrow N'$  are  $R$ -module homomorphisms.*

- (1) *There is a unique group homomorphism, denoted  $\phi \otimes \psi$ , mapping  $M \otimes_R N$  into  $M' \otimes_R N'$  such that  $(\phi \otimes \psi)(m \otimes n) = \phi(m) \otimes \psi(n)$  for all  $m \in M$  and  $n \in N$ .*
- (2) *If  $M$  and  $M'$  are also  $(S, R)$ -bimodules for some ring  $S$ , and  $\phi$  is also an  $S$ -module homomorphism, then  $\phi \otimes \psi$  is a homomorphism of left  $S$ -modules. In particular, if  $R$  is commutative, then  $\phi \otimes \psi$  is always an  $R$ -module homomorphism for the standard  $R$ -module structures.*
- (3) *If  $\lambda : M' \rightarrow M''$  and  $\mu : N' \rightarrow N''$  are  $R$ -module homomorphisms, then  $(\lambda \otimes \mu) \circ (\phi \otimes \psi) = (\lambda \circ \phi) \otimes (\mu \circ \psi)$ .*

*Proof.* The map  $(m, n) \mapsto \phi(m) \otimes \psi(n)$  from  $M \times N$  to  $M' \otimes_R N'$  is clearly  $R$ -balanced, so (1) is immediate. Now, the definition of the left action of  $S$  on  $M$ , along with the assumption that  $\phi$  is an  $S$ -module homomorphism imply that

$$(\phi \otimes \psi)(s(m \otimes n)) = (\phi \otimes \psi)(sm \otimes n) = \phi(sm) \otimes \psi(n) = s\phi(m) \otimes \psi(n).$$

Since  $\phi \otimes \psi$  is additive, this extends to show that  $\phi \otimes \psi$  is an  $S$ -module homomorphism, giving (2). Finally, (3) follows from uniqueness. ■

**Theorem 3.1.68.** *Suppose that  $M$  is a right  $R$ -module,  $N$  is an  $(R, T)$ -bimodule, and  $L$  is a left  $T$ -module. then, there is a unique isomorphism*

$$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$$

*of abelian groups such that  $(m \otimes n) \otimes \ell \mapsto m \otimes (n \otimes \ell)$ . If  $M$  is an  $(S, R)$ -bimodule, then this is an isomorphism of  $S$ -modules.*

*Proof.* Note that the  $(R, T)$ -bimodule structure on  $N$  makes  $M \otimes_R N$  into a right  $T$ -module and  $N \otimes_T L$  into a left  $R$ -module, so both sides are well-defined. Fix  $\ell \in L$ . Then, the map  $(m, n) \mapsto m \otimes (n \otimes \ell)$  is  $R$ -balanced, so there is a homomorphism  $M \otimes_R N \rightarrow M \otimes_R (N \otimes_T L)$  with  $m \otimes n \mapsto m \otimes (n \otimes \ell)$ . Thus, the map from  $(M \otimes_R N) \times L$  to  $M \otimes_R (N \otimes_T L)$  given by  $(m \otimes n, \ell) \mapsto m \otimes (n \otimes \ell)$  is well defined, and is  $T$ -balanced. Thus, there is a homomorphism from  $(M \otimes_R N) \otimes_T L$  to  $M \otimes_R (N \otimes_T L)$  such that  $(m \otimes n) \otimes \ell \mapsto m \otimes (n \otimes \ell)$ . In a similar

way, we can construct a homomorphism in the opposite direction that is an inverse, so this is an isomorphism.

Suppose that  $M$  is an  $(S, R)$ -bimodule. Then, for  $s \in S$  and  $t \in T$ , we have

$$s((m \otimes n)t) = s(m \otimes nt) = sm \otimes nt = (sm \otimes n)t = (s(m \otimes n))t,$$

so that  $M \otimes_R N$  is an  $(S, T)$ -bimodule. Hence,  $(M \otimes_R N) \otimes_T L$  is a left  $S$ -module. Since  $N \otimes_T L$  is a left  $R$ -module, we also have that  $M \otimes_R (N \otimes_T L)$  is a left  $S$ -module. The group isomorphism established is clearly a homomorphism of left  $S$ -modules. ■

**Corollary 3.1.69.** *Suppose  $R$  is commutative and  $M$ ,  $N$  and  $L$  are  $R$ -modules with the standard  $R$ -module structure. Then,*

$$(M \otimes N) \otimes L \cong M \otimes (N \otimes L).$$

**Definition 3.1.70.** Let  $R$  be a commutative ring with 1, and let  $M_1, \dots, M_n$  and  $L$  be  $R$ -modules with the standard  $R$ -module structures. A map  $\phi : M_1 \times \dots \times M_n \rightarrow L$  is called  $n$ -multilinear over  $R$  (or simply multilinear if  $n$  and  $R$  are clear) if it is an  $R$ -module homomorphism in each component when the other component entries are kept constant. When  $n = 2$ , we say  $\phi$  is bilinear, and when  $n = 3$ , we say that  $\phi$  is trilinear.

**Notation.** Let  $R$  be a commutative ring, and let  $M_1, \dots, M_n$  be given the standard  $R$ -module structure. We write  $M_1 \otimes M_2 \otimes \dots \otimes M_n$  for

$$M_1 \otimes (M_2 \otimes (\dots (M_{n-1} \otimes M_n) \dots)).$$

*Remark 3.1.71.* The order of bracketing in the definition above does not matter, as we obtain isomorphic  $R$ -modules.

**Corollary 3.1.72.** *Let  $R$  be a commutative ring and let  $M_1, \dots, M_n, L$  be  $R$ -modules. Let  $\iota : M_1 \times \dots \times M_n \rightarrow M_1 \otimes \dots \otimes M_n$  be the map defined by  $(m_1, \dots, m_n) \mapsto m_1 \otimes \dots \otimes m_n$ .*

- (1) *For every  $R$ -module homomorphism  $\Phi : M_1 \otimes \dots \otimes M_n \rightarrow L$ , the map  $\phi = \Phi \circ \iota$  is  $n$ -multilinear from  $M_1 \times \dots \times M_n$  to  $L$ .*
- (2) *If  $\phi : M_1 \times \dots \times M_n \rightarrow L$  is an  $n$ -multilinear map, then there is a unique  $R$ -module homomorphism  $\Phi : M_1 \otimes \dots \otimes M_n \rightarrow L$  such that  $\phi = \Phi \circ \iota$ .*

**Theorem 3.1.73.** *Let  $M$  and  $M'$  be right  $R$ -modules, and let  $N$  and  $N'$  be left  $R$ -modules. Then there are unique group isomorphisms*

$$\begin{aligned} (M \oplus M') \otimes_R N &\cong (M \otimes_R N) \oplus (M' \otimes_R N) \\ M \otimes_R (N \oplus N') &\cong (M \otimes_R N) \oplus (M \otimes_R N') \end{aligned}$$

*such that  $(m, m') \otimes n \mapsto (m \otimes n, m' \otimes n)$  and  $m \otimes (n, n') \mapsto (m \otimes n, m \otimes n')$  respectively. If  $M$  and  $M'$  are also  $(S, R)$ -bimodules, then these are isomorphisms of left  $S$ -modules.*

*Proof.* The map  $(M \oplus M') \times N$  to  $(M \otimes_R N) \oplus (M' \otimes_R N)$  defined by  $((m, m'), n) \mapsto (m \otimes n, m' \otimes n)$  is well defined and  $R$ -balanced, so we have a homomorphism  $f$  from  $(M \oplus M') \otimes_R N$  to  $(M \otimes_R N) \oplus (M' \otimes_R N)$  with  $f((m, m') \otimes n) = (m \otimes n, m' \otimes n)$ . Now, the  $R$ -balanced maps from  $M \times N$  to  $(M \oplus M') \otimes_R N$  and  $M' \times N$  to  $(M \oplus M') \otimes_R N$  given by  $(m, n) \mapsto (m, 0) \otimes n$  and  $(m', n) \mapsto (0, m') \otimes n$  respectively, give a homomorphism  $g$  from  $(M \otimes_R N) \oplus (M' \otimes_R N)$  to  $(M \oplus M') \otimes_R N$  with

$$g((m \otimes n_1, m' \otimes n_2)) = (m, 0) \otimes n_1 + (0, m') \otimes n_2.$$

Then,  $f$  and  $g$  are inverses, and are  $S$ -module isomorphisms when  $M$  and  $M'$  are  $(S, R)$ -bimodules. ■

**Corollary 3.1.74.** *Let  $R$  and  $S$  be rings, and let  $f : R \rightarrow S$  be a ring homomorphism with  $f(1_R) = 1_S$ . Give  $S$  the structure of an  $(S, R)$ -bimodule by  $s \cdot r = sf(r)$ . Then,  $S \otimes_R R^n \cong S^n$  as left  $S$ -modules.*

*Proof.* By the theorem, it suffices to show that  $S \otimes_R R \cong S$ . The map  $\phi : S \times R \rightarrow S$  defined by  $(s, r) \mapsto sf(r)$  is  $R$ -balanced. Thus, there is a group homomorphism  $\Phi : S \otimes_R R \rightarrow S$  with  $\Phi(s \otimes r) = sf(r)$ . Since  $\Phi(s'(s \otimes r)) = s'sf(r) = s'\Phi(s \otimes r)$ ,  $\Phi$  is also an  $S$ -module homomorphism. Let  $\Phi' : S \rightarrow S \otimes_R R$  be given by  $s \mapsto s \otimes 1$ . Then,  $(\Phi \circ \Phi')(s) = \Phi(s \otimes 1) = s$  and  $(\Phi' \circ \Phi)(s \otimes r) = \Phi'(sf(r)) = sf(r) \otimes 1 = s \otimes r$ , so  $\Phi$  is an isomorphism. ■

**Definition 3.1.75.** Let  $f : R \rightarrow S$  be a ring homomorphism with  $f(1_R) = 1_S$ , and give  $S$  the structure of a right  $R$ -module by  $s \cdot r = sf(r)$ . For any left  $R$ -module  $N$ , the left  $S$ -module  $S \otimes_R N$  is said to be the left  $S$ -module obtained by changing the base from  $R$  to  $S$ .

**Corollary 3.1.76.** *Let  $R$  be a commutative ring. Then,  $R^s \otimes_R R^t \cong R^{st}$ .*

*Proof.* It suffices to prove that, if  $N$  is a left  $R$ -module,  $R \otimes_R N \cong N$ . The map from  $R \times N$  to  $N$  given by  $(r, n) \mapsto rn$  is  $R$ -balanced, so there is a group homomorphism from  $R \otimes_R N$  to  $N$  with  $r \otimes n \mapsto rn$ . This is clearly a homomorphism of  $R$ -modules. The map  $n \mapsto 1 \otimes n$  provides an inverse. ■

**Proposition 3.1.77.** *Suppose that  $R$  is a commutative ring, and  $M$  and  $N$  are  $R$ -modules with the standard  $R$ -module structure. Then there is a unique  $R$ -module isomorphism  $M \otimes_R N \cong N \otimes_R M$  mapping  $m \otimes n$  to  $n \otimes m$ .*

*Proof.* The map  $M \times N \rightarrow N \otimes_R M$  defined by  $(m, n) \mapsto n \otimes m$  is  $R$ -balanced. Thus, there is a unique homomorphism  $f$  from  $M \otimes_R N$  to  $N \otimes_R M$  with  $f(m \otimes n) = n \otimes m$ . Similarly, we have a unique homomorphism  $g$  from  $N \otimes_R M$  to  $M \otimes_R N$  with  $g(n \otimes m) = m \otimes n$ . This gives the inverse of  $f$ , and both maps are clearly  $R$ -module isomorphisms. ■

**Proposition 3.1.78.** *Let  $R$  be a commutative ring and let  $A$  and  $B$  be  $R$ -algebras. Then the multiplication  $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$  is well defined and makes  $A \otimes_R B$  into an  $R$ -algebra.*

*Proof.* Notice that

$$r(a \otimes b) = ra \otimes b = ar \otimes b = a \otimes rb = a \otimes br = (a \otimes b)r.$$

Thus, if the multiplication is well defined, the homomorphism  $r \mapsto r(1 \otimes 1)$  makes  $A \otimes_R B$  an  $R$ -algebra. Now, the map  $\phi : A \times B \times A \times B \rightarrow A \otimes_R B$  given by  $f(a, b, a', b') = aa' \otimes bb'$  is multilinear over  $R$ . Thus, there is a corresponding  $R$ -module homomorphism  $\Phi$  from  $A \otimes B \otimes A \otimes B$  to  $A \otimes B$  with  $\Phi(a \otimes b \otimes a' \otimes b') = aa' \otimes bb'$ . Writing  $A \otimes B \otimes A \otimes B = (A \otimes B) \otimes (A \otimes B)$ , we get a well defined  $R$ -bilinear map  $\phi' : (A \otimes B) \times (A \otimes B) \rightarrow A \otimes B$  given by  $\phi'(a \otimes b, a' \otimes b') = aa' \otimes bb'$ . Thus, the multiplication is well defined. ■

## 3.2 Vector Spaces

### 3.2.1 Definitions and Basic Theory

**Notation.** When  $R$  is a field, we use the following terminology.

| $R$ is a Ring                                           | $R$ is a field                                      |
|---------------------------------------------------------|-----------------------------------------------------|
| $M$ is an $R$ -module                                   | $M$ is a vector space over $R$                      |
| $m$ is an element of $M$                                | $m$ is a vector in $M$                              |
| $\alpha$ is a ring element                              | $\alpha$ is a scalar                                |
| $N$ is a submodule of $M$                               | $N$ is a subspace of $M$                            |
| $M/N$ is a quotient module                              | $M/N$ is a quotient space                           |
| $M$ is a free module of rank $n$                        | $M$ is a vector space of dimension $n$              |
| $M$ is a finitely generated module                      | $M$ is a finite dimensional vector space            |
| $M$ is a nonzero cyclic module                          | $M$ is a 1-dimensional vector space                 |
| $\phi : M \rightarrow N$ is an $R$ -module homomorphism | $\phi : M \rightarrow N$ is a linear transformation |
| $M$ and $N$ are isomorphic as $R$ -modules              | $M$ and $N$ are isomorphic as vector spaces         |
| the subset $A$ of $M$ generates $M$                     | the subset $A$ of $M$ spans $M$                     |
| $M = RA$                                                | $M = \text{Span}(A)$                                |

*Remark 3.2.1.* For the rest of this section,  $F$  is a field, and  $V$  is a vector space over  $F$ .

**Definition 3.2.2.** A subset  $S$  of  $V$  is called a set of linearly independent vectors if, for any  $\alpha_1, \dots, \alpha_n \in F$  and  $v_1, \dots, v_n \in S$ ,  $\alpha_1 v_1 + \dots + \alpha_n v_n = 0$  if and only if  $\alpha_1 = \dots = \alpha_n = 0$ .

**Definition 3.2.3.** A basis of a vector space  $V$  is an ordered set of linearly independent vectors which span  $V$ . Two bases are considered different even if one is a rearrangement of the other.

**Proposition 3.2.4.** Suppose the set  $\mathcal{A} = \{v_1, v_2, \dots, v_n\}$  spans  $V$ , but no proper subset of  $\mathcal{A}$  spans  $V$ . Then  $\mathcal{A}$  is a basis of  $V$ . In particular, any finitely generated (i.e. finitely spanned) vector space over  $F$  is a free  $F$ -module.

*Proof.* We only need to show that  $v_1, v_2, \dots, v_n$  are linearly independent. Suppose that  $\alpha_1 v_1 + \dots + \alpha_n v_n = 0$ , where not all of the  $\alpha_i$  are 0. Without loss of generality, suppose that  $\alpha_1 \neq 0$ . Then,

$$v_1 = -\frac{1}{\alpha_1}(\alpha_2 v_2 + \dots + \alpha_n v_n).$$

Thus,  $\{v_2, \dots, v_n\}$  also spans  $V$ , since any linear combination of  $v_1, \dots, v_n$  can be written as a linear combination of  $v_2, \dots, v_n$  using the equation above. This is a contradiction. ■

**Corollary 3.2.5.** *Suppose that the finite set  $\mathcal{A}$  spans the vector space  $V$ . Then  $\mathcal{A}$  contains a basis of  $V$*

*Proof.* Any subset  $\mathcal{B}$  of  $\mathcal{A}$  spanning  $V$  such that no proper subset of  $\mathcal{B}$  also spans  $V$  (there clearly must be such subsets) is a basis for  $V$ . ■

**Theorem 3.2.6** (Replacement Theorem). *Suppose that  $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$  is a basis for  $V$  containing  $n$  elements and  $\{b_1, \dots, b_m\}$  is a set of linearly independent vectors in  $V$ . Then there is an ordering  $a_1, a_2, \dots, a_n$  such that, for each  $1 \leq k \leq m$ , the set  $\{b_1, \dots, b_k, a_{k+1}, \dots, a_n\}$  is a basis of  $V$ . In particular,  $n \geq m$ .*

*Proof.* We show this by induction on  $k$ , with  $k = 0$  being trivial. Suppose that  $\{b_1, \dots, b_k, a_{k+1}, \dots, a_n\}$  is a basis for  $V$ . Then, we can write

$$b_{k+1} = \beta_1 b_1 + \dots + \beta_k b_k + \alpha_{k+1} a_{k+1} + \dots + \alpha_n a_n.$$

There must be some nonzero  $\alpha_i$ , so reorder to take  $\alpha_{k+1} \neq 0$ . Solving the equation for  $a_{k+1}$  in terms of  $b_1, \dots, b_{k+1}, a_{k+2}, \dots, a_n$  shows that

$$\text{Span}\{b_1, \dots, b_{k+1}, a_{k+2}, \dots, a_n\} = \text{Span}\{b_1, \dots, b_k, a_{k+1}, \dots, a_n\} = V.$$

It remains to show that  $b_1, \dots, b_{k+1}, a_{k+2}, \dots, a_n$  are linearly independent. Suppose

$$\gamma_1 b_1 + \dots + \gamma_{k+1} b_{k+1} + \gamma_{k+2} a_{k+2} + \dots + \gamma_n a_n = 0.$$

Writing  $b_{k+1}$  as a linear combination of  $b_1, \dots, b_k, a_{k+1}, \dots, a_n$ , we get a linear dependence relation where the coefficient of  $a_{k+1}$  is  $\gamma_{k+1} \alpha_{k+1}$ . Thus,  $\gamma_{k+1} \alpha_{k+1} = 0$ , so  $\gamma_{k+1} = 0$ . Now, we have

$$\gamma_1 b_1 + \dots + \gamma_k b_k + \gamma_{k+2} a_{k+2} + \dots + \gamma_n a_n = 0,$$

so  $\gamma_i = 0$  for all  $i$ . Thus,  $\{b_1, \dots, b_{k+1}, a_{k+2}, \dots, a_n\}$  is a basis for  $V$ . ■

**Corollary 3.2.7.**

- (1) *Suppose  $V$  has a finite basis with  $n$  elements. Any set of linearly independent vectors has at most  $n$  elements, and any spanning set has at least  $n$  elements.*
- (2) *If  $V$  has some finite basis, then any two bases of  $V$  have the same cardinality.*

*Proof.*

- (1) All linearly independent sets have at most  $n$  elements by the previous theorem. By (2), all bases have the same cardinality. As all finite spanning sets contain a basis, they have size at least  $n$ .
- (2) If two bases have sizes  $m$  and  $n$ , then  $m \leq n$  and  $n \leq m$ , as they are both linearly independent.

■

**Definition 3.2.8.** If  $V$  is a finitely generated  $F$ -module (i.e. has a finite basis), the cardinality of any basis is called the dimension of  $V$ , denoted  $\dim_F V$ , or just  $\dim V$  when  $F$  is clear. We say  $V$  is finite dimensional over  $F$ . If  $V$  is not finitely generated,  $V$  is said to be infinite dimensional, and we write  $\dim V = \infty$ .

**Corollary 3.2.9** (Building-Up Lemma). *If  $A$  is a set of linearly independent vectors in the finite dimensional vector space  $V$ , then there exists a basis of  $V$  containing  $A$ .*

*Proof.* Use the elements of  $A$  to successively replace the elements of any given basis for  $V$ . ■

**Theorem 3.2.10.** *If  $V$  is an  $n$  dimensional vector space over  $F$ , then  $V \cong F^n$ . In particular, any two finite dimensional vector spaces over  $F$  of the same dimension are isomorphic.*

*Proof.* Let  $v_1, \dots, v_n$  be a basis for  $V$ . Define the map  $\phi : F^n \rightarrow V$  by  $\phi(\alpha_1, \dots, \alpha_n) = \alpha_1 v_1 + \dots + \alpha_n v_n$ . Then,  $\phi$  is clearly  $F$ -linear and surjective. Since the  $v_i$  are linearly independent, it is injective, and so an isomorphism. ■

**Theorem 3.2.11.** *Let  $V$  be a vector space over  $F$  and let  $W$  be a subspace of  $V$ . Then  $V/W$  is a vector space with  $\dim V = \dim W + \dim V/W$  (if one side is infinite, the other is as well).*

*Proof.* Suppose that  $W$  has dimension  $m$  and  $V$  has dimension  $n$ . Let  $w_1, \dots, w_m$  be a basis for  $W$ . Extend this to a basis  $w_1, \dots, w_m, v_{m+1}, \dots, v_n$  of  $V$ . The natural projection from  $V$  to  $V/W$  maps each  $w_i$  to 0. No linear combination of the  $v_i$  is mapped to 0, as it would then be in  $W$ , and so expressible as a linear combination of the  $w_i$ . Hence, the image  $V/W$  of this map is isomorphic to the subspace of  $V$  spanned by the  $v_i$ , and so  $\dim V/W = n - m$ . Now, if either side is infinite, it is easy to find infinitely many linearly independent vectors showing the other side is also infinite. ■

**Corollary 3.2.12.** *Let  $\phi : V \rightarrow U$  be a linear transformation. Then  $\ker \phi$  is a subspace of  $V$ ,  $\text{im } \phi$  is a subspace of  $U$ , and  $\dim V = \dim \ker \phi + \dim \text{im } \phi$ .*

**Corollary 3.2.13.** *Let  $\phi : V \rightarrow W$  be a linear transformation of vector spaces of the same finite dimension. Then the following are equivalent.*

- (1)  $\phi$  is an isomorphism.
- (2)  $\phi$  is injective.
- (3)  $\phi$  is surjective.
- (4)  $\phi$  sends a basis of  $V$  to a basis of  $W$ .

*Proof.* Count dimensions using the previous corollary. ■

**Definition 3.2.14.** If  $\phi : V \rightarrow U$  is a linear transformation,  $\ker \phi$  is sometimes called the null space of  $\phi$ , and the dimension of  $\ker \phi$  is called the nullity of  $\phi$ . The dimension of  $\text{im } \phi$  is called the rank of  $\phi$ . If  $\ker \phi = 0$ , the transformation is said to be nonsingular.

### 3.2.2 The Matrix of a Linear Transformation

**Definition 3.2.15.** Let  $V$  and  $W$  be vector spaces over  $F$ ,  $\mathcal{B} = \{v_1, \dots, v_n\}$  be an (ordered) basis of  $V$ ,  $\mathcal{E} = \{w_1, \dots, w_m\}$  be an (ordered) basis of  $W$ , and let  $\phi \in \text{Hom}(V, W)$ . For each  $1 \leq j \leq n$ , write

$$\phi(v_j) = \sum_{i=1}^m \alpha_{i,j} w_i.$$

The matrix  $M_{\mathcal{B}}^{\mathcal{E}}(\phi)$  whose  $i, j$  entry is given by  $\alpha_{i,j}$  is called the matrix of  $\phi$  with respect to the bases  $\mathcal{B}$  and  $\mathcal{E}$ .

*Remark 3.2.16.* Given  $M_{\mathcal{B}}^{\mathcal{E}}(\phi)$ , we can reconstruct  $\phi$ . To compute  $\phi(v)$ , write  $v = \sum_{i=1}^n \alpha_i v_i$ . Let

$$M_{\mathcal{B}}^{\mathcal{E}}(\phi) \times \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}.$$

Then,  $\phi(v) = \sum_{i=1}^m \beta_i w_i$ .

**Definition 3.2.17.** The  $m \times n$  matrix  $A = (a_{i,j})$  associated to the linear transformation  $\phi$  is said to represent the linear transformation  $\phi$  with respect to the bases  $\mathcal{B}$  and  $\mathcal{E}$ . We say,  $\phi$  is the linear transformation represented by  $A$  with respect to the bases  $\mathcal{B}$  and  $\mathcal{E}$ .

**Theorem 3.2.18.** Let  $V$  be a vector space over  $F$  of dimension  $n$ , and let  $W$  be a vector space over  $F$  of dimension  $m$ , with bases  $\mathcal{B}$  and  $\mathcal{E}$  respectively. Then the map  $\text{Hom}_F(V, W) \rightarrow M_{m \times n}(F)$  given by  $\phi \mapsto M_{\mathcal{B}}^{\mathcal{E}}(\phi)$  is a vector space isomorphism.

*Proof.* The columns of  $M_{\mathcal{B}}^{\mathcal{E}}(\phi)$  are determined by the action of  $\phi$  on  $\mathcal{B}$ . Thus, the map  $\phi \mapsto M_{\mathcal{B}}^{\mathcal{E}}(\phi)$  is an  $F$ -linear map. It is surjective, as each matrix defines the map  $\phi$  on a basis, and is injective, since two linear transformations that agree on a basis are the same. ■

**Corollary 3.2.19.** The dimension of  $\text{Hom}_F(V, W)$  is  $(\dim V)(\dim W)$ .

*Proof.* The dimension of  $M_{m \times n}(F)$  is  $mn$ . ■

**Definition 3.2.20.** An  $m \times n$  matrix  $A$  is called nonsingular if  $Ax = 0$  with  $x \in F^n$  implies  $x = 0$ .

**Theorem 3.2.21.** Let  $U, V$  and  $W$  be finite dimensional vector spaces, with ordered bases  $\mathcal{D}, \mathcal{B}$ , and  $\mathcal{E}$  respectively. Let  $\psi : U \rightarrow V$  and  $\phi : V \rightarrow W$  be linear. Then,  $M_{\mathcal{D}}^{\mathcal{E}}(\phi \circ \psi) = M_{\mathcal{B}}^{\mathcal{E}}(\phi)M_{\mathcal{D}}^{\mathcal{B}}(\psi)$ .

*Proof.* Write  $\mathcal{D} = \{u_1, \dots, u_k\}$ ,  $\mathcal{B} = \{v_1, \dots, v_n\}$ , and  $\mathcal{E} = \{w_1, \dots, w_m\}$ . Let

$$(\phi \circ \psi)(u_j) = \sum_{i=1}^m \gamma_{i,j} w_i.$$

We compute

$$\psi(u_j) = \sum_{p=1}^n \alpha_{p,j} v_p \quad \phi(v_p) = \sum_{i=1}^m \beta_{i,p} w_i.$$



Now,

$$(\phi \circ \psi)(u_j) = \phi \left( \sum_{p=1}^n \alpha_{p,j} v_p \right) = \sum_{p=1}^n \alpha_{p,j} \phi(v_p) = \sum_{p=1}^n \alpha_{p,j} \sum_{i=1}^m \beta_{i,p} w_i = \sum_{p=1}^n \sum_{i=1}^m \alpha_{p,j} \beta_{i,p} w_i.$$

Interchanging the sums,

$$\gamma_{i,j} = \sum_{p=1}^n \beta_{i,p} \alpha_{p,j}.$$

■

**Corollary 3.2.22.** *Matrix multiplication is associative and distributive. An  $n \times n$  matrix  $A$  is nonsingular if and only if it is invertible.*

*Proof.* Let  $A$ ,  $B$  and  $C$  be matrices such that the products  $(AB)C$  and  $A(BC)$  are defined, and let  $S$ ,  $T$  and  $R$  denote the associated linear transformations. Then, the linear transformations corresponding to the products are  $(S \circ T) \circ R$  and  $S \circ (T \circ R)$  respectively. These are the same linear transformation, so  $(AB)C = A(BC)$ . Distributivity is proved similarly.

If  $A$  is invertible, then  $Ax = 0$  implies  $x = A^{-1}Ax = A^{-1}0 = 0$ , so  $A$  is nonsingular. Conversely, if  $A$  is nonsingular, fix bases  $\mathcal{B}$  and  $\mathcal{E}$  for  $V$  and let  $\phi$  be the linear transformation from  $V$  to itself represented by  $A$  with respect to these bases.  $\phi$  is nonsingular, and so an isomorphism. Let  $B$  be the matrix representing  $\phi^{-1}$  with respect to the matrices  $\mathcal{E}$  and  $\mathcal{B}$  (note that the order has changed). Then,  $AB = M_{\mathcal{B}}^{\mathcal{E}}(\phi)M_{\mathcal{E}}^{\mathcal{B}}(\phi^{-1}) = M_{\mathcal{E}}^{\mathcal{E}}(\phi \circ \phi^{-1}) = M_{\mathcal{E}}^{\mathcal{E}}(1) = I$ . Similarly,  $BA = I$ , so  $B$  is the inverse of  $A$ . ■

**Corollary 3.2.23.**

- (1) *If  $\mathcal{B}$  is a basis of the  $n$ -dimensional space  $V$ , the map  $\phi \mapsto M_{\mathcal{B}}^{\mathcal{B}}(\phi)$  is a ring and vector space isomorphism of  $\text{Hom}_F(V, V)$  onto the space  $M_n(F)$  of  $n \times n$  matrices with coefficients in  $F$ .*
- (2)  $\text{GL}(V) \cong \text{GL}_n(F)$ , where  $\dim V = n$ .

*Proof.*

- (1) We have already seen that this is an isomorphism of vector spaces over  $F$ . We have shown that  $M_n(F)$  is a ring and that multiplication is preserved under this map in the last two results.
- (2) A ring isomorphism sends units to units. ■

**Definition 3.2.24.** If  $A$  is any  $m \times n$  matrix with entries from  $F$ , the row rank (column rank) of  $A$  is the maximal number of linearly independent rows (columns) of  $A$ .

**Definition 3.2.25.** Two  $n \times n$  matrices  $A$  and  $B$  are said to be similar if there is an invertible (i.e. nonsingular)  $n \times n$  matrix  $P$  such that  $P^{-1}AP = B$ . Two linear transformations  $\phi$  and  $\psi$  from a vector space  $V$  to itself are said to be similar if there is a nonsingular linear transformation  $\xi : V \rightarrow V$  with  $\xi^{-1}\phi\xi = \psi$ .

**Proposition 3.2.26.** *Suppose  $A$  and  $B$  are  $n \times n$  matrices. Then  $A$  and  $B$  are similar if and only if they represent the same linear transformation using appropriate bases of the  $n$ -dimensional vector space  $V$ .*

*Proof.* Let  $\mathcal{B}$  and  $\mathcal{E}$  be bases of  $V$ , and let  $\phi \in \text{Hom}_F(V, V)$ . Let  $I$  denote the identity map from  $V$  to itself, and let  $P = M_{\mathcal{E}}^{\mathcal{B}}(I)$ . If  $\mathcal{B} \neq \mathcal{E}$ , then  $P$  is not the identity matrix. In this case,  $P^{-1}M_{\mathcal{B}}^{\mathcal{B}}(\phi)P = M_{\mathcal{E}}^{\mathcal{E}}(\phi)$ .

Now suppose that  $P^{-1}AP = B$ , and let  $\mathcal{B}$  be a basis for  $V$ . Define  $\phi : V \rightarrow V$  so that  $A = M_{\mathcal{B}}^{\mathcal{B}}(\phi)$ . Let  $\mathcal{E}$  be a basis of  $V$  such that the  $i$ th column of  $P$  expresses  $w_i \in \mathcal{E}$  in terms of  $\mathcal{B}$ . By definition,  $P = M_{\mathcal{E}}^{\mathcal{B}}(I)$ . Now,  $B = P^{-1}AP = P^{-1}M_{\mathcal{B}}^{\mathcal{B}}(\phi)P = M_{\mathcal{E}}^{\mathcal{E}}(\phi)$ . ■

*Remark 3.2.27.* The following two propositions are restatements of previous results in the context of vector spaces.

**Proposition 3.2.28.** *Let  $F$  be a subfield of the field  $K$ . If  $W$  is an  $m$ -dimensional vector space over  $F$  with basis  $w_1, \dots, w_m$ , then  $K \otimes_F W$  is an  $m$ -dimensional vector space over  $K$  with basis  $1 \otimes w_1, \dots, 1 \otimes w_m$ .*

**Proposition 3.2.29.** *Let  $V$  and  $W$  be finite dimensional vector spaces over the field  $F$  with bases  $v_1, \dots, v_n$  and  $w_1, \dots, w_m$  respectively. Then  $V \otimes_F W$  is a vector space over  $F$  of dimension  $nm$  with basis  $v_i \otimes w_j$  for  $1 \leq i \leq n$  and  $1 \leq j \leq m$ .*

**Definition 3.2.30.** Let  $A = (\alpha_{i,j})$  and  $B$  be  $r \times n$  and  $s \times m$  matrices, respectively, with coefficients from any commutative ring. The Kronecker product or tensor product of  $A$  and  $B$ , denoted  $A \otimes B$ , is the  $rs \times nm$  matrix consisting of an  $r \times n$  block matrix whose  $i, j$  block is the  $s \times m$  matrix  $\alpha_{i,j}B$ .

**Proposition 3.2.31.** *Let  $\phi : V \rightarrow X$  and  $\psi : W \rightarrow Y$  be linear transformations of finite dimensional vector spaces. Then the Kronecker product of the matrices representing  $\phi$  and  $\psi$  is a matrix representation of  $\phi \otimes \psi$ .*

*Proof.* Let  $\mathcal{B}_1 = \{v_1, \dots, v_n\}$  and  $\mathcal{B}_2 = \{w_1, \dots, w_m\}$  be (ordered) bases of  $V$  and  $W$  respectively, and let  $\mathcal{E}_1 = \{x_1, \dots, x_r\}$  and  $\mathcal{E}_2 = \{y_1, \dots, y_s\}$  be (ordered) bases of  $X$  and  $Y$  respectively. Let  $\mathcal{B} = \{v_i \otimes w_j\}$  and  $\mathcal{E} = \{x_i \otimes y_j\}$  be the bases of  $V \otimes W$  and  $X \otimes Y$ . Suppose

$$\phi(v_i) = \sum_{p=1}^r \alpha_{p,i} x_p \quad \psi(w_j) = \sum_{q=1}^s \beta_{q,j} y_q.$$

Then,

$$(\phi \otimes \psi)(v_i \otimes w_j) = \phi(v_i) \otimes \psi(w_j) = \left( \sum_{p=1}^r \alpha_{p,i} x_p \right) \otimes \left( \sum_{q=1}^s \beta_{q,j} y_q \right) = \sum_{p=1}^r \sum_{q=1}^s \alpha_{p,i} \beta_{q,j} (x_p \otimes y_q).$$

We order the basis  $\mathcal{E}$  into  $r$  ordered sets, with the  $p$ th list being  $x_p \otimes y_1, x_p \otimes y_2, \dots, x_p \otimes y_s$ , and similarly order  $\mathcal{B}$ . The equation above determines the columns for the corresponding matrix of  $\phi \otimes \psi$ . This ordering gives the Kronecker product of the matrices for  $\phi$  and  $\psi$ . ■

### 3.2.3 Dual Vector Spaces

#### Definition 3.2.32.

- (1) For any vector space  $V$ , let  $V^* = \text{Hom}_F(V, F)$ . Then  $V^*$  is called the dual space of  $V$ , and elements of  $V^*$  are called linear functionals.
- (2) If  $\mathcal{B} = \{v_1, \dots, v_n\}$  is a basis for the finite dimensional vector space  $V$ , define  $v_i^* \in V^*$  by

$$v_i^*(v_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

**Proposition 3.2.33.** *Let  $\{v_1, \dots, v_n\}$  be a basis for the finite dimensional vector space  $V$ . Then  $\{v_1^*, \dots, v_n^*\}$  is a basis for  $V^*$ .*

*Proof.* Since  $V$  is finite dimensional,  $\dim V^* = \dim \text{Hom}_F(V, F) = \dim V = n$ . Thus, it suffices to show that the  $v_i^*$ 's are linearly independent. If

$$\alpha_1 v_1^* + \alpha_2 v_2^* + \dots + \alpha_n v_n^* = 0,$$

then applying this to  $v_i$ , we get that  $\alpha_i = 0$ . ■

**Definition 3.2.34.** The basis  $\{v_1^*, v_2^*, \dots, v_n^*\}$  of  $V^*$  is called the dual basis to  $\{v_1, v_2, \dots, v_n\}$ .

*Example 3.2.35.* In the infinite dimensional case, this does not form a basis. To see this, pick a basis  $\mathcal{B}$  for  $V$ , and let  $\phi$  be 1 on each of the basis vectors. As each element is the sum of finitely many basis vectors, this is well defined. However, this cannot be written as the sum of finitely many elements of the form  $v_i^*$ . Suppose that  $\phi = \alpha_1 v_1 + \dots + \alpha_n v_n$ , and let  $v \in \mathcal{B} \setminus \{v_1, \dots, v_n\}$ . Applying both sides to  $v$ , we get  $1 = 0$ .

**Definition 3.2.36.** The dual of  $V^*$ , denoted  $V^{**}$ , is called the double dual or second dual of  $V$ .

**Theorem 3.2.37.** *There is a natural injective linear transformation from  $V$  to  $V^{**}$ . If  $V$  is finite dimensional, then this is an isomorphism.*

*Proof.* Let  $v \in V$ , and define

$$E_v : V^* \rightarrow F \quad E_v(f) = f(v).$$

Then,

$$E_v(f + \alpha g) = (f + \alpha g)(v) = f(v) + \alpha g(v) = E_v(f) + \alpha E_v(g),$$

so  $E_v \in V^{**}$ . Define  $\phi : V \rightarrow V^{**}$  by  $v \mapsto E_v$ . Now, for every  $v, w \in V$  and  $\alpha \in F$ ,

$$E_{v+\alpha w}(f) = f(v + \alpha w) = f(v) + \alpha f(w) = E_v(f) + \alpha E_w(f)$$

for all  $f \in V^*$ , so

$$\phi(v + \alpha w) = E_{v+\alpha w} = E_v + \alpha E_w = \phi(v) + \alpha \phi(w).$$

Now, let  $v \in V$  be nonzero, and let  $\mathcal{B}$  be a basis containing  $v$ . Define  $f$  to send  $v$  to 1 and every element of  $\mathcal{B} \setminus \{v\}$  to 0. Then  $f \in V^*$  and  $E_v(f) = f(v) = 1$ . Thus,  $\phi(v) = E_v \neq 0$ , so  $\phi$  is injective.

In the case  $V$  is finite dimensional,  $\dim V^{**} = \dim(V^*)^* = \dim V^* = \dim V$ , so  $\phi$  is an isomorphism. ■

**Theorem 3.2.38.** *Let  $V$  and  $W$  be finite dimensional vector spaces with bases  $\mathcal{B}$  and  $\mathcal{E}$  respectively. Fix  $\phi \in \text{Hom}_F(V, W)$ , and define  $\phi^* : W^* \rightarrow V^*$  by  $f \mapsto f \circ \phi$ . Then  $\phi^*$  is linear, and  $M_{\mathcal{E}^*}^{\mathcal{B}^*}(\phi^*)$  is the transpose of the matrix  $M_{\mathcal{B}}^{\mathcal{E}}(\phi)$ .*

*Proof.* The map  $\phi^*$  is clearly linear. Let

$$\phi(v_j) = \sum_{i=1}^m \alpha_{i,j} w_i.$$

Then,

$$\phi^*(w_k^*)(v_j) = (w_k^* \circ \phi)(v_j) = w_k^* \left( \sum_{i=1}^m \alpha_{i,j} w_i \right) = \alpha_{k,j} = \left( \sum_{i=1}^n \alpha_{k,i} v_i^* \right) (v_j),$$

so these two linear transformations agree on a basis of  $V$ , and so are the same. We have

$$\phi^*(w_k^*) = \sum_{i=1}^n \alpha_{k,i} v_i^*.$$

■

**Corollary 3.2.39.** *For any matrix  $A$ , the row rank of  $A$  equals the column rank of  $A$ .*

*Proof.* Let  $\phi : V \rightarrow W$  be a linear transformation whose matrix with respect to some fixed bases of  $V$  and  $W$  is  $A$ . The matrix of  $\phi^* : W^* \rightarrow V^*$  with respect to the dual bases is the transpose of  $A$ . The column rank of  $A$  is the rank of  $\phi$ , and the row rank of  $A$  is the rank of  $\phi^*$ . Thus, it suffices to show that  $\phi$  and  $\phi^*$  have the same rank. Now,<sup>5</sup>

$$f \in \ker \phi^* \iff \phi^*(f) = 0 \iff \forall v \in V [f \circ \phi(v) = 0] \iff \phi(V) \subseteq \ker f \iff f \in \text{Ann}(\phi(V)),$$

where  $\text{Ann}(S)$  denotes the subspace of linear functionals mapping every element of  $S$  to 0. Thus,  $\text{Ann}(\phi(V)) = \ker \phi^*$ .

We first show that  $\dim \text{Ann}(\phi(V)) = \dim W - \dim \phi(V)$ . Let  $\mathcal{B} = \{v_1, \dots, v_n, w_{n+1}, \dots, w_m\}$  be a basis for  $W$ , where  $\{v_1, \dots, v_n\}$  are a basis for  $\phi(V)$ . We claim that  $w_{n+1}^*, \dots, w_m^*$  form a basis for  $\text{Ann} \phi(V)$ . They are a subset of a basis of  $W^*$ , and so linearly independent. Thus we only need to show they are spanning. The function

$$\sum_{i=n+1}^m f(w_i) w_i^*$$

agrees with  $f$  on a basis for any  $f \in \text{Ann} \phi(V)$ . Thus,  $\dim \text{Ann}(\phi(V)) = \dim W - \dim \phi(V)$ .

We have that  $\dim \ker \phi^* + \dim \text{im} \phi^* = \dim W^*$ , so  $\dim \ker \phi^* = \dim W^* - \dim \phi^*(W^*)$ . Since  $W$  and  $W^*$  have the same dimension,  $\dim \phi(V) = \dim \phi^*(W^*)$ . ■

### 3.2.4 Determinants

*Remark 3.2.40.* In this subsection,  $R$  is a commutative ring with 1, and  $V_1, \dots, V_n, V, W$  are  $R$ -modules.

**Definition 3.2.41.**

- (1) A map  $\phi : V_1 \times \cdots \times V_n \rightarrow W$  is called multilinear if, for each fixed  $i$  and fixed  $v_j \in V_j$  for  $j \neq i$ , the map

$$V_i \rightarrow W \quad x \mapsto \phi(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_n)$$

is an  $R$ -module homomorphism. If  $V_i = V$  for all  $i$ , then  $\phi$  is called an  $n$ -multilinear function on  $V$ , and if in addition  $W = R$ ,  $\phi$  is called an  $n$ -multilinear form on  $V$ .

- (2) An  $n$ -multilinear function  $\phi$  on  $V$  is called alternating if  $\phi(v_1, v_2, \dots, v_n) = 0$  whenever  $v_i = v_{i+1}$  for some  $1 \leq i \leq n-1$ . The function  $\phi$  is called symmetric if interchanging  $v_i$  and  $v_j$  for any  $i$  and  $j$  does not alter the value of  $\phi$ .

**Proposition 3.2.42.** *Let  $\phi$  be an  $n$ -multilinear alternating function on  $V$ .*

- (1)  $\phi(v_1, \dots, v_{i-1}, v_{i+1}, v_i, v_{i+2}, \dots, v_n) = -\phi(v_1, v_2, \dots, v_n)$  for any  $1 \leq i \leq n-1$ .
- (2) For each  $\sigma \in S_n$ ,  $\phi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)}) = \epsilon(\sigma)\phi(v_1, v_2, \dots, v_n)$ .
- (3) If  $v_i = v_j$  for any pair of distinct  $i$  and  $j$ , then  $\phi(v_1, v_2, \dots, v_n) = 0$ .
- (4) If  $v_i$  is replaced by  $v_i + \alpha v_j$  in  $(v_1, \dots, v_n)$  for any  $j \neq i$  and  $\alpha \in R$ , then the value of  $\phi$  is not changed.

*Proof.*

- (1) Let  $\psi(x, y)$  be the function  $\phi$  with entries  $x$  and  $y$  in positions  $i$  and  $i+1$ , with the fixed entry  $v_j$  in position  $j$  for all  $j \neq i, i+1$ . We must show that  $\psi(y, x) = -\psi(x, y)$ . Since  $\phi$  is alternating,  $\psi(x+y, x+y) = 0$ . Thus,

$$0 = \psi(x, x) + \psi(x, y) + \psi(y, x) + \psi(y, y) = \psi(x, y) + \psi(y, x).$$

- (2) Every permutation can be written as a product of transpositions, and every transposition can be written as a product of transpositions which interchange two consecutive integers.
- (3) Choose  $\sigma$  to be any permutation fixing  $i$  and moving  $i+1$  to  $j$ . Then,  $\phi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)}) = 0$ . Now,  $\phi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)}) = \pm \phi(v_1, v_2, \dots, v_n)$ .
- (4) Expand in the  $i$ th position, and use (3).

■

**Proposition 3.2.43.** Suppose that  $\phi$  is an  $n$ -multilinear alternating function on  $V$ , and that for some  $v_1, v_2, \dots, v_n$  and  $w_1, w_2, \dots, w_n \in V$ , and some  $\alpha_{i,j} \in R$ , we have

$$\begin{aligned} w_1 &= \alpha_{1,1}v_1 + \alpha_{2,1}v_2 + \cdots + \alpha_{n,1}v_n \\ w_2 &= \alpha_{1,2}v_1 + \alpha_{2,2}v_2 + \cdots + \alpha_{n,2}v_n \\ &\vdots \\ w_n &= \alpha_{1,n}v_1 + \alpha_{2,n}v_2 + \cdots + \alpha_{n,n}v_n \end{aligned}$$

Then,

$$\phi(w_1, w_2, \dots, w_n) = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1),1} \alpha_{\sigma(2),2} \cdots \alpha_{\sigma(n),n} \phi(v_1, v_2, \dots, v_n).$$

*Proof.* If we expand  $\phi(w_1, w_2, \dots, w_n)$  by multilinearity we obtain a sub of  $n^n$  terms of the form  $\alpha_{i_1,1} \alpha_{i_2,2} \cdots \alpha_{i_n,n} \phi(v_{i_1}, v_{i_2}, \dots, v_{i_n})$ . Since  $\phi$  is zero when two or more of the  $i_j$ 's are equal, we only need to consider the terms where  $i_1, \dots, i_n$  are distinct. We can then write each nonzero term as  $\alpha_{\sigma(1),1} \alpha_{\sigma(2),2} \cdots \alpha_{\sigma(n),n} \phi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)})$ , for some  $\sigma \in S_n$ . Using (2) from the previous proposition gives the desired expression. ■

**Definition 3.2.44.** An  $n \times n$  determinant function on  $R$  is any function

$$\det : M_{n \times n}(R) \rightarrow R$$

that satisfies the following two conditions.

- (1)  $\det$  is an  $n$ -multilinear alternating form on  $R^n$ , where the  $n$ -tuples are the  $n$  columns of the matrices in  $M_{n \times n}(R)$ .
- (2)  $\det(I) = 1$ , where  $I$  is the  $n \times n$  identity matrix.

We may write  $\det(A_1, A_2, \dots, A_n)$  for  $\det A$ , where  $A_1, A_2, \dots, A_n$  are the columns of  $A$ .

**Theorem 3.2.45.** There is a unique  $n \times n$  determinant function on  $R$ , and it can be computed for any  $n \times n$  matrix  $(\alpha_{i,j})$  by the formula

$$\det(\alpha_{i,j}) = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1),1} \alpha_{\sigma(2),2} \cdots \alpha_{\sigma(n),n}.$$

*Proof.* Let  $A_1, A_2, \dots, A_n$  be the column vectors in a general  $n \times n$  matrix  $(\alpha_{i,j})$ . It is simple to see that the function above satisfies the requirements of a determinant function. Now, let  $e_i$  be the column  $n$ -tuple with 1 in the position  $i$  and 0 in all other positions. Then,

$$\begin{aligned} A_1 &= \alpha_{1,1}e_1 + \alpha_{2,1}e_2 + \cdots + \alpha_{n,1}e_n \\ A_2 &= \alpha_{1,2}e_1 + \alpha_{2,2}e_2 + \cdots + \alpha_{n,2}e_n \\ &\vdots \\ A_n &= \alpha_{1,n}e_1 + \alpha_{2,n}e_2 + \cdots + \alpha_{n,n}e_n \end{aligned}$$

Thus,

$$\det A = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1),1} \alpha_{\sigma(2),2} \cdots \alpha_{\sigma(n),n} \det(e_1, e_2, \dots, e_n) = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1),1} \alpha_{\sigma(2),2} \cdots \alpha_{\sigma(n),n}.$$

■

**Corollary 3.2.46.** *The determinant is an  $n$ -multilinear function of the rows of  $M_{n \times n}(R)$ , and for any  $n \times n$  matrix  $A$ ,  $\det A = \det(A^t)$ .*

*Proof.* The first statement follows immediately from the second, so we only prove that. Let  $A = (\alpha_{i,j})$ . Then,

$$\det A^t = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{1,\sigma(1)} \alpha_{2,\sigma(2)} \cdots \alpha_{n,\sigma(n)}.$$

Each number from 1 to  $n$  appears exactly once among  $\sigma(1), \dots, \sigma(n)$ , so we may rearrange the product  $\alpha_{1,\sigma(1)} \alpha_{2,\sigma(2)} \cdots \alpha_{n,\sigma(n)}$  as  $\alpha_{\sigma^{-1}(1),1} \alpha_{\sigma^{-1}(2),2} \cdots \alpha_{\sigma^{-1}(n),n}$ . Notice that  $\epsilon(\sigma) = \epsilon(\sigma^{-1})$ . Thus, the sum for  $\det A^t$  can be rewritten as

$$\sum_{\sigma \in S_n} \epsilon(\sigma^{-1}) \alpha_{\sigma^{-1}(1),1} \alpha_{\sigma^{-1}(2),2} \cdots \alpha_{\sigma^{-1}(n),n}.$$

Since the sum is over all permutations, the index  $\sigma^{-1}$  may be replaced by  $\sigma$ , giving the sum for  $\det A$ . ■

**Theorem 3.2.47** (Cramer's Rule). *If  $A_1, A_2, \dots, A_n$  are the columns of an  $n \times n$  matrix  $A$ , and  $B = \beta_1 A_1 + \beta_2 A_2 + \cdots + \beta_n A_n$  for some  $\beta_1, \dots, \beta_n \in R$ , then*

$$\beta_i \det A = \det(A_1, \dots, A_{i-1}, B, A_{i+1}, \dots, A_n).$$

*Proof.* Expand the right hand side linearly. ■

**Corollary 3.2.48.** *If  $R$  is an integral domain, then  $\det A = 0$  for  $A \in M_n(R)$  if and only if the columns of  $A$  are  $R$ -linearly dependent as elements of the free  $R$ -module of rank  $n$ .*

*Proof.* Suppose that the columns of  $A$  are linearly dependent, and that

$$0 = \beta_1 A_1 + \beta_2 A_2 + \cdots + \beta_n A_n$$

is a nontrivial dependence relation with  $\beta_i \neq 0$ . By Cramer's Rule,  $\beta_i \det A = 0$ . Since  $R$  is an integral domain and  $\beta_i \neq 0$ ,  $\det A = 0$ .

Now suppose that the columns of  $A$  are independent. Consider  $R$  as embedded in its quotient field  $F$ , so that  $M_{n \times n}(R)$  can be considered a subring of  $M_{n \times n}(F)$ . The columns of  $A$  in this way become elements of  $F^n$ . Any nonzero  $F$ -linear combination of the columns of  $A$  which is zero in  $F^n$  gives, clearing denominators, a nonzero  $R$ -linear dependence relation. Thus, the columns of  $A$  are independent vectors in  $F^n$ . Since  $A$  has  $n$  columns, these form a basis of  $F^n$ , so there are elements  $\beta_{i,j}$  of  $F$  such that

$$e_i = \beta_{1,i} A_1 + \beta_{2,i} A_2 + \cdots + \beta_{n,i} A_n.$$

The  $n \times n$  identity matrix has columns  $e_1, e_2, \dots, e_n$ . By Proposition 3.2.43 with  $\phi = \det$ , the determinant of the identity matrix is some  $F$ -multiple of  $\det A$ . Since  $\det I = 1$ ,  $\det A \neq 0$ . ■

**Theorem 3.2.49.** For matrices  $A, B \in M_{n \times n}(R)$ ,  $\det AB = (\det A)(\det B)$ .

*Proof.* Let  $B = (\beta_{i,j})$  and let  $A_1, A_2, \dots, A_n$  be the columns of  $A$ . Then  $C = AB$  has  $j$ th column  $C_j = \beta_{1,j}A_1 + \beta_{2,j}A_2 + \dots + \beta_{n,j}A_n$ . Now,

$$\det C = \det(C_1, \dots, C_n) = \left[ \sum_{\sigma \in S_n} \epsilon(\sigma) \beta_{\sigma(1),1} \beta_{\sigma(2),2} \cdots \beta_{\sigma(n),n} \right] \det(A_1, \dots, A_n) = (\det A)(\det B).$$

■

**Definition 3.2.50.** Let  $A = (\alpha_{i,j})$  be an  $n \times n$  matrix. For each  $i, j$ , let  $A_{i,j}$  be the  $(n-1) \times (n-1)$  matrix obtained from  $A$  by deleting its  $i$ th row and  $j$ th column. Then  $(-1)^{i+j} \det(A_{i,j})$  is called the  $ij$  cofactor of  $A$ .

**Theorem 3.2.51** (Cofactor Expansion Formula Along the  $i$ th Row). *Let  $A = (\alpha_{i,j})$  be an  $n \times n$  matrix, and fix  $1 \leq i \leq n$ . Then,*

$$\det A = (-1)^{i+1} \alpha_{i,1} \det A_{i,1} + (-1)^{i+2} \alpha_{i,2} \det A_{i,2} + \cdots + (-1)^{i+n} \alpha_{i,n} \det A_{i,n}.$$

*Proof.* For each  $A$ , let  $D(A)$  denote the element of  $R$  obtained from the cofactor expansion formula above. We show that  $D$  satisfies the axioms of a determinant function, and so is the determinant function. We use induction on  $n$ . If  $n = 1$ ,  $D((\alpha)) = \alpha$  for all  $1 \times 1$  matrices  $(\alpha)$ , and the result holds. Now assume  $n \geq 2$ . Fix an index  $k$ , and let the  $k$ th column vary while all other columns are fixed. If  $j \neq k$ ,  $\alpha_{i,j}$  does not depend on  $k$  and  $D(A_{i,j})$  is linear in the  $k$ th column by induction. Now, as the  $k$ th column varies linearly, so does  $\alpha_{i,k}$ , while  $D(A_{i,k})$  is constant. Thus, each term in the formula for  $D$  varies linearly in the  $k$ th column.

Now, assume that columns  $k$  and  $k+1$  of  $A$  are equal. If  $j \neq k, k+1$ , the two equal columns of  $A$  become two equal columns of the matrix  $A_{i,j}$ . By induction,  $D(A_{i,j}) = 0$ . The matrices  $A_{i,k}$  and  $A_{i,k+1}$  are identical, and  $\alpha_{i,k} = \alpha_{i,k+1}$ . Thus,  $D(A) = (-1)^{i+k} \alpha_{i,k} D(A_{i,k}) + (-1)^{i+k+1} \alpha_{i,k+1} D(A_{i,k+1}) = 0$ , so  $D$  is alternating.

It follows easily by induction that  $D(I) = 1$ . ■

**Theorem 3.2.52** (Cofactor Formula for the Inverse of a Matrix). *Let  $A = (\alpha_{i,j})$  be an  $n \times n$  matrix, and let  $B$  be the transpose of its matrix of cofactors;  $B = (\beta_{i,j})$ , where  $\beta_{i,j} = (-1)^{j+i} \det A_{j,i}$ . Then  $AB = BA = (\det A)I$ . Moreover,  $\det A$  is a unit in  $R$  if and only if  $A$  is a unit in  $M_{n \times n}(R)$ , and in this case the matrix  $(\det A)^{-1}B$  is the inverse of  $A$ .*

*Proof.* The  $i, j$  entry of  $AB$  is  $\alpha_{i,1}\beta_{1,j} + \alpha_{i,2}\beta_{2,j} + \cdots + \alpha_{i,n}\beta_{n,j}$ . By definition, this is

$$\alpha_{i,1}(-1)^{j+1} D(A_{j,1}) + \alpha_{i,2}(-1)^{j+2} D(A_{j,2}) + \cdots + \alpha_{i,n}(-1)^{j+n} D(A_{j,n}).$$

If  $i = j$ , this is the cofactor expansion for  $\det A$  along the  $i$ th row. If  $i \neq j$ , let  $\bar{A}$  be the matrix  $A$  with the  $j$ th row replaced by the  $i$ th row, so  $\det \bar{A} = 0$ . Then  $\bar{A}_{j,k} = A_{j,k}$  and  $\alpha_{i,k} = \bar{\alpha}_{j,k}$  for every  $1 \leq k \leq n$ . Plugging this in, the  $i, j$  entry in  $AB$  is

$$\bar{\alpha}_{j,1}(-1)^{1+j} D(\bar{A}_{j,1}) + \bar{\alpha}_{j,2}(-1)^{2+j} D(\bar{A}_{j,2}) + \cdots + \bar{\alpha}_{j,n}(-1)^{n+j} D(\bar{A}_{j,n}),$$

which is the cofactor expansion for  $\det \bar{A}$  along the  $j$ th row, and so 0.



By the definition of  $B$ , the pair  $(A^t, B^t)$  satisfies the same hypothesis as the pair  $(A, B)$ . Thus,  $(BA)^t = A^t B^t = (\det A^t)I = (\det A)I$ . Hence,  $BA = ((\det A)I)^t = (\det A)I$ .

Suppose that  $d = \det A$  is a unit in  $R$ . Then  $d^{-1}B$  is a matrix with entries in  $R$  whose product with  $A$  (on either side) is the identity, so that  $A$  is a unit in  $M_{n \times n}(R)$ . Conversely, assume that  $A$  is a unit in  $R$  with (2-sided) inverse  $C$ . Since  $\det C \in R$  and

$$1 = \det A = \det AC = (\det A)(\det C) = (\det C)(\det A),$$

$\det A$  has a 2-sided inverse in  $R$ , as needed. ■

### 3.2.5 Tensor, Symmetric and Exterior Algebras

*Remark 3.2.53.* In this subsection,  $R$  is a commutative ring with 1, and the left and right actions of  $R$  on each  $R$ -module are the same.

**Definition 3.2.54.** Let  $M$  be an  $R$ -module. Define  $\mathcal{T}^0(M) = R$  and  $\mathcal{T}^k(M) = \mathcal{T}^{k-1}(M) \otimes_R M$  for  $k \geq 1$ , identifying  $M$  with  $\mathcal{T}^1(M)$ . The elements of  $\mathcal{T}^k(M)$  are called  $k$ -tensors. Define

$$\mathcal{T}(M) = \mathcal{T}^0(M) \oplus \mathcal{T}^1(M) \oplus \mathcal{T}^2(M) \oplus \mathcal{T}^3(M) \oplus \cdots = \bigoplus_{k=0}^{\infty} \mathcal{T}^k(M).$$

Every element of  $\mathcal{T}(M)$  is a finite linear combination of  $k$  tensors for various  $k \geq 0$ .

**Theorem 3.2.55.** *Let  $M$  be any  $R$ -module.*

(1)  $\mathcal{T}(M)$  is an  $R$ -algebra containing  $M$  with multiplication defined by mapping

$$(m_1 \otimes \cdots \otimes m_i)(m'_1 \otimes \cdots \otimes m'_j) = m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j$$

and extended to sums via the distributive laws. With respect to this multiplication,  $\mathcal{T}^i(M)\mathcal{T}^j(M) \subseteq \mathcal{T}^{i+j}(M)$ .

(2) If  $A$  is any  $R$ -algebra and  $\phi : M \rightarrow A$  is an  $R$ -module homomorphism, then there is a unique  $R$ -algebra homomorphism  $\Phi : \mathcal{T}(M) \rightarrow A$  such that  $\Phi|_M = \phi$ .

*Proof.*

(1) The map

$$\underbrace{M \times \cdots \times M}_{i \text{ times}} \times \underbrace{M \times \cdots \times M}_{j \text{ times}} \rightarrow \mathcal{T}^{i+j}(M)$$

defined by

$$(m_1, \dots, m_i, m'_1, \dots, m'_j) \mapsto m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j$$

is  $R$ -multilinear, so there is an induced bilinear map from  $\mathcal{T}^i(M) \times \mathcal{T}^j(M)$  to  $\mathcal{T}^{i+j}(M)$ . This gives a well defined multiplication satisfying (1).

(2) Suppose that  $\phi : M \rightarrow A$  is an  $R$ -module homomorphism. Then,

$$(m_1, m_2, \dots, m_k) \mapsto \phi(m_1)\phi(m_2) \cdots \phi(m_k)$$

is an  $R$ -multilinear map from  $M \times \cdots \times M$  to  $A$ . Thus, there is a unique  $R$ -module homomorphism  $\Phi$  from  $\mathcal{T}^k(M)$  to  $A$  mapping  $m_1 \otimes \cdots \otimes m_k$  to  $\phi(m_1) \cdots \phi(m_k)$ . By the definition of multiplication above, this uniquely defined map is an  $R$ -algebra homomorphism. ■

**Definition 3.2.56.** The ring  $\mathcal{T}(M)$  is called the tensor algebra of  $M$ .

**Proposition 3.2.57.** Let  $V$  be a finite dimensional vector space over the field  $F$  with basis  $\mathcal{B} = \{v_1, \dots, v_n\}$ . Then the  $k$ -tensors

$$v_{i_1} \otimes v_{i_2} \otimes \cdots \otimes v_{i_k} \quad v_{i_j} \in \mathcal{B}$$

are a vector space basis of  $\mathcal{T}^k(V)$  over  $F$  (with the understanding that the basis vector is  $1 \in F$  when  $k = 0$ ). In particular,  $\dim_F(\mathcal{T}^k(V)) = n^k$ .

*Proof.*  $\mathcal{T}^k(V)$  is just the tensor product of  $V$  with itself  $k$  times. ■

**Definition 3.2.58.**

- (1) A ring  $S$  is said to be a graded ring if it is the direct sum of additive subgroups:  $S = S_0 \oplus S_1 \oplus S_2 \oplus \cdots$  such that  $S_i S_j \subseteq S_{i+j}$  for all  $i, j \geq 0$ . The elements of  $S_k$  are said to be homogeneous of degree  $k$ , and  $S_k$  is called the homogeneous component of  $S$  of degree  $k$ .
- (2) An ideal  $I$  of the graded ring  $S$  is called a graded ideal if  $I = \bigoplus_{k=0}^{\infty} (I \cap S_k)$ .
- (3) A ring homomorphism  $\phi : S \rightarrow T$  between two graded rings is called a homomorphism of graded rings if it respects the grading structures of  $S$  and  $T$  (i.e. if  $\phi(S_k) \subseteq T_k$  for  $k \geq 0$ ).

*Remark 3.2.59.* We have that  $S_0 S_0 \subseteq S_0$ , so  $S_0$  is a subring of  $S$ . Thus,  $S$  is an  $S_0$ -module. If  $S_0$  is in the center of  $S$  and contains an identity of  $S$ , then  $S$  is an  $S_0$ -algebra.

*Remark 3.2.60.* An ideal  $I$  is graded if, whenever  $i_{k_1} + \cdots + i_{k_n} \in I$  is a sum of homogeneous elements with distinct degrees  $k_1, \dots, k_n$ , each of the  $i_{k_j}$  is in  $I$ .

*Example 3.2.61.* The polynomial ring  $R[x_1, x_2, \dots, x_n]$  in  $n$  variables over the commutative ring  $R$  is a graded ring. Here,  $S_0 = R$ , and  $S_k$  contains all  $R$ -linear combinations of monomials of degree  $k$ .

**Proposition 3.2.62.** Let  $S$  be a graded ring,  $I$  be a graded ideal in  $S$ , and let  $I_k = I \cap S_k$ . Then  $S/I$  is a naturally graded ring whose homogeneous component of degree  $k$  is isomorphic to  $S_k/I_k$ .

*Proof.* The map

$$S = \bigoplus_{k=0}^{\infty} S_k \rightarrow \bigoplus_{k=0}^{\infty} (S_k/I_k)$$

$$(\dots, s_k, \dots) \mapsto (\dots, s_k \bmod I_k, \dots)$$

is a surjective map, and has kernel  $I = \bigoplus_{k=0}^{\infty} I_k$ . Moreover, this is a homomorphism of graded rings, and the induced isomorphism is an isomorphism of graded rings. ■

**Definition 3.2.63.** The symmetric algebra of an  $R$ -module  $M$  is the  $R$ -algebra obtained by taking the quotient of the tensor algebra  $\mathcal{T}(M)$  by the ideal  $\mathcal{C}(M)$  generated by all elements of the form  $m_1 \otimes m_2 - m_2 \otimes m_1$  for  $m_1, m_2 \in M$ . The symmetric algebra  $\mathcal{T}(M)/\mathcal{C}(M)$  is denoted  $\mathcal{S}(M)$ .

*Remark 3.2.64.*  $\mathcal{T}(M)$  is generated as a ring by  $R = \mathcal{T}^0(M)$  and  $M = \mathcal{T}^1(M)$ , and by definition these commute in  $\mathcal{S}(M)$ . Thus,  $\mathcal{S}(M)$  is commutative. The ideal  $\mathcal{C}(M)$  is generated by homogeneous tensors of degree 2, and so it follows that  $\mathcal{C}(M)$  is a graded ideal. Thus,  $\mathcal{S}(M)$  is a graded ring whose homogeneous component of degree  $k$  is  $\mathcal{S}^k(M) = \mathcal{T}^k(M)/\mathcal{C}^k(M)$ . Since  $\mathcal{C}(M) \cap M = 0$ , the image of  $M = \mathcal{T}^1(M)$  in  $\mathcal{S}(M)$  is isomorphic to  $M$ . Identifying this with  $M$ , we have that  $\mathcal{S}^1(M) = M$ . Similarly,  $\mathcal{S}^0(M) = R$ , so  $\mathcal{S}(M)$  is an  $R$ -algebra. We call  $\mathcal{S}^k(M)$  the  $k$ th symmetric power of  $M$ .

**Theorem 3.2.65.** Let  $M$  be an  $R$ -module over the commutative ring  $R$ .

- (1) The  $k$ th symmetric power  $\mathcal{S}^k(M)$  is  $\underbrace{M \otimes \cdots \otimes M}_{k \text{ times}}$  modulo the submodule generated by all elements of the form

$$m_1 \otimes m_2 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes m_{\sigma(2)} \otimes \cdots \otimes m_{\sigma(k)} \quad m_i \in M, \sigma \in S_k.$$

- (2) If  $\phi : M \times \cdots \times M \rightarrow N$  is a symmetric  $k$ -multilinear map over  $R$ , then there is a unique  $R$ -module homomorphism  $\Phi : \mathcal{S}^k(M) \rightarrow N$  such that  $\phi = \Phi \circ \iota$ , where  $\iota$  is the natural map from  $M \times \cdots \times M$  to  $\mathcal{S}^k(M)$  by  $(m_1, \dots, m_k) \mapsto m_1 \otimes \cdots \otimes m_k \bmod \mathcal{C}(M)$ .
- (3) If  $A$  is any commutative  $R$ -algebra and  $\phi : M \rightarrow A$  is an  $R$ -module homomorphism, then there is a unique  $R$ -algebra homomorphism  $\Phi : \mathcal{S}(M) \rightarrow A$  such that  $\Phi|_M = \phi$ .

*Proof.*

- (1) The  $k$ -tensors  $\mathcal{C}^k(M)$  in  $\mathcal{C}(M)$  are finite sums of elements of the form

$$m_1 \otimes \cdots \otimes m_{i-1} \otimes (m_1 \otimes m_{i+1} - m_{i+1} \otimes m_i) \otimes m_{i+2} \otimes \cdots \otimes m_k.$$

This gives a difference of two  $k$ -tensors which are equal except that two entries (in positions  $i$  and  $i+1$ ) has been swapped. Using  $\sigma = (i \ i+1)$ , we have one inclusion. To get the other, every permutation in  $S_k$  can be written as a product of transpositions.

- (2) Let  $\varphi : M \otimes \cdots \otimes M \rightarrow N$  be the unique induced map. Since  $\phi$  is symmetric,  $\mathcal{C}^k(M) \subseteq \ker \varphi$ . Let  $\Phi : \mathcal{S}^k(M) = \mathcal{T}^k(M)/\mathcal{C}^k(M) \rightarrow N$  send  $m_1 \otimes \cdots \otimes m_n \bmod \mathcal{C}(M)$  to  $\varphi(m_1 \otimes \cdots \otimes m_n)$ . This is well defined, and satisfies (2).
- (3) Let  $\varphi : \mathcal{T}(M) \rightarrow A$  be the induced  $R$ -algebra homomorphism. As  $A$  is commutative,  $\mathcal{C}(M) \subseteq \ker \varphi$ . Let  $\Phi : \mathcal{S}(M) \rightarrow A$  send  $m_1 \otimes \cdots \otimes m_n \bmod \mathcal{C}(M)$  to  $\varphi(m_1 \otimes \cdots \otimes m_n)$ . Restricting to  $M$ , this satisfies (3). ■

**Corollary 3.2.66.** *Let  $V$  be an  $n$ -dimensional vector space over  $F$ . Then  $\mathcal{S}(V)$  is isomorphic as a graded  $F$ -algebra to the ring of polynomials in  $n$  variables over  $F$  (i.e. the isomorphism is also a vector space isomorphism from  $\mathcal{S}^k(V)$  onto the space of all homogeneous polynomials of degree  $k$ ). In particular,  $\dim_F(\mathcal{S}^k(V)) = \binom{k+n-1}{n-1}$ .*

*Proof.* Let  $\mathcal{B} = \{v_1, \dots, v_n\}$  be a basis of  $V$ . There is a bijection between a basis of  $\mathcal{T}^k(V)$  and  $\mathcal{B}^k$ . Define two  $k$ -tuples in  $\mathcal{B}^k$  to be equivalent if there is some permutation of the entries of one that gives the other, giving an equivalence relation on  $\mathcal{B}^k$ . Let  $S(\mathcal{B}^k)$  denote the corresponding set of equivalence classes. Any symmetric  $k$ -multilinear function from  $V^k$  to a vector space over  $F$  will be constant on basis tensors corresponding to  $k$ -tuples lying in the same equivalence class, and any function from  $S(\mathcal{B}^k)$  can be uniquely extended to a symmetric  $k$ -multilinear function on  $V^k$ . Thus, the vector space over  $F$  with basis  $S(\mathcal{B}^k)$  satisfies the same property of  $\mathcal{S}^k(V)$  as (2) above, and so is isomorphic to  $\mathcal{S}(V)$ . Each equivalence class has a unique representative of the form  $(v_1^{a_1}, v_2^{a_2}, \dots, v_n^{a_n})$ , where  $v_i^{a_i}$  denotes the sequence  $v_i, \dots, v_i$  taken  $a_i$  times, where each  $a_i \geq 0$  and  $a_1 + \cdots + a_n = k$ . Thus, there is a bijection between  $S(\mathcal{B}^k)$  and the set  $x_1^{a_1} \cdots x_n^{a_n}$  of monic monomials of degree  $k$ . This extends to an isomorphism of graded  $F$ -algebras. ■

**Definition 3.2.67.** The exterior algebra of an  $R$ -module  $M$  is the  $R$ -algebra obtained by taking the quotient of  $\mathcal{T}(M)$  by the ideal  $\mathcal{A}(M)$  generated by all elements of the form  $m \otimes m$ , for  $m \in M$ . The exterior algebra  $\mathcal{T}(M)/\mathcal{A}(M)$  is denoted  $\bigwedge(M)$ , and the image of  $m_1 \otimes m_2 \otimes \cdots \otimes m_k$  in  $\bigwedge(M)$  is denoted by  $m_1 \wedge m_2 \wedge \cdots \wedge m_k$ .

*Remark 3.2.68.* The ideal  $\mathcal{A}(M)$  is generated by homogeneous elements, and is thus a graded ideal. The exterior algebra is graded, and the  $k$ th homogeneous component is  $\bigwedge^k(M) = \mathcal{T}^k(M)/\mathcal{A}^k(M)$ . We can identify  $R$  with  $\bigwedge^0(M)$  and  $M$  with  $\bigwedge^1(M)$ , and so consider  $M$  as a submodule of the  $R$ -algebra  $\bigwedge(M)$ . The  $R$ -module  $\bigwedge^k(M)$  is called the  $k$ th exterior power of  $M$ .

**Definition 3.2.69.** The multiplication

$$(m_1 \wedge \cdots \wedge m_i) \wedge (m'_1 \wedge \cdots \wedge m'_j) = m_1 \wedge \cdots \wedge m_i \wedge m'_1 \wedge \cdots \wedge m'_j$$

in the exterior algebra is called the wedge (or exterior) product. By definition, the product  $m_1 \wedge \cdots \wedge m_k$  is 0 in  $\bigwedge(M)$  if  $m_i = m_{i+1}$  for any  $1 \leq i < k$ .

*Remark 3.2.70.* The wedge product is anticommutative on simple tensors, as

$$0 = (m + m') \wedge (m + m') = (m \wedge m) + (m \wedge m') + (m' \wedge m) + (m' \wedge m') = (m \wedge m') + (m' \wedge m).$$

However, this does not extend to arbitrary products. For example,

$$(m \wedge m') \wedge m'' = m \wedge (-m'' \wedge m') = -(-m'' \wedge m) \wedge m' = m'' \wedge (m \wedge m').$$

**Theorem 3.2.71.** *Let  $M$  be an  $R$ -module over the commutative ring  $R$ .*

- (1)  $\bigwedge^k(M)$  is equal to  $\underbrace{M \otimes \cdots \otimes M}_{k \text{ times}}$  modulo the submodule generated by all elements of the form  $m_1 \otimes \cdots \otimes m_k$ , where  $m_i = m_j$  for some  $i \neq j$ .
- (2) If  $\phi : M \times \cdots \times M \rightarrow N$  is an alternating  $k$ -multilinear map, then there is a unique  $R$ -module homomorphism  $\Phi : \bigwedge^k(M) \rightarrow N$  such that  $\phi = \Phi \circ \iota$ , where  $\iota : M \times \cdots \times M \rightarrow \bigwedge^k(M)$  sends  $(m_1, \dots, m_k)$  to  $m_1 \wedge \cdots \wedge m_k$ .

*Proof.*

- (1) The  $k$ -tensors in  $\mathcal{A}(M)$  are finite sums of elements of the form  $m_1 \otimes \cdots \otimes m_{i-1} \otimes (m \otimes m) \otimes m_{i+2} \otimes \cdots \otimes m_k$ , which is a  $k$ -tensor with two equal entries, and is so in the correct form. For the reverse inclusion, notice that since  $m' \otimes m \equiv -m \otimes m' \pmod{\mathcal{A}(M)}$ , interchanging any two consecutive entries and multiplying by  $-1$  gives an equivalent tensor modulo  $\mathcal{A}(M)$ . Thus, we can arrange for  $m_i$  and  $m_j$  to be adjacent, giving an element of  $\mathcal{A}(M)$ . Thus, the generators are contained in  $\mathcal{A}(M)$ .
- (2) Let  $\varphi : M \otimes \cdots \otimes M \rightarrow N$  be the unique induced map. Since  $\phi$  is alternating,  $\mathcal{A}(M) \subseteq \ker \varphi$ . Let  $\Phi : \bigwedge^k(M) \rightarrow N$  be the corresponding map. Then  $\Phi(m_1 \wedge \cdots \wedge m_k) = \phi(m_1, \dots, m_k)$ , and  $\Phi$  is well defined.

■

**Corollary 3.2.72.** *Let  $V$  be a finite dimensional vector space over  $F$  with basis  $\mathcal{B} = \{v_1, \dots, v_n\}$ . Then the vectors*

$$v_{i_1} \wedge v_{i_2} \wedge \cdots \wedge v_{i_k} \quad 1 \leq i_1 < i_2 < \cdots < i_k \leq n$$

*are a basis of  $\bigwedge^k(V)$ , and  $\bigwedge^k(V) = 0$  when  $k > n$  (when  $k = 0$ , the basis vector is  $1 \in F$ ). In particular,  $\dim_F(\bigwedge^k(V)) = \binom{n}{k}$ .*

*Proof.* We have seen that, modulo  $\mathcal{A}^k(M)$ , the order of terms in any simple  $k$ -tensor can be rearranged up to a sign change. Thus, the  $k$ -tensors above are generators for  $\bigwedge^k(V)$ , so we only have to show that they are linearly independent. It suffices to find an alternating  $k$ -multilinear function from  $V^k$  to  $F$  which is 1 on a given  $v_{i_1} \wedge v_{i_2} \wedge \cdots \wedge v_{i_k}$ , and 0 on all other generators. We define  $f$  on the basis of  $\mathcal{T}^k(V)$  by  $f(v_{j_1} \otimes v_{j_2} \otimes \cdots \otimes v_{j_k}) = \epsilon(\sigma)$ , if  $\sigma$  is the unique permutation of  $j_1, j_2, \dots, j_k$  into  $i_1, i_2, \dots, i_k$ , and  $f$  is zero on every basis tensor whose  $k$ -tuple of indices cannot be permuted to  $i_1, i_2, \dots, i_k$ . Extended to all elements of  $\mathcal{T}^k(V)$ ,  $f$  is alternating, so  $f$  factors through  $\mathcal{A}^k(V)$ . Thus,  $f$  is the desired function. ■

**Definition 3.2.73.** Let  $\phi : M \rightarrow N$  be an  $R$ -module homomorphism. Define  $\mathcal{T}^k(\phi) : \mathcal{T}^k(M) \rightarrow \mathcal{T}^k(N)$  by  $m_1 \otimes \cdots \otimes m_k \mapsto \phi(m_1) \otimes \cdots \otimes \phi(m_k)$ . This sends the generators of  $\mathcal{C}(M)$  and  $\mathcal{A}(M)$  to themselves, so we also have induced  $R$ -module homomorphisms on the quotients:  $\mathcal{S}^k(\phi) : \mathcal{S}^k(M) \rightarrow \mathcal{S}^k(N)$  and  $\bigwedge^k(\phi) : \bigwedge^k(M) \rightarrow \bigwedge^k(N)$ .

**Proposition 3.2.74.** *If  $\phi$  is an endomorphism on a  $n$ -dimensional vector space  $V$ , then  $\bigwedge^n(\phi)(w) = \det(\phi)w$  for all  $w \in \bigwedge^n(V)$ .*

*Proof.* Let  $\{v_1, \dots, v_n\}$  be a basis for  $V$ , so  $v_1 \wedge \dots \wedge v_n$  is a basis of  $\bigwedge^n(V)$ . Let

$$\bigwedge^n(\phi)(v_1 \wedge \dots \wedge v_n) = \phi(v_1) \wedge \dots \wedge \phi(v_n) = D(\phi)v_1 \wedge \dots \wedge v_n,$$

for some  $D(\phi) \in F$ . Let  $A$  be the associated matrix, and let  $D(A) = D(\phi)$ . Then  $D(A)$  satisfies the properties of the determinant function, and so is equal to  $\det A = \det \phi$ . ■

*Remark 3.2.75.* For any  $R$ -module  $M$ , there is a natural left group action of  $S_k$  on  $M \times \dots \times M$  by

$$\sigma(m_1, m_2, \dots, m_k) = (m_{\sigma^{-1}(1)}, m_{\sigma^{-1}(2)}, \dots, m_{\sigma^{-1}(k)}).$$

This is clearly  $R$ -multilinear, so there is a well defined  $R$ -linear left group action of  $S_k$  on  $\mathcal{T}^k(M)$ , defined on simple tensors by

$$\sigma(m_1 \otimes m_2 \otimes \dots \otimes m_k) = m_{\sigma^{-1}(1)} \otimes m_{\sigma^{-1}(2)} \otimes \dots \otimes m_{\sigma^{-1}(k)}.$$

**Definition 3.2.76.** An element  $z \in \mathcal{T}^k(M)$  is called a symmetric  $k$ -tensor if  $\sigma z = z$  for all  $\sigma$  in  $S_k$ . It is called an alternating  $k$ -tensor if  $\sigma z = \epsilon(\sigma)z$  for all  $\sigma \in S_k$ .

**Proposition 3.2.77.** *Let  $\sigma \in S_k$ . For every  $w \in \mathcal{S}^k(M)$ ,  $\sigma w = w$ , and for every  $w \in \bigwedge^k(M)$ ,  $\sigma w = \epsilon(\sigma)w$ .*

*Proof.* Both of these have already been shown to hold for simple products. Since both sides (in either statement) are  $R$ -linear in  $w$ , they hold for all  $w$ . ■

**Definition 3.2.78.** Let  $z \in \mathcal{T}^k(M)$ . Define

$$\begin{aligned} \text{Sym}(z) &= \sum_{\sigma \in S_k} \sigma z \\ \text{Alt}(z) &= \sum_{\sigma \in S_k} \epsilon(\sigma) \sigma z. \end{aligned}$$

For any  $z$ ,  $\text{Sym}(z)$  is symmetric and  $\text{Alt}(z)$  is alternating. These are sometimes called the symmetrization of  $z$  and the skew-symmetrization of  $z$  respectively.

*Remark 3.2.79.* If  $z$  is a symmetric (alternating) tensor,  $\text{Sym}(z)$  ( $\text{Alt}(z)$ ) is just  $k!z$ , where  $k! \in R$  denotes the sum of  $k!$  1s. Thus,  $\text{Sym}$  ( $\text{Alt}$ ) is an  $R$ -module endomorphism of  $\mathcal{T}^k(M)$  whose image lies in the submodule of symmetric (alternating) tensors. In general, these are not surjective, but if  $k!$  is a unit in  $R$ , then  $\text{Sym}(z)/k! = z$  ( $\text{Alt}(z)/k! = z$ ) for any symmetric (alternating) tensor  $z$ . Thus, these give surjective  $R$ -module homomorphisms from  $\mathcal{T}^k(M)$  to the submodule of symmetric (alternating) tensors.

**Proposition 3.2.80.** *Suppose that  $k!$  is a unit in  $R$ , and  $M$  is an  $R$ -module.*

- (1) *The map  $\text{Sym}/k!$  induces an  $R$ -module isomorphism between the  $k$ th symmetric power of  $M$  and the  $R$ -submodule of symmetric  $k$ -tensors.*

- (2) The map  $\text{Alt}/k!$  induces an  $R$ -module isomorphism between the  $k$ th exterior power of  $M$  and the  $R$ -submodule of alternating  $k$ -tensors.

*Proof.*

- (1) We have seen that this is surjective, so we only need to show that its kernel is  $\mathcal{C}^k(M)$ . Clearly,  $\text{Sym}$  is 0 on any difference of two  $k$ -tensors which differ only in the order of their factors, so  $\mathcal{C}^k$  is contained in the kernel. Now,

$$z - \frac{1}{k!} \text{Sym}(z) = \frac{1}{k!} \sum_{\sigma \in S_k} (z - \sigma z)$$

for any  $k$ -tensor  $z$ . If  $z \in \ker \text{Sym}$ , then the left hand side is  $z$ , and the right hand side is an element of  $\mathcal{C}^k(M)$ , as  $z - \sigma z \in \mathcal{C}^k(M)$  for each  $\sigma$ .

- (2) We only need to show that the kernel of this map is  $\mathcal{A}^k(M)$ . Clearly,  $\text{Alt}$  is 0 on any element of  $\mathcal{A}^k(M)$ , so  $\mathcal{A}^k(M)$  is contained in the kernel. Now,

$$z - \frac{1}{k!} \text{Alt}(z) = \frac{1}{k!} \sum_{\sigma \in S_k} (z - \epsilon(\sigma)\sigma z)$$

for any  $k$ -tensor  $z$ . If  $z$  is in the kernel of  $\text{Alt}$ , the left hand side is  $z$ , and the right hand side is an element of  $\mathcal{A}^k(M)$ , as  $z - \epsilon(\sigma)\sigma z \in \mathcal{A}^k(M)$  for each  $\sigma$  (this is verifiable for transpositions, and splitting permutations into transpositions verifies this for the remaining elements of  $S_k$ ).

■

## 3.3 Modules over Principal Ideal Domains

### 3.3.1 The Basic Theory

**Definition 3.3.1.**

- (1) The left  $R$ -module  $M$  is said to be a Noetherian  $R$ -module, or to satisfy the ascending chain condition on submodules (ACC on submodules) if there are no infinite increasing chains of submodules. In other words, whenever  $M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$  is an increasing chain of submodules of  $M$ , there is some  $m$  such that, for all  $k \geq m$ ,  $M_k = M_m$ .
- (2) The ring  $R$  is said to be Noetherian if it is Noetherian as a left module over itself.

**Theorem 3.3.2.** *Let  $R$  be a ring and  $M$  be a left  $R$ -module. Then the following are equivalent.*

- (1)  $M$  is a Noetherian  $R$ -module.
- (2) Every nonempty set of submodules of  $M$  contains a maximal element under inclusion.
- (3) Every submodule of  $M$  is finitely generated.

*Proof.* Suppose (1), and let  $\Sigma$  be any nonempty collection of submodules of  $M$ . Pick  $M_1 \in \Sigma$ . If  $M_1$  is maximal, we are done. If not, pick  $M_2 \in \Sigma$  with  $M_1 \subsetneq M_2$ . If  $M_2$  is maximal, we are done, and if not pick  $M_3$  in  $\Sigma$  properly containing  $M_2$ . If (2) fails, we can in this way construct an infinite strictly increasing chain of elements of  $\Sigma$ .

Suppose (2), and let  $N$  be a submodule of  $M$ . Let  $\Sigma$  be the collection of all finitely generated submodules of  $N$ . Since  $\{0\} \in \Sigma$ , it is nonempty. Thus,  $\Sigma$  contains a maximal element  $N'$ . If  $N' \neq N$ , let  $x \in N \setminus N'$ . Since  $N'$  is finitely generated, the submodule generated by  $N'$  and  $x$  is finitely generated, contradicting the maximality of  $N'$ . Thus,  $N = N'$  is finitely generated.

Suppose (3), and let  $M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$  be a chain of submodules of  $M$ . Let  $N = \bigcup_{i=1}^{\infty} M_i$ . Note that  $N$  is a submodule of  $M$ .  $N$  is finitely generated, say by  $a_1, a_2, \dots, a_n$ . Since  $a_i \in N$  for all  $i$ , each  $a_i$  is in one of the submodules, say  $M_{j_i}$ . Let  $m = \max\{j_1, j_2, \dots, j_n\}$ . Then  $a_i \in M_m$  for all  $i$ , so  $N \subseteq M_m$ . Thus,  $M_m = N = M_k$  for all  $k \geq m$ . ■

**Corollary 3.3.3.** *If  $R$  is a PID, every nonempty set of ideals of  $R$  has a maximal element and  $R$  is a Noetherian ring.*

*Proof.* Every submodule of  $R$  is finitely generated. ■

**Proposition 3.3.4.** *Let  $R$  be an integral domain and let  $M$  be a free  $R$ -module of rank  $n < \infty$ . Then any  $n + 1$  elements of  $M$  are  $R$ -linearly dependent.*

*Proof.* Embed  $R$  in its quotient field  $F$ . Since  $M \cong R \oplus R \oplus \cdots \oplus R$ ,  $M \subseteq F \oplus F \oplus \cdots \oplus F$ . The latter is an  $n$ -dimensional vector space over  $F$ , so any  $n + 1$  elements of  $M$  are  $F$ -linearly dependent. Clearing denominators, any  $n + 1$  elements of  $M$  are  $R$ -linearly dependent. ■

**Definition 3.3.5.** For any integral domain  $R$ , the rank of an  $R$ -module  $M$  is the maximum number of  $R$ -linearly independent elements of  $M$ .

**Definition 3.3.6.** Let  $R$  be an integral domain and  $M$  be an  $R$ -module.

- (1) The torsion submodule of  $M$  is

$$\text{Tor}(M) = \{x \in M : \exists r \neq 0 \in R, rx = 0\}.$$

If  $N$  is a submodule of  $\text{Tor}(M)$ ,  $N$  is called a torsion submodule of  $M$ . If  $\text{Tor}(M) = 0$ ,  $M$  is said to be torsion free.

- (2) If  $N$  is a submodule of  $M$ , the annihilator of  $N$  is the ideal of  $R$  given by

$$\text{Ann}(N) = \{r \in R : \forall n \in N, rn = 0\}.$$

If  $N$  is not a torsion submodule of  $M$ , then  $\text{Ann}(N) = (0)$ .

**Theorem 3.3.7.** *Let  $R$  be a PID, let  $M$  be a free  $R$ -module of finite rank  $n$ , and let  $N$  be a submodule of  $M$ .*

- (1)  $N$  is free of rank  $m \leq n$ .



- (2) There exists a basis  $y_1, y_2, \dots, y_n$  of  $M$  such that  $a_1y_1, a_2y_2, \dots, a_my_m$  is a basis of  $N$ , where  $a_1, a_2, \dots, a_m$  are nonzero elements of  $R$  with divisibility relations  $a_1|a_2|\dots|a_m$ .

*Proof.* This is trivial for  $N = \{0\}$ , so suppose  $N \neq \{0\}$ . For each  $R$ -module homomorphism  $\phi : M \rightarrow R$ ,  $\phi(N)$  must be an ideal in  $R$ , say  $\phi(N) = (a_\phi)$ . Let  $\Sigma = \{(a_\phi) : \phi \in \text{Hom}_R(M, R)\}$  be the collection of ideals obtained in this way. This is nonempty, as we can take  $\phi$  to be the trivial homomorphism to get that  $(0) \in \Sigma$ . Thus,  $\Sigma$  has at least one maximal element, so there is at least one homomorphism  $\nu : M \rightarrow R$  such that  $\nu(N) = (a_\nu)$  is not properly contained in any other element of  $\Sigma$ . Let  $a_1 = a_\nu$ , and let  $y \in N$  be an element mapping to  $a_1$  under  $\nu$ .

We first show  $a_1 \neq 0$ . Let  $x_1, x_2, \dots, x_n$  be a basis of  $M$ , and let  $\pi_i \in \text{Hom}_R(M, R)$  be the natural projection into  $x_i$ . Since  $N \neq \{0\}$ , there is some  $i$  such that  $\pi_i(N) \neq 0$ , so  $\Sigma$  contains more than just  $(0)$ . Since  $(a_1)$  is maximal,  $a_1 \neq 0$ .

We now show that  $a_1|\phi(y)$  for each  $\phi \in \text{Hom}_R(M, R)$ . Let  $d$  be a generator for the principal ideal generated by  $a_1$  and  $\phi(y)$ . Then  $d|a_1, \phi(y)$ , and we can write  $d = r_1a_1 + r_2\phi(y)$  for some  $r_1, r_2 \in R$ . Let  $\psi = r_1\nu + r_2\phi$ . Then  $\psi(y) = d$ , so  $d \in \psi(N)$ , and so  $(d) \subseteq \psi(N)$ . Since  $d|a_1$ , we have  $(a_1) \subseteq (d) \subseteq \psi(N)$ , so  $(a_1) = (d) = \psi(N)$ . In particular,  $a_1|\phi(y)$ , as  $d|\phi(y)$ .

In particular, we have that  $a_1|\pi_1(y)$  for all  $i$ . Let  $\pi_i(y) = a_1b_i$ , and define

$$y_1 = \sum_{i=1}^n b_i x_i.$$

Then  $a_1y_1 = y$ . Since  $a_1 = \nu(y) = \nu(a_1y_1) = a_1\nu(y_1)$  and  $a_1 \neq 0$ , we have that  $\nu(y_1) = 1$ .

We now show that  $y_1$  can be taken as an element in a basis for  $M$ , and that  $a_1y_1$  can be taken as an element in a basis for  $N$ . We show that  $M = Ry_1 \oplus \ker \nu$  and  $N = Ra_1y_1 \oplus (N \cap \ker \nu)$ . Let  $x \in M$ , and write  $x = \nu(x)y_1 + (x - \nu(x)y_1)$ . Since  $\nu(x - \nu(x)y_1) = \nu(x) - \nu(x)\nu(y_1) = 0$ , this is an element of  $\ker \nu$ . Thus,  $M = Ry_1 + \ker \nu$ . If  $ry_1 \in \ker \nu$ , then  $0 = \nu(ry_1) = r\nu(y_1) = r$ , so we have a direct sum. Now, notice that  $\nu(x')$  is divisible by  $a_1$  for all  $x' \in N$  by the definition of  $a_1$ . Write  $\nu(x') = ba_1$ . Then,  $x' = \nu(x')y_1 + (x' - \nu(x')y_1) = ba_1y_1 + (x' - ba_1y_1)$ . The second summand is in  $\ker \nu$  and in  $N$ . Thus,  $N = Ra_1y_1 + (N \cap \ker \nu)$ . If  $ra_1y_1 \in N \cap \ker \nu$ , then  $0 = \nu(ra_1y_1) = ra_1\nu(y_1) = ra_1$ , and  $a_1 \neq 0$  so that  $r = 0$ .

We prove (1) by induction on  $m$ , the rank of  $N$ . If  $m = 0$ ,  $N$  is a torsion module, and so  $N = 0$  since a free module is torsion free. Thus, (1) holds trivially in this case. Suppose  $m > 0$ . Since  $N = Ra_1y_1 \oplus (N \cap \ker \nu)$  is a direct sum,  $N \cap \ker \nu$  has rank  $m - 1$ . By induction, it is a free  $R$ -module of rank  $m - 1$ . Adjoining  $a_1y_1$  to a basis of  $N \cap \ker \nu$  gives a basis of  $N$ , so  $N$  is also free.

We prove (2) by induction on  $n$ , the rank of  $M$ . The submodule  $\ker \nu$  is free of rank  $n - 1$ . By induction, there is a basis  $y_2, y_3, \dots, y_n$  of  $\ker \nu$  such that  $a_2y_2, a_3y_3, \dots, a_my_m$  is a basis for  $N \cap \ker \nu$  for some  $a_2, a_3, \dots, a_m \in R$  with  $a_2|a_3|\dots|a_m$ . Now,  $y_1, y_2, \dots, y_n$  is a basis of  $M$ , and  $a_1y_1, a_2y_2, \dots, a_my_m$  is a basis of  $N$ . It remains to show that  $a_1|a_2$ . Define  $\phi : M \rightarrow R$  by  $\phi(y_1) = \phi(y_2) = 1$  and  $\phi(y_i) = 0$  for  $i > 2$ . Then,  $a_1 = \phi(a_1y_1)$ , so  $a_1 \in \phi(N)$ , and thus  $(a_1) \subseteq \phi(N)$ . Since  $(a_1)$  is maximal in  $\Sigma$ ,  $(a_1) = \phi(N)$ . As  $a_2 = \phi(a_2y_2) \in \phi(N)$ , we have that  $a_2 \in (a_1)$ , so  $a_1|a_2$ . ■

**Definition 3.3.8.** The left  $R$ -module  $C$  is said to be cyclic if there is  $x \in C$  with  $C = Rx$ .

*Remark 3.3.9.* If  $C$  is a cyclic  $R$ -module, we can define  $\pi : R \rightarrow C$  by  $\pi(r) = rx$ . This is surjective by definition, so  $R/\ker \pi \cong C$ . If  $R$  is a PID,  $\ker \pi$  is a principal ideal  $(a) = \text{Ann}(C)$ . Thus, the cyclic  $R$ -modules are of the form  $R/(a)$ .

**Theorem 3.3.10.** *Let  $R$  be a PID and  $M$  be a finitely generated  $R$ -module.*

(1)  *$M$  is isomorphic to the direct sum of finitely many cyclic modules. More precisely,*

$$M \cong R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

*for some  $r \geq 0$  and nonzero elements  $a_1, a_2, \dots, a_m \in R \setminus R^\times$  satisfying  $a_1 | a_2 | \cdots | a_m$ .*

(2)  *$M$  is torsion free if and only if  $M$  is free.*

(3) *In the decomposition in (1),*

$$\text{Tor}(M) \cong R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m).$$

*In particular,  $M$  is a torsion module if and only if  $r = 0$ , and in this case the annihilator of  $M$  is  $(a_m)$ .*

*Proof.* Let  $x_1, x_2, \dots, x_n$  be a set of generators of  $M$  of minimal cardinality. Let  $R^n$  be the free  $R$ -module of rank  $n$  with basis  $b_1, b_2, \dots, b_n$ , and define  $\pi : R^n \rightarrow M$  by  $\pi(b_i) = x_i$  for all  $i$ . This is surjective, so  $R^n/\ker \pi \cong M$ . Pick a basis  $y_1, y_2, \dots, y_n$  of  $R^n$  so that  $a_1 y_1, a_2 y_2, \dots, a_m y_m$  is a basis of  $\ker \pi$  for some  $a_1, a_2, \dots, a_m \in R$  with  $a_1 | a_2 | \cdots | a_m$ . Thus,

$$M \cong R^n / \ker \pi = (Ry_1 \oplus Ry_2 \oplus \cdots \oplus Ry_n) / (Ra_1 y_1 \oplus Ra_2 y_2 \oplus \cdots \oplus Ra_m y_m).$$

Consider the natural surjective homomorphism from  $Ry_1 \oplus Ry_2 \oplus \cdots \oplus Ry_n$  to  $R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}$  sending  $(\alpha_1 y_1, \dots, \alpha_n y_n)$  to  $(\alpha_1 \bmod (a_1), \dots, \alpha_m \bmod (a_m), \alpha_{m+1}, \dots, \alpha_n)$ . The kernel of this map is the set of elements where  $a_i | \alpha_i$  for  $1 \leq i \leq m$ , which is  $Ra_1 y_1 \oplus Ra_2 y_2 \oplus \cdots \oplus Ra_m y_m$ . Thus,

$$M \cong R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}.$$

If  $A$  is a unit in  $R$ ,  $R/(a) = 0$ , so we may remove any of the initial  $a_i$  that are units, giving the decomposition in (1).

Since  $R/(a)$  is a torsion  $R$ -module for any  $a \neq 0$ , (1) immediately implies that  $M$  is a torsion free module if and only if  $M \cong R^r$ . Part (3) is immediate from definitions, as the annihilator of  $R/(a)$  is  $(a)$ . ■

**Definition 3.3.11.** The integer  $r$  is called the free rank or the Betti number of  $M$ , and the elements  $a_1, a_2, \dots, a_m$  (defined up to units) are called the invariant factors of  $M$ .

*Remark 3.3.12.* We have not yet proven that these values are unique, so we should refer to a set of invariant factors of  $M$ , rather than *the* set of invariant factors of  $M$ .

**Theorem 3.3.13.** *Let  $R$  be a PID and  $M$  be a finitely generated  $R$ -module. Then  $M$  is the direct sum of a finite number of cyclic modules whose annihilators are either  $(0)$  or generated by powers of primes in  $R$ . In other words,*

$$M \cong R^r \oplus R/(p_1^{\alpha_1}) \oplus R/(p_2^{\alpha_2}) \oplus \cdots \oplus R/(p_t^{\alpha_t}),$$

where  $r \geq 0$  is an integer and  $p_1^{\alpha_1}, \dots, p_t^{\alpha_t}$  are positive powers of (not necessarily distinct) primes in  $R$ .

*Proof.* Suppose that  $a \in R$  is nonzero. Since  $R$  is a PID, it is a UFD, and we can write  $a = up_1^{\alpha_1}p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ , where the  $p_i$  are distinct primes in  $R$  and  $u$  is a unit. This is unique up to units, so the ideals  $(p_i^{\alpha_i})$  for  $1 \leq i \leq s$  are uniquely defined. For  $i \neq j$ , we have  $(p_i^{\alpha_i}) + (p_j^{\alpha_j}) = R$ , since the sum of the two ideals is generated by a greatest common divisor, which is 1 for distinct primes  $p_i$  and  $p_j$ . The intersection of these ideals is  $(a)$ , since  $a = \text{lcm}(p_1^{\alpha_1}, p_2^{\alpha_2}, \dots, p_s^{\alpha_s})$ . Thus, by the Chinese Remainder Theorem,

$$R/(a) \cong R/(p_1^{\alpha_1}) \oplus R/(p_2^{\alpha_2}) \oplus \cdots \oplus R/(p_s^{\alpha_s})$$

as rings, and also as  $R$ -modules. Applying this to each of the cyclic modules in the previous theorem, we are done. ■

**Definition 3.3.14.** Let  $R$  be a PID and  $M$  be a finitely generated  $R$ -module. The prime powers  $p_1^{\alpha_1}, \dots, p_t^{\alpha_t}$  (defined up to units) are called the elementary divisors of  $M$ .

*Remark 3.3.15.* We have not yet proven that these values are unique, so we should refer to a set of elementary divisors of  $M$ , rather than *the* set of elementary divisors of  $M$ .

**Theorem 3.3.16** (Primary Decomposition Theorem). *Let  $R$  be a PID and  $M$  be a nonzero torsion  $R$ -module (not necessarily finitely generated) with nonzero annihilator  $a$ . Suppose that the factorization of  $a$  into distinct prime powers in  $R$  is  $a = up_1^{\alpha_1}p_2^{\alpha_2} \cdots p_n^{\alpha_n}$ , and let  $N_i = \{x \in M : p_i^{\alpha_i}x = 0\}$  for  $1 \leq i \leq n$ . Then  $N_i$  is a submodule of  $M$  with annihilator  $p_i^{\alpha_i}$ , and is the submodule of  $M$  of all elements annihilated by some power of  $p_i$ . We have  $M = N_1 \oplus N_2 \oplus \cdots \oplus N_n$ .*

*Proof.* We have already shown these in the case where  $M$  is finitely generated. In the general case, clearly  $N_i$  is a submodule of  $M$  annihilator dividing  $p_i^{\alpha_i}$ . Since  $R$  is a PID,  $(p_i^{\alpha_i})$  and  $(p_j^{\alpha_j})$  are comaximal for  $i \neq j$ , so we can easily modify the argument in the proof of the Chinese Remainder Theorem to apply it to modules. Using this, it is easy to see that the annihilator of  $N_i$  is exactly  $p_i^{\alpha_i}$ . ■

**Definition 3.3.17.** The submodule  $N_i$  is called the  $p_i$ -primary component of  $M$ .

**Lemma 3.3.18.** *Let  $R$  be a PID, and let  $p$  be a prime in  $R$ . Let  $F = R/(p)$ .*

(1) *Let  $M = R^r$ . Then  $M/pM \cong F^r$ .*

(2) Let  $M = R/(a)$ , where  $a \neq 0 \in R$ . Then,

$$M/pM \cong \begin{cases} F & p|a \text{ in } R \\ 0 & p \nmid a \text{ in } R \end{cases}.$$

(3) Let  $M = R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_k)$ , where each  $a_i$  is divisible by  $p$ . Then  $M/pM \cong F^k$ .

*Proof.*

- (1) There is a natural map from  $R^r$  to  $(R/(p))^r$  defined by sending  $(\alpha_1, \dots, \alpha_r)$  to  $(\alpha_1 \bmod (p), \dots, \alpha_r \bmod (p))$ . This is surjective, and has a kernel consisting of all  $r$ -tuples whose coordinates are divisible by  $p$ . This is precisely  $pR^r$ , so  $R^r/pR^r \cong (R/(p))^r$ .
- (2) Note that  $p(R/(a))$  is the image of  $(p)$  in the quotient  $R/(a)$ , and thus is  $(p) + (a)/(a)$ . The ideal  $(p) + (a)$  is generated by a greatest common divisor of  $p$  and  $a$ . Thus,  $(p) + (a) = (p)$  if  $p|a$  and  $(p) + (a) = (1) = R$  otherwise. Hence,  $pM = (p)/(a)$  if  $p|a$  and  $pM = R/(a) = M$  if  $p \nmid a$ . Now, if  $p|a$ ,  $M/pM = (R/(a))/(p)/(a) \cong R/(p)$ , and if  $p \nmid a$ ,  $M/pM = M/M = 0$ .
- (3) This follows from (2), using a similar method of splitting the quotient of sums as we used earlier.

■

**Theorem 3.3.19.** *Let  $R$  be a PID.*

- (1) *Two finitely generated  $R$ -modules  $M_1$  and  $M_2$  are isomorphic if and only if they have the same free rank and the same list of invariant factors.*
- (2) *Two finitely generated  $R$ -modules  $M_1$  and  $M_2$  are isomorphic if and only if they have the same free rank and the same list of elementary divisors.*

*Proof.* If  $M_1$  and  $M_2$  have the same free rank and list of invariant factors, or the same free rank and list of elementary divisors, then they are clearly isomorphic.

Suppose that  $M_1$  and  $M_2$  are isomorphic. Any isomorphism between  $M_1$  and  $M_2$  maps the torsion in  $M_1$  to the torsion in  $M_2$ , so we must have  $\text{Tor}(M_1) \cong \text{Tor}(M_2)$ . Then  $R^{r_1} \cong M_1/\text{Tor}(M_1) \cong M_2/\text{Tor}(M_2) \cong R^{r_2}$ , where  $r_1$  is the free rank of  $M_1$  and  $r_2$  is the free rank of  $M_2$ . Let  $p \neq 0$  be prime in  $R$ . Then,  $R^{r_1}/pR^{r_1} \cong R^{r_2}/pR^{r_2}$ . Thus,  $F^{r_1} \cong F^{r_2}$ , where  $F$  is the field  $R/pR$ . We therefore have that  $r_1 = r_2$ .

We now only need to show that  $M_1$  and  $M_2$  have the same lists of invariant factors and elementary divisors. We only need to work with the isomorphic torsion modules  $\text{Tor}(M_1)$  and  $\text{Tor}(M_2)$ , so we will assume that both  $M_1$  and  $M_2$  are torsion  $R$ -modules.

We first show that they have the same elementary divisors. It suffices to show that, for a fixed prime  $p$ , the elementary divisors which are a power of  $p$  are the same for both  $M_1$  and  $M_2$ . If  $M_1 \cong M_2$ , then the  $p$ -primary submodule of  $M_1$  is isomorphic to the  $p$ -primary

submodule of  $M_2$ , as these are the submodules of elements which are annihilated by some power of  $p$ . We have thus reduced to the case of proving that if two modules  $M_1$  and  $M_2$  which have annihilator a power of  $p$  are isomorphic, then they have the same elementary divisors.

We use induction on the power of  $p$  in the annihilator of  $M_1$ . If this is 0, both  $M_1$  and  $M_2$  are 0, so we are done. Otherwise, they have nontrivial elementary divisors. Suppose that the elementary divisors of  $M_1$  are given by  $\underbrace{p, p, \dots, p}_{m \text{ times}}, p^{\alpha_1}, p^{\alpha_2}, \dots, p^{\alpha_s}$ , where

$2 \leq \alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_s$ . Write  $M_1$  as the direct sum of cyclic modules with generators  $x_1, \dots, x_m, x_{m+1}, \dots, x_{m+s}$  with annihilators  $(p), \dots, (p), (p^{\alpha_1}), \dots, (p^{\alpha_s})$  respectively. Then,  $pM_1$  has elementary divisors  $p^{\alpha_1-1}, p^{\alpha_2-1}, \dots, p^{\alpha_s-1}$ , since  $pM_1$  is the direct sum of cyclic modules with generators  $px_1, px_2, \dots, px_m, px_{m+1}, \dots, px_{m+s}$ , whose annihilators are  $(1), \dots, (1), (p^{\alpha_1-1}), \dots, (p^{\alpha_s-1})$ , respectively. Similarly, if the elementary divisors of  $M_2$  are  $\underbrace{p, p, \dots, p}_{n \text{ times}}, p^{\beta_1}, p^{\beta_2}, \dots, p^{\beta_t}$ , where  $2 \leq \beta_1 \leq \beta_2 \leq \dots \leq \beta_t$ , then  $pM_2$  has elementary divisors  $p^{\beta_1-1}, p^{\beta_2-1}, \dots, p^{\beta_t-1}$ . Since  $M_1 \cong M_2$ ,  $pM_1 \cong pM_2$ , and the power of  $p$  in the annihilator of  $pM_1$  is one less than the power of  $p$  in the annihilator of  $M_1$ . By induction,  $s = t$  and  $\alpha_i - 1 = \beta_i - 1$  for  $1 \leq i \leq s = t$ . Now, since  $M_1/pM_1 \cong M_2/pM_2$ , we have that  $F^{m+s} \cong F^{n+t}$  by (3) of the lemma above. Since  $s = t$ ,  $m = n$ .

We now show that  $M_1$  and  $M_2$  have the same invariant factors. Suppose  $a_1|a_2|\dots|a_m$  are the invariant factors for  $M_1$ . We obtain a set of elementary divisors for  $M_1$  by taking the prime power factors of these elements. The divisibility restrictions imply that  $a_m$  is the product of the largest of the prime powers among the elementary divisors,  $a_{m-1}$  is the product of the largest prime powers once those for  $a_m$  have been removed, and so on. If  $b_1|b_2|\dots|b_n$  are invariant factors for  $M_2$ , then we similarly obtain a set of elementary divisors for  $M_2$ . However, the elementary divisors of  $M_1$  and  $M_2$  are the same, and it follows that the same is true of the invariant factors. ■

**Corollary 3.3.20.** *Let  $R$  be a PID, and let  $M$  be a finitely generated  $R$ -module.*

- (1) *The elementary divisors of  $M$  are the prime power factors of the invariant factors of  $M$ .*
- (2) *The largest invariant factor of  $M$  is the product of the largest of the distinct prime powers among the elementary divisors of  $M$ , the next largest invariant factor is the product of the largest of the distinct prime powers among the remaining elementary divisors of  $M$ , and so on.*

**Corollary 3.3.21.** *Theorem 1.5.7.*

*Proof.* Take  $R = \mathbb{Z}$  previously. ■

### 3.3.2 The Rational Canonical Form

*Remark 3.3.22.* Recall that, given a vector space over  $F$   $V$  and a linear transformation  $T : V \rightarrow V$ , we can consider  $V$  as a  $F[x]$ -module, where  $x$  acts as  $T$ . If  $V$  is finite dimensional over  $F$ , then it is finitely generated over  $F$ , and so finitely generated over  $F[x]$ .

A nonzero free  $F[x]$ -module is an infinite dimensional vector space over  $F$ , so if  $V$  is finite dimensional, it must be a torsion  $F[x]$ -module. Thus,  $V$  is isomorphic as an  $F[x]$ -module to the direct sum of cyclic, torsion  $F[x]$ -modules.

**Definition 3.3.23.**

- (1) An element  $\lambda \in F$  is called an eigenvalue of the linear transformation  $T$  if there is some nonzero  $v \in V$  with  $T(v) = \lambda v$ . In this case,  $v$  is called an eigenvector of  $T$  with eigenvalue  $\lambda$ .
- (2) If  $A$  is an  $n \times n$  matrix with coefficients in  $F$ , an element  $\lambda$  is called an eigenvalue of  $A$  with corresponding eigenvector  $v$  if  $v$  is a nonzero  $n \times 1$  column vector with  $Av = \lambda v$ .
- (3) If  $\lambda$  is an eigenvalue of  $T$ , the set  $\{v \in V : Tv = \lambda v\}$  is called the eigenspace of  $T$  corresponding to the eigenvalue  $\lambda$ . Similarly, if  $\lambda$  is an eigenvalue of the  $n \times n$  matrix  $A$ , the set of  $n \times 1$  matrices  $v$  with  $Av = \lambda v$  is called the eigenspace of  $A$  corresponding to the eigenvalue  $\lambda$ .

**Definition 3.3.24.** The determinant of a linear transformation from  $V$  to  $V$  is the determinant of any matrix representing the linear transformation (which does not depend on the choice of basis used).

**Proposition 3.3.25.** *The following are equivalent.*

- (1)  $\lambda$  is an eigenvalue of  $T$ .
- (2)  $\lambda I - T$  is a singular linear transformation.
- (3)  $\det(\lambda I - T) = 0$ .

*Proof.*  $\lambda$  is an eigenvalue of  $T$  with eigenvector  $v$  if and only if  $v \in \ker(\lambda I - T)$ , so (1) and (2) are equivalent. (2) and (3) are equivalent by previous results. ■

**Definition 3.3.26.** Let  $x$  be an indeterminate over  $F$ . The polynomial  $\det(xI - T)$  is called the characteristic polynomial of  $T$ , denoted  $c_T(x)$ . If  $A$  is an  $n \times n$  matrix with coefficients in  $F$ ,  $\det(xI - A)$  is called the characteristic polynomial of  $A$ , denoted  $c_A(x)$ .

*Remark 3.3.27.* Expanding the determinant, the characteristic polynomial is monic of degree  $n = \dim V$ . We have that the set of eigenvalues of  $T$  is precisely the set of roots of the characteristic polynomial of  $T$ . In particular,  $T$  has at most  $n$  distinct eigenvalues.

**Definition 3.3.28.** The unique monic polynomial generating  $\text{Ann}(V)$  in  $F[x]$ , where  $V$  is viewed as an  $F[x]$ -module with  $x$  acting as  $T$ , is called the minimal polynomial of  $T$ , denoted  $m_T(x)$ . Equivalently,  $m_T(x)$  is the unique monic polynomial of smallest degree which is the zero transformation when evaluated at  $T$ . The unique monic polynomial of smallest degree which, when evaluated at the matrix  $A$  is the zero matrix, is called the minimal polynomial of  $A$ , denoted  $m_A(x)$ .

*Remark 3.3.29.* In the invariant factor decomposition of  $V$ , we make the invariant factors unique by requiring that they be monic.

**Proposition 3.3.30.** *The minimal polynomial  $m_T(x)$  is the largest invariant factor  $V$ . All the invariant factors of  $V$  divide  $m_T(x)$ .*

*Proof.* The annihilator of  $V$  is the ideal generated by the largest invariant factor of  $V$ . As we require the invariant factors be monic, this is precisely  $m_T(x)$ . ■

**Definition 3.3.31.** Let  $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1x + b_0$  be any monic polynomial in  $F[x]$ . The companion matrix of  $a(x)$  is

$$\mathcal{C}_{a(x)} = \begin{pmatrix} 0 & 0 & \cdots & \cdots & \cdots & -b_0 \\ 1 & 0 & \cdots & \cdots & \cdots & -b_1 \\ 0 & 1 & \cdots & \cdots & \cdots & -b_2 \\ 0 & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & \cdots & 1 & -b_{k-1} \end{pmatrix}.$$

**Definition 3.3.32.**

- (1) A matrix is said to be in rational canonical form if it is of the form

$$\begin{pmatrix} \mathcal{C}_{a_1(x)} & 0 & \cdots & 0 \\ 0 & \mathcal{C}_{a_2(x)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathcal{C}_{a_m(x)} \end{pmatrix},$$

where  $a_1(x), \dots, a_m(x)$  are monic of degree at least 1, with  $a_1(x)|a_2(x)|\cdots|a_m(x)$ . The polynomials  $a_i(x)$  are called the invariant factors of the matrix.

- (2) A rational canonical form for a linear transformation  $T$  is a matrix representing  $T$  which is in rational canonical form.

**Theorem 3.3.33.** *Any linear transformation  $T : V \rightarrow V$  has a unique rational canonical form.*

*Proof.* Make  $V$  into a  $F[x]$ -module, where  $x$  acts as  $T$ . Write

$$V \cong F[x]/(a_1(x)) \oplus F[x]/(a_2(x)) \oplus \cdots \oplus F[x]/(a_m(x)),$$

where the  $a_i(x)$  are the invariant factors of  $V$ , which we require to be monic. Notice that  $1, \bar{x}, \bar{x}^2, \dots, \bar{x}^{k-1}$  are a basis for  $F[x]/(a(x))$ , where  $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1x + b_0$  is a monic polynomial in  $F[x]$ , and  $\bar{x} = x \pmod{(a(x))}$ . Thus, the linear transformation of multiplication by  $x$  (which corresponds to acting  $T$  on  $V$ ) on this basis sends  $\bar{x}^a$  to  $\bar{x}^{a+1}$  for  $0 \leq a < k-1$ , and sends  $\bar{x}^{k-1}$  to  $\bar{x}^k = -b_0 - b_1\bar{x} = \cdots b_{k-1}\bar{x}^{k-1}$ . Thus, the matrix for multiplication by  $x$  is  $\mathcal{C}_{a(x)}$ . Applying this to each of the cyclic modules on the right hand side, we get a rational canonical form for  $T$ .

Now, notice that this process is reversible. Suppose  $b_1(x), b_2(x), \dots, b_t(x)$  are monic polynomials in  $F[x]$  of degree at least 1 such that  $b_i(x) | b_{i+1}(x)$  for all  $i$  and  $\mathcal{E}$  is a basis for  $V$  such that the matrix of  $T$  with respect to  $\mathcal{E}$  is a block matrix with the companion matrices of the  $b_i(x)$  along the diagonal. Then,  $V$  is a direct sum of  $T$ -stable subspaces  $D_i$ , such that the matrix of  $T$  on  $D_i$  is  $C_{b_i(x)}$ . Let  $\mathcal{E}_i$  be the corresponding ordered basis of  $D_i$ , and let  $e_i$  be the first basis element in  $\mathcal{E}_i$ . Then  $D_i$  is a cyclic  $F[x]$ -module with generator  $e_i$ , and the annihilator of  $D_i$  is  $(b_i(x))$ . Thus,  $V$  decomposes into a direct sum of cyclic  $F[x]$ -modules two ways, which both give lists of invariant factors. Since invariant factors are unique,  $a_i(x)$  and  $b_i(x)$  differ by a unit, and since they are both monic, they are equal for all  $i$ . ■

**Theorem 3.3.34.** *Let  $S$  and  $T$  be linear transformations of  $V$ . Then the following are equivalent.*

- (1)  $S$  and  $T$  are similar linear transformations.
- (2) The  $F[x]$ -modules obtained from  $V$  via  $S$  and via  $T$  are isomorphic.
- (3)  $S$  and  $T$  have the same rational canonical form.

*Proof.* Suppose (1), so that there is a nonsingular linear transformation  $U$  such that  $S = UTU^{-1}$ . The isomorphism  $U : V \rightarrow V$  is an  $F[x]$ -module homomorphism, where  $x$  acts on the first  $V$  via  $T$ , and the second via  $S$ . This is because  $U(xv) = U(Tv) = UT(v) = SU(v) = x(Uv)$ . Thus, this is an  $F[x]$ -module isomorphism of the two modules in (2).

Now suppose (2). Let  $V_1$  be  $V$  made into a  $F[x]$ -module via  $S$ , and  $V_2$  be  $V$  made into a  $F[x]$ -module via  $T$ . Since  $V_1 \cong V_2$ , they have the same list of invariant factors. Thus,  $S$  and  $T$  have the same rational canonical form.

Finally, assume (3). Since  $S$  and  $T$  have the same matrix representation with respect to some choice of (possibly different) bases of  $V$ , they are, up to change of basis, the same linear transformation of  $V$ , and so are similar. ■

**Theorem 3.3.35.** *Any  $n \times n$  matrix  $A$  is similar to a unique matrix in rational canonical form.*

**Definition 3.3.36.** The invariant factors of an  $n \times n$  matrix over a field  $F$  are the invariant factors of its rational canonical form.

**Theorem 3.3.37.** *Let  $A$  and  $B$  be  $n \times n$  matrices over  $F$ . Then  $A$  and  $B$  are similar if and only if  $A$  and  $B$  have the same rational canonical form.*

**Corollary 3.3.38.** *Let  $A$  and  $B$  be two  $n \times n$  matrices over a field  $F$ , and suppose  $F$  is a subfield of the field  $K$ .*

- (1) *The rational canonical form, minimal polynomial, characteristic polynomial, and invariant factors of  $A$  is the same computed over  $K$  and over  $F$ .*
- (2)  *$A$  and  $B$  are similar over  $K$  if and only if they are similar over  $F$ .*

*Proof.*



- (1) Let  $M$  be the rational canonical form of  $A$  when computed over  $F$ . Then,  $M$  is in rational canonical form when viewed over  $K$ , so the uniqueness of rational canonical form implies that  $M$  is also the rational canonical form of  $A$  over  $K$ . In particular, the invariant factors and the minimal polynomial (the largest invariant factor) are the same over  $F$  and  $K$ . By the definition of the determinant, it is clear that the characteristic polynomial of  $A$  only depends on its entries.
- (2) If  $A$  and  $B$  are similar over  $F$ , they are clearly similar over  $K$ . If  $A$  and  $B$  are similar over  $K$ , they have the same rational canonical form over  $K$ . Since this is the same as their rational canonical form over  $F$ , they are similar over  $F$ .

■

**Lemma 3.3.39.** *Let  $a(x) \in F[x]$  be monic.*

- (1) *The characteristic polynomial of  $\mathcal{C}_{a(x)}$  is  $a(x)$ .*
- (2) *If  $M$  is given by*

$$M = \begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_k \end{pmatrix},$$

*then the characteristic polynomial of  $M$  is the product of the characteristic polynomials of  $A_1, A_2, \dots, A_k$ .*

*Proof.*

- (1) We prove this by induction on  $k = \deg a(x)$ . When  $k = 1$ , this is clearly true. Now,

$$\begin{aligned} \det \begin{pmatrix} x & 0 & \cdots & \cdots & \cdots & b_0 \\ -1 & x & \cdots & \cdots & \cdots & b_1 \\ 0 & -1 & \cdots & \cdots & \cdots & b_2 \\ 0 & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & \cdots & -1 & x + b_{k-1} \end{pmatrix} = x \det \begin{pmatrix} x & \cdots & \cdots & \cdots & b_1 \\ -1 & \cdots & \cdots & \cdots & b_2 \\ 0 & \ddots & & & \vdots \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & \cdots & -1 & x + b_{k-1} \end{pmatrix} \\ + (-1)^{k+1} b_0 \det \begin{pmatrix} -1 & x & 0 & \cdots & 0 \\ 0 & -1 & x & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & x \end{pmatrix}. \end{aligned}$$

By induction, the first determinant on the right hand side is  $b_1 + b_2x + \cdots + b_{k-1}x^{k-1}$ . It is easy to see that the second determinant is  $(-1)^{k+1}$ , so we get  $a(x)$ .

- (2) This follows from the corresponding property of determinants (which can be easily proven by induction).

■

**Proposition 3.3.40.** *Let  $A$  be an  $n \times n$  matrix over  $F$ .*

- (1) *The characteristic polynomial of  $A$  is the product of all the invariant factors of  $A$ .*
- (2) *The minimal polynomial of  $A$  divides the characteristic polynomial of  $A$ .*
- (3) *The characteristic polynomial of  $A$  divides some power of the minimal polynomial of  $A$ . In particular, these polynomials have the same roots, not counting multiplicities.*

*The same statements are true if the matrix  $A$  is replaced by a linear transformation  $T$  of an  $n$ -dimensional vector space over  $F$ .*

*Proof.* Let  $B$  be the rational canonical form of  $A$ . Then, the characteristic polynomial of  $B$  is the product of the characteristic polynomials of the companion matrices of the invariant factors of  $A$ . This is just the product of the invariant factors of  $A$ . Since  $A$  and  $B$  are similar, they have the same characteristic polynomial, proving (1). Since the minimal polynomial of  $A$  is the largest invariant factor of  $A$ , (2) is immediate. As all invariant factors divide the largest invariant factor, we get (3). ■

### 3.3.3 The Jordan Canonical Form

**Definition 3.3.41.** The  $k \times k$  matrix

$$\begin{pmatrix} \lambda & 1 & & & \\ & \lambda & \ddots & & \\ & & \ddots & 1 & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

is called the  $k \times k$  elementary Jordan matrix with eigenvalue  $\lambda$  or the Jordan block of size  $k$  with eigenvalue  $\lambda$ .

**Definition 3.3.42.**

- (1) A matrix is said to be in Jordan canonical form if it is a block diagonal matrix with Jordan blocks along the diagonal.
- (2) A Jordan canonical form for a linear transformation  $T$  is a matrix representing  $T$  which is in Jordan canonical form.

**Theorem 3.3.43.** *Let  $T : V \rightarrow V$  be a linear transformation such that  $F$  contains all eigenvalues of  $T$ . Then  $T$  has a Jordan canonical form that is unique up to rearrangement of the Jordan blocks.*

*Proof.* Make  $V$  into an  $F[x]$ -module, where  $x$  acts as  $T$ . As  $F$  contains all eigenvalues of  $T$ , the invariant factors factor completely into linear factors. As the elementary divisors of  $V$  are the prime power divisors of the invariant factors, the elementary divisors are powers  $(x - \lambda)^k$  of linear polynomials, where  $\lambda$  is an eigenvalue of  $T$ . Now,  $V$  is the direct sum of finitely many cyclic  $F[x]$ -modules of the form  $F[x]/(x - \lambda)^k$ . Consider the elements  $(\bar{x} - \lambda)^{k-1}, (\bar{x} - \lambda)^{k-2}, \dots, \bar{x} - \lambda, 1$  in the quotient  $F[x]/(x - \lambda)^k$ . Expanding each of these in  $\bar{x}$ , the matrix relating these to  $\bar{x}^{k-1}, \bar{x}^{k-2}, \dots, \bar{x}, 1$  is upper triangular with 1's along the diagonal. As this is invertible (it has determinant 1), these elements are an  $F$ -basis for  $F[x]/(x - \lambda)^k$ . Multiplying by  $x$  sends  $(\bar{x} - \lambda)^a$  to  $\lambda(\bar{x} - \lambda)^a + (\bar{x} - \lambda)^{a+1}$  for  $1 \leq a < k - 1$ , and sends  $(\bar{x} - \lambda)^{k-1}$  to  $\lambda(\bar{x} - \lambda)^{k-1}$  (this is obtained using the fact that  $x = \lambda + (x - \lambda)$ ). Thus, with respect to this basis, the matrix for multiplication for  $x$  is the Jordan block of size  $k$  with eigenvalue  $\lambda$ . Applying this to each cyclic factor, we get a Jordan canonical form for  $T$ .

This process is reversible. Suppose that we have another Jordan canonical form for  $T$ , with respect to the basis  $\mathcal{E}$ . Then  $V$  is the direct sum of  $T$ -stable subspaces  $D_i$  such that the matrix of  $T$  on  $D_i$  is the corresponding Jordan block. Let  $\mathcal{E}_i$  be the corresponding ordered basis of  $D_i$ , and let  $e_i$  be the last element of  $\mathcal{E}_i$ . Then  $D_i$  is a cyclic  $F[x]$ -module with generator  $e_i$ , and the annihilator of  $D_i$  is  $(x - \lambda)^k$ , where  $k$  is the size of the Jordan block and  $\lambda$  is the eigenvalue. We thus have two ways to decompose  $V$  into a direct sum of cyclic  $F[x]$ -modules, which both give lists of invariant factors. Since these are unique, we only have ambiguity in ordering. ■

**Theorem 3.3.44.** *Let  $A$  be an  $n \times n$  matrix over  $F$  such that  $F$  contains all eigenvalues of  $A$ . Then  $A$  is similar to a matrix in Jordan canonical form that is unique up to rearrangement of the Jordan blocks.*

**Corollary 3.3.45.**

- (1) *If a matrix  $A$  is similar to a diagonal matrix  $D$ , then  $D$  is the Jordan canonical form of  $A$ .*
- (2) *Two diagonal matrices are similar if and only if their diagonal entries are the same up to a permutation.*

*Proof.* As diagonal matrices are in Jordan canonical form, (1) is immediate from the uniqueness of Jordan canonical form. The uniqueness also gives (2). ■

**Corollary 3.3.46.** *If  $A$  is an  $n \times n$  matrix with entries from  $F$  and  $F$  contains all the eigenvalues of  $A$ , then  $A$  is similar to a diagonal matrix over  $F$  if and only if the minimal polynomial of  $A$  has no repeated roots.*

*Proof.* Suppose that  $A$  is similar to a diagonal matrix. The minimal polynomial of a diagonal matrix has no repeated roots (its roots are precisely the distinct elements along the diagonal). Since similar matrices have the same minimal polynomial, the minimal polynomial for  $A$  has no repeated roots.

Suppose that the minimal polynomial for  $A$  has no repeated roots, and let  $B$  be the Jordan canonical form of  $A$ . The matrix  $B$  is a block diagonal matrix with elementary

Jordan matrices along the diagonal. It is easy to see that the minimal polynomial for  $B$  is the least common multiple of the minimal polynomials of the Jordan blocks. Notice that a Jordan block of size  $k$  with eigenvalue  $\lambda$  has minimal polynomial  $(x - \lambda)^k$ . Since  $A$  and  $B$  have the same minimal polynomial, the least common multiple of the  $(x - \lambda)^k$  cannot have any repeated roots, so  $k = 1$ . Thus, each Jordan block is of size 1 and  $B$  is diagonal. ■